

SKG 12 Mechanical Engineering

DOMI of Compressor and Blower (1of7)

Introduction

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Introduction

Program Introduction

- Introduction of self.
- Introduction of participants.
- Introduction of program.

Program Introduction

Objectives :

- To provide participants with specialist knowledge about compressors
 - Develop specification for compressor & associated systems.
 - Review detail design in accordance with project requirement.
 - Perform necessary inspection during equipment fabrication and witness testing.
 - Supervise the installation & commissioning of compressor.
 - Develop and execute maintenance programs of compressor.
 - Perform troubleshooting.

- Methodology :
 - Presentation / facilitation
 - Discussions / Sharing of Experiences
 - Group exercise
 - Q & A

Program Introduction

- Code of Training :
 - Mobile phones in meeting or silent mode
 - Respect other participants, be on time
 - No whispering or meeting within meeting
 - Allowed to go out when feeling restless
 - Physically and mentally present
 - Ask as many questions as possible
 - No question is silly

SKG 12 Mechanical Engineering

DOMI of Compressor and Blower (2of7)

-Type of Compressor

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Sec 2 : Type of compressor

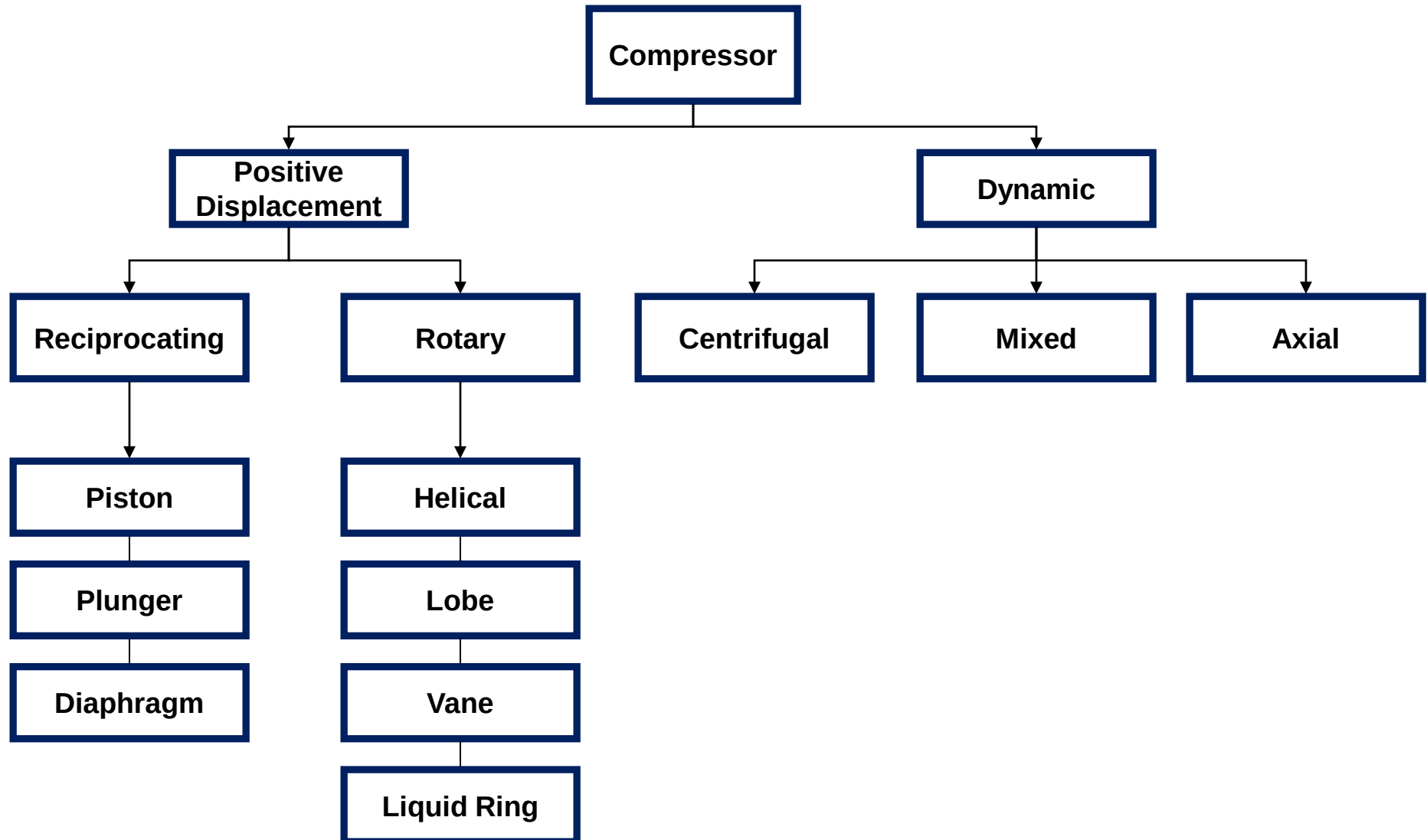
- 2.1 Compressor Design and Selection
- 2.2 Special application

Overview of Gas Compressor Machinery

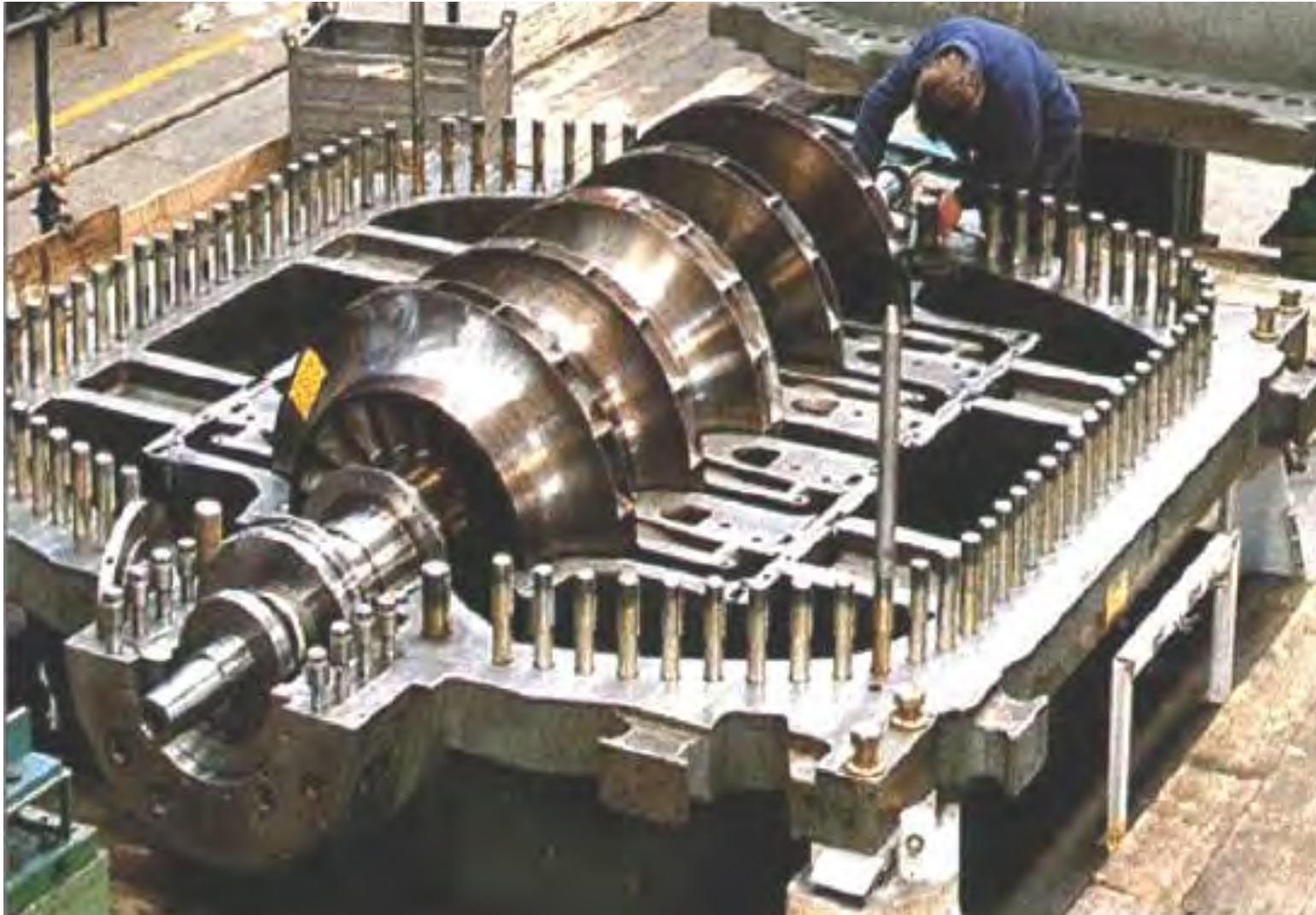
- What is compressor
 - A machinery or equipment used to raise gas or air pressures from one level to another for purpose of transportation or process demands are called compressors.

- What is Fan
 - Essentially belonging to the same group & working on same principle. Fans & blowers are employed for high flow & low pressures.
 - When discharge pressures are lower than 0.3 Kg/cm²(g) it is referred as blower or Fans.

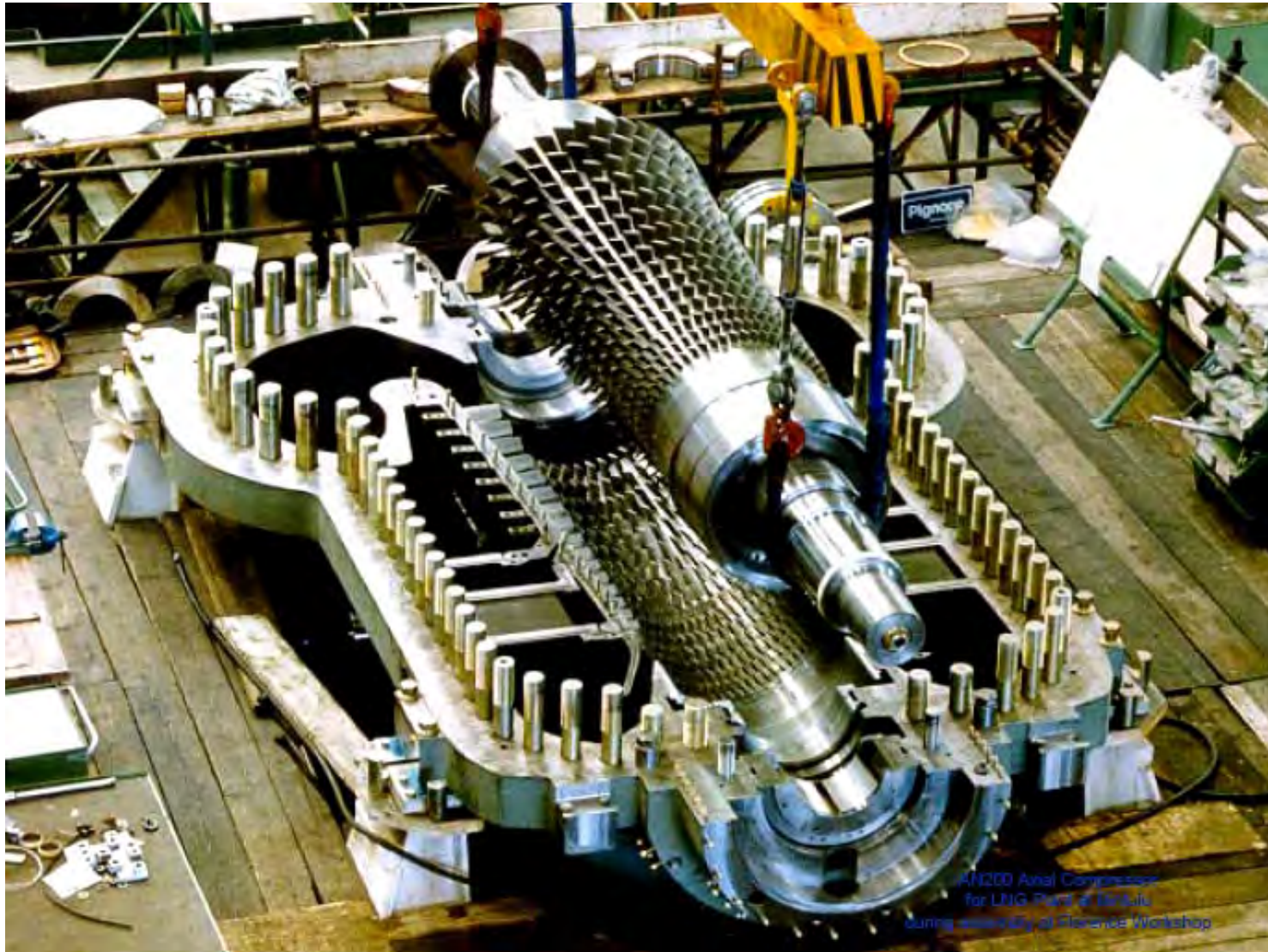
Compressor Classification



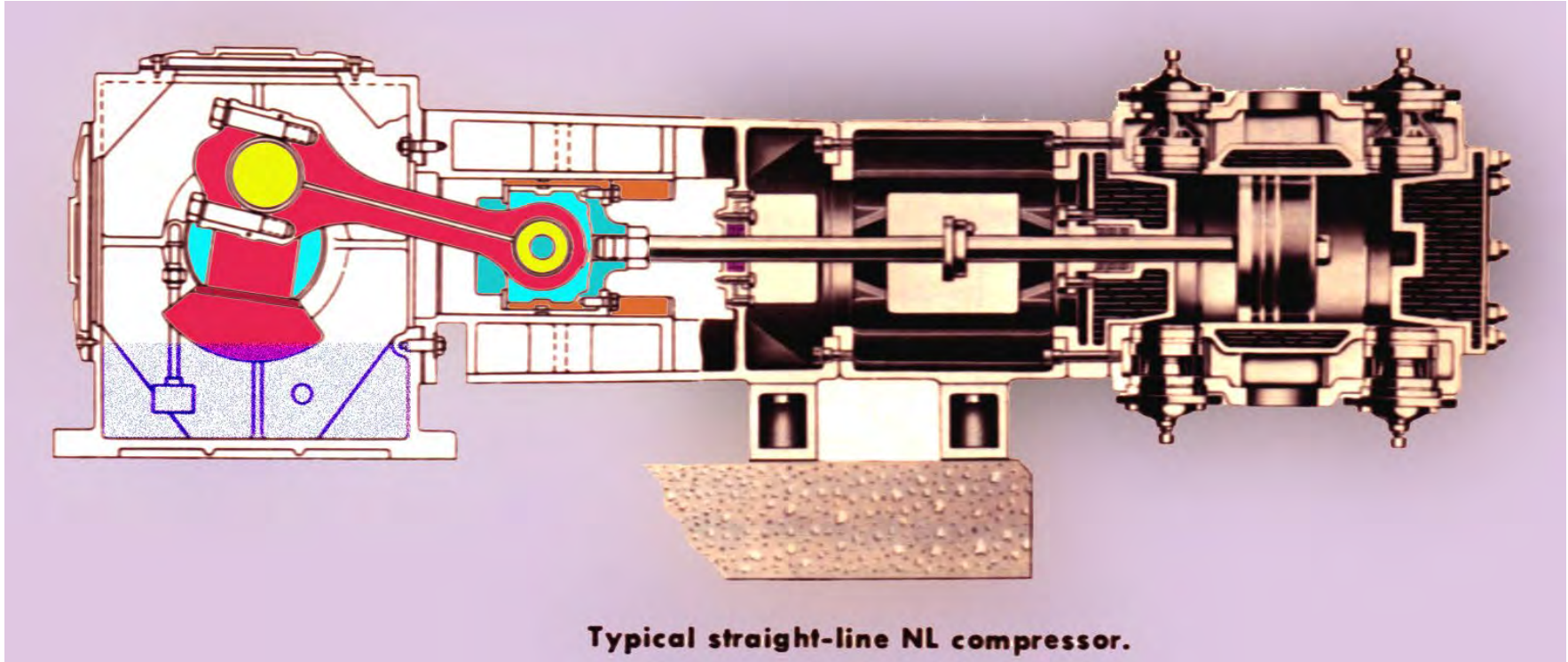
Centrifugal Compressor (API617)



Axial Compressor (API617)

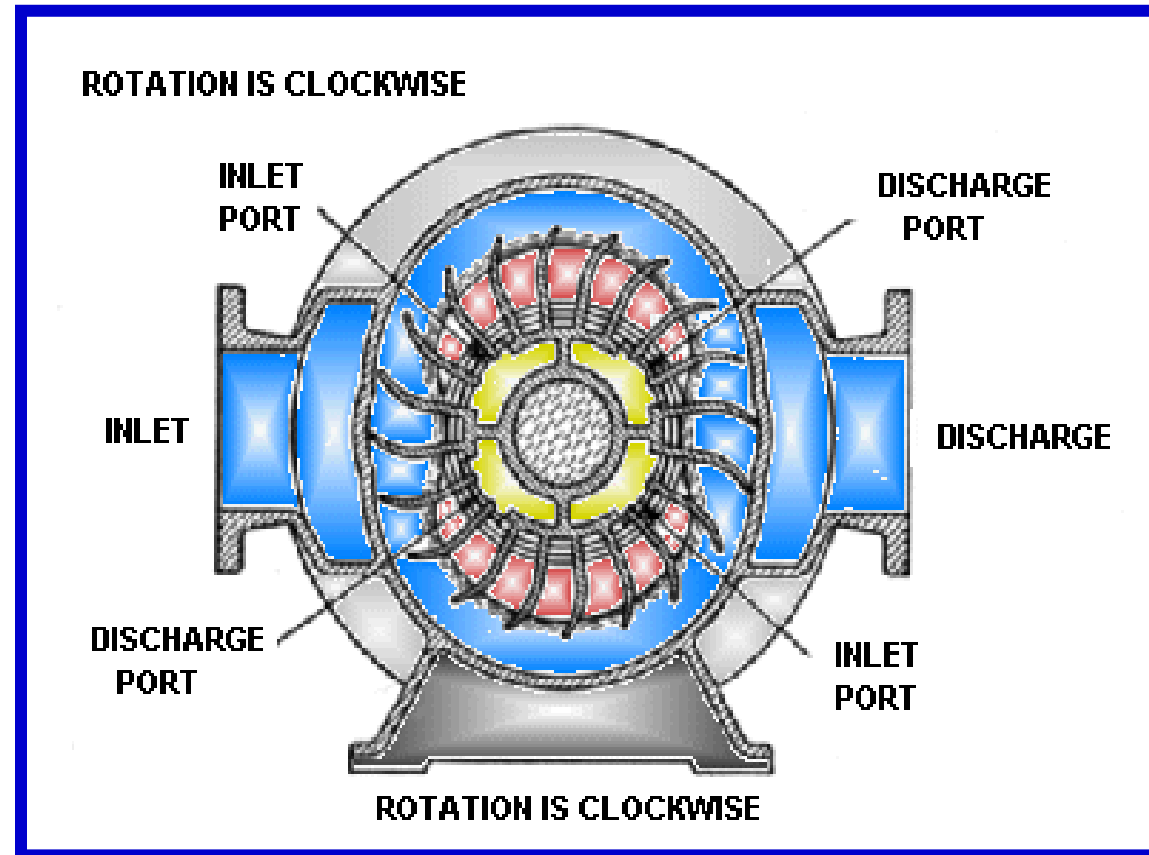


Reciprocating/Piston Compressor (API618)



Liquid Ring Type Compressor (API681)

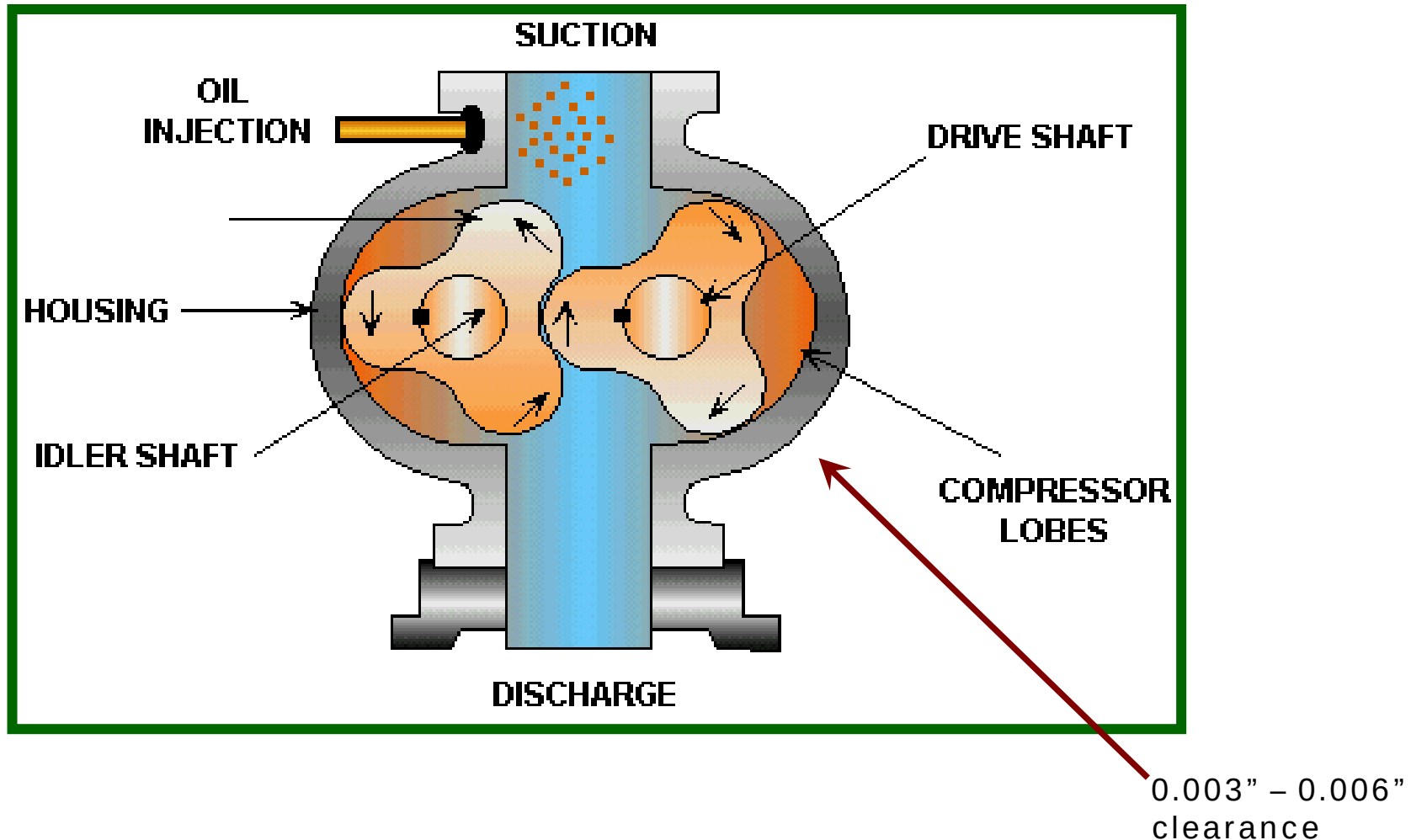
Virtually no gas
temperature rise



- For explosive gases
- When gases tend to polymerise or otherwise react under rising compression temperatures

Eq : For Ozone compression (ozone dissociate at T above 40C)

Lobe Compressor (API619)

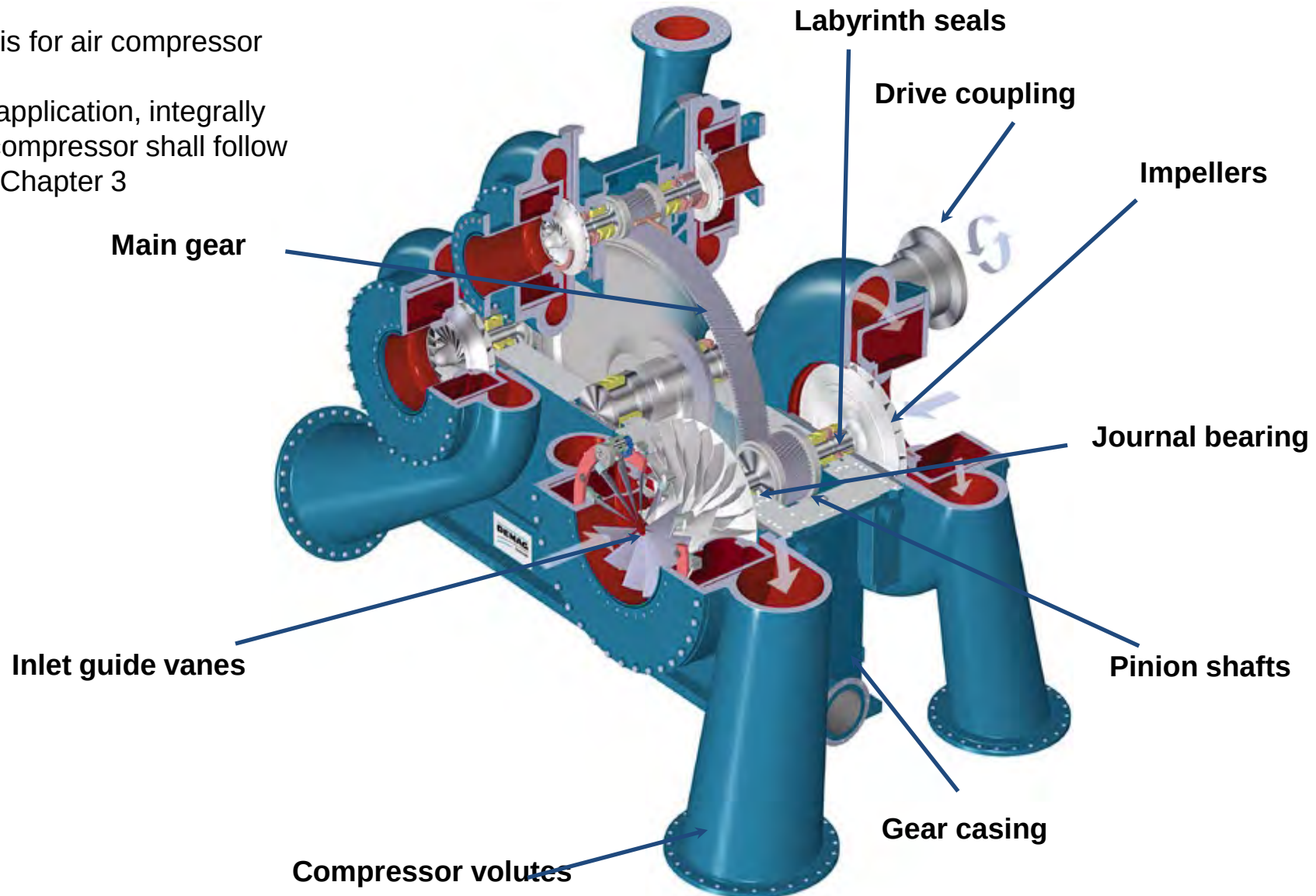


Integrally Geared Compressor (API672)

Note:

API 672 is for air compressor only.

For gas application, integrally geared compressor shall follow API 617 Chapter 3



Gas Compressor Application



Field Boost



Vapor Recovery



Gas Transmission

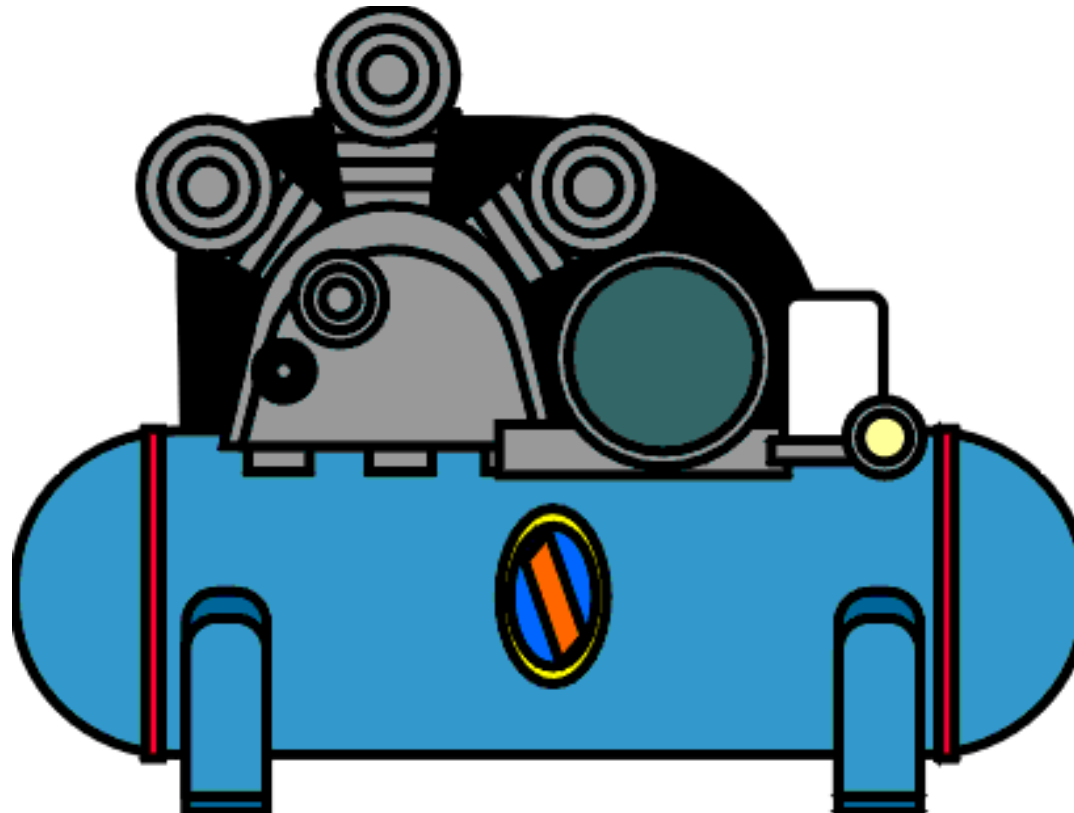


Transportation/
Storage

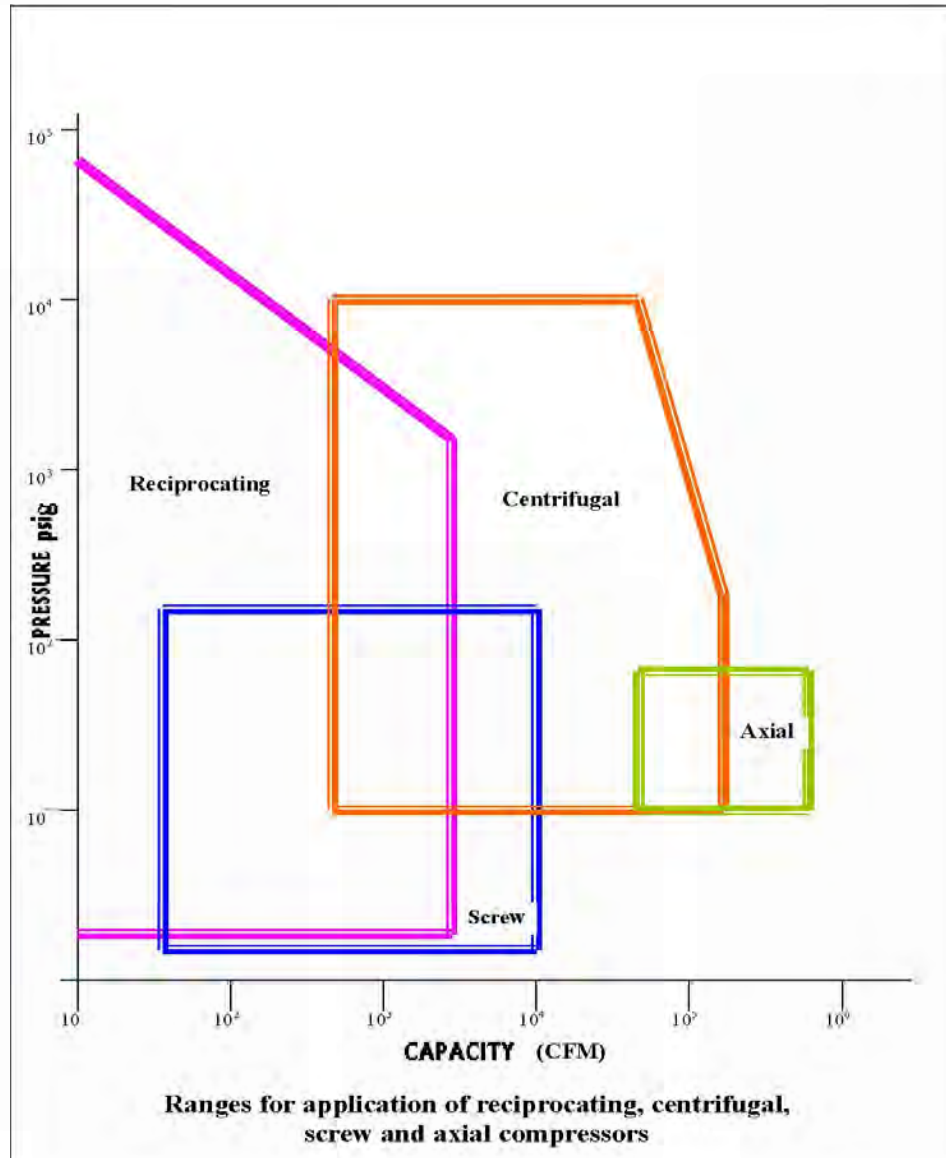


Product
Recovery

Compressed Air



Compressor Selection



Positive Displacement

< 3000 m³/hr

Axial > 30,000

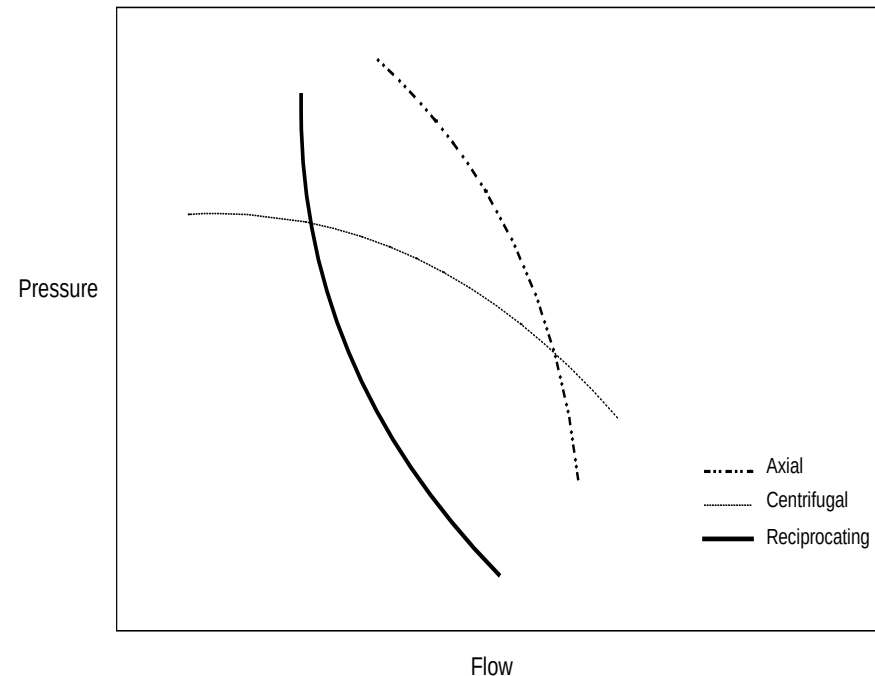
3000 -12,000 overlapping
between PD & Centrifugal

30,000 -120,000 overlapping
between Centrifugal & Axial

Compressor Selection

- The governing criteria for compressor type is volume flow & pressure.
- The other criteria which controls selection when you have a choice are:
 - Operational flexibility.
 - Required reliability.
 - Life cycle cost.
 - Licensor's experience.
 - Process gas conditions etc.

*Compressor Performance
Characteristics*



Gas Compressors

- Site elevation above sea level
- Gas inlet pressure
- Gas composition (for molecular weight (MW)), heat capacity at constant pressure C_p , k -value, compressibility Z (for budgetary considerations); also indicate the water content in the gas
- Gas flow
- Gas discharge pressure required
- Discharge temperature limitations
- Drive selection:
 - For gas engines or turbines, provide the fuel gas lower heating value (LHV), temperature, and pressure
 - For steam turbines, provide the steam temperature, inlet pressure, and acceptable backpressure
- For air or water cooling, provide the design temperature of the medium
- Level of specification:
 - Manufacturer's standard
 - "Near" API
 - API specifications with client inputs
 - Client's specifications
 - Oil level permitted in discharge gas

Example for selection

COMPRESSOR CASE STUDY 1 SMALL CENTRIFUGAL VS. RECIPROCATING

Service – Crude Unit Offgas

Driver(s) – Electric Motor

Comparison is between one 100% motor and gear driven centrifugal and two 100% motor and gear driven reciprocating compressors.

Gas Composition (Mol Percent)	
Water Vapor	4.82
H ₂ S	1.09
C ₁	3.77
C ₂	19.99
C ₃	32.77
C ₄	<u>37.56</u>
	100.00
Molecular Weight	44.0
Process Requirements	
10 ⁶ std ft ³ /d	1.77
A ft ³ /min	1239
T ₁ , °F	100
P ₁ , psia	15.7
P ₂ , psia	69.7
P	54
Comp. Ratio	4.44
C _p /C _v	1.115
Z _{avg}	1.0
M, lb/min	143
Polytropic Head, ft.	32,540
Centrifugal bhp	200
Reciprocating bhp	175

Example for selection

SUMMARY

	Reciprocating	Centrifugal
Total Field Cost	\$450,000	\$735,000
Delivery	12 months	14 months
Horsepower	175	200
Utility Penalty (3.5¢/kWh)	\$1750/a	\$6,600/a
Maintenance	15 x 20 (each)	12 x 22 (including console)

The above study was done in 1975 for a Canadian grass roots refinery. Based on the economics, the reciprocating machine would be the obvious choice. UOP has ample operating experience with reciprocating compressors for this service which presents some difficulties for the reciprocating machine, namely a potentially large variation in flow, a dirty and wet gas and, in this case, a very harsh climate. The client elected to spend the extra money and buy the centrifugal compressor based on his desire to avoid reciprocating machinery.

Example for selection

COMPRESSOR CASE STUDY II MEDIUM CENTRIFUGAL VS. RECIPROCATING

Service – Platforming Net Gas Booster

Driver – Motor

Comparison is between one 100% motor/gear driven two casing centrifugal compressor train and two 60% single-stage multi-cylinder slow speed motor driven reciprocating compressors.

Gas Composition (Mol Percent)	
Hydrogen	85.27
Methane	3.72
Ethane	3.60
Propane	2.92
i-Butane	0.85
n-Butane	1.11
i-Pentane	0.62
n-Pentane	0.35
Hexanes +	1.56
	100.00
Molecular Weight	7.84
Note: Molecular weight may vary from approximately 6.6 to 9.3	
Process Requirements	
10 ⁶ std ft ³ /d	46.40
A ft ³ /min	4315
T ₁ , °F	100
P ₁ , psia	118.2
P ₂ , psia	318.9
P	200.7
Comp. Ratio	2.7
C _p /C _v	1.316
Z _{avg}	1.00
M, lb/min	666
Polytropic Head, ft.	130,000
Centrifugal bhp	(110% of normal process duty) 3900
Reciprocating bhp	(60% of normal process duty) Each 1780

Example for selection (Cont'd)

Summary		
	Reciprocating	Centrifugal
Total Field Cost	1,013,800	1,163,940
Delivery	12 months	12 months
Horsepower	3,560	3,900
Utility Penalty (3.5¢/kWh)		77,766
Maintenance	\$18,000/yr	
Plot Area	20 x 30 (Each)	14 x 35

Based on the above economics, it is a close call and the decision must be made in consideration of other factors. For the Platformer net gas booster service, a machine must have the ability to handle a range of molecular weights. Defining this range can be difficult or impossible. The compressor train must have sufficient flexibility to handle any flow and molecular weight which may result from the plant's operating severity, catalyst condition, feedstock, and octane number required, etc. A turbine driven centrifugal would have a better chance to cover all potential operating points fairly efficiently. The poorer the definition of "range of operation," the more attractive the reciprocating machine becomes. In most cases, UOP ultimately recommends reciprocating machinery for this service but, on occasion, the centrifugal has been chosen.

End of presentation

THANK YOU

SKG 12 Mechanical Engineering

DOMI of Compressor and Blower (3of7)

-Construction of Compressor (Centrifugal Compressor)

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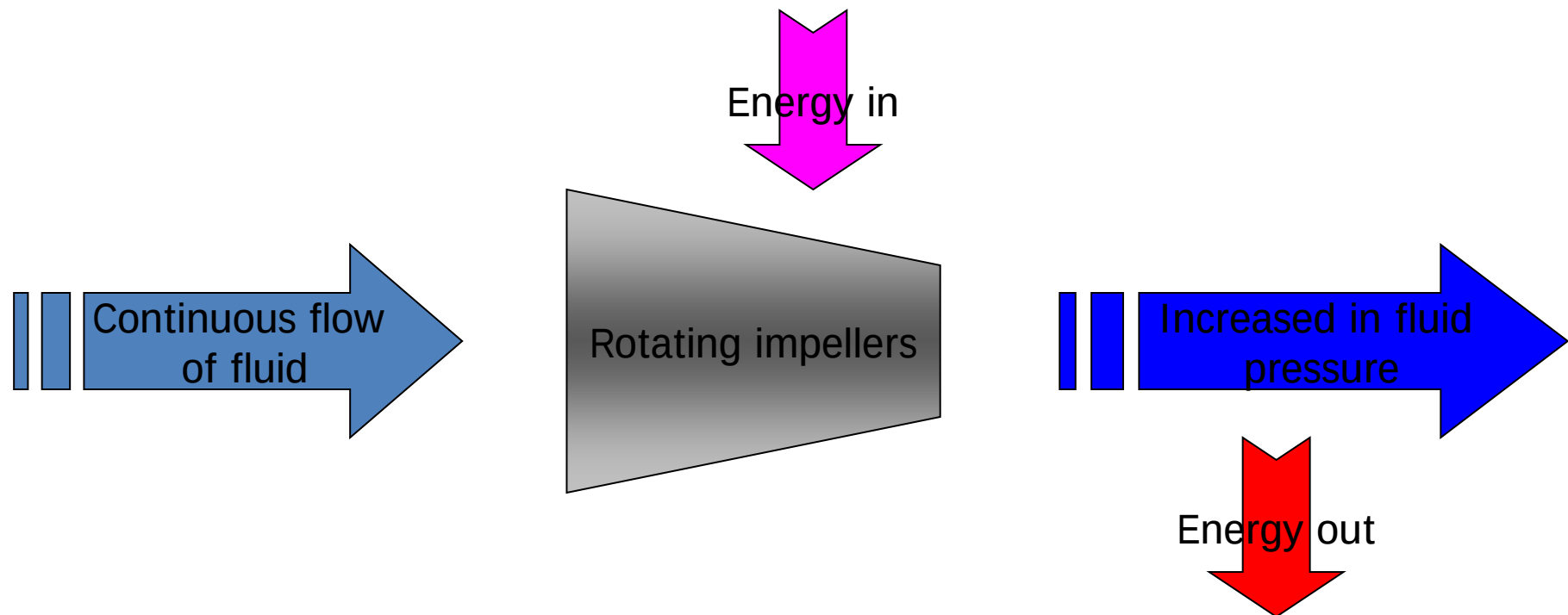
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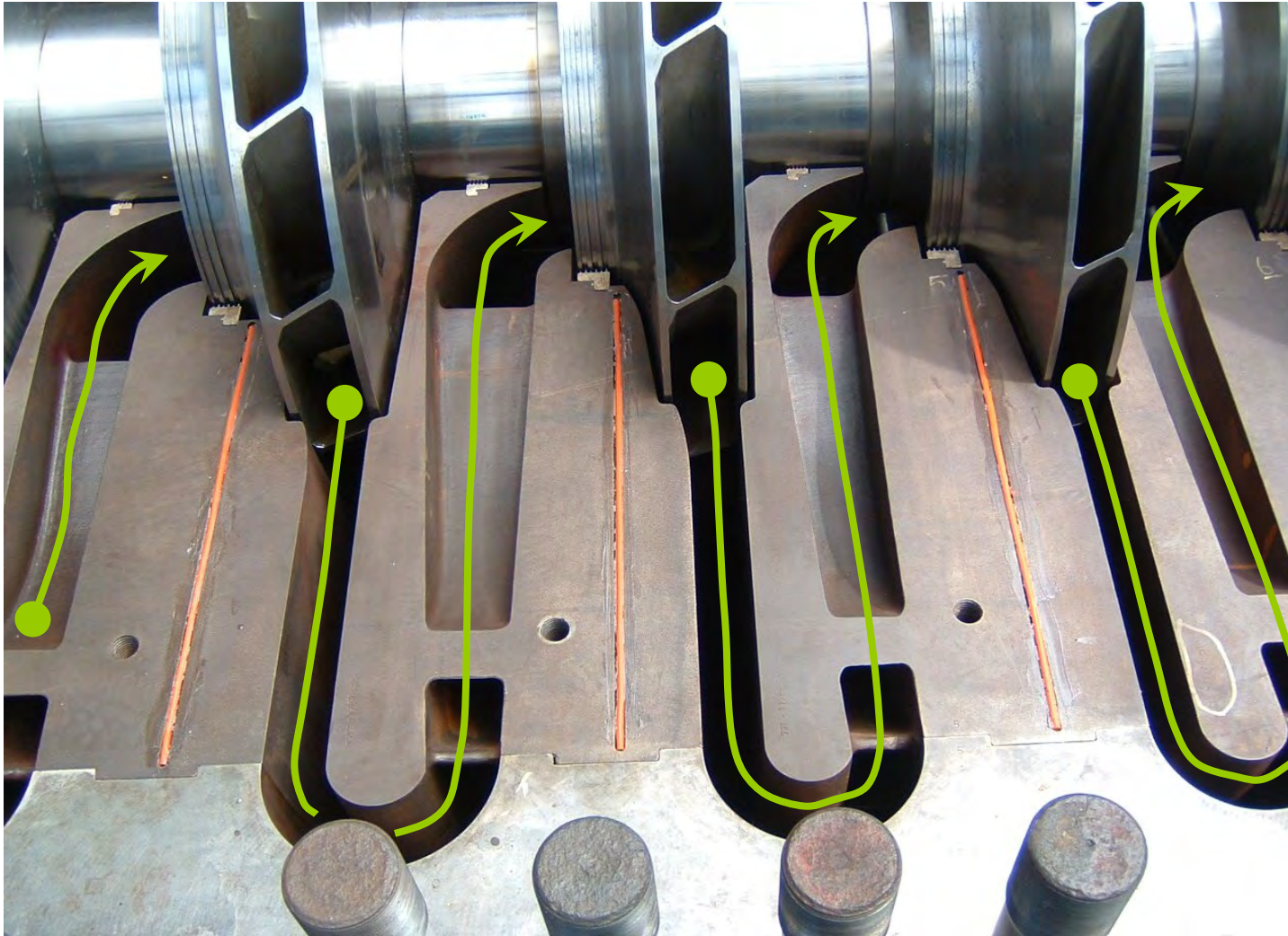
How it works : Centrifugal Compressor

Operating Principles

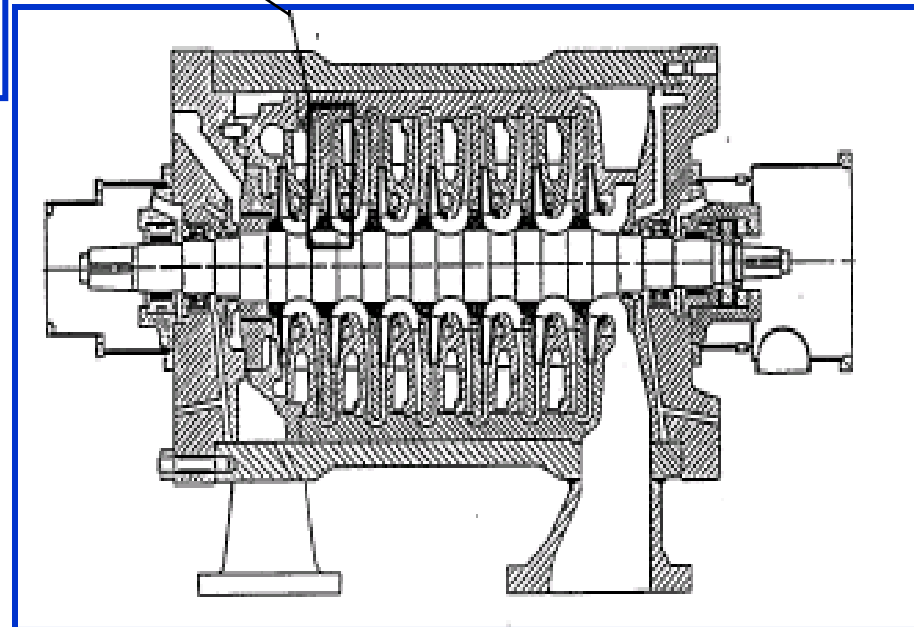
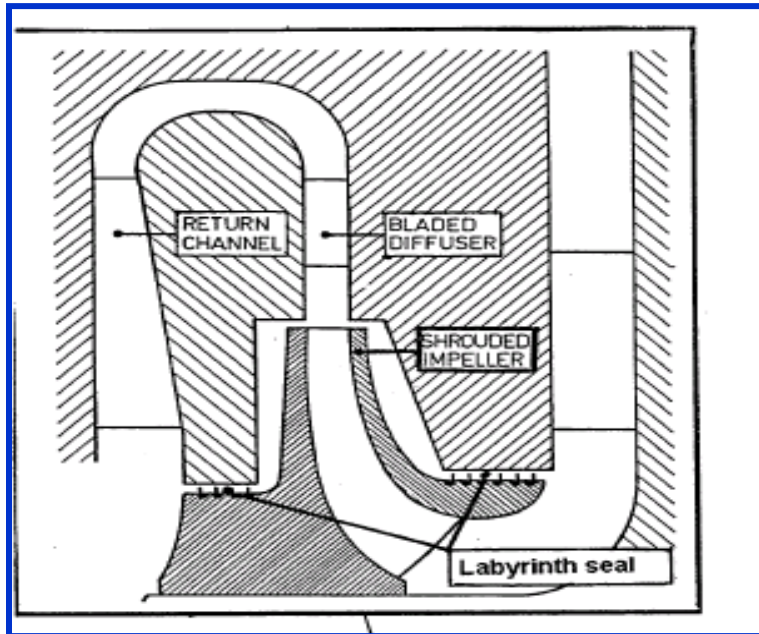
- Operating principle for a centrifugal compressor
- Transformation of kinetic energy exerted to the gas into potential energy



Operating Principles

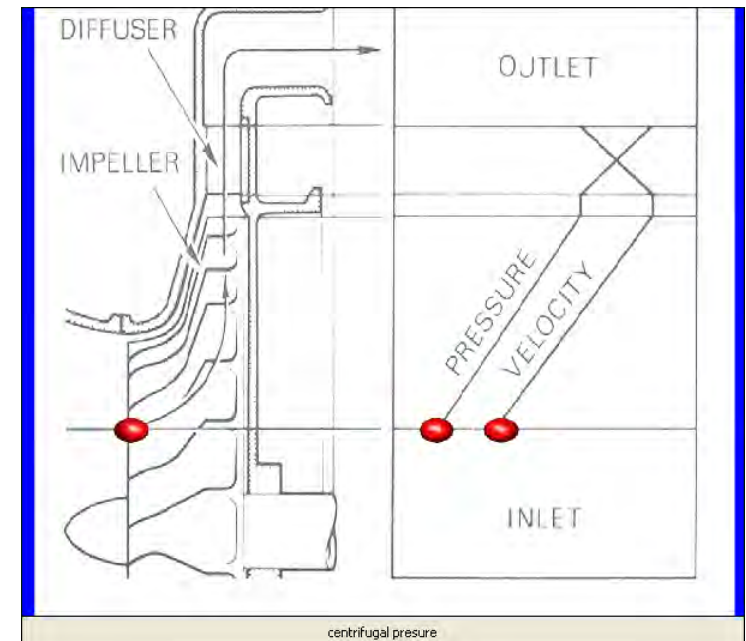
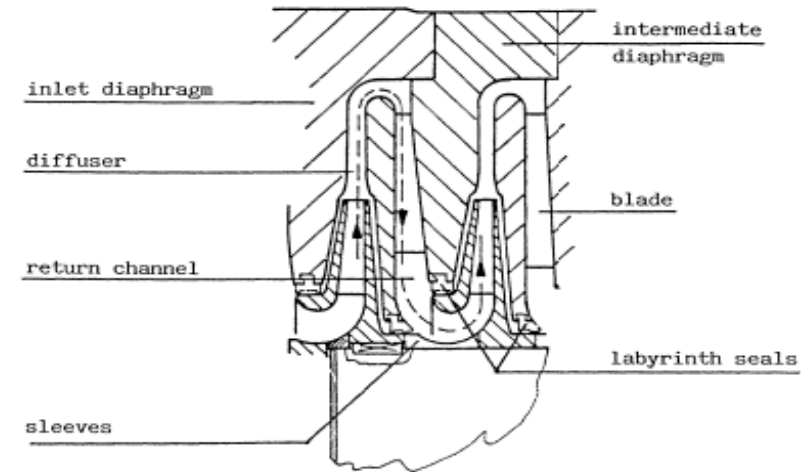


Centrifugal Compressor

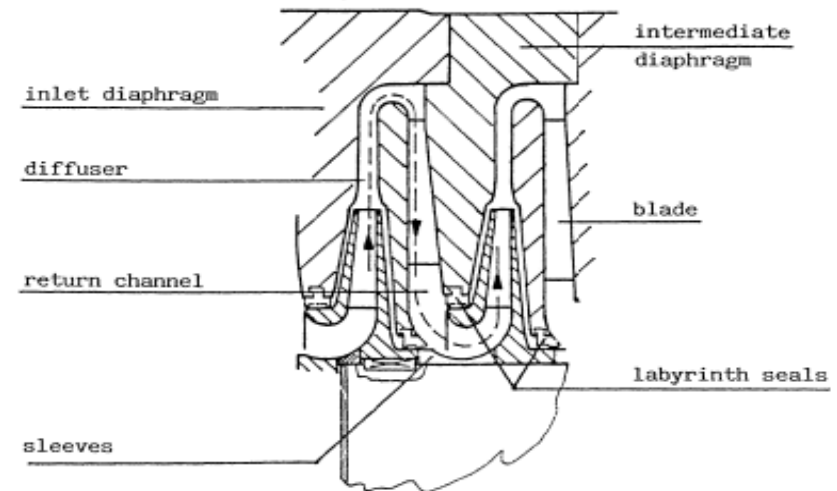


Operating Principles

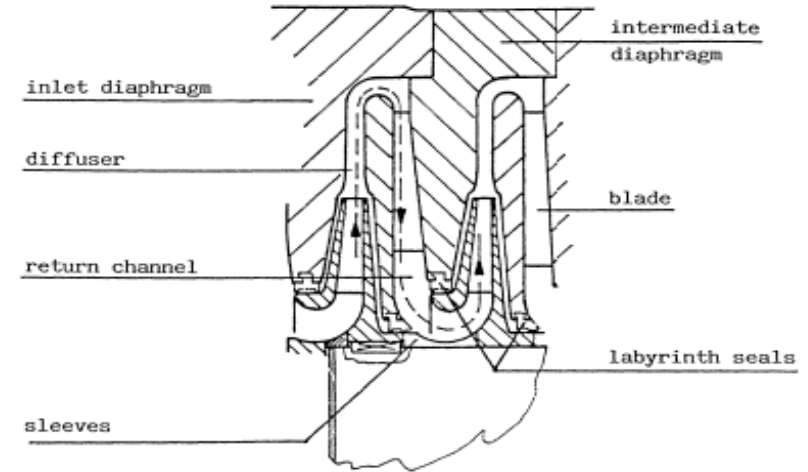
1. Gas is drawn into the compressor through a suction nozzle
2. Then, gas is drawn into the 1st impeller through its suction eye.
3. The impeller consists of a shroud, a disc and radial blades, pushes the gas outwards raising its pressure and velocity.



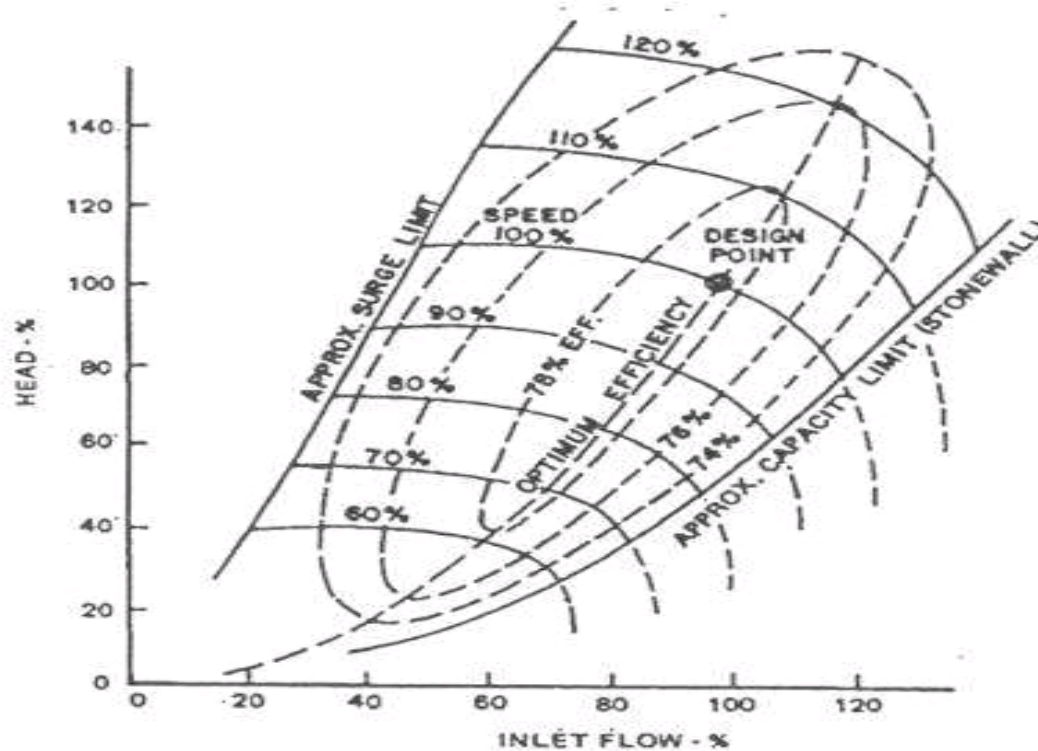
4. The gas then flows through a diffuser, where it loses velocity and gains pressure further.
5. Then the gas flows into return channel (diaphragm) where it is then directed into the next impeller.
6. Process 1-5 continues along each set of impeller in the compressor.
7. Amount of head increase per stage depend on impeller diameter and speed.



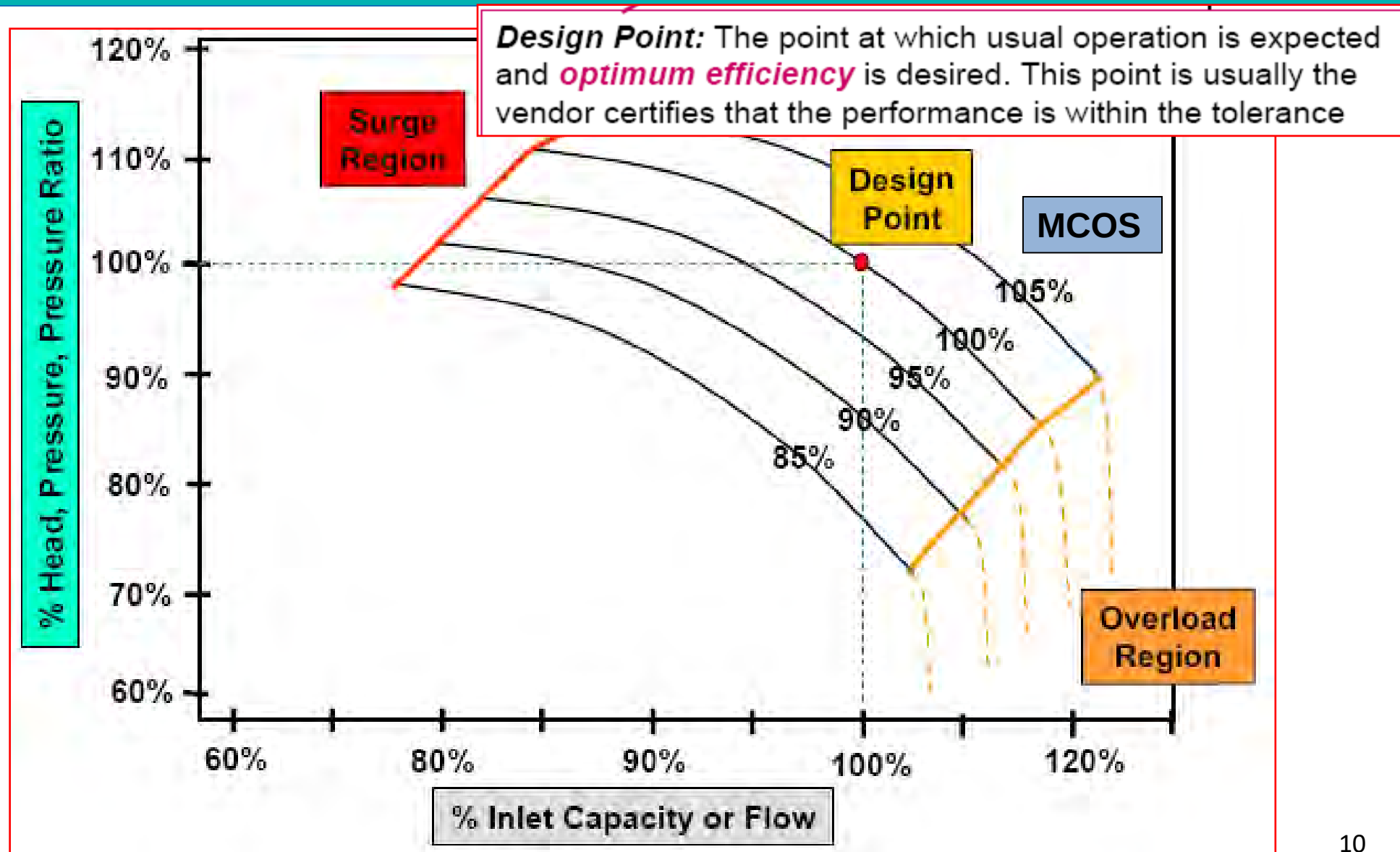
- 8. Labyrinth seals are installed on the diaphragms to minimize internal gas leakage.
- 9. Compressed gas is sent out through discharge nozzle
- 10. Maximum number of stage allowed in PTS is 9 stages in a single casing.



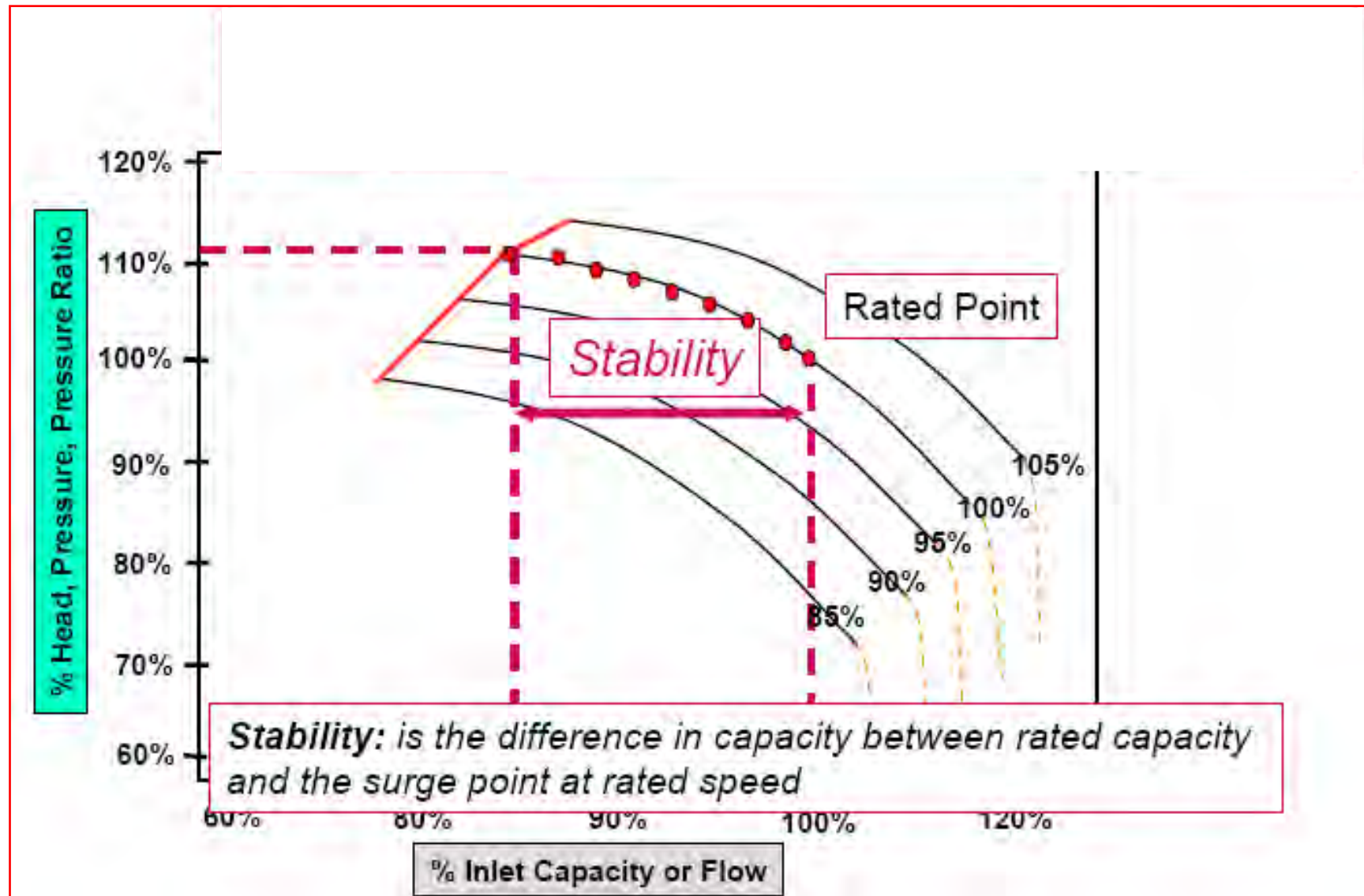
Performance Curve



Performance Curve

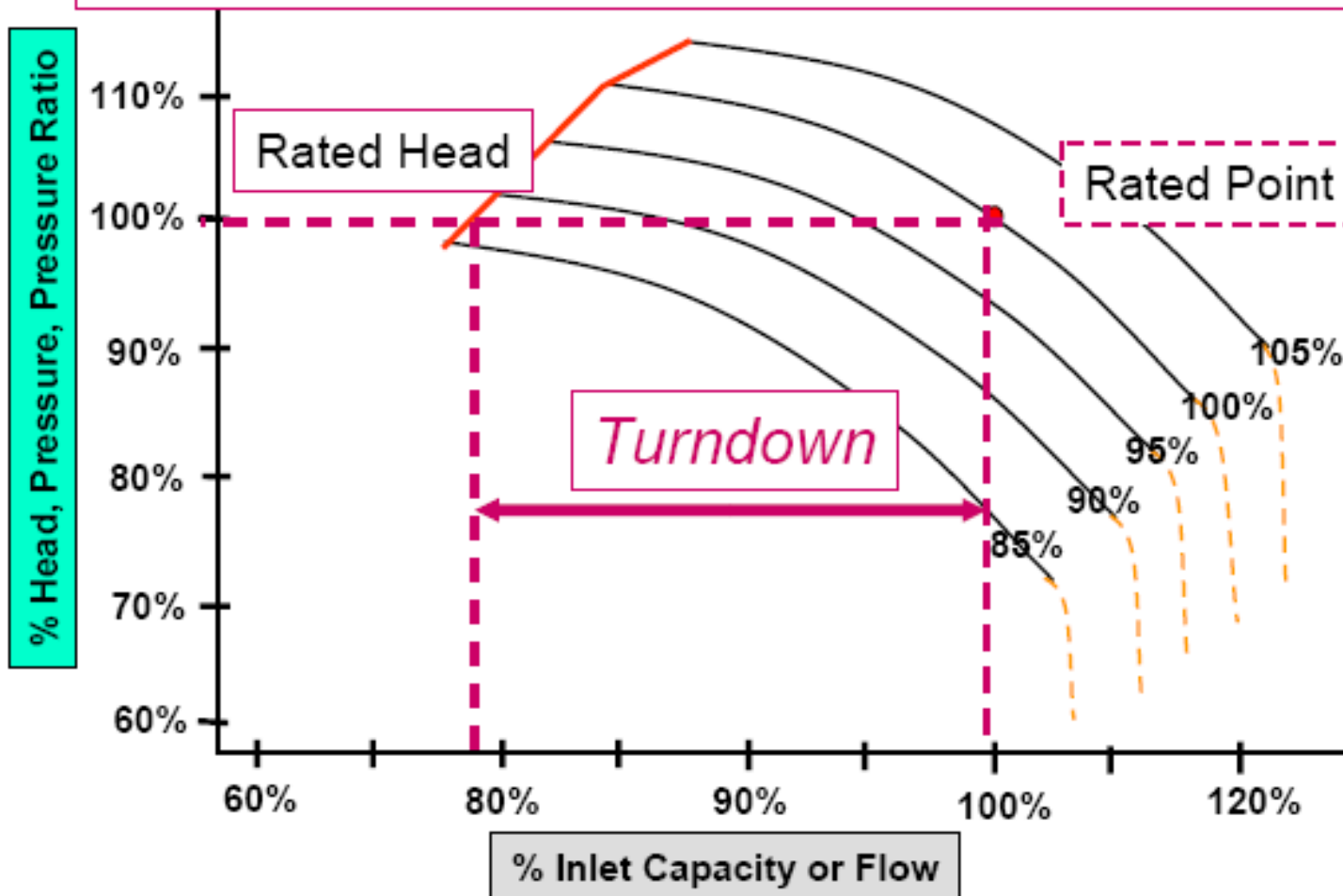


Compressor Stability Range



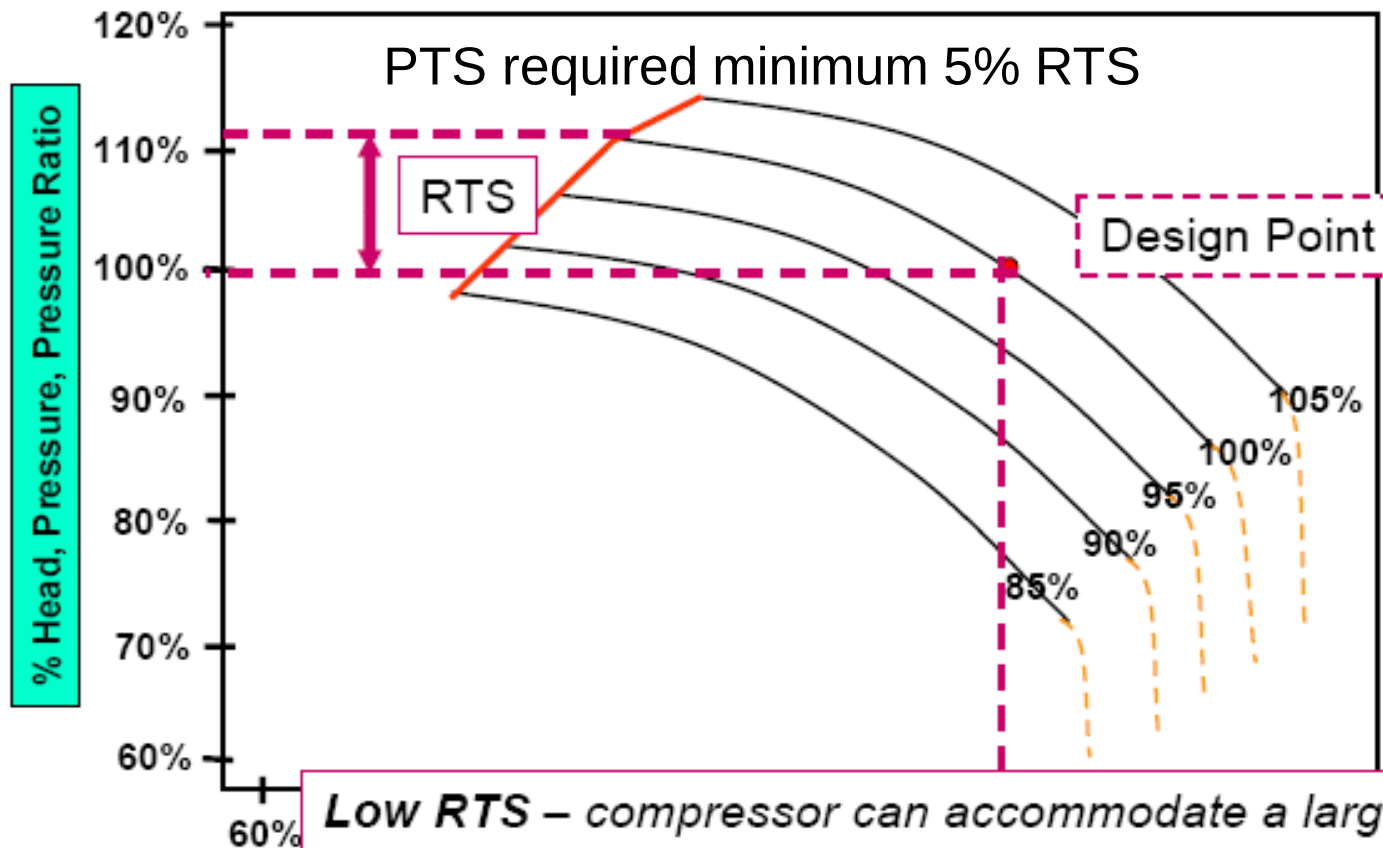
Compressor Turndown Range

Turndown: is the percentage of change in capacity between the rated capacity and surge point capacity at the rated head when unit is operated at rated suction head and gas MW



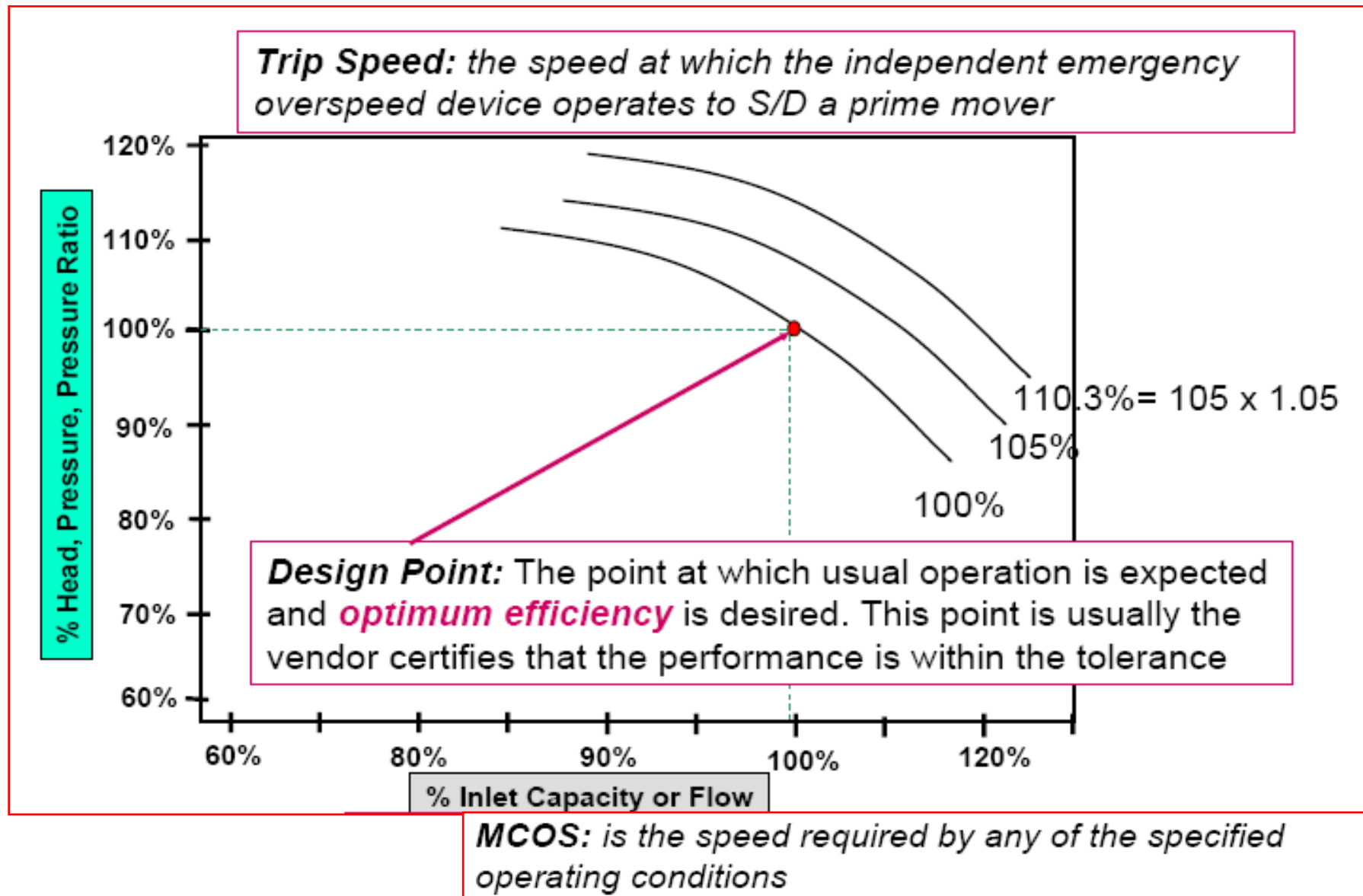
Compressor Rise To Surge Range

High RTS – compressor can accommodate a modest increase in discharge pressure with a little change in flow. The curve is relatively steep.



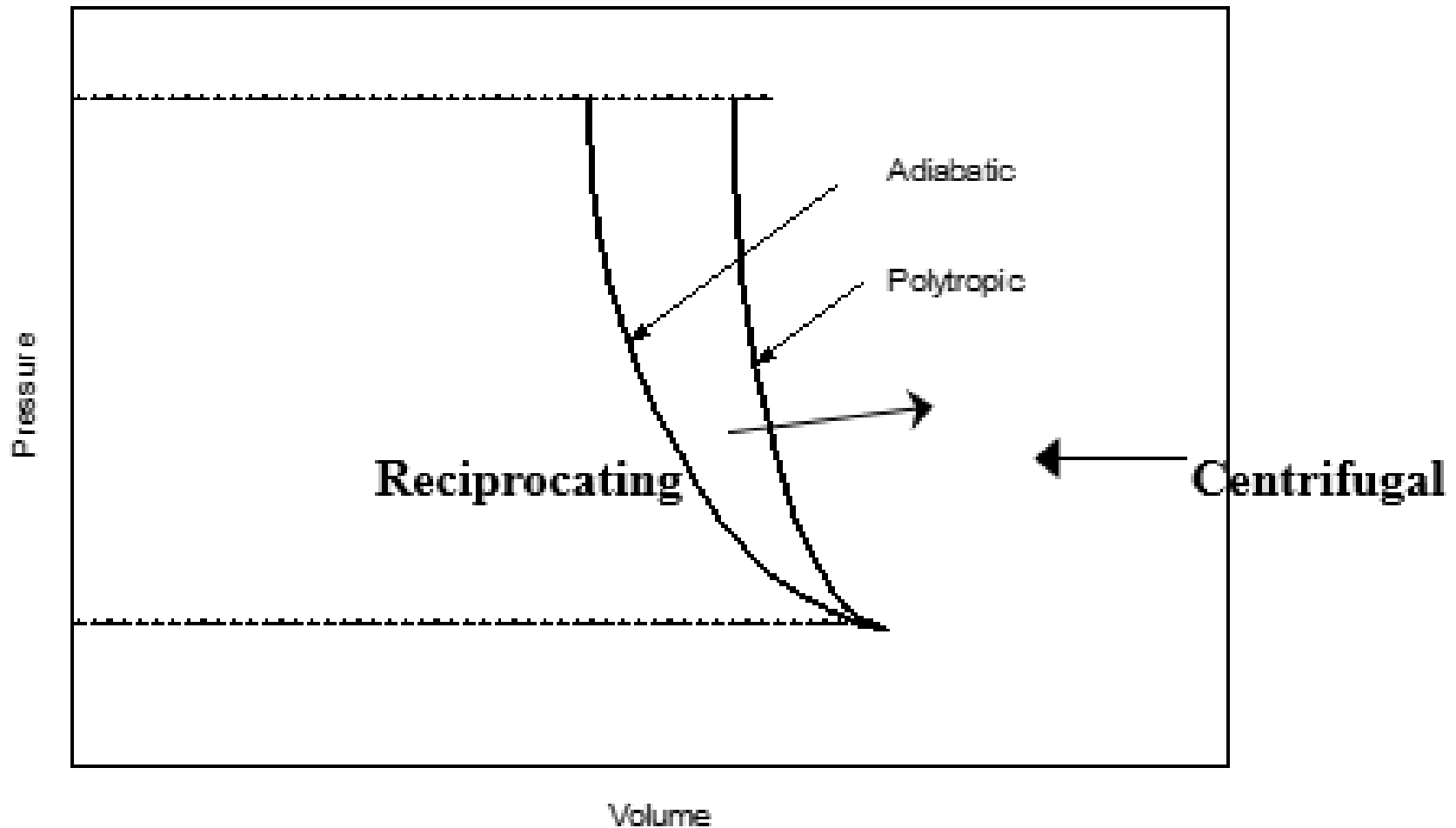
Low RTS – compressor can accommodate a large reduction in inlet flow corresponding to a very small increase of discharge pressure. The curve is relatively flat.

Basic Curve Components



Basic Thermodynamic

Defined on PV Diagram



Isothermal Compression

Occurs when temperature is kept constant as the pressure increase

Requires continuous removal of heat of compression

$$P_1V_1 = P_2V_2 = \text{Constant}$$

Adiabatic Compression

During compression stage, no heat flow takes place across the boundaries.

Closely approached by most of reciprocating compressor

$$P_1 V_1^k = P_2 V_2^k = \text{Constant}$$

Where

k = ratio of specific heat (dimensionless)
= c_p/c_v

Polytropic Compression

Real gas compression, other than isothermal and adiabatic compression

Dynamic compressor (centrifugal and axial) are designed based on this process

$$P_1 V_1^n = P_2 V_2^n = \text{Constant}$$

Where

n = polytropic exponent, experimental value

$$\frac{n-1}{n} = \frac{k-1}{k}$$

n $k\eta_p$ where η_p is polytropic efficiency

Polytropic Compression

Determination of n (polytropic exponent)

$$\frac{n-1}{n} = \frac{k-1}{k\eta_p} \quad \text{where } \eta_p \text{ is polytropic efficiency} \quad (1)$$

Or

If the suction and discharge temperature / pressure are known,

$$\frac{T_2}{T_1} = \frac{P_2}{P_1}^{(n-1)/n} \quad (2)$$

From (2)

$$\left(\frac{T_2}{T_1}\right) = \left(\frac{P_2}{P_1}\right)^{(n-1)/n}$$

$$\text{Log}\left(\frac{T_2}{T_1}\right) = \frac{(n-1)}{n} \text{Log}\left(\frac{P_2}{P_1}\right)$$

$$\text{Log}\left(\frac{T_2}{T_1}\right) = \frac{(k-1)}{k\eta_p} \text{Log}\left(\frac{P_2}{P_1}\right)$$

$$\eta_p = \frac{(k-1)}{k} \frac{\text{Log}\left(\frac{P_2}{P_1}\right)}{\text{Log}\left(\frac{T_2}{T_1}\right)}$$

Adiabatic Head

$$H_{ad} = \frac{Z.R.T_1}{M_w} \cdot \left(\frac{k}{k-1} \right) \cdot \left\{ \left(\frac{P_2}{P_1} \right)^{(k-1)/k} - 1 \right\}$$

		Metric	English
Had	Adiabatic Head		
T1	Suction Temperature	K	R
Z	Average Compressibility Factor (Z1+Z2)/2	-	-
R	Gas Law Constant		
k	Ratio of specific Heat (cp/cv)	-	-
Mw	Gas Molecular Weight	kg/kmol	Lb/lb-mol
(P2/P1)	Compression Ratio	-	-

Head	R
kJ/kg	8.314 kJ/kmol.K
Btu/lbm	1.99 Btu/lb-mol.R

Polytropic Head

$$H_p = \frac{Z.R.T_1}{M_w} \cdot \left(\frac{n}{n-1} \right) \cdot \left\{ \left(\frac{P_2}{P_1} \right)^{(n-1)/n} - 1 \right\}$$

$$H_p = \frac{Z.R.T_1}{M_w} \cdot \left(\frac{kn_p}{k-1} \right) \cdot \left\{ \left(\frac{P_2}{P_1} \right)^{(k-1)/kn_p} - 1 \right\}$$

- Mass Flow (lb/min)
- Inlet (Actual) Volume Flow (ACFM)
- Standard Flow (MMSCFD)

Standard Flow (MMSCFD)

ISO STANDARD (NORMAL) CONDITIONS

Flow: Normal Cubic meters per hour (Nm^3/h)
Normal Cubic meters per min. ($\text{Nm}^3/\text{min.}$)
Pressure: 1.013 bar absolute
Temperature: 0°C

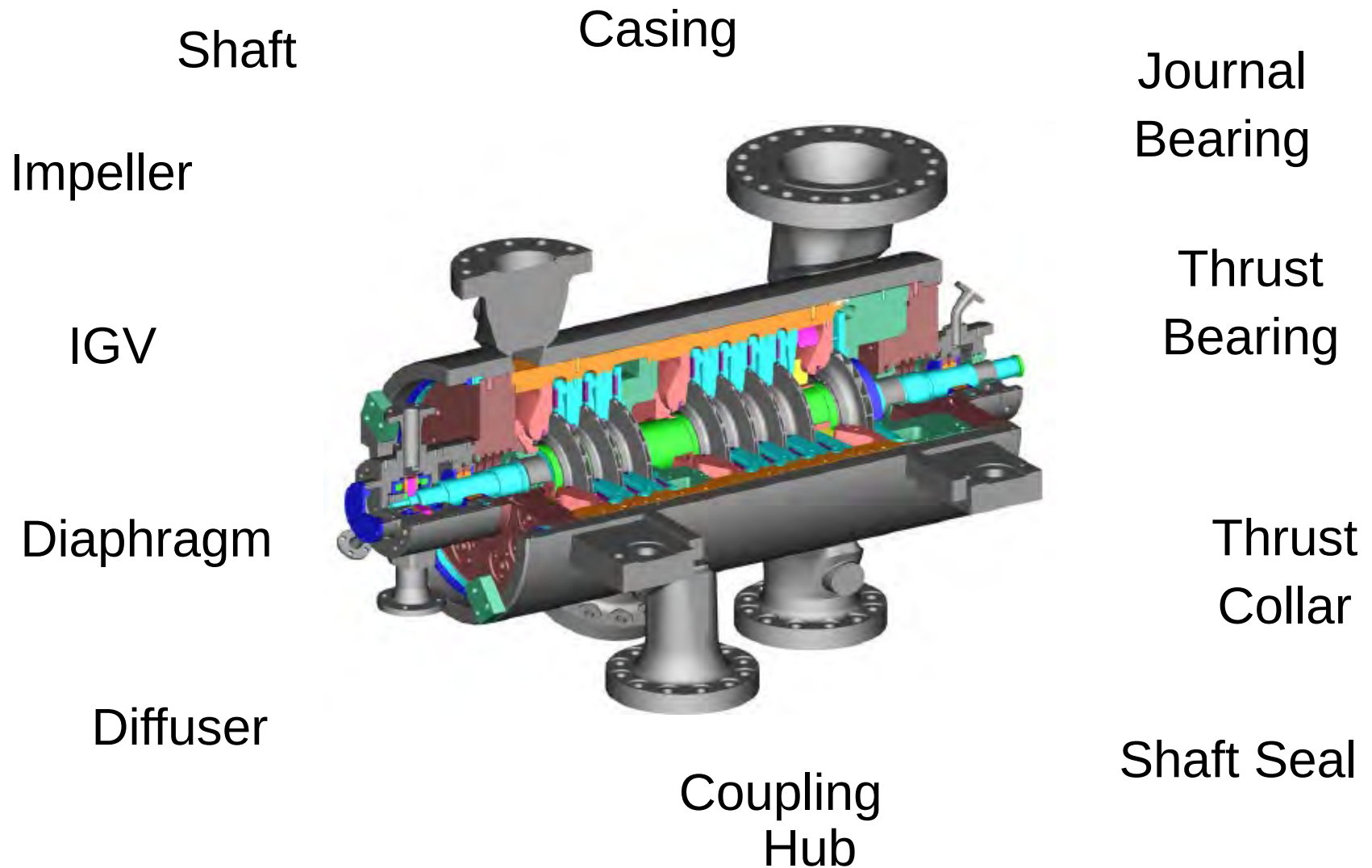
U.S. CUSTOMARY STANDARD CONDITIONS

Flow: Standard cubic ft per min. (scfm)
Million standard cubic ft per day (mmscfd)
Pressure: 14.7 PSIA
Temperature: 60°F

API 617

Centrifugal Compressor Major Component

Centrifugal Compressor Major Component



Casing : Horizontal Split



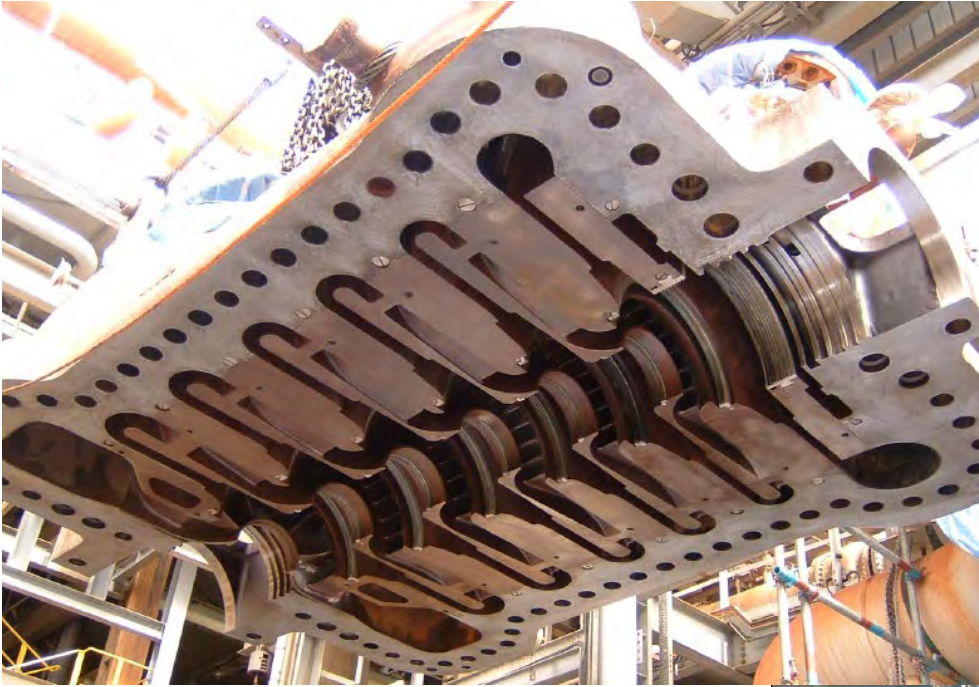
Normally used for discharge pressures up to 50 bar.

Do not make leak tight joint at high pressures (use sealant,
no gasket permitted)

Both up ward & downward nozzle configuration possible.

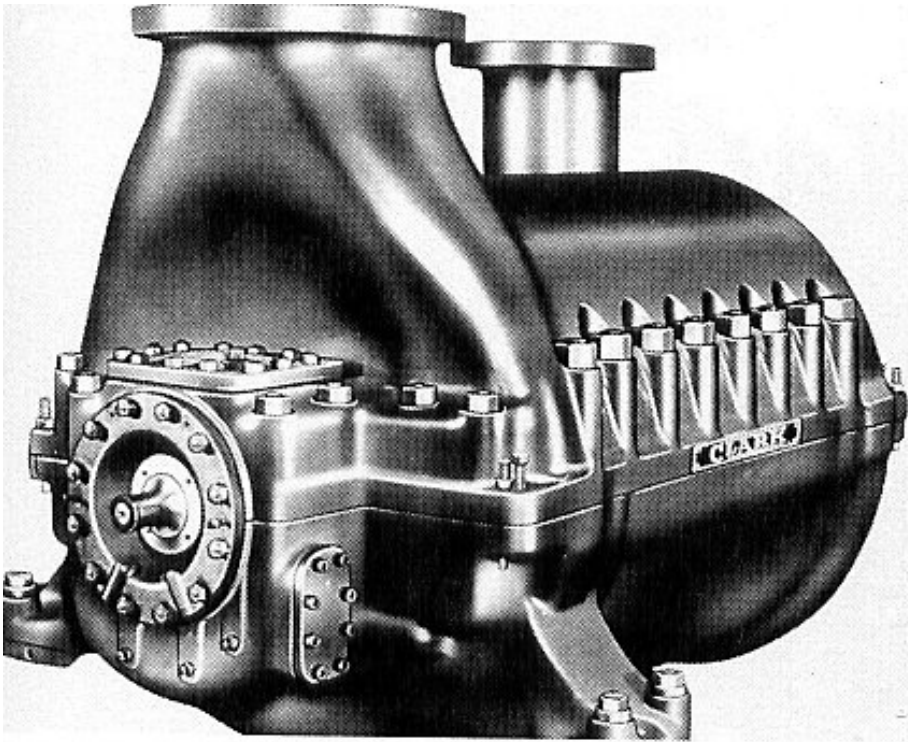
Ease of maintenance & Inspection for downward nozzle

Casing

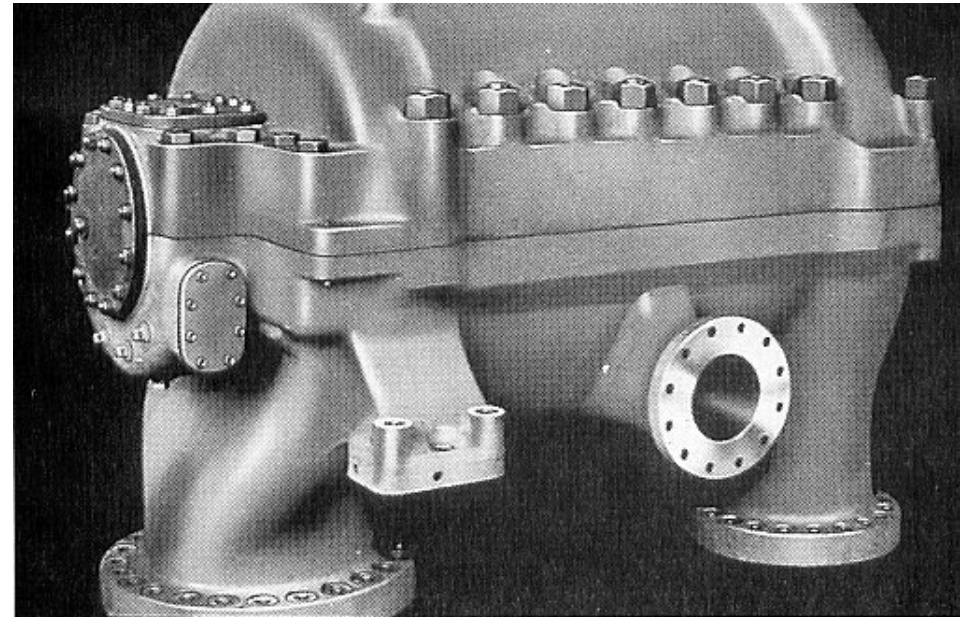


Horizontally Split





Top Connected



Bottom Connected

Casing (Horizontally Split)



Casing (Horizontally Split)

Advantages

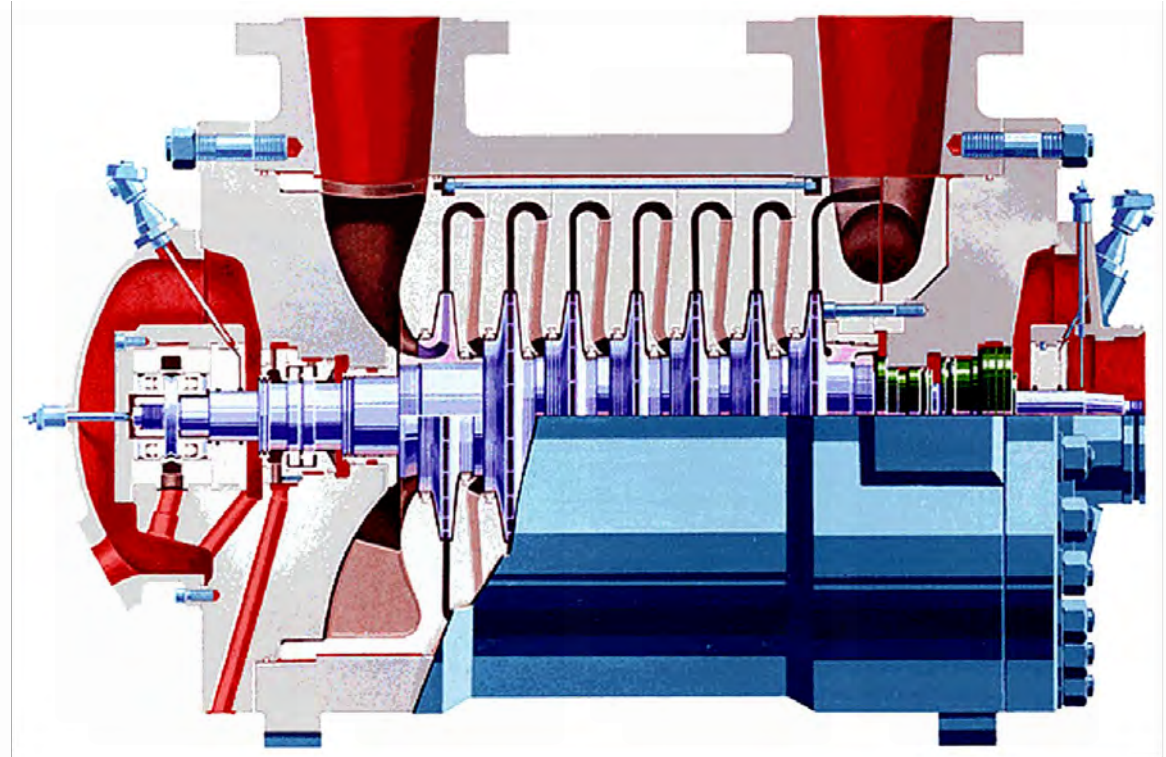
- Able to inspect, clean and replace the compressor internal by simply lifting off the top half of the case

Disadvantages

- Possible leak between mating
- Inability to handle high pressure

Casing (Radially Split)

- Casing split is vertical & makes high integrity joint due to better stress distribution & no pipe load unlike HS.
- As per API-617 this type is mandatory for service where partial pressure of Hydrogen is more than 200 psi.
- Common in many petrochemical applications.



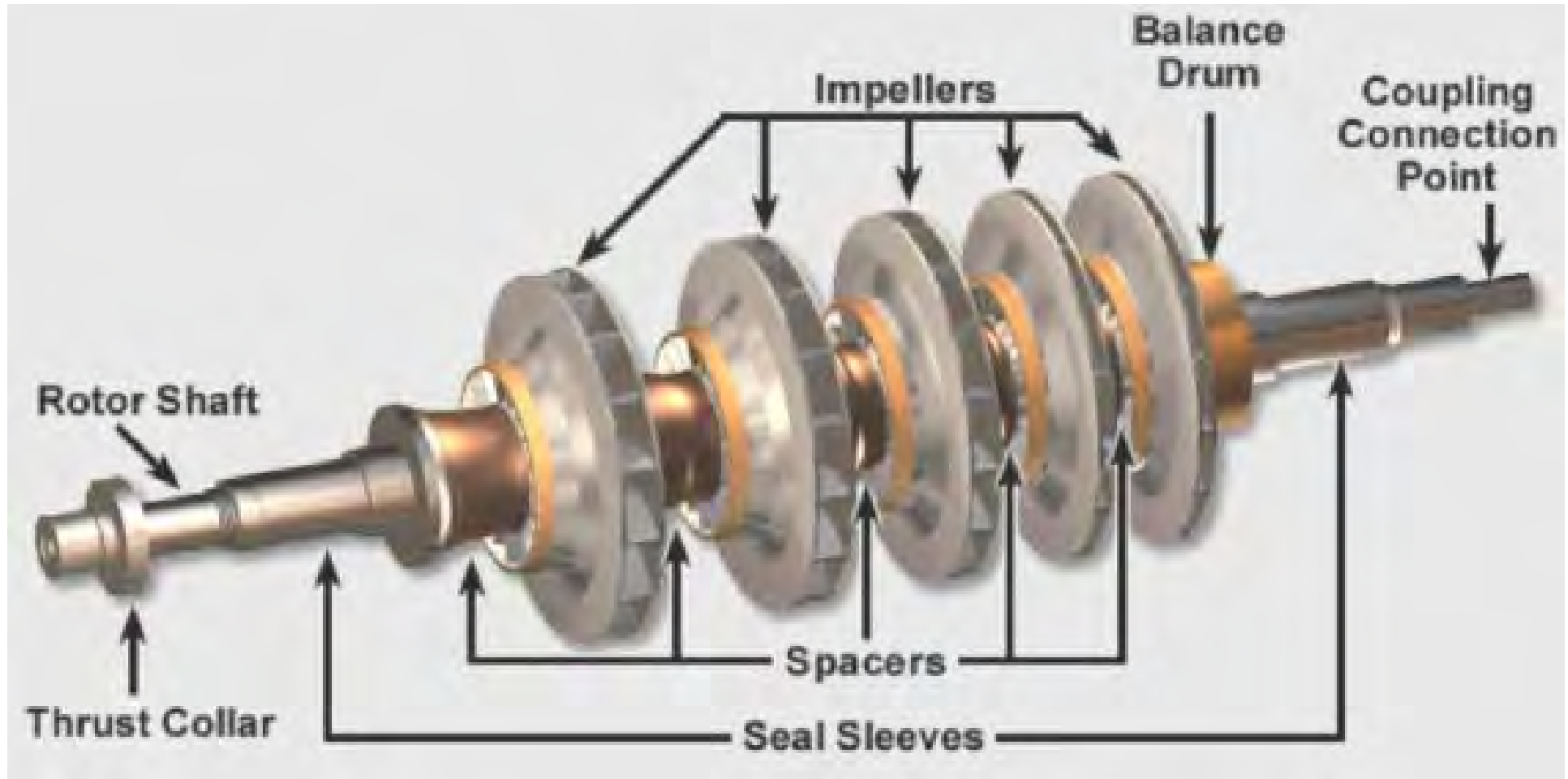
Advantages

- High pressure capability due to the one piece construction and conventional, proven flange type seals at each end
- Makes high integrity joint due to better stress distribution & no pipe load unlike horizontal split.

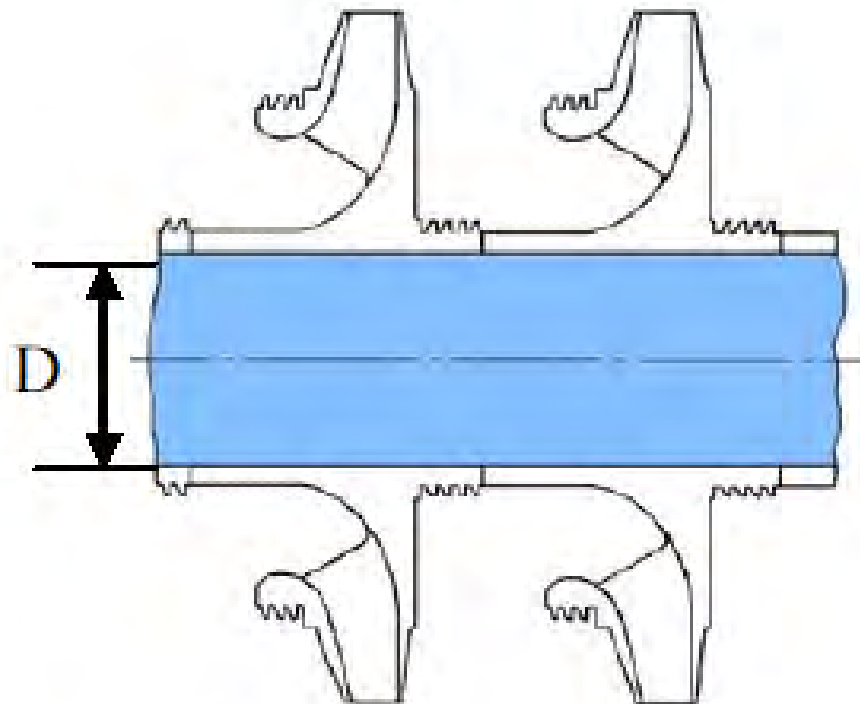
Disadvantages

- Internal must be removed (or unstacked) for inspection and cleaning
- Sufficient space must be provided for removing of the rotor assembly

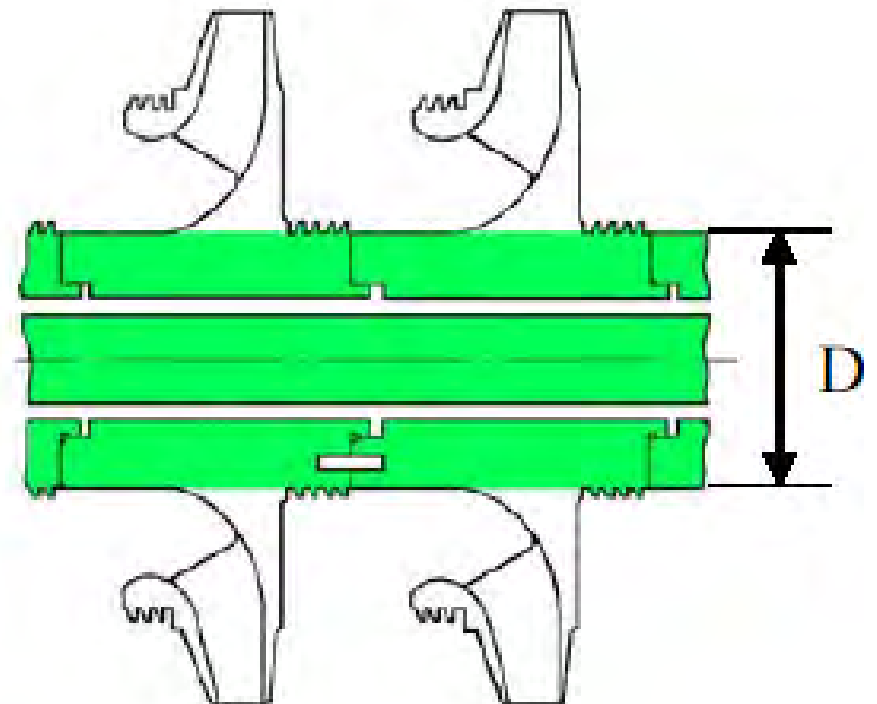
Rotor Assembly



Shaft (Two basic rotor design)

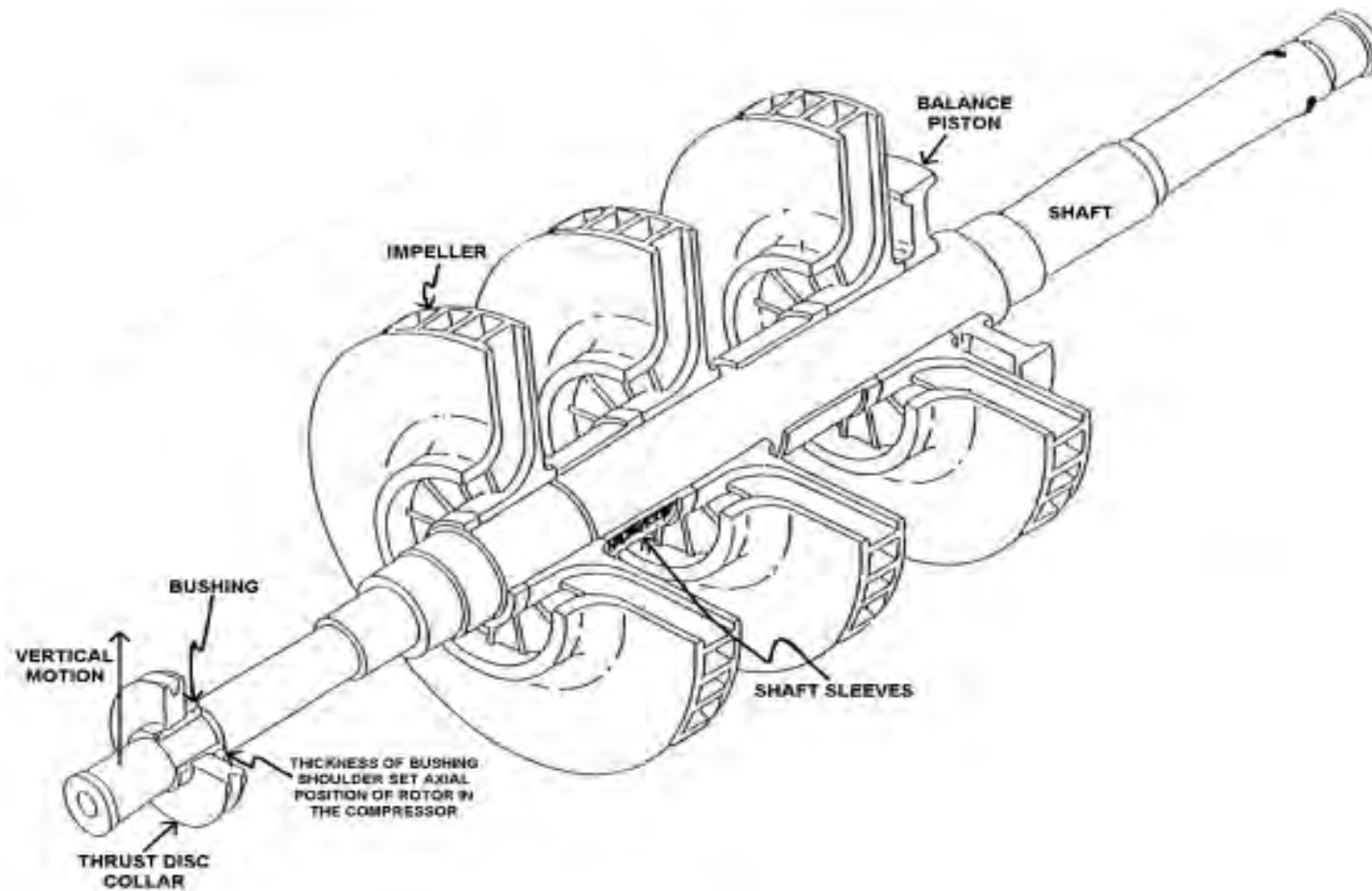


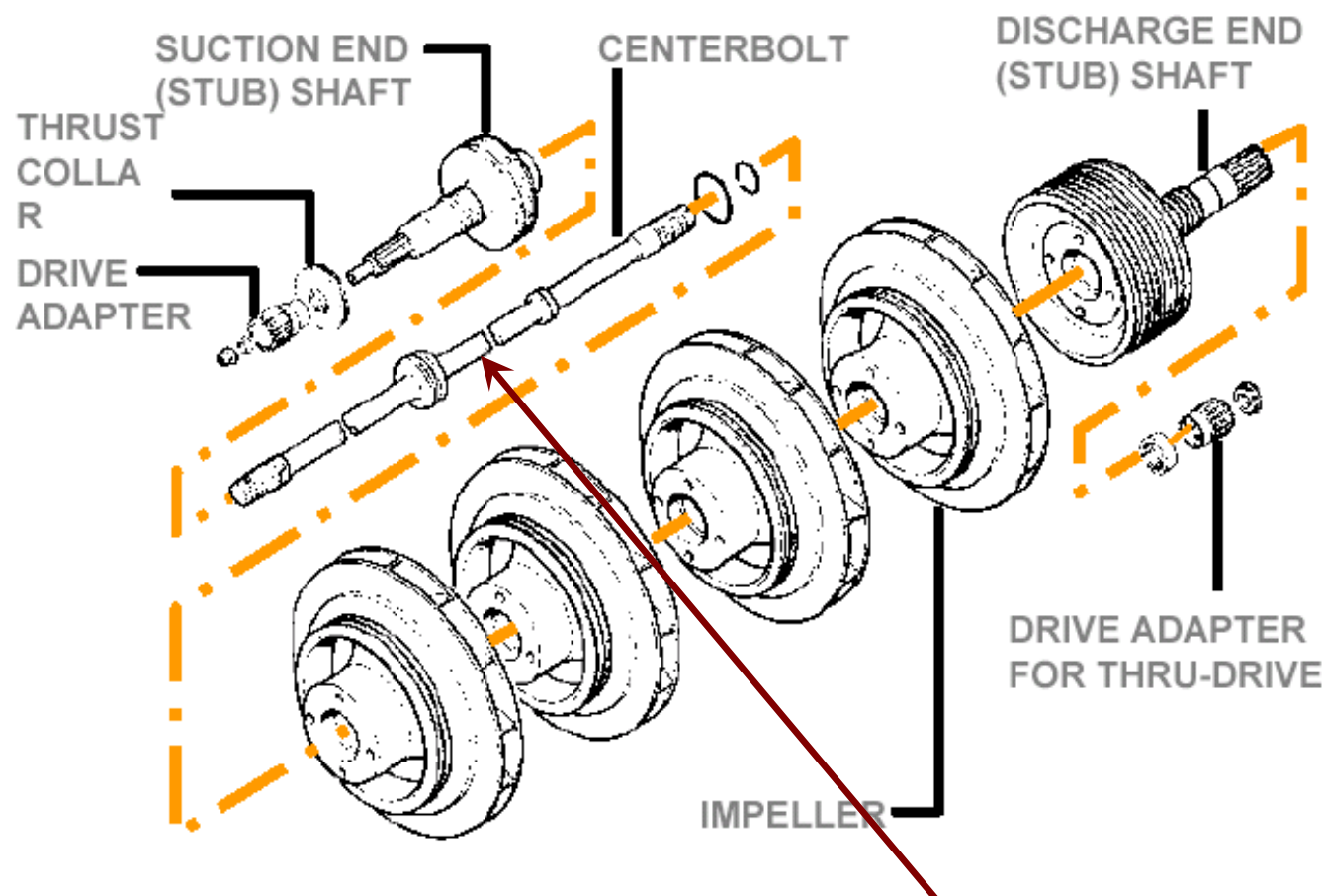
Solid-Shaft Rotor



Modular Rotor

Shaft





2.5.3.2.1 The studs or tie-bolts used to clamp a built-up rotor shall be made from bar or forgings. Threads shall be formed by rolling. Each tie-bolt shall be tested with a proof load corresponding to at least 110% of maximum stretch that occurs during assembly or in operation.

Shaft Material

Part	Material ^a	Specification ^b	Form	Temperature Limits ^c	
				Minimum °C (°F)	Maximum °C (°F)
Shaft	Steel	ASTM A 470, Class 1	Forged	– 28 (– 20)	345 (650)
		ASTM A 470, Class 7	Forged	– 115 (– 175)	400 (750)
		AISI Types 1040 – 1050 ^c	Bar or Forged	– 28 (– 20)	345 (650)
		AISI Types 4140 – 4150 ^c	Bar or Forged	– 28 (– 20)	400 (750)
		AISI Type 2320 ^c	Bar or Forged	– 110 (– 170)	345 (650)
		ASTM A 522, Type 1	Forged	– 195 (– 320)	345 (650)
	Stainless steel	ASTM 4340 – 4345	Forged	– 115 (– 175)	345 (650)
		ASTM A 336, Grade F6	Forged	– 60 (– 75)	345 (650)
		ASTM A 473, Type 410	Forged	– 60 (– 75)	345 (650)
	Precipitation hardening stainless steel	ASTM A 705, Types 630 or XM-12	Forged	– 75 (– 100)	345 (650)
		ASTM A 564, Type 630 or XM-12	Bar	– 270 (– 454)	345 (650)

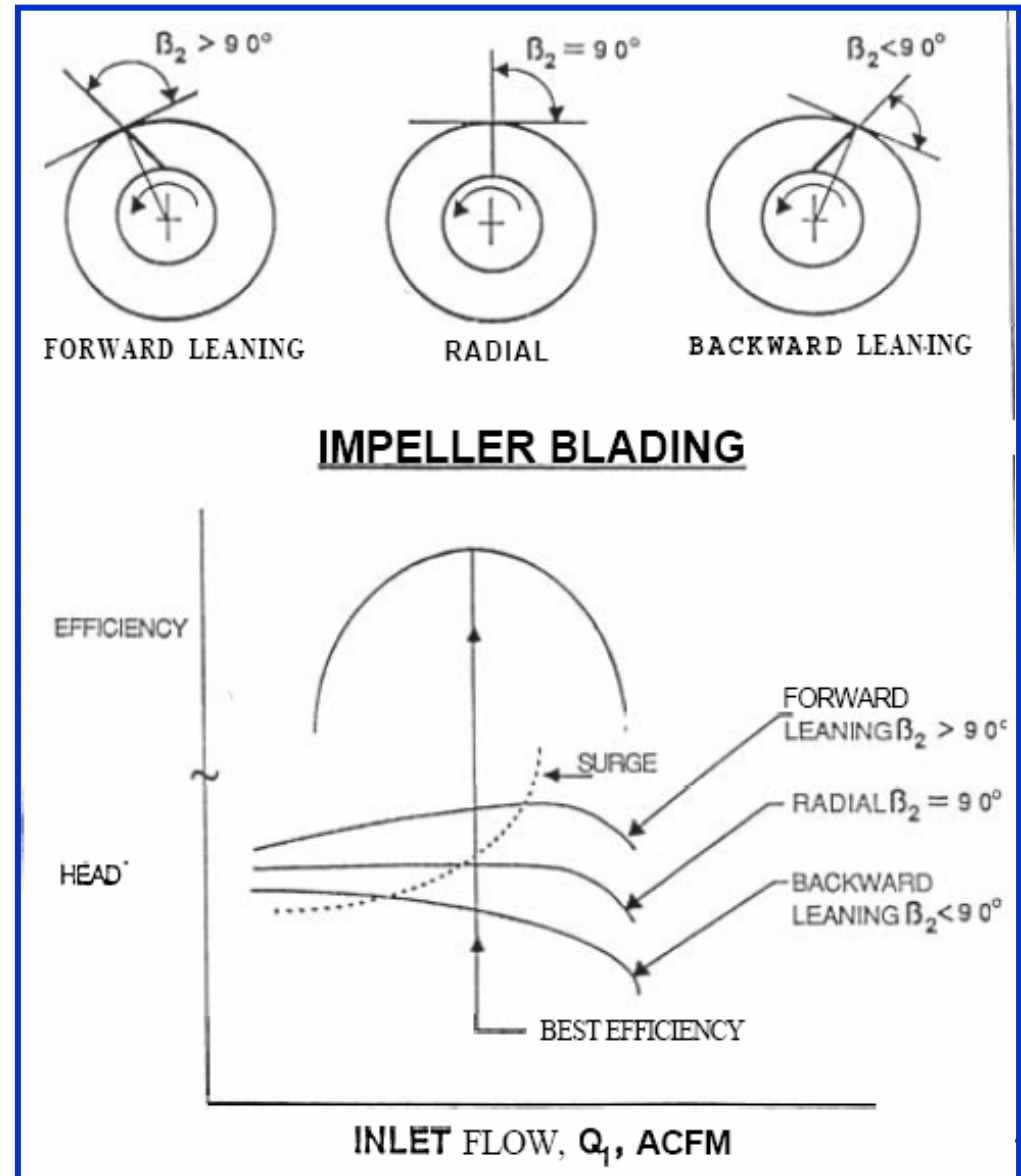
Reference :API 617

Impeller

Impeller Shape

- Forward Leaning
- Radial
- Backward Leaning

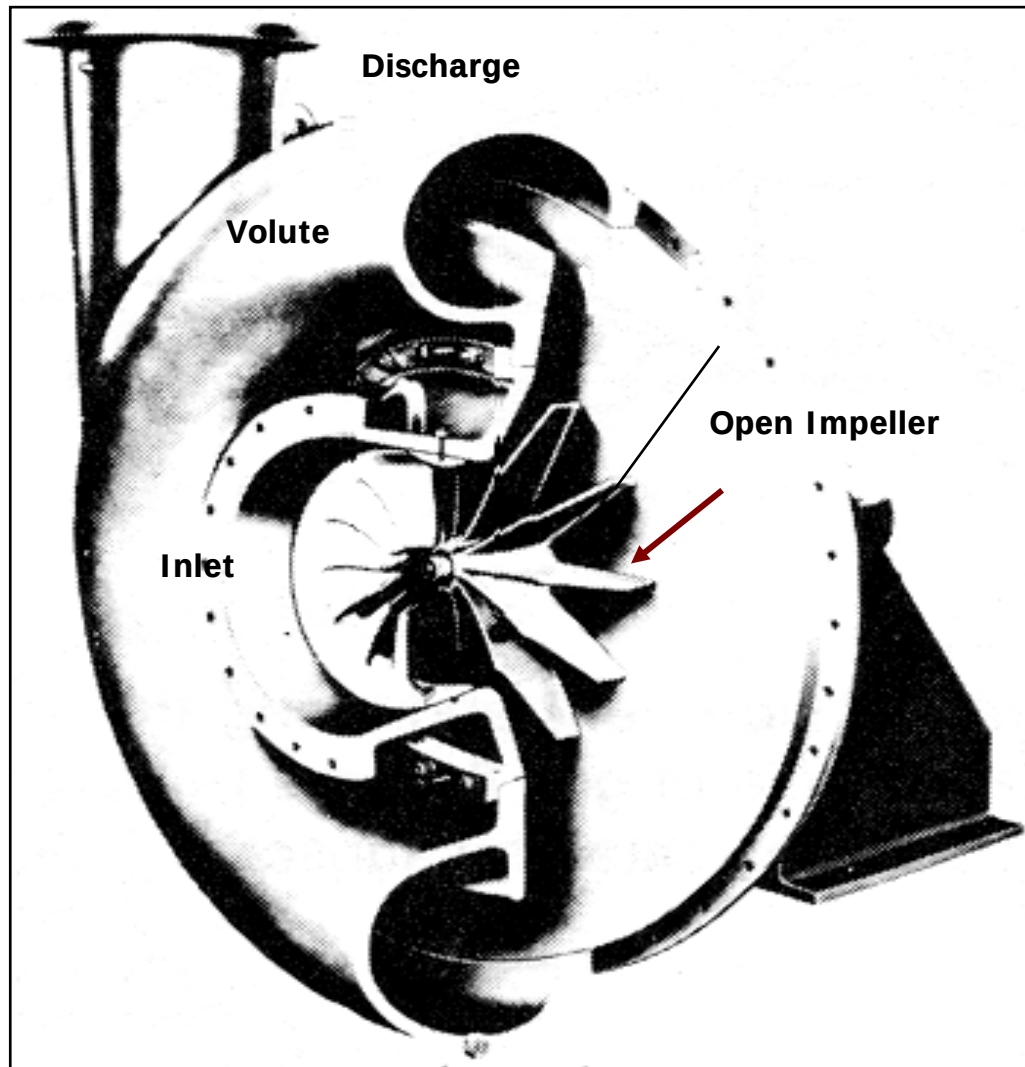
Backward Leaning Impeller is most commonly used since it produces the greatest operating range and offers the best efficiency.



Impeller Classification

- Open
- Semi-Enclosed
- Closed / Shrouded

Impeller



Open Impeller

Impeller



Semi-Enclosed

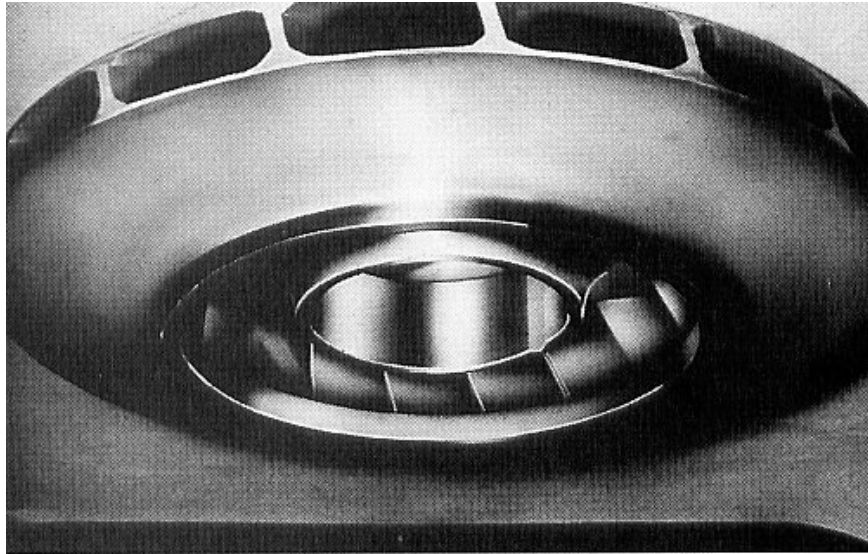
- For large flow rate in single stage compressors
- Also normally found as 1st stage in multi-stage compressors

Impeller



Closed / Shrouded

- Mainly used in multi-stage compressors
- Backward leaning, closed impeller configuration commonly used due to its wide flow range capability

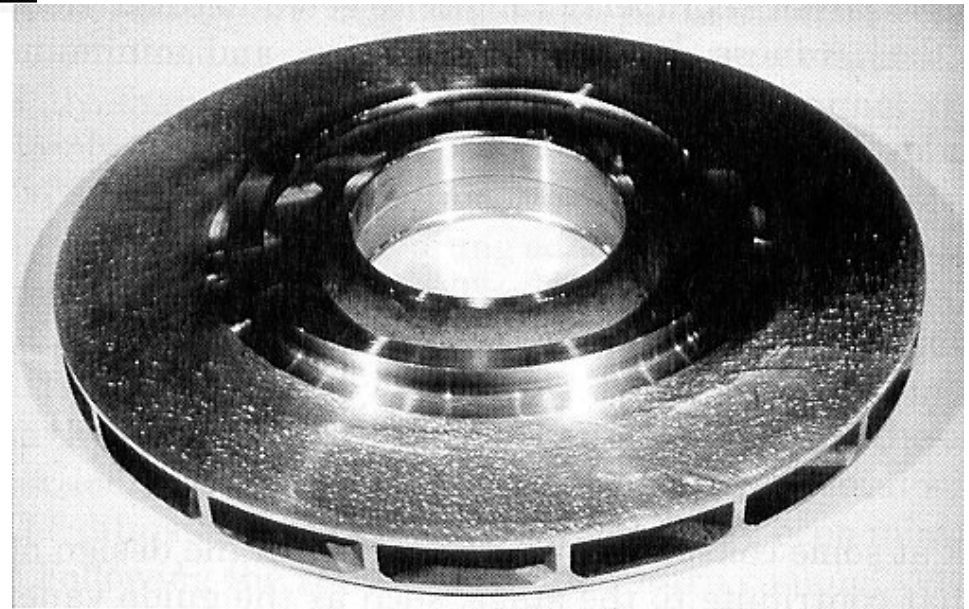


Three Piece Impeller Design

- Disc and Cover are machined from forgings
- Blades are die formed
- Blades are assembled and welded to disc by fillet weld
- Cover is fitted and blades are fillet welded to cover

Two Piece Impeller Design

- Blades are milled integral with disc or cover from forging
- The mating cover or disc is assembled and the blades are fillet welded



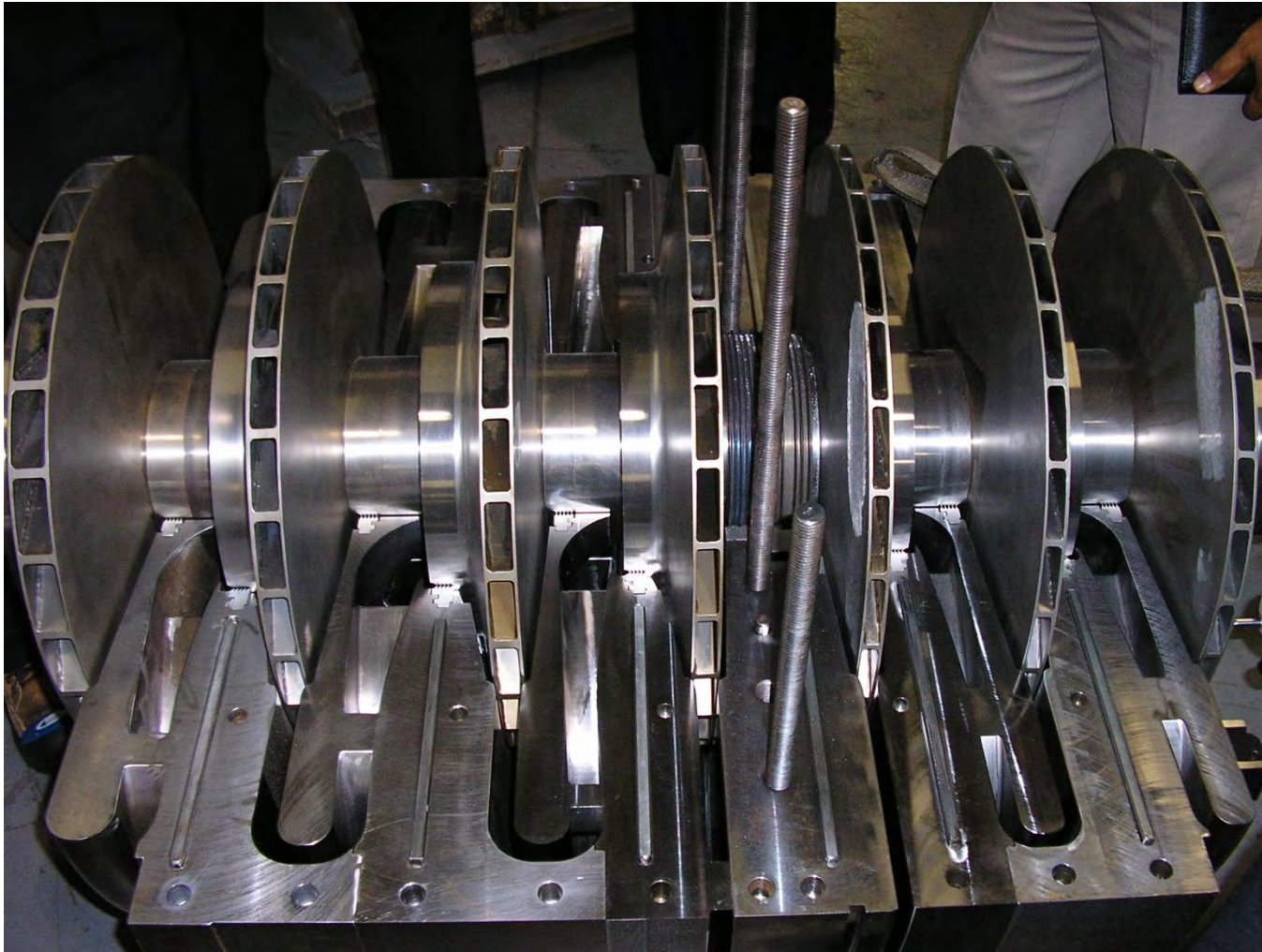
Impeller



Scalloped Impeller



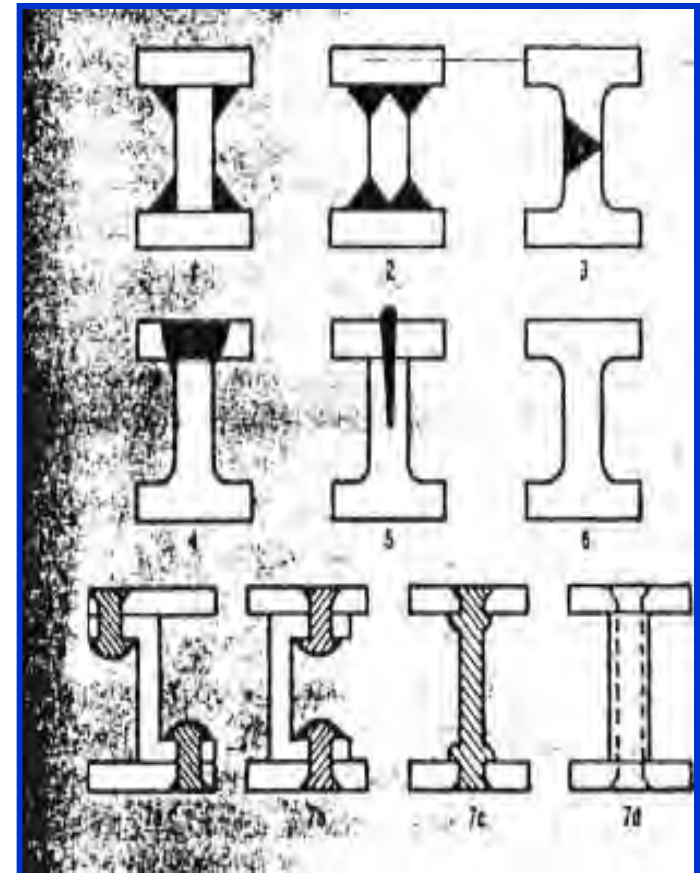
Impeller



Back-To-Back Impeller

Impeller are manufactured using five basic techniques:

- Milling & Welding (Fig.1 -5)
 - Governed by access to flow passages & cost.
- Single piece by casting or EDM (Fig. 6)
 - Casting used for lower size impeller.
 - EDM is slow & results in recast layer.
- Milling & Riveting (Fig.7)
 - Cheap & Fast method but lesser strength.
 - Vulnerable to fatigue cracking due to stress concentration around hole.
- Milling & Brazing.
 - Most used methods for modern impellers.



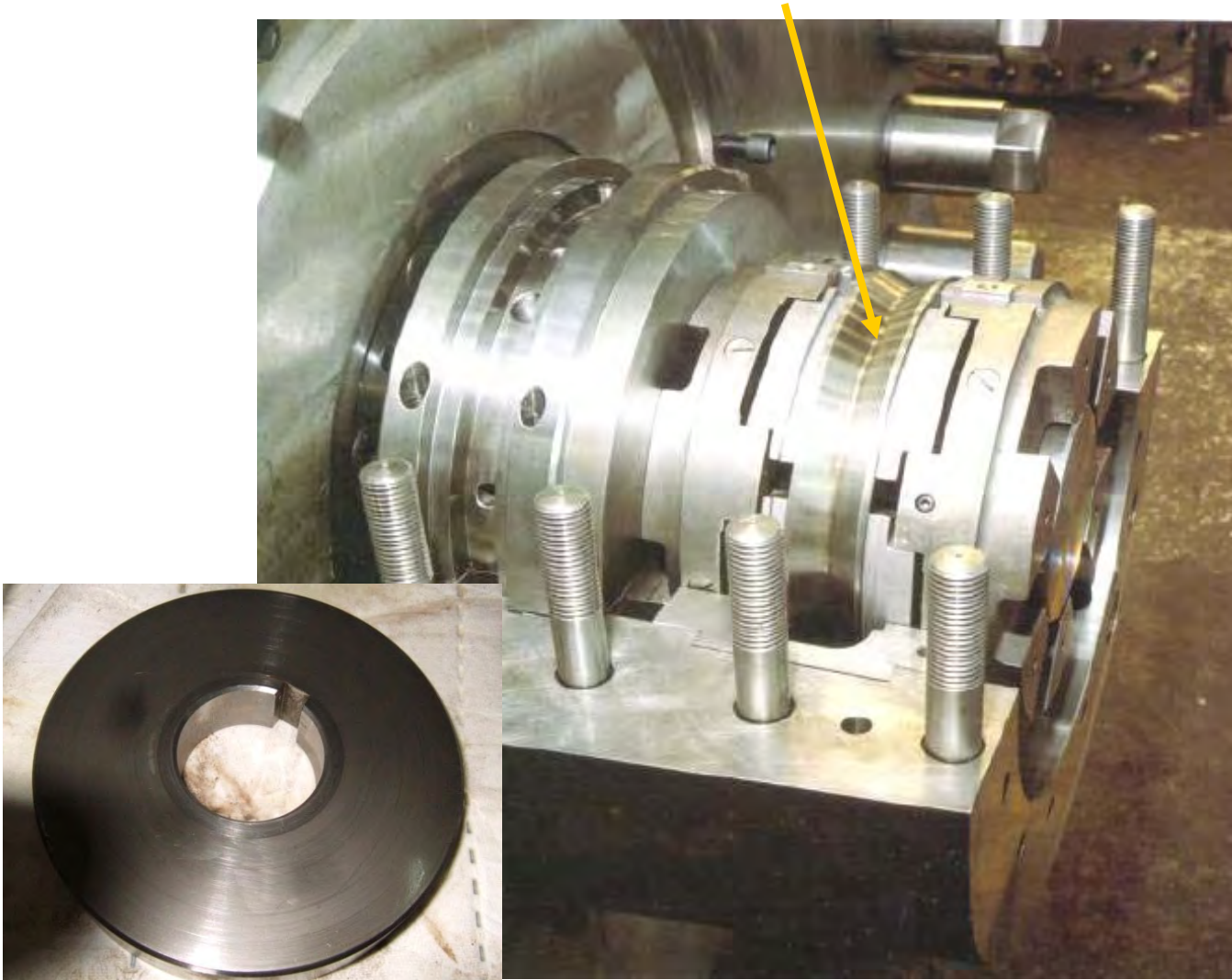
Common Impeller materials:

Key properties

- Weldability
- Strength at operating temperature
- Hardenability
- Corrosion Resistance
- Impeller disc & cover are low alloy steel forgings : AISI 4140 or 4340.
- For mild corrosive services involving moisture etc. AISI Type 410 (12% Cr steel).
- For sever corrosive service 17-4 PH or 15-5 PH
- Austenitic SS (SS 304 or 316) used very rarely because of low strength.

Material	Ultimate Tensile Strength	Yield Strength	BHN
AISI 4140	100,000	80 –90K	212-235
AISI 4340	120,000	105K	248-302
AISI 410	100,000	80 K	212-235
AISI 304	75,000	30K	205
AISI 316	75,000	30K	217
17-4 PH	130,000	105 K	277-341
Monel (K-500)	130,000	90K	250

Thrust Collar



API 617
Requirement
Both side surface
finish to 0.4um Ra
Both Face Run-Out
13um TIR

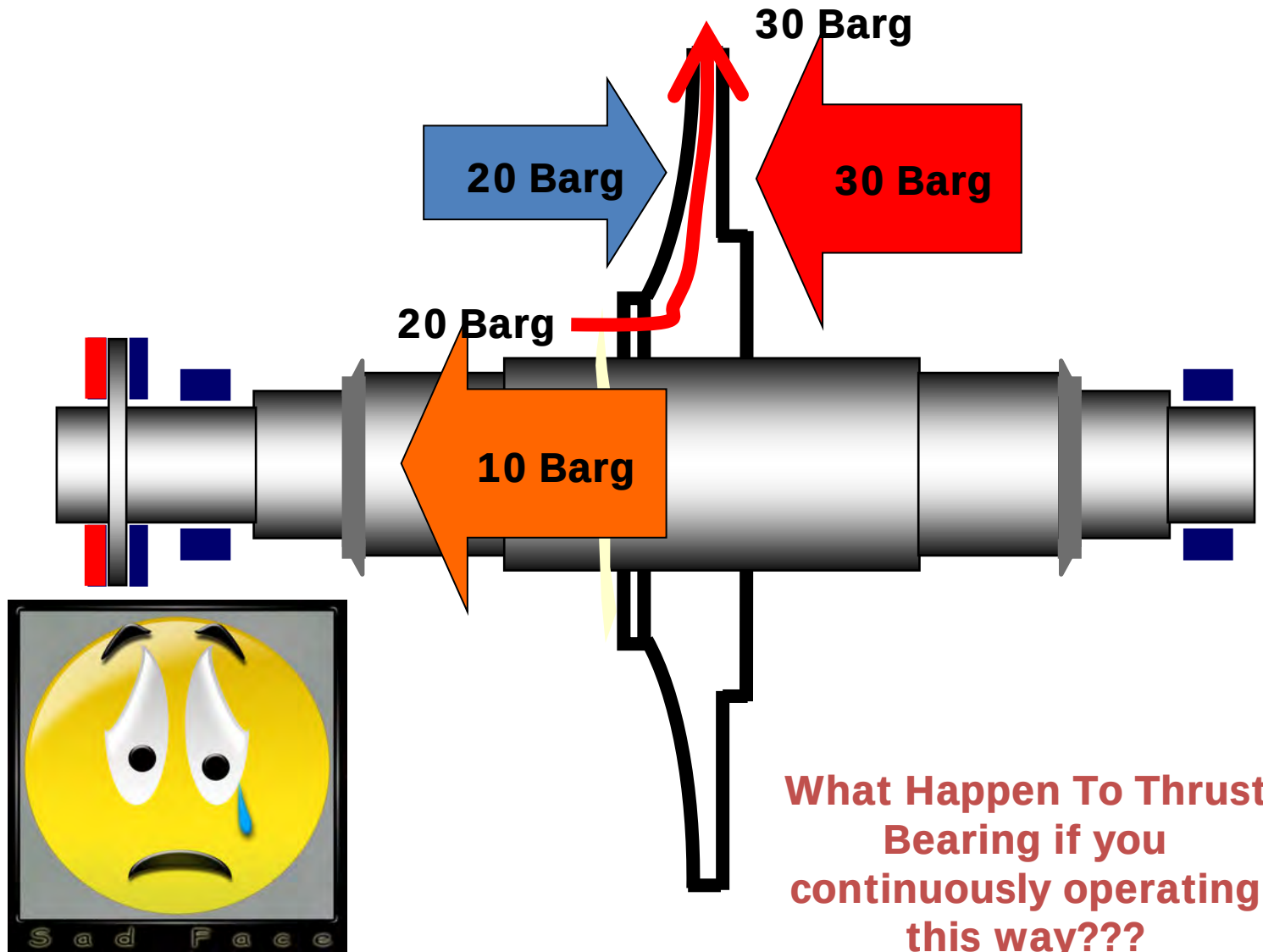
Balance Piston

- In centrifugal compressor due to accumulated differential pressures across impellers , there is huge axial thrust towards suction.
- To compensate part of the thrust balance piston are used.
- Normally balance piston compensate 70-75% of thrust & rest is handled by thrust bearing.



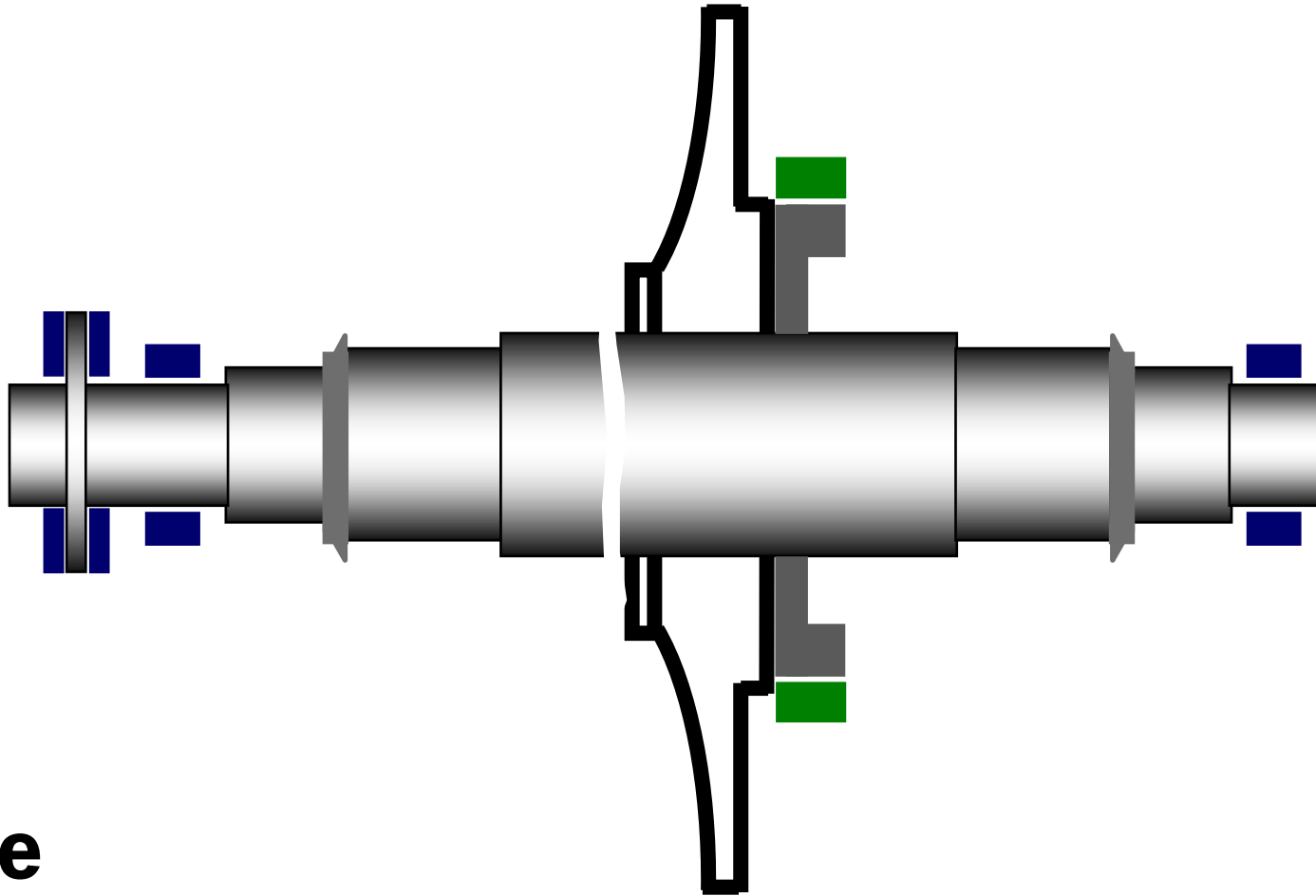
Thrust Load Control

Thrust Load Control



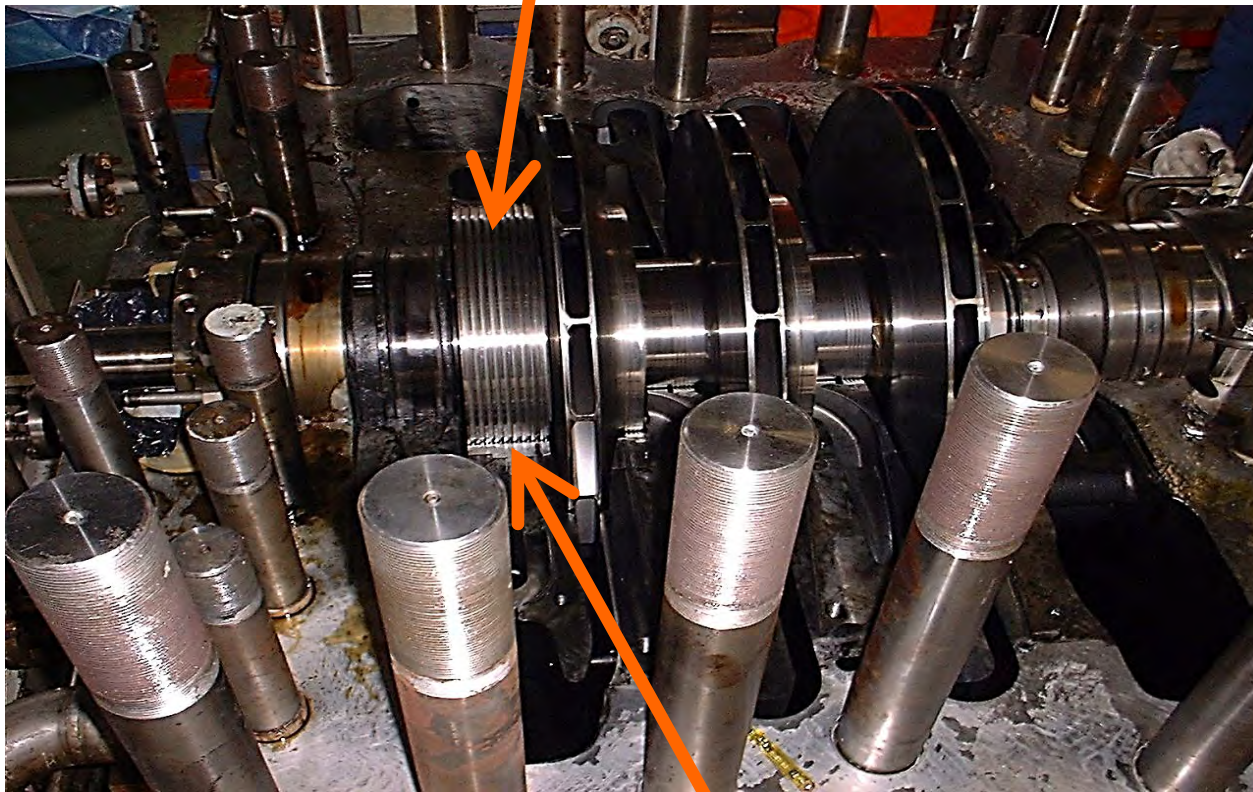


**Bearing Damaged Due to
Excessive Thrust Load**



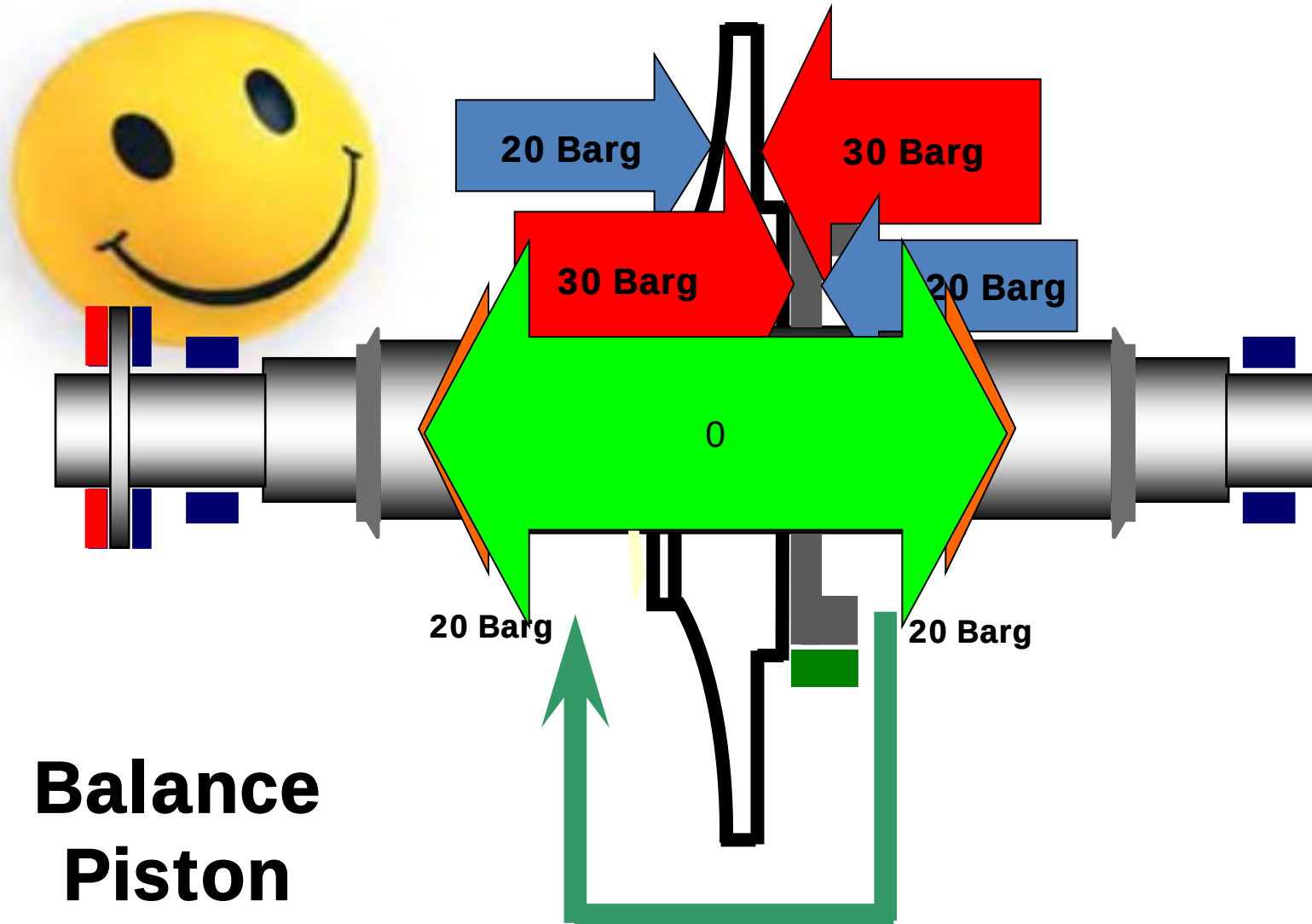
**Balance
Piston**

Balance Piston

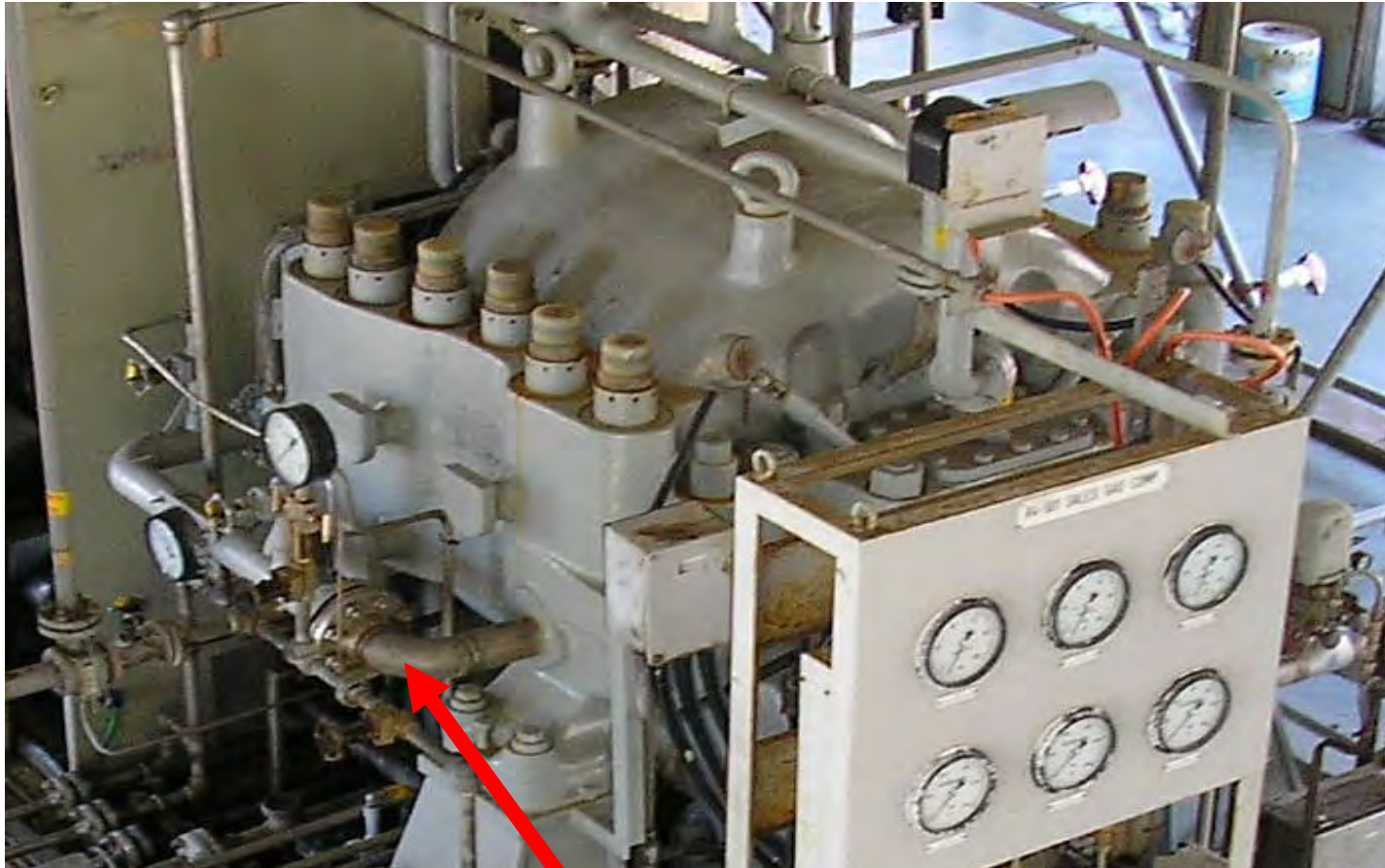


Balance Piston Labyrinth

Thrust Load Control



Thrust Load Control



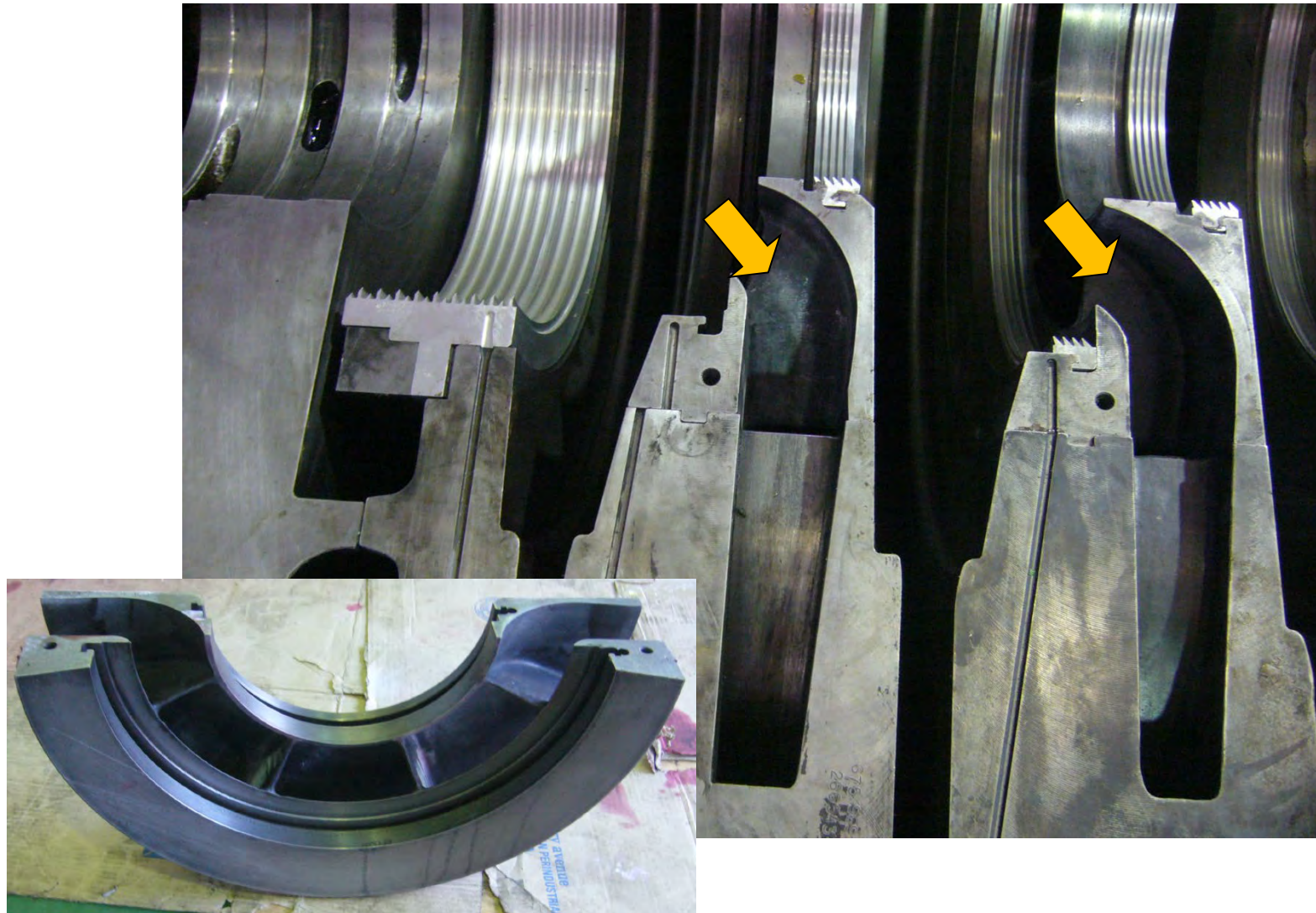
Balance Piston Line

Inlet Guide Vane

Located on the inlet side of the impeller reduces the turbulence in the gas stream & direct the flow to the eyes of the impeller

Labyrinth seals prevent gas compressed by impeller from recirculating back into the guide vane / impeller eye

Inlet Guide Vane



Diaphragm

A stationary aerodynamic element which serves as the dividing wall between individual impellers of a multi-stage compressor

Diffuser

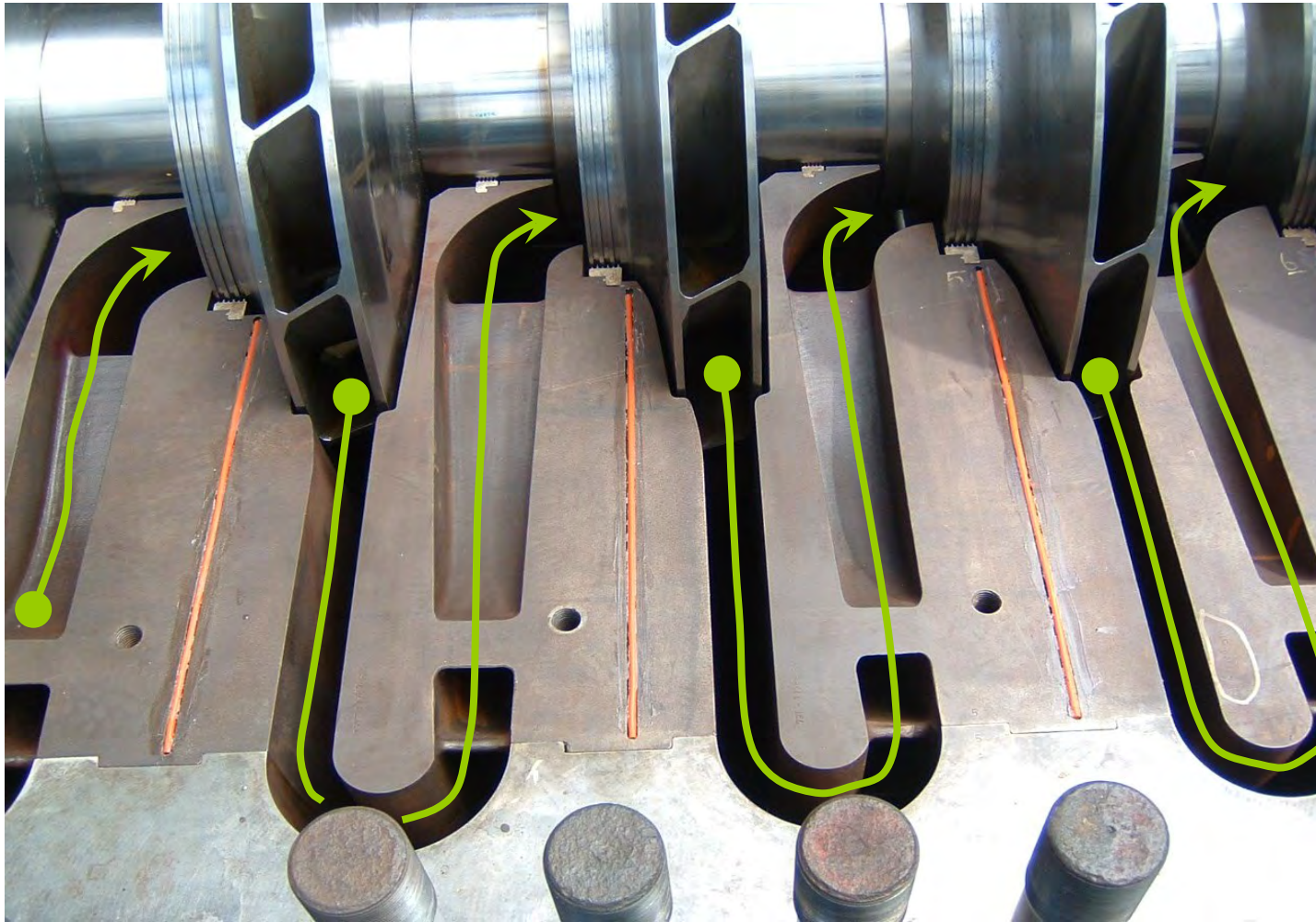
Stationary passage surrounding an impeller in which velocity imparted to the gas by impeller is converted into static pressure

The diffuser is formed by the backside of the diaphragm of the proceeding stage and front side of the diaphragm of the following stage

Diaphragm / Diffuser



Diaphragm / Diffuser

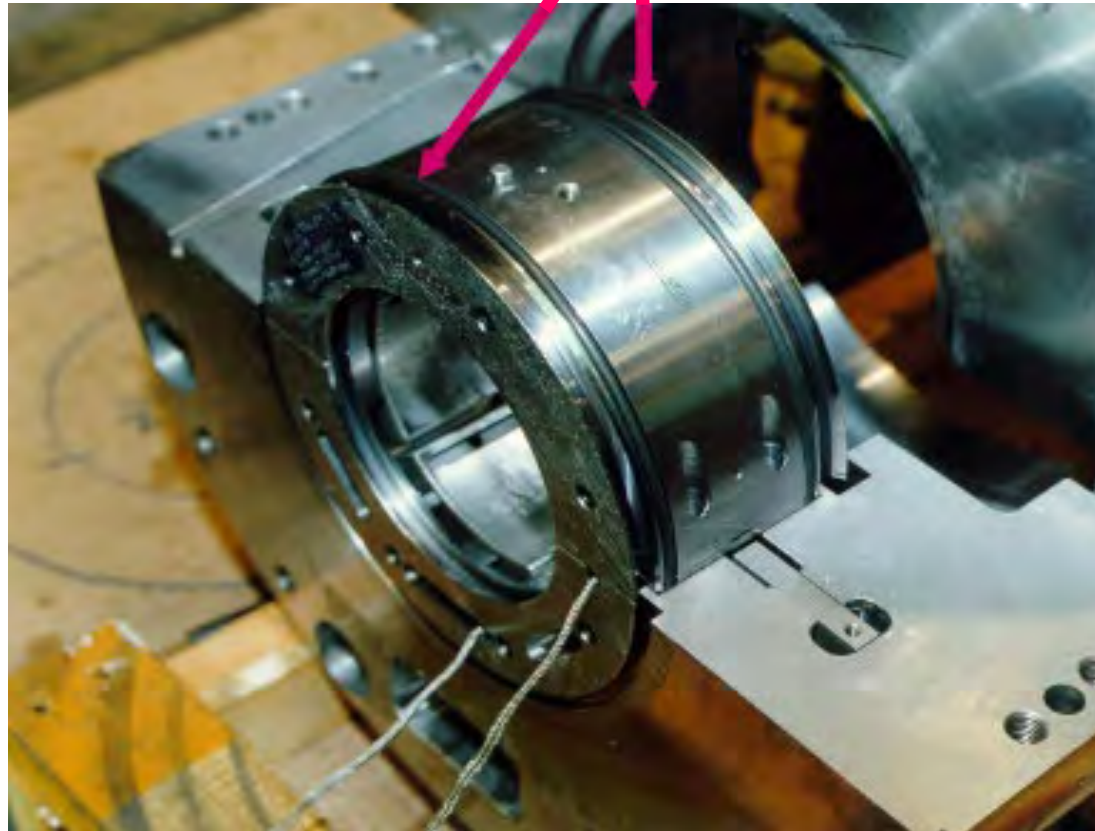


- Compressor rotor is radially located wrt to stationary components by radial bearing or radial load generated by unequal pressure distribution around impeller is supported by radial bearing.
- As per API-617 radial bearings are pressure lubricated & horizontal split for ease of maintenance.
- Most common material is steel backed liners with white metal lining.
- Because of higher speeds of centrifugal compressor & relatively light load, tilting pad bearings are used to suppress rotor instability.
- Clearance : 1-1.5 mils/inch of journal diameter



Journal Bearing

O-Ring



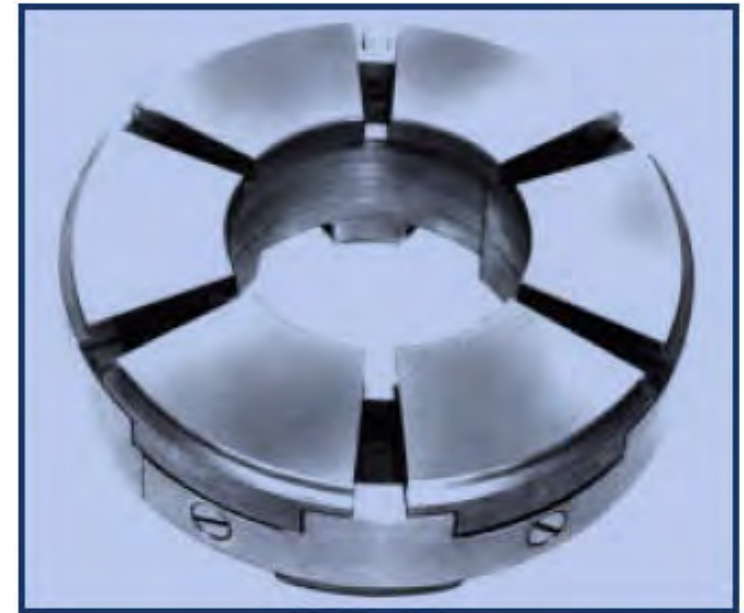
Damper Journal Bearing



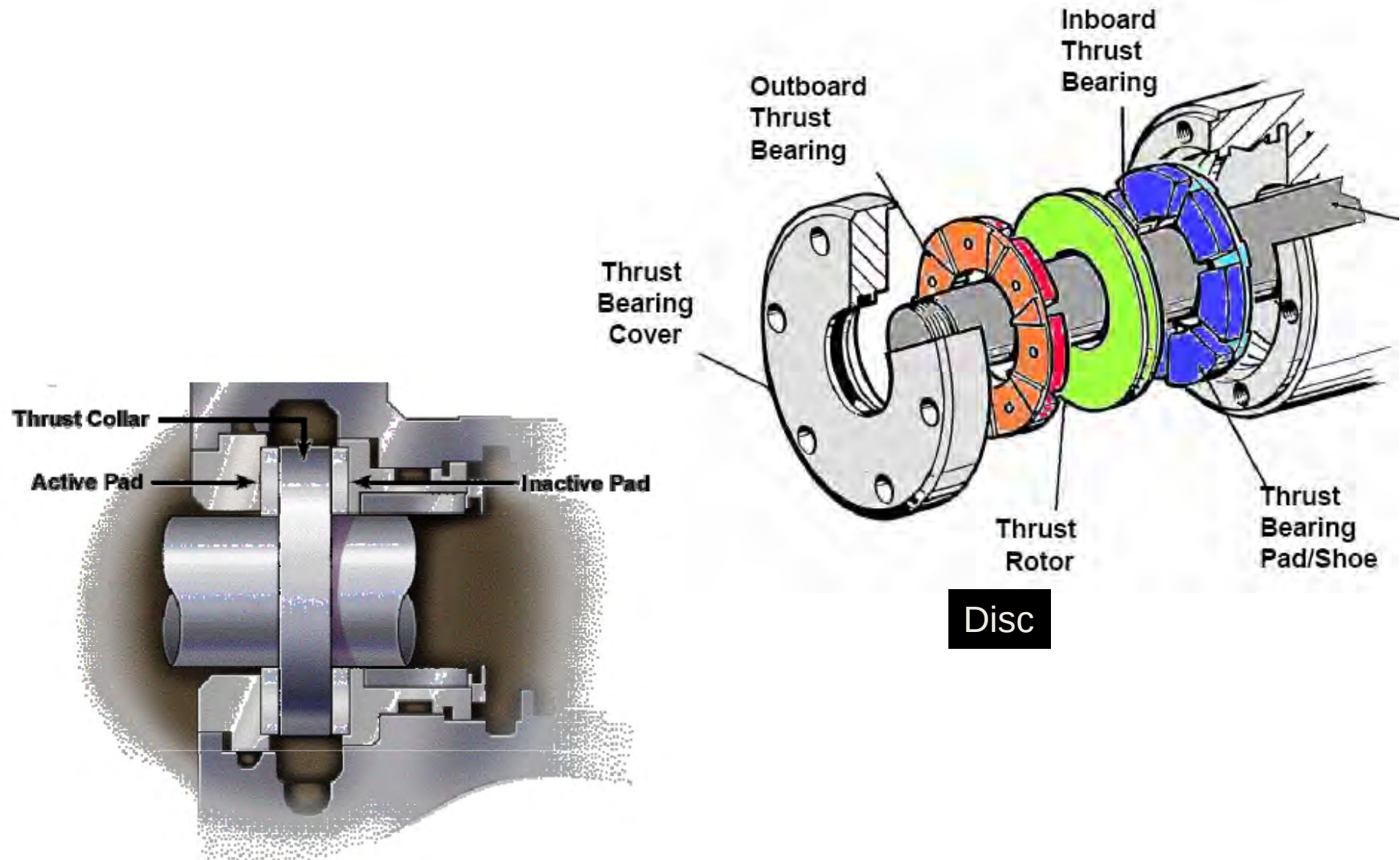
Journal Bearing with Thermocouple

Thrust Bearing

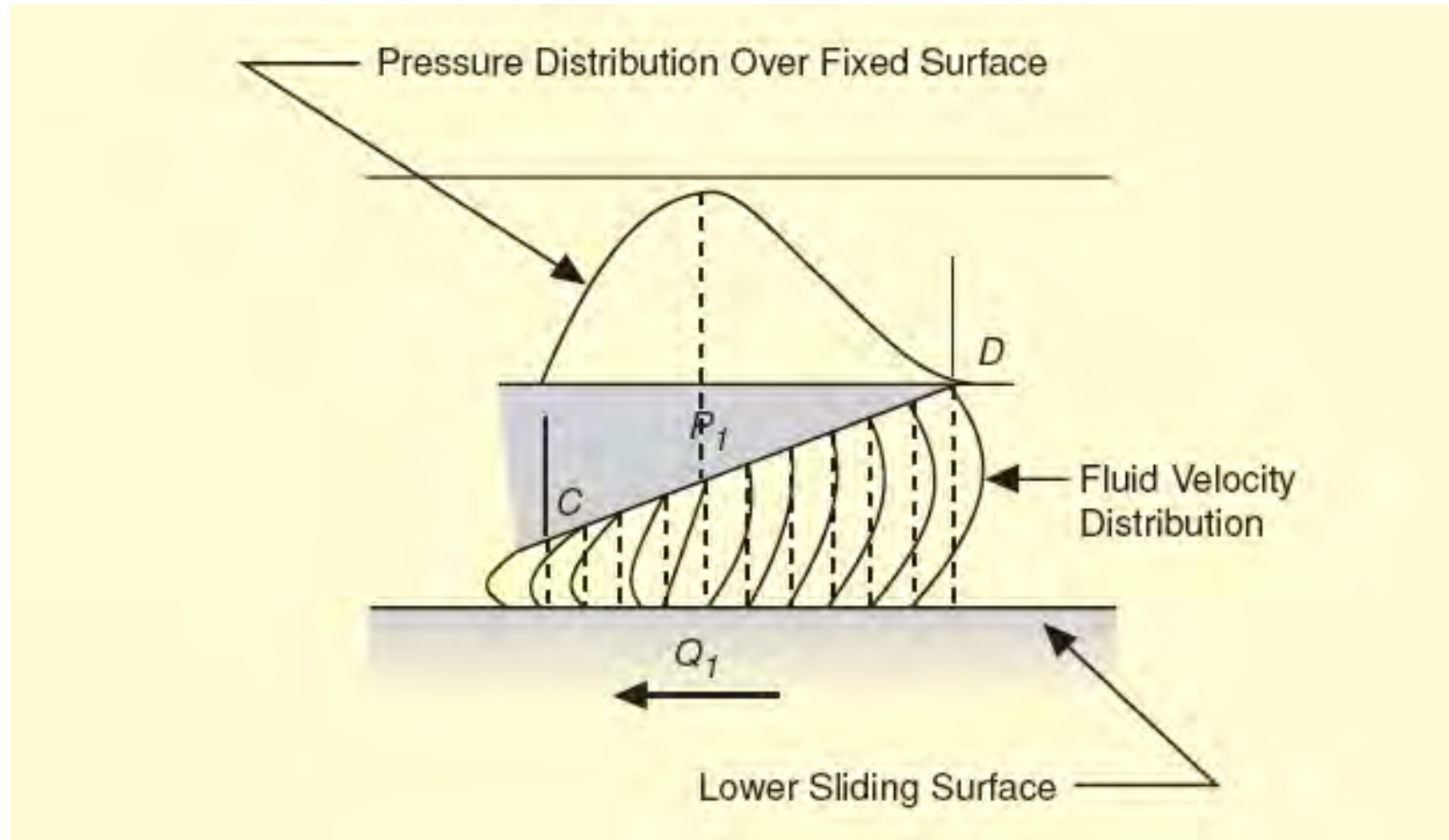
- Compressor rotor is axially located wrt to stationary components by thrust bearing. Axial thrust generated by pressure differential across impeller is taken by axial bearing.
- Most common material is steel backed liners with white metal lining.
- As per API-617, thrust bearing shall be designed for **double** than maximum possible thrust.
- Both thrust bearing, active and non-active shall have the same capacity.
- Tilting pad thrust bearings are common for compressors.
- Self equalizing thrust bearings are preferred.



Thrust Bearing



Thrust Bearing



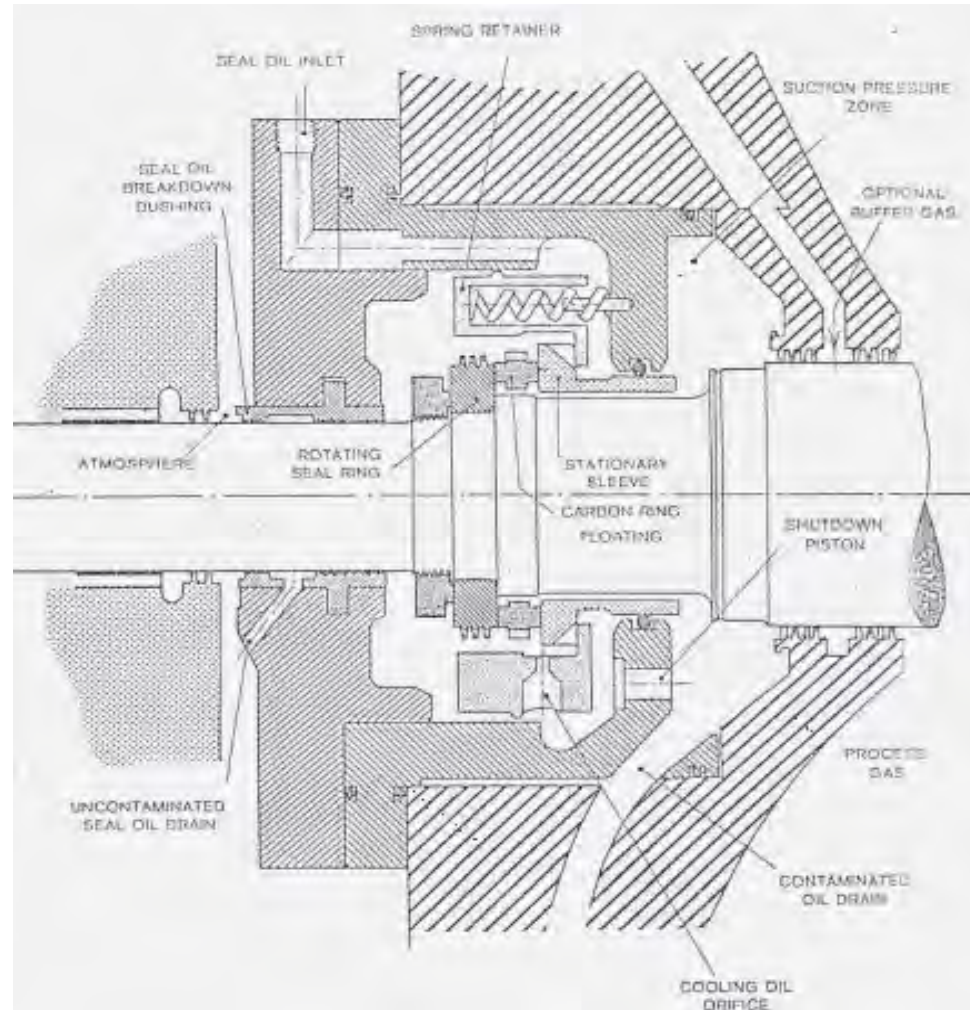
Shaft Sealing

- To Prevent or Restrict The Escape of Process Gas to Atmosphere or to Bearing Chamber Through Shaft-Casing Interface

Type Of Seal

- Clearance Seals
- Wet Seal
- Dry Gas Seal

Mechanical Contact Seal

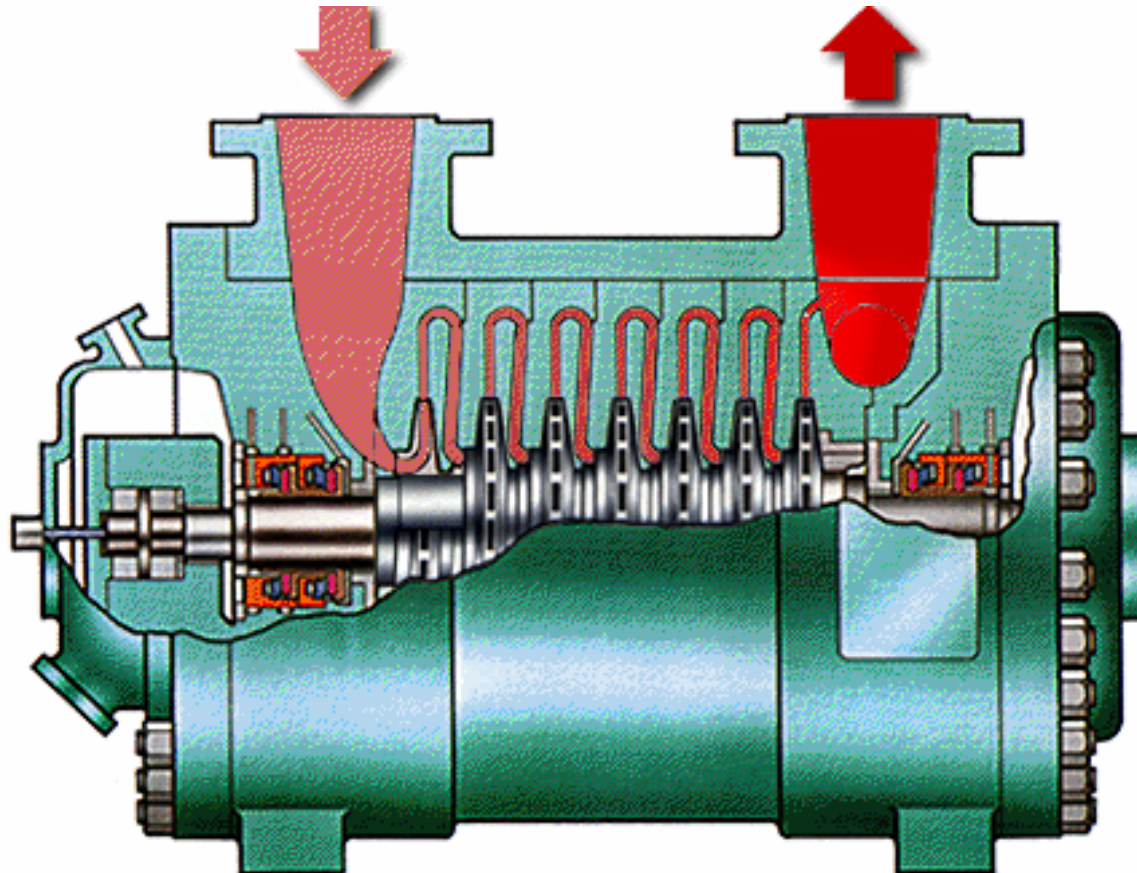


Dry Gas Seal



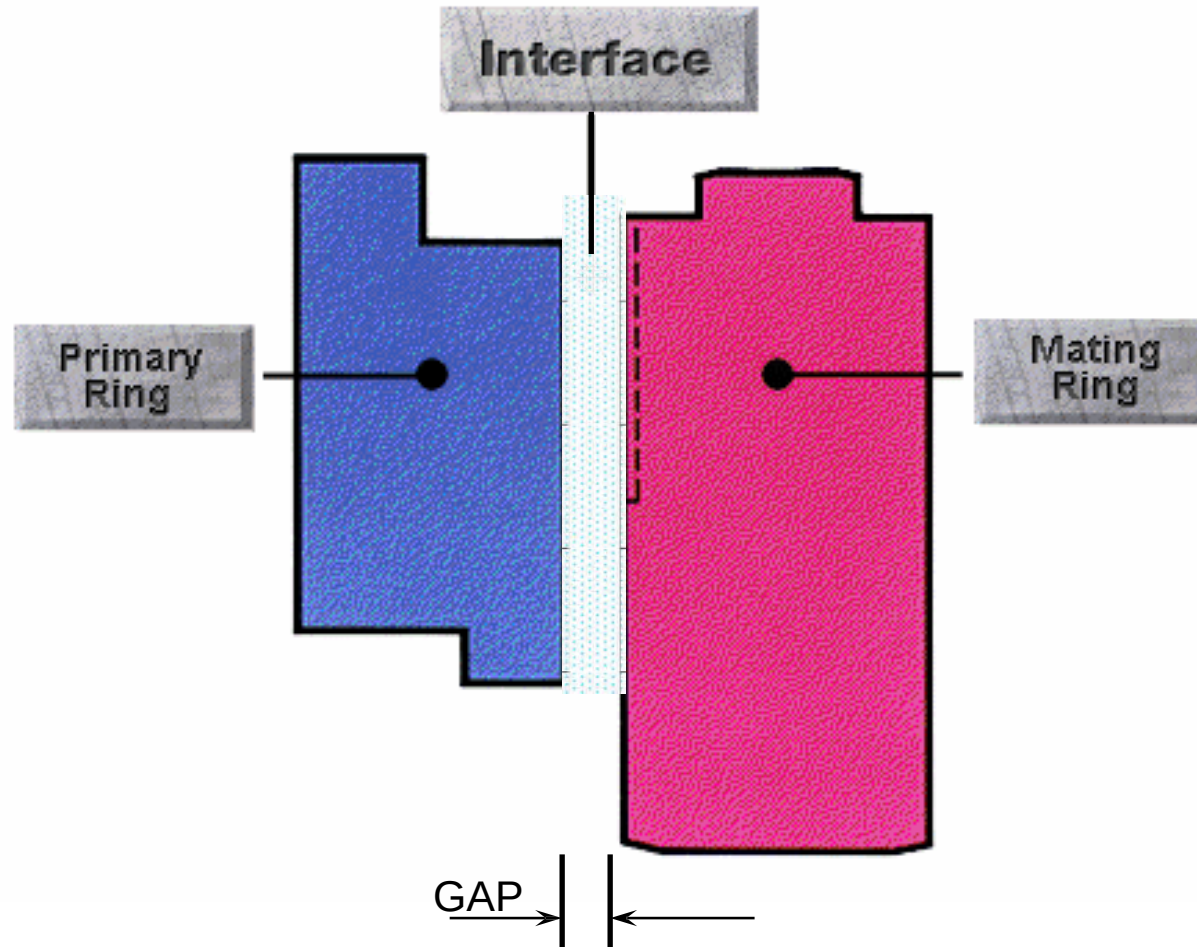
Single, Tandem, Double

Dry Gas Seal Overview



- Basic Concept
 - It is a non contact end face seal.
 - It could use process gas as a sealing medium.
 - Consists of a rotating part (mating ring) and stationary part (primary ring) .
 - Has primary and secondary sealing stage in a capsule.
 - Each stage is capable of sealing to full line pressure.

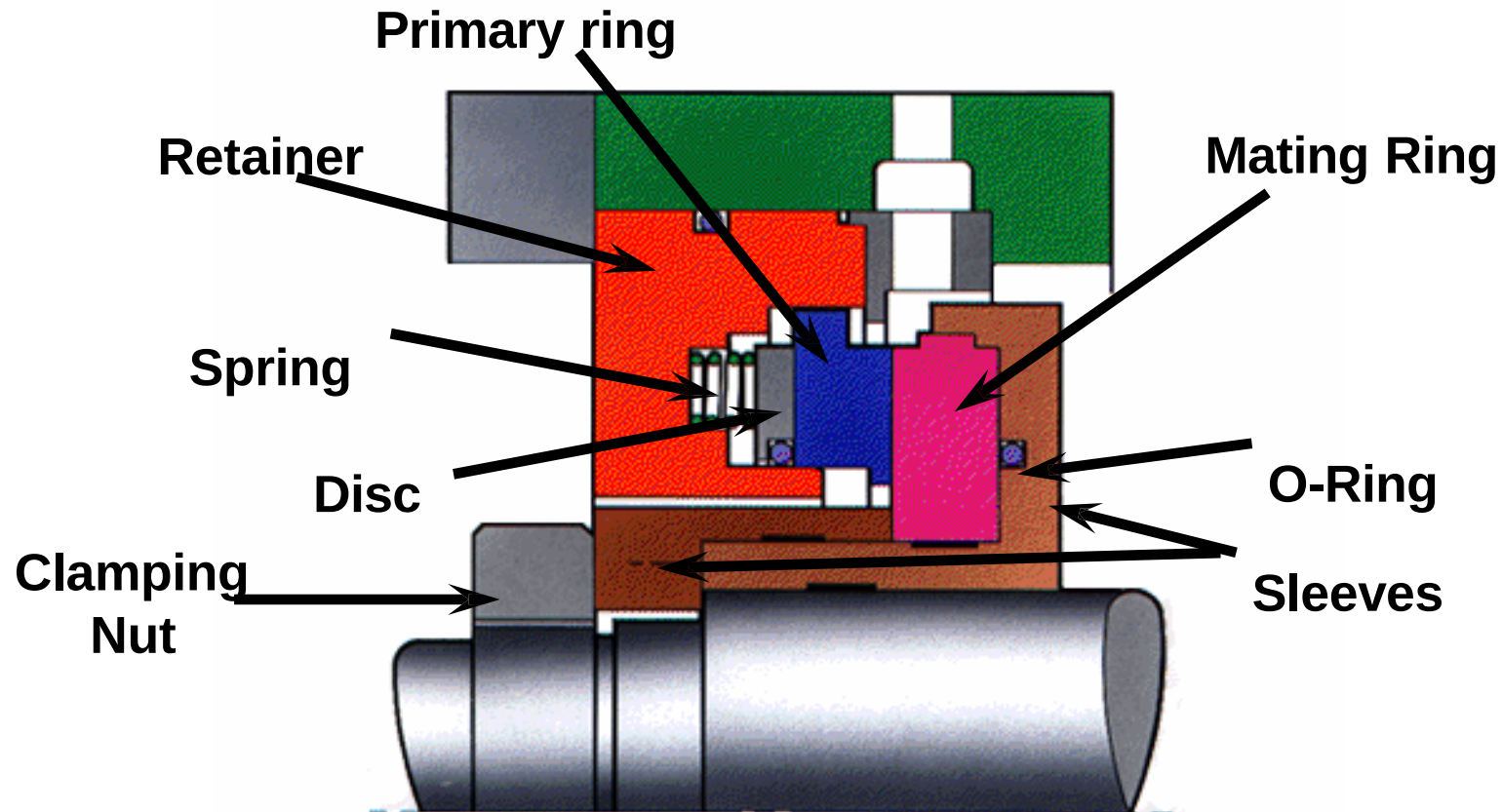
Dry Gas Seal Overview



Basic Components

- Primary Ring (stationary part).
- Mating Ring (rotating part).
- Springs.
- O-Rings.
- Retainers.
- Sleeves.
- Clamping Nut.

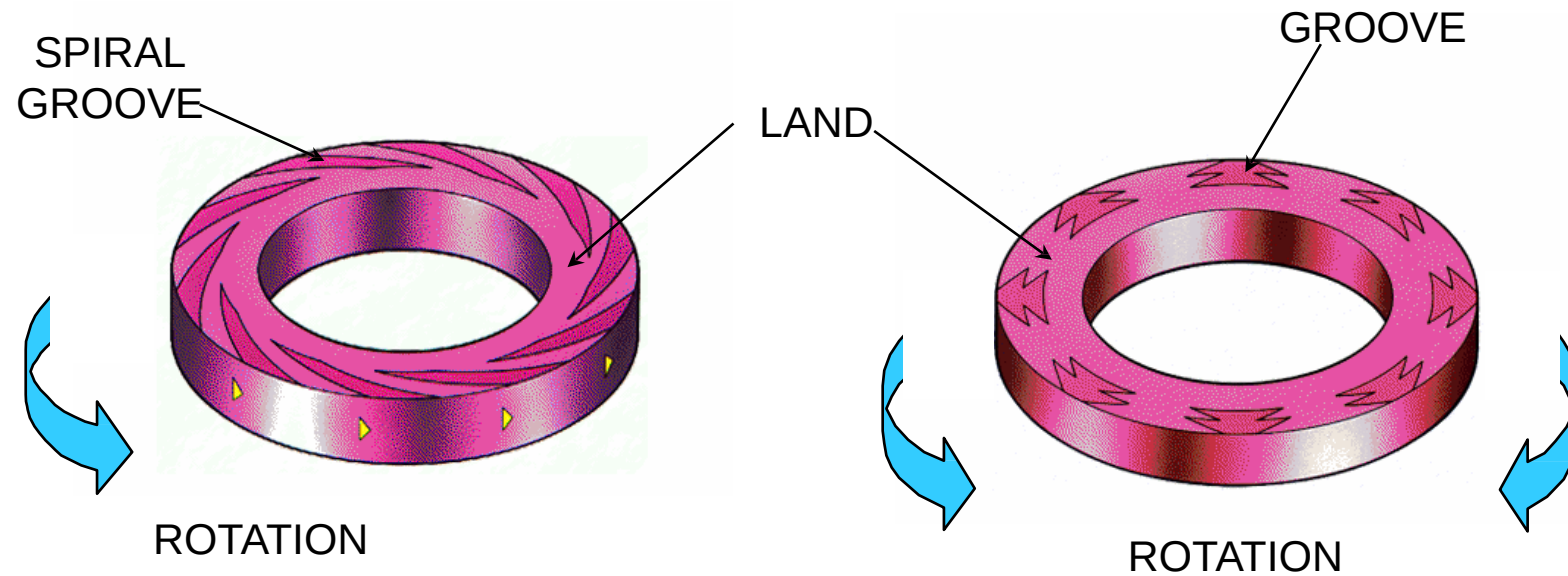
Dry Gas Seal Overview



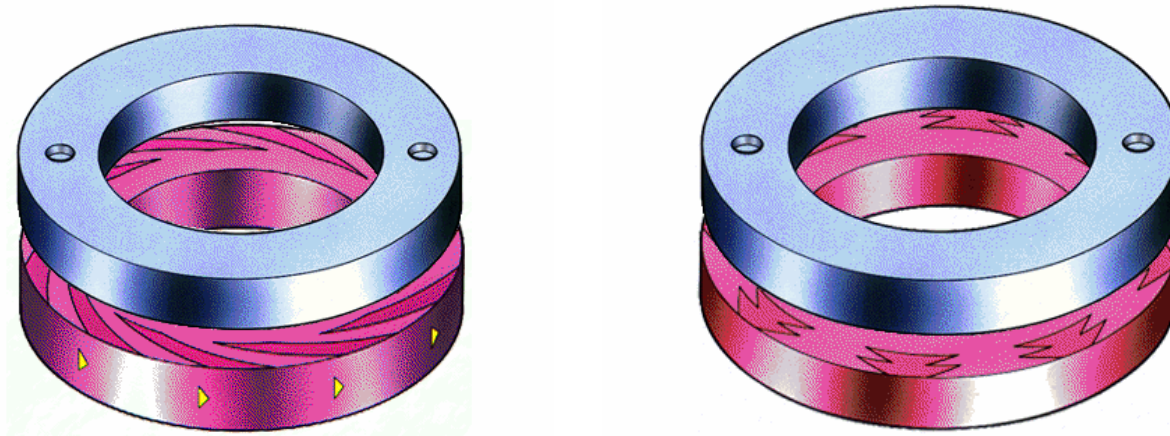
- General Concept
 - Under normal operating conditions, the primary stage seals against full process gas pressure while the secondary stage is unloaded.
 - The secondary stage only receives high pressure in the event of a failure of the first seal stage.

- General Concept (Cont'd)
 - The Labyrinth seals located in the seal capsule restrict the amount of gas which would leak out in the event of a total failure.
 - An air barrier is used to separated the seal and bearing cavities

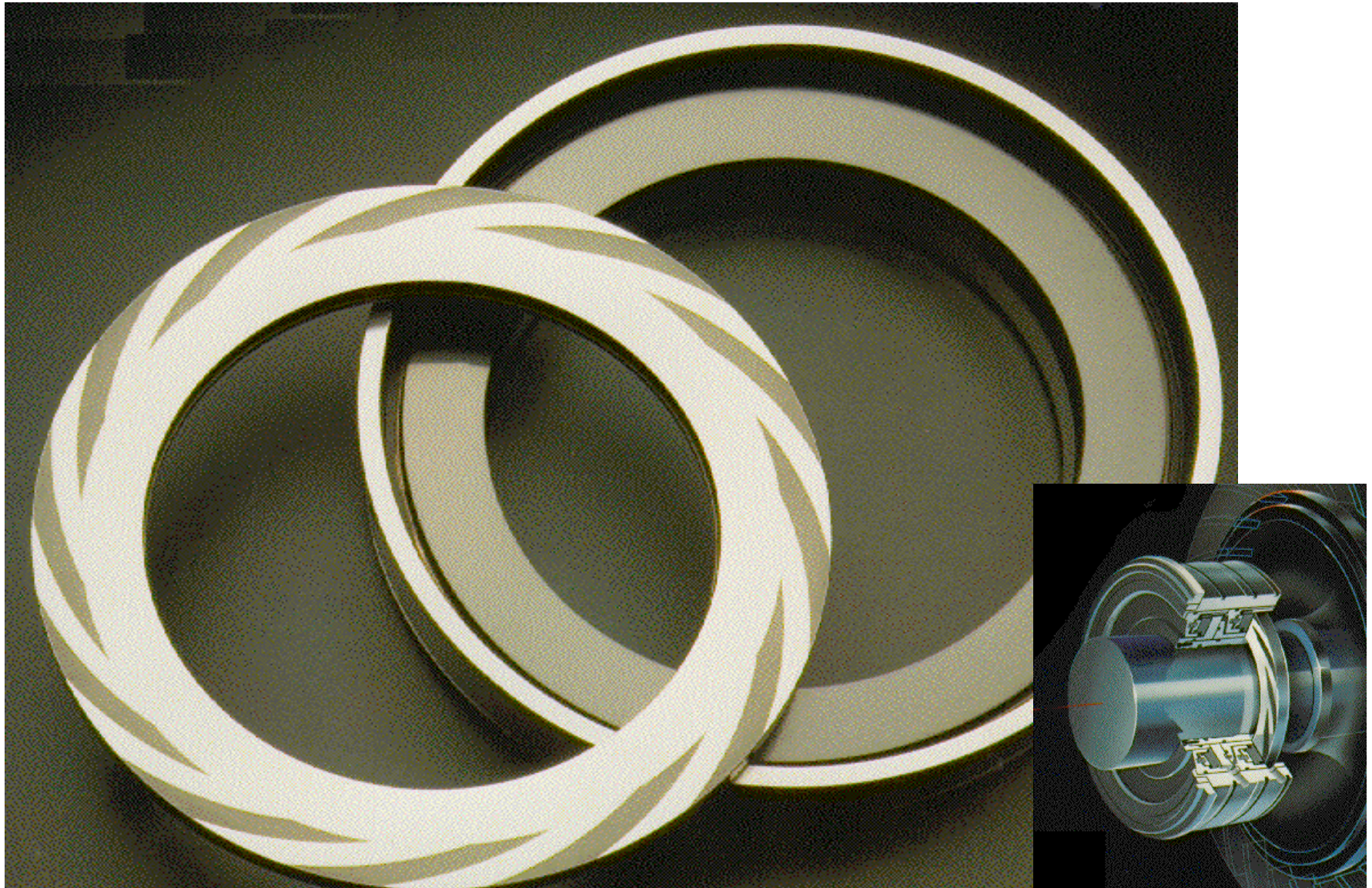
Groove Pattern



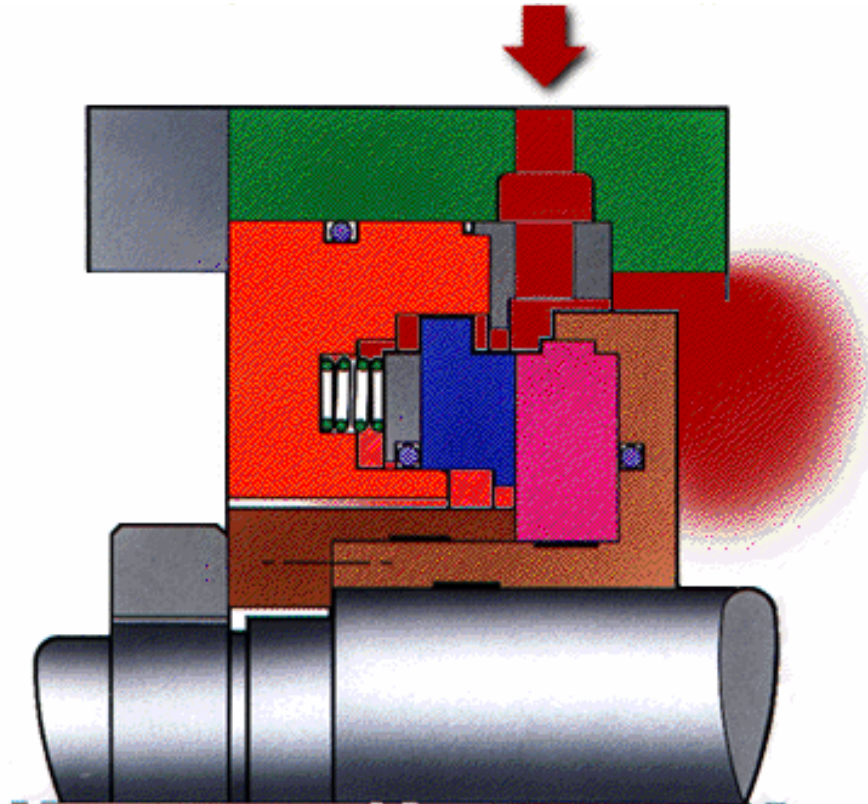
Groove Pattern



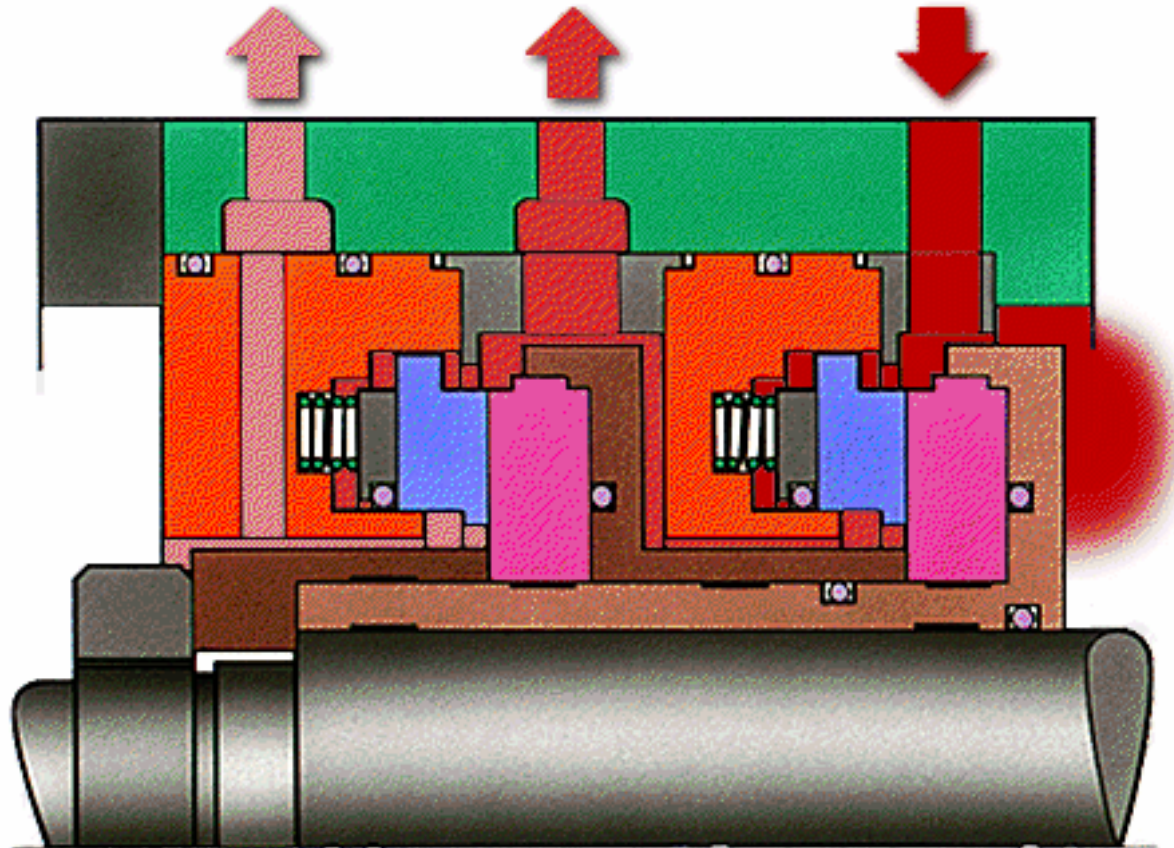
Dry Gas Seal Overview



Dry Gas Seal Overview



Dry Gas Seal Overview

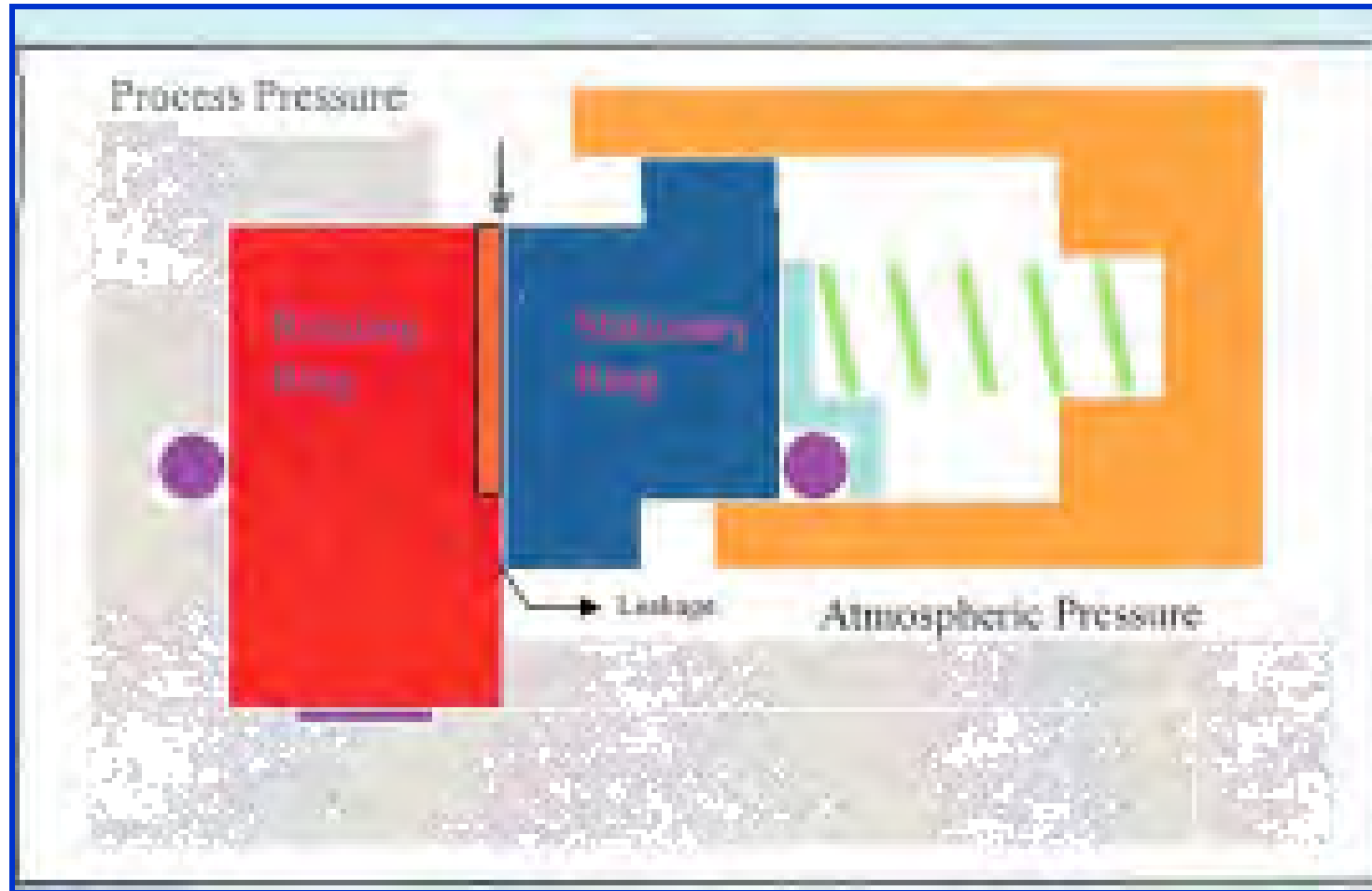


Dry Gas Seal

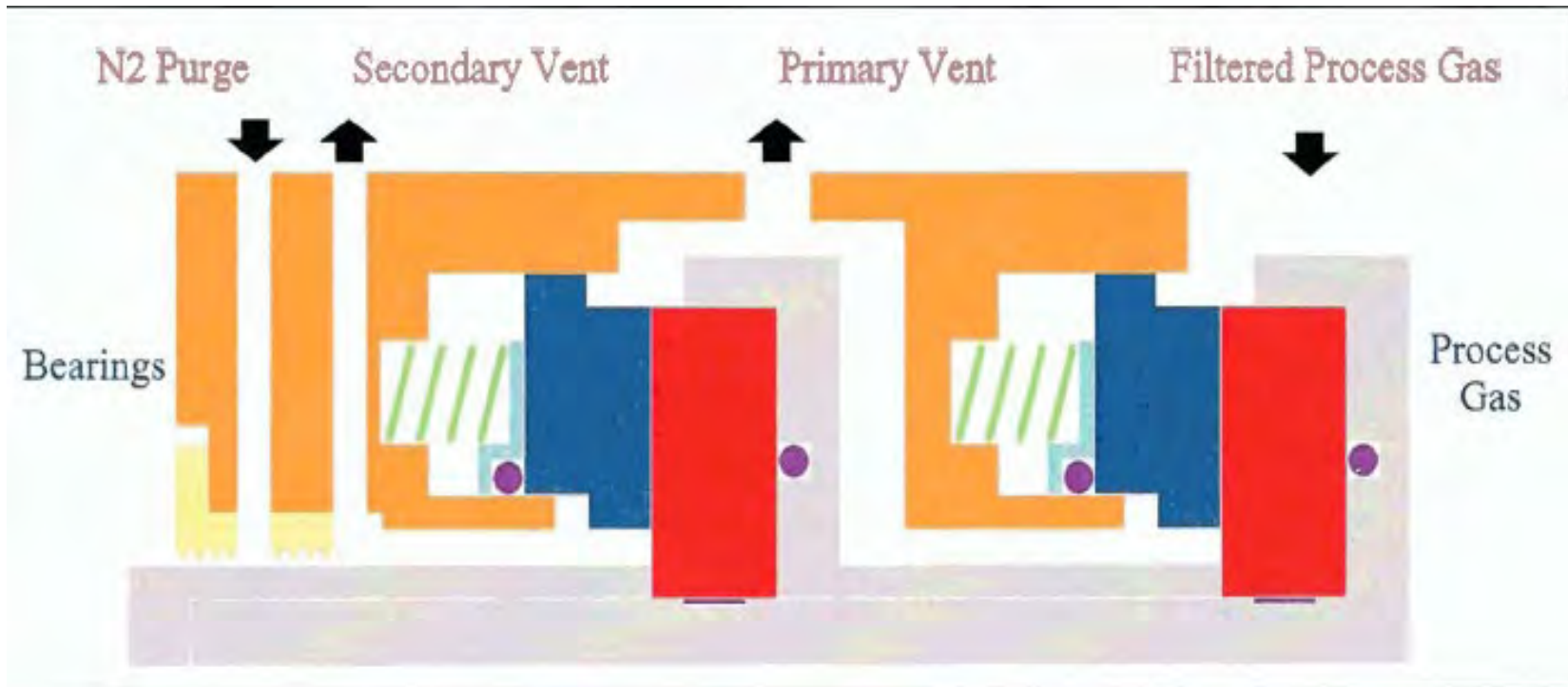
- Dry gas seal is the latest technology in shaft sealing, as name implies it runs dry & no liquid is required for sealing/cooling.
- The spiral groove technology is patented by John Crane for dry gas seal.
- With invention of dry gas seal technology the reliability of centrifugal compressors is greatly enhanced.
- The reliability of the seal is much superior than earlier versions of seal i.e liquid film seal, Mechanical seal, Iso-carbon seal etc.
- Some facts :
 - The dry gas operates on very fine gap of 2-3 microns:
 - Very sensitive to dirt , liquid & Sustained reverse rotation.
 - Dry gas always have minor leakage.



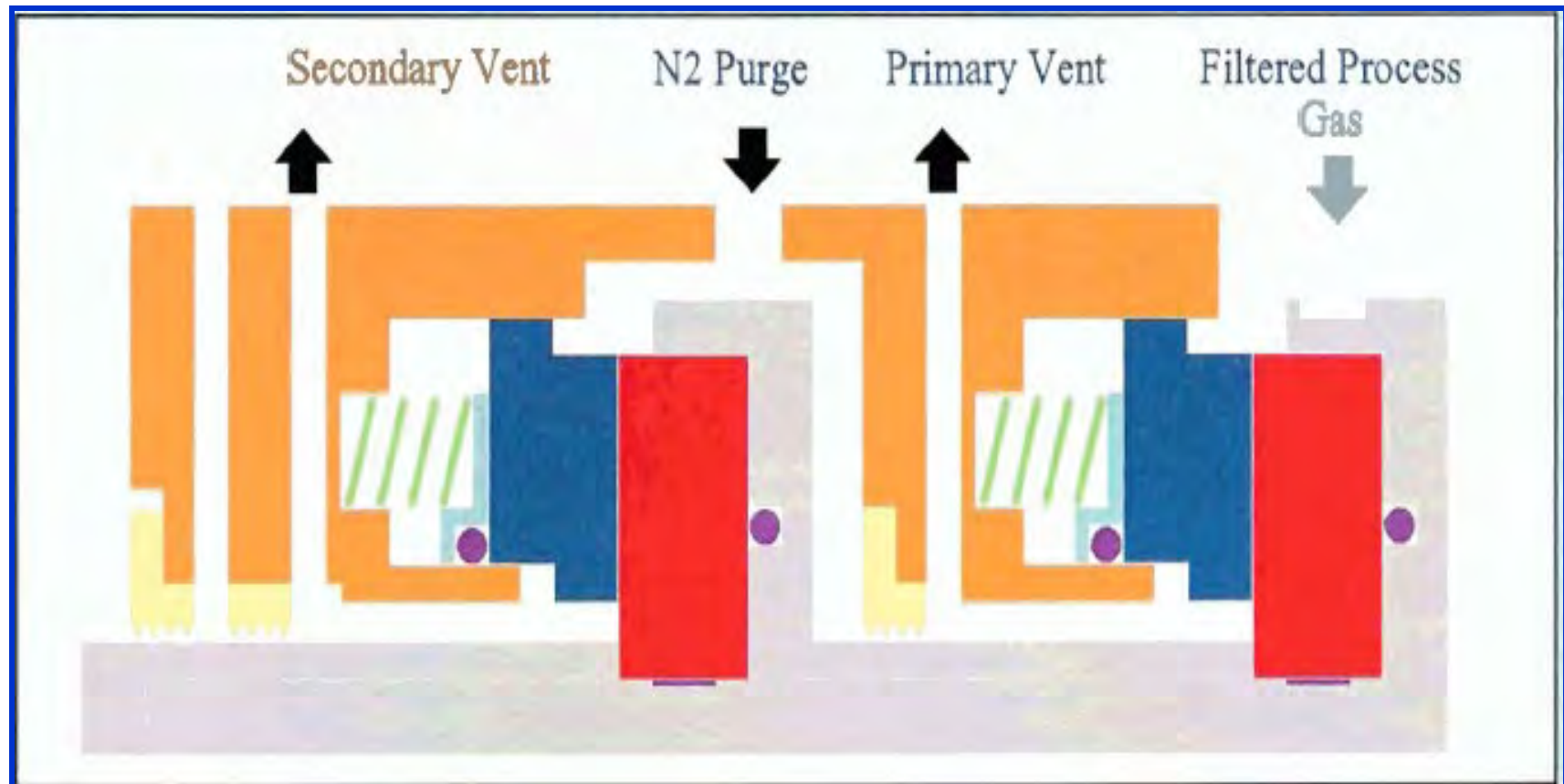
Single Dry gas seal



Tandem Seal

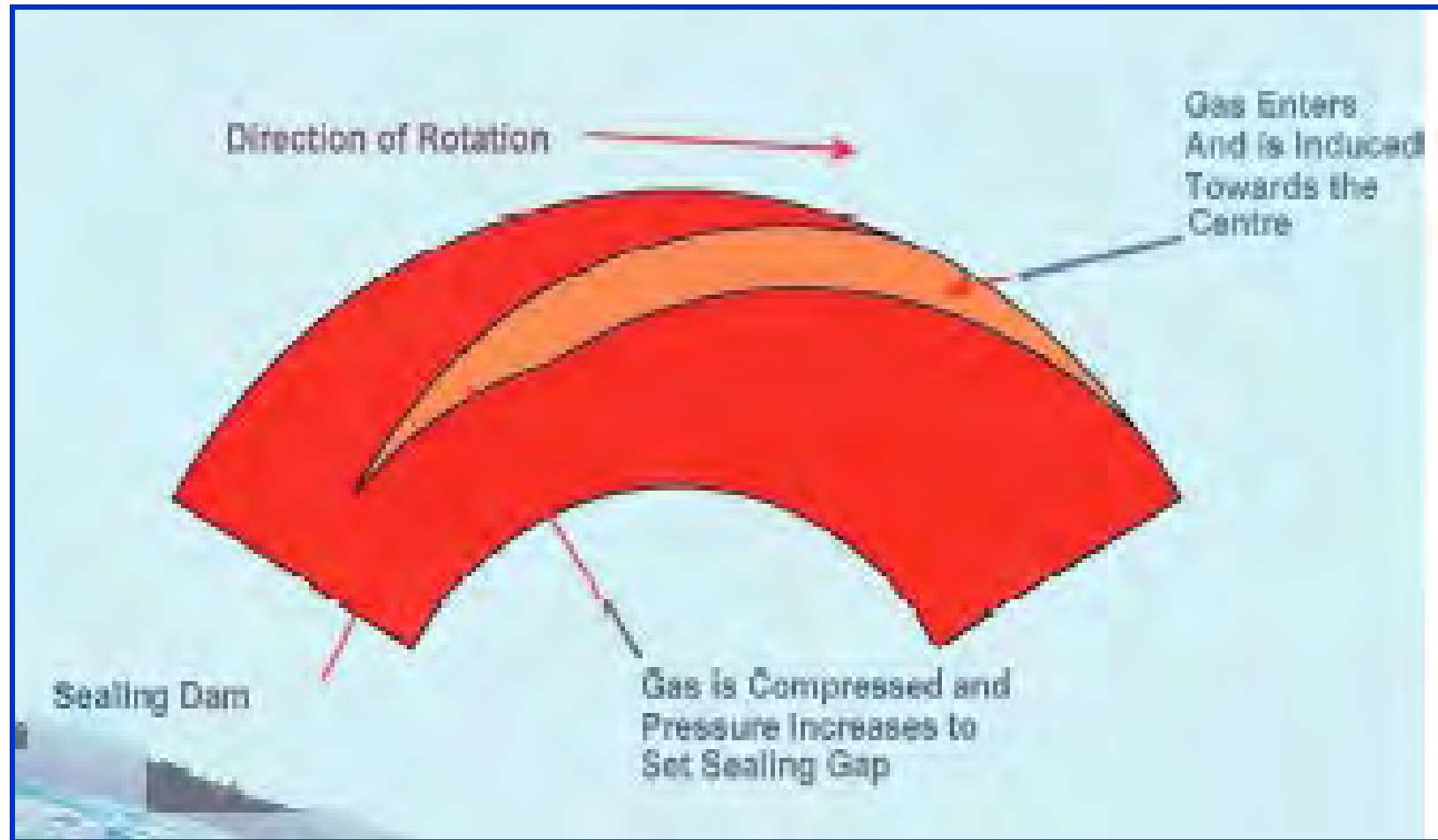


Tandem Seal with Labyrinth

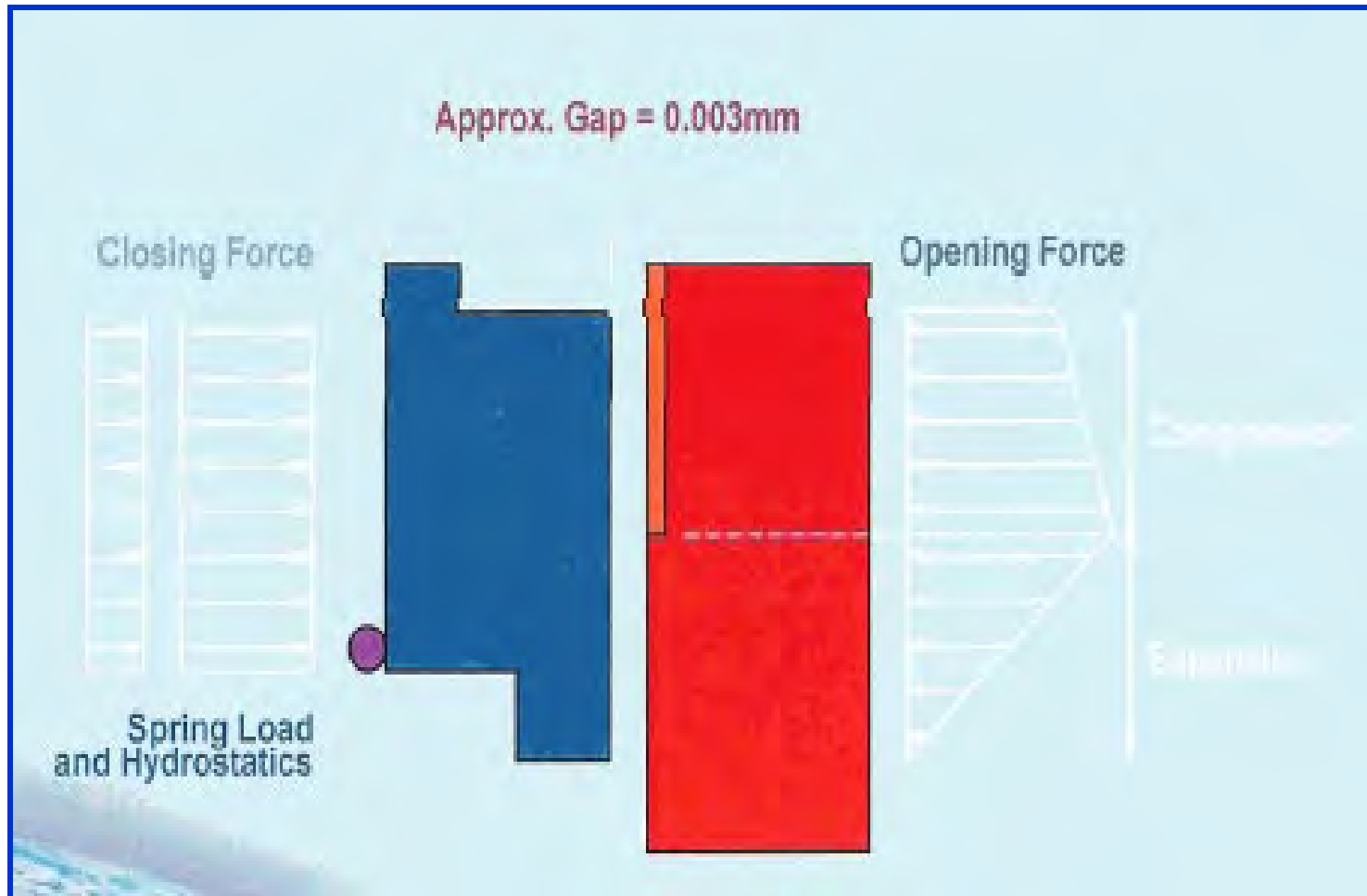




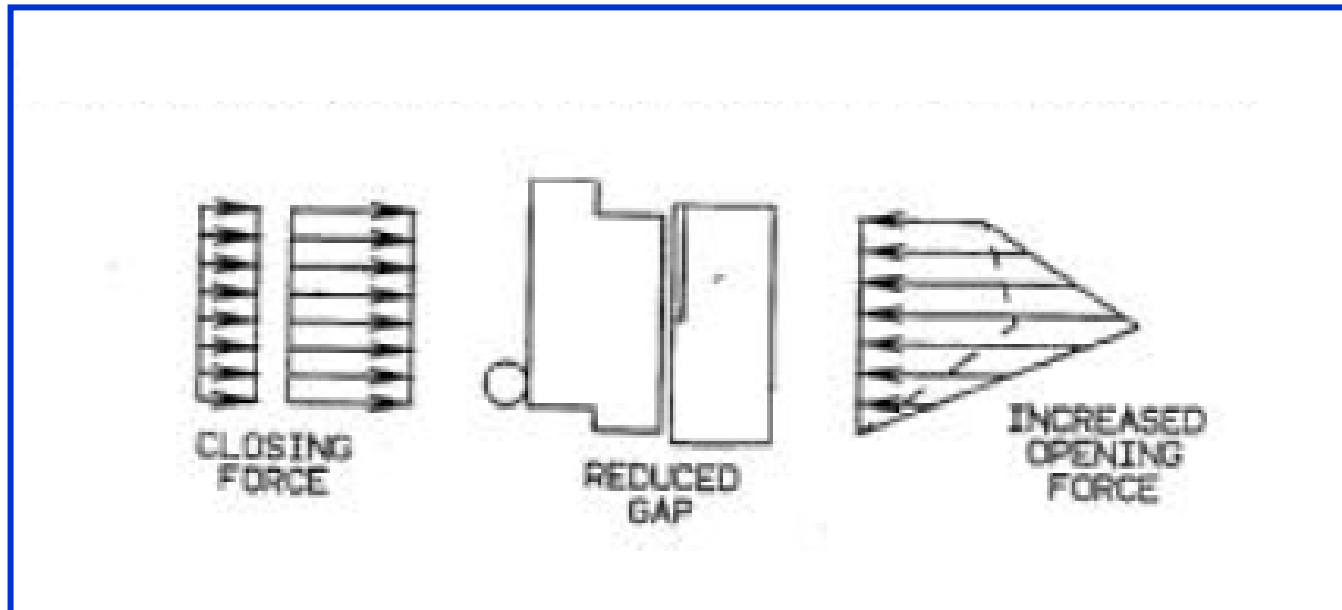
Dry gas Seal Operation

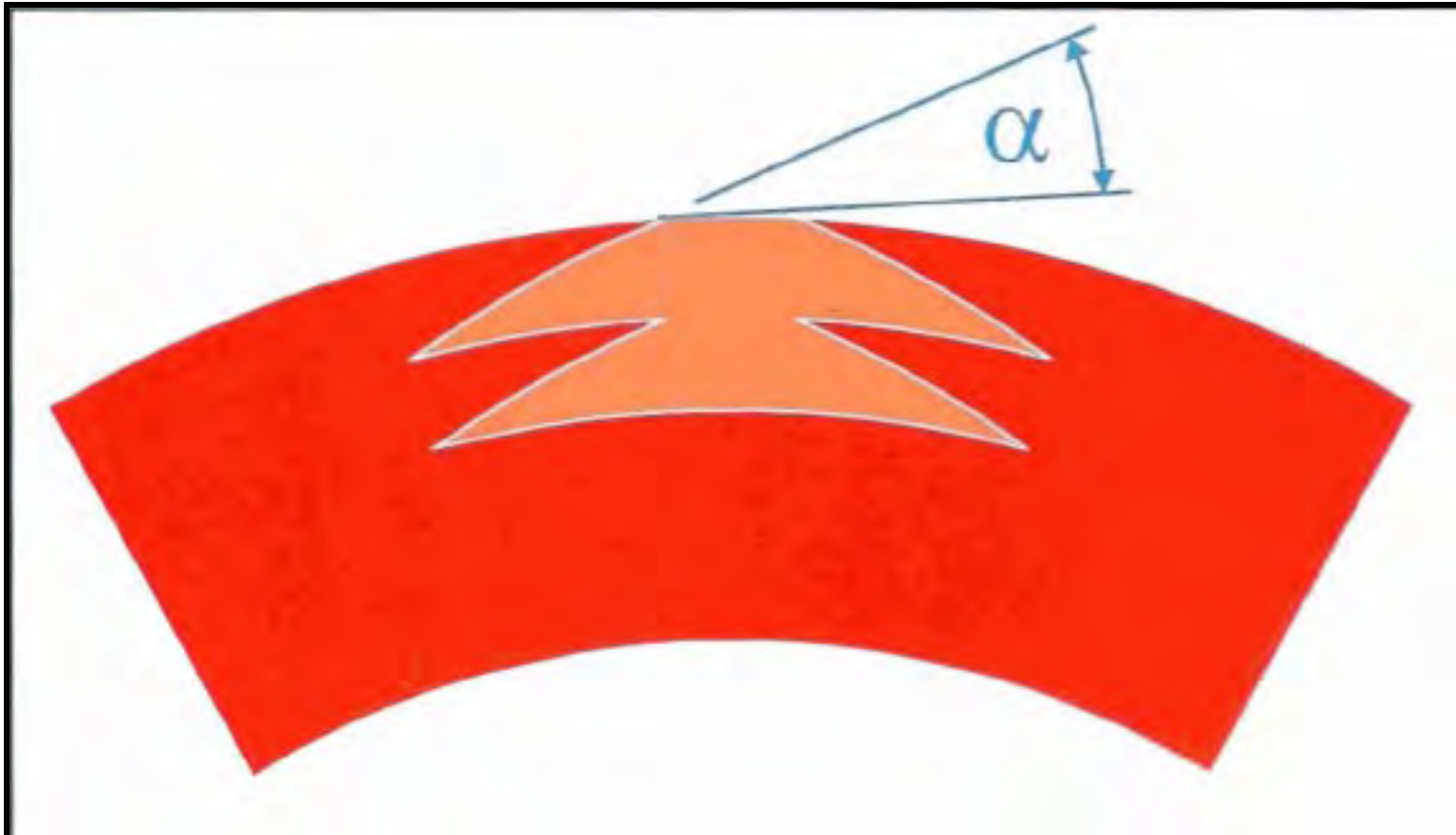


Seal Normal Operation



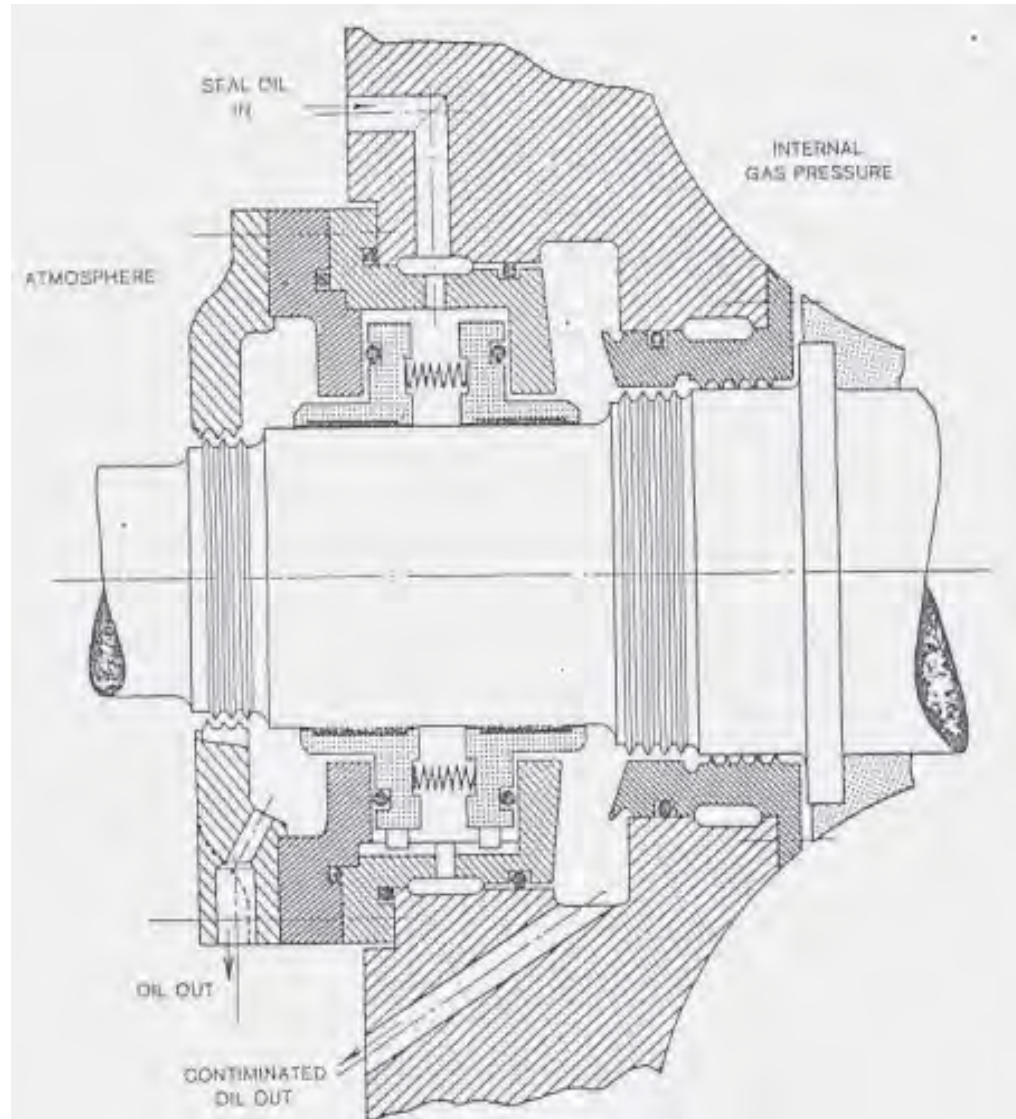
Self Adjusting Feature

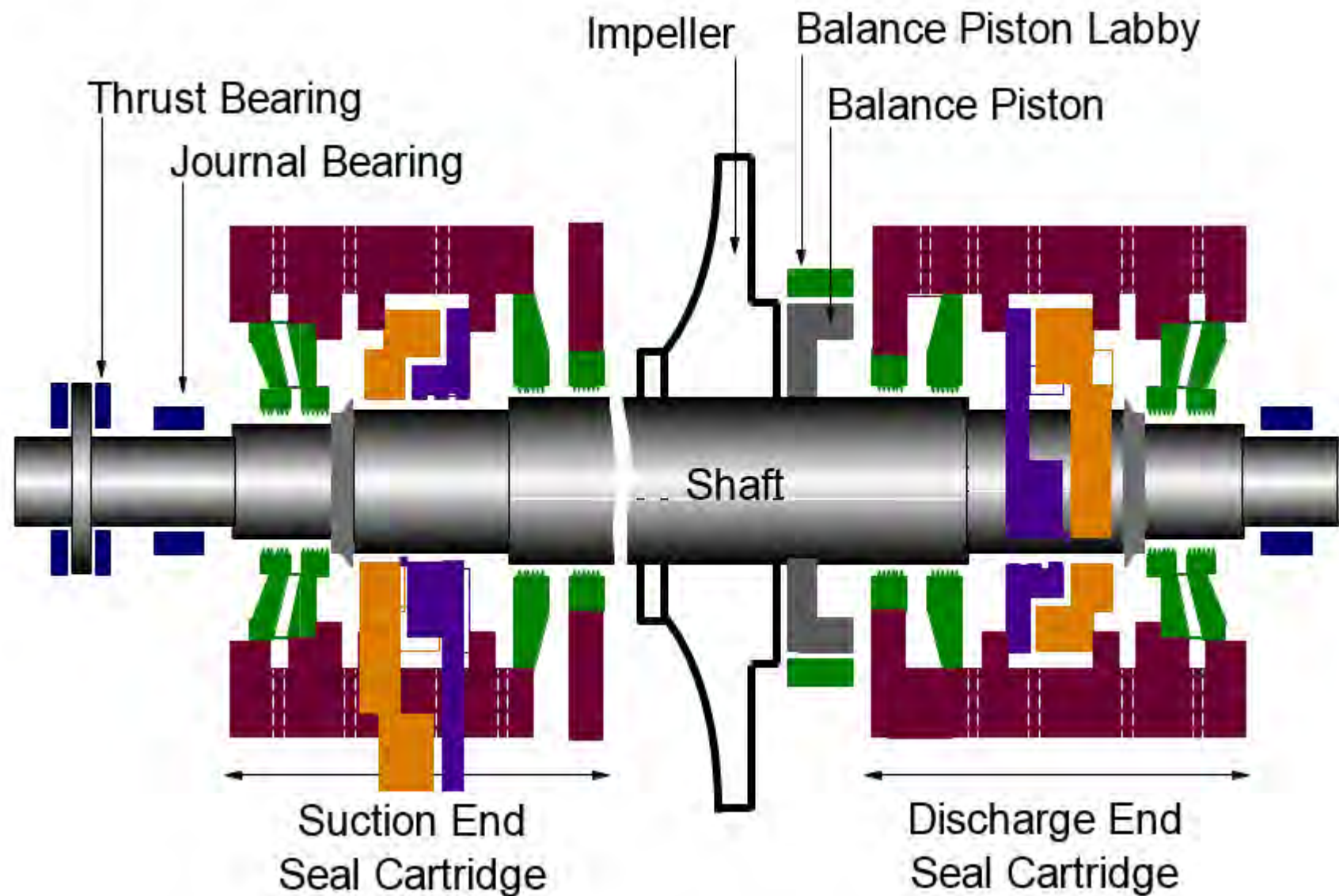




Bidirectional Groove Design

Wet Seal



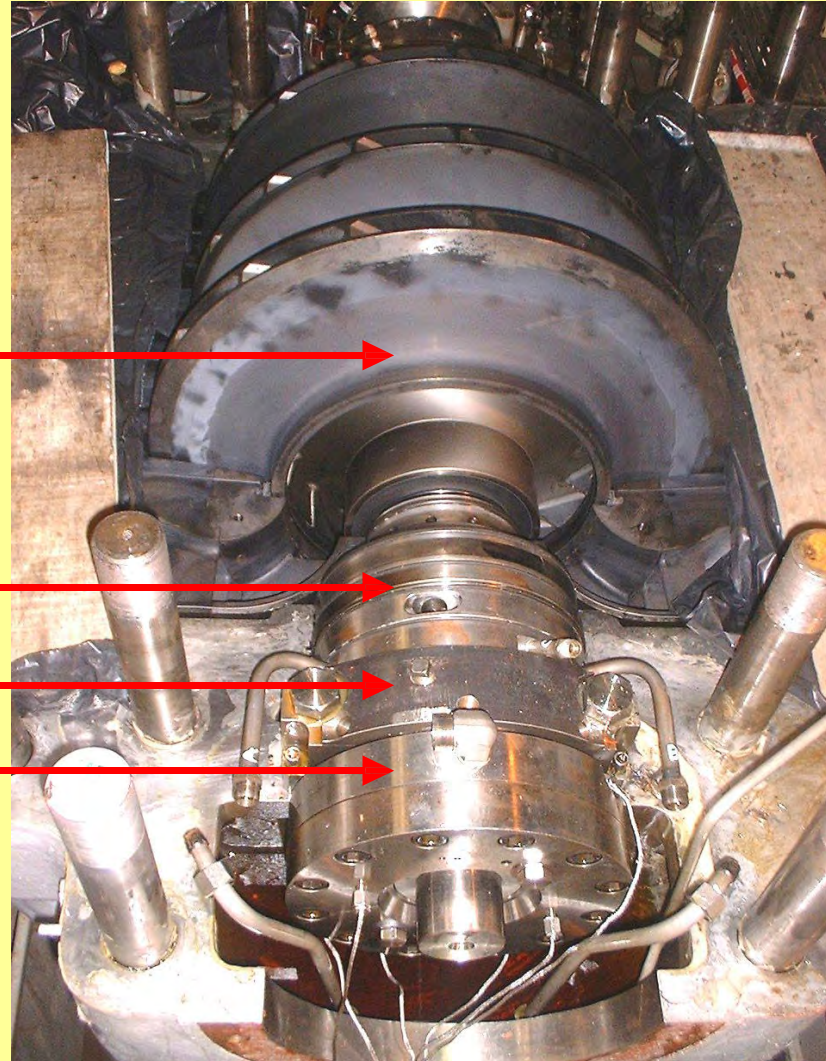


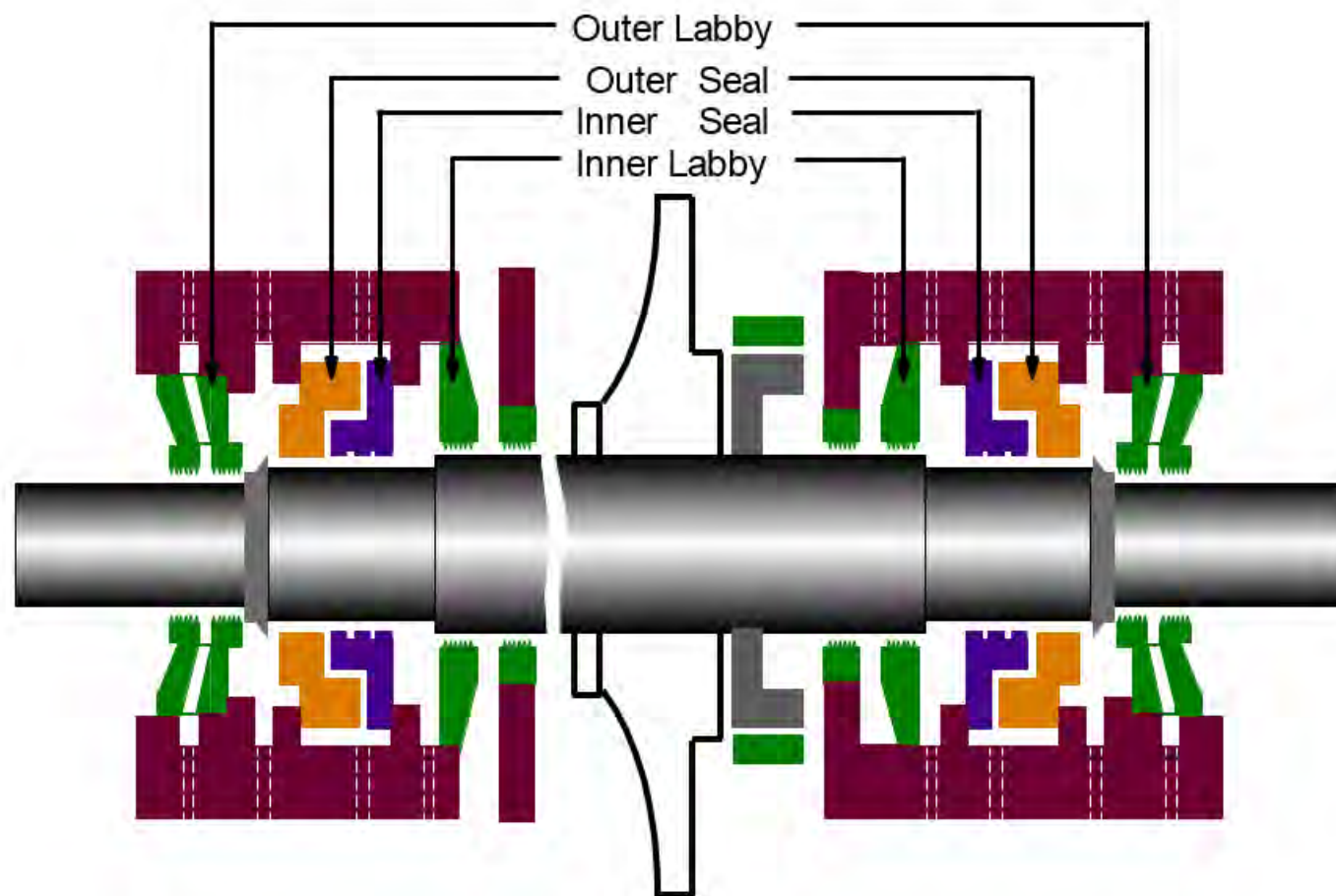
Impeller

Seal (Wet)

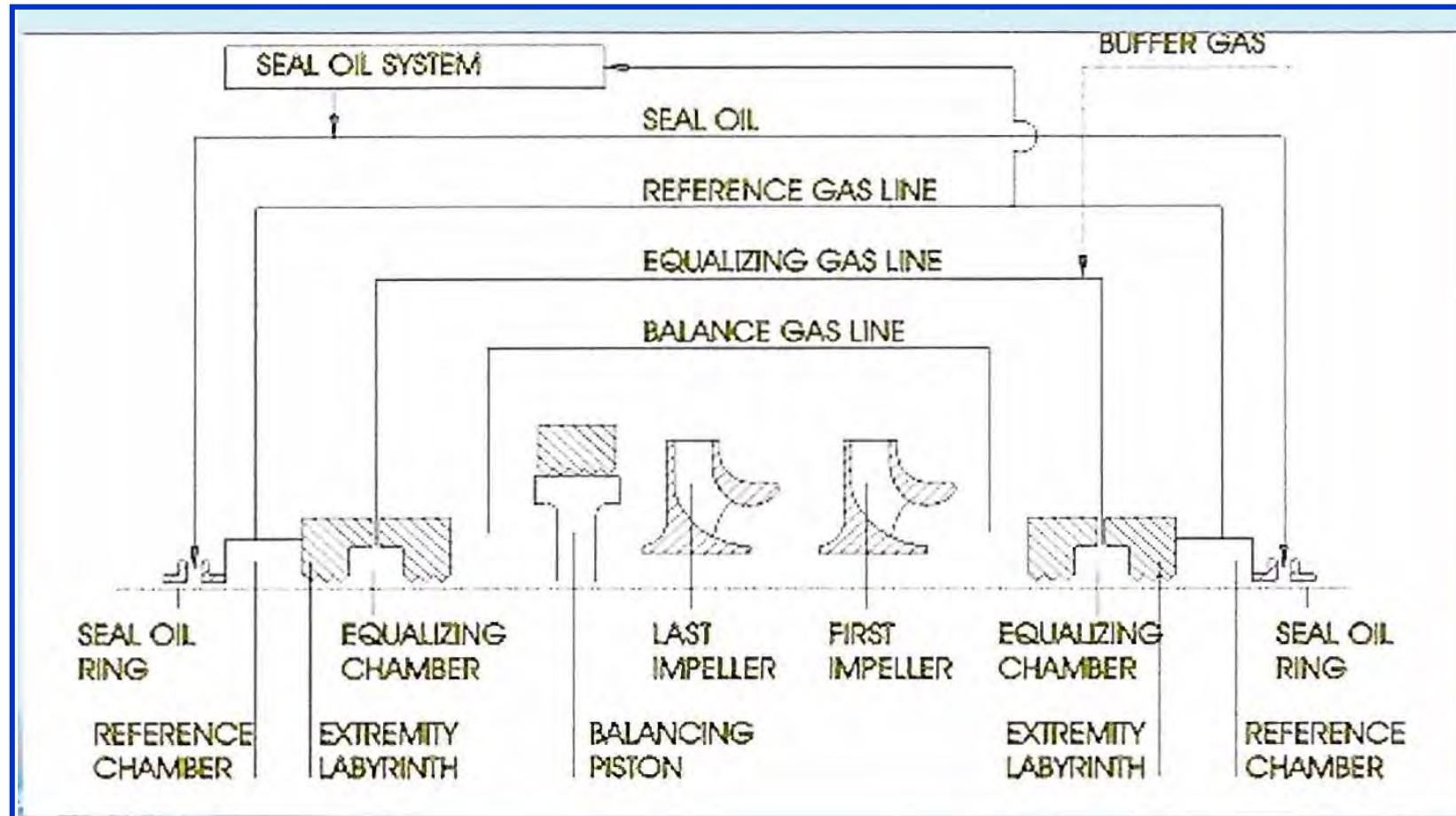
Journal Bearing

Thrust Bearing

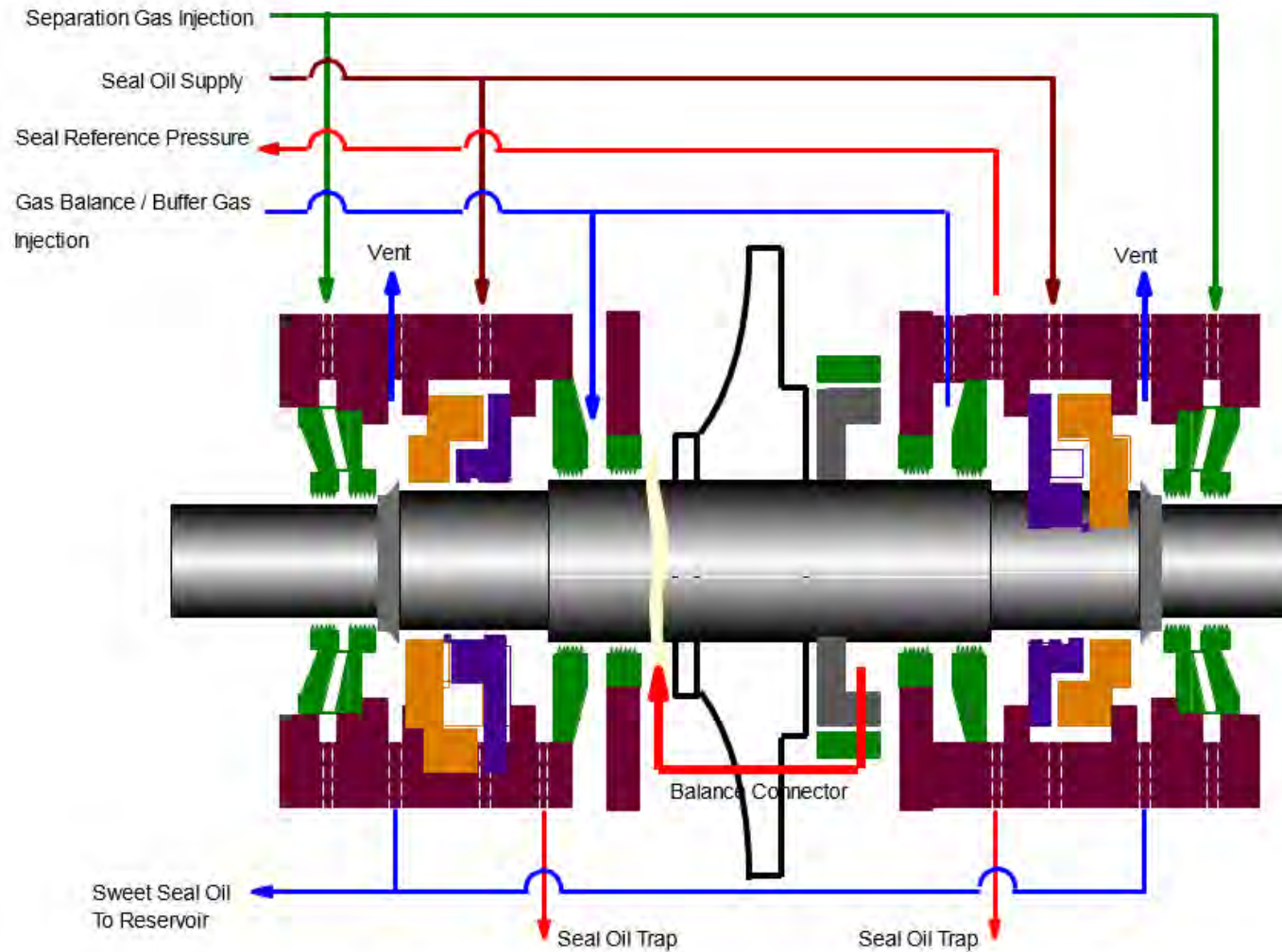




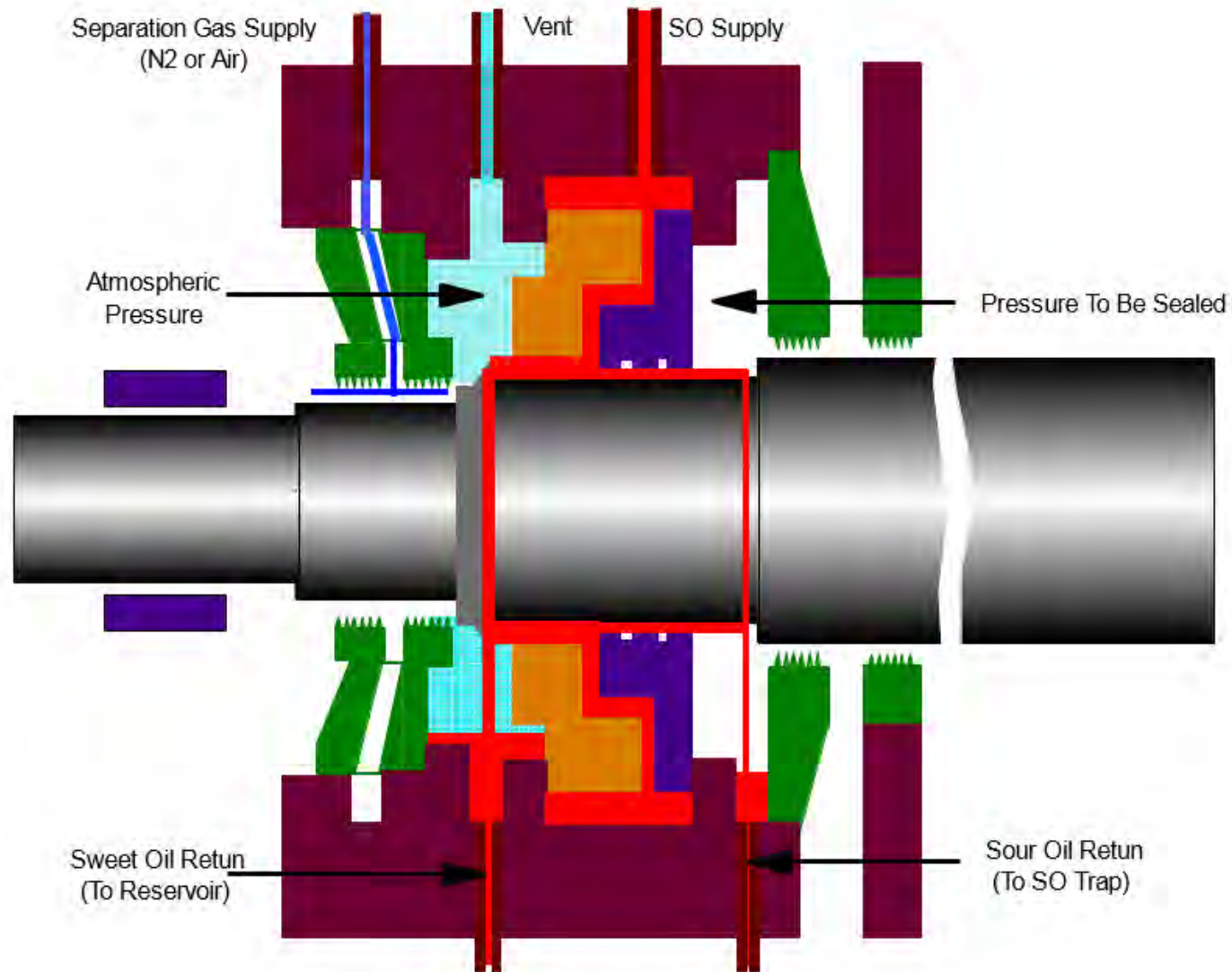
Seal Piping



Seal Piping



Seal Piping

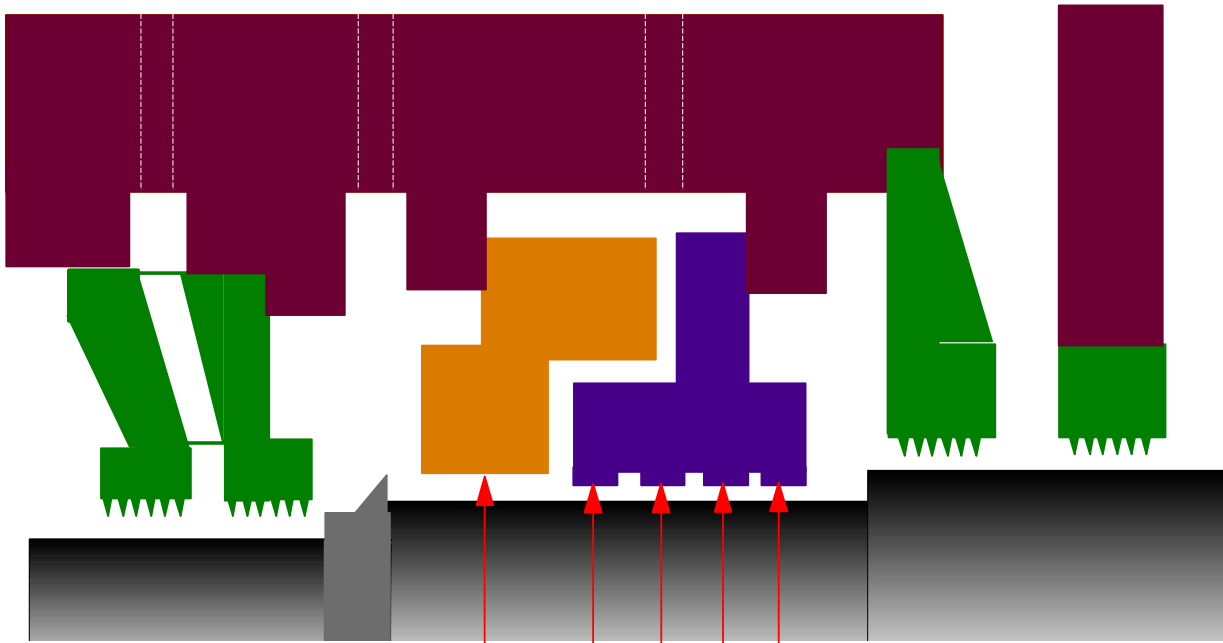


Why lots of seal oil supplied to seal chamber through outer seal than inner seal

Two Reasons

- Pressure Differential Across Seal
- Seal Clearance

Clearance



Diametral Clearance

0.0015" / 0.0025"

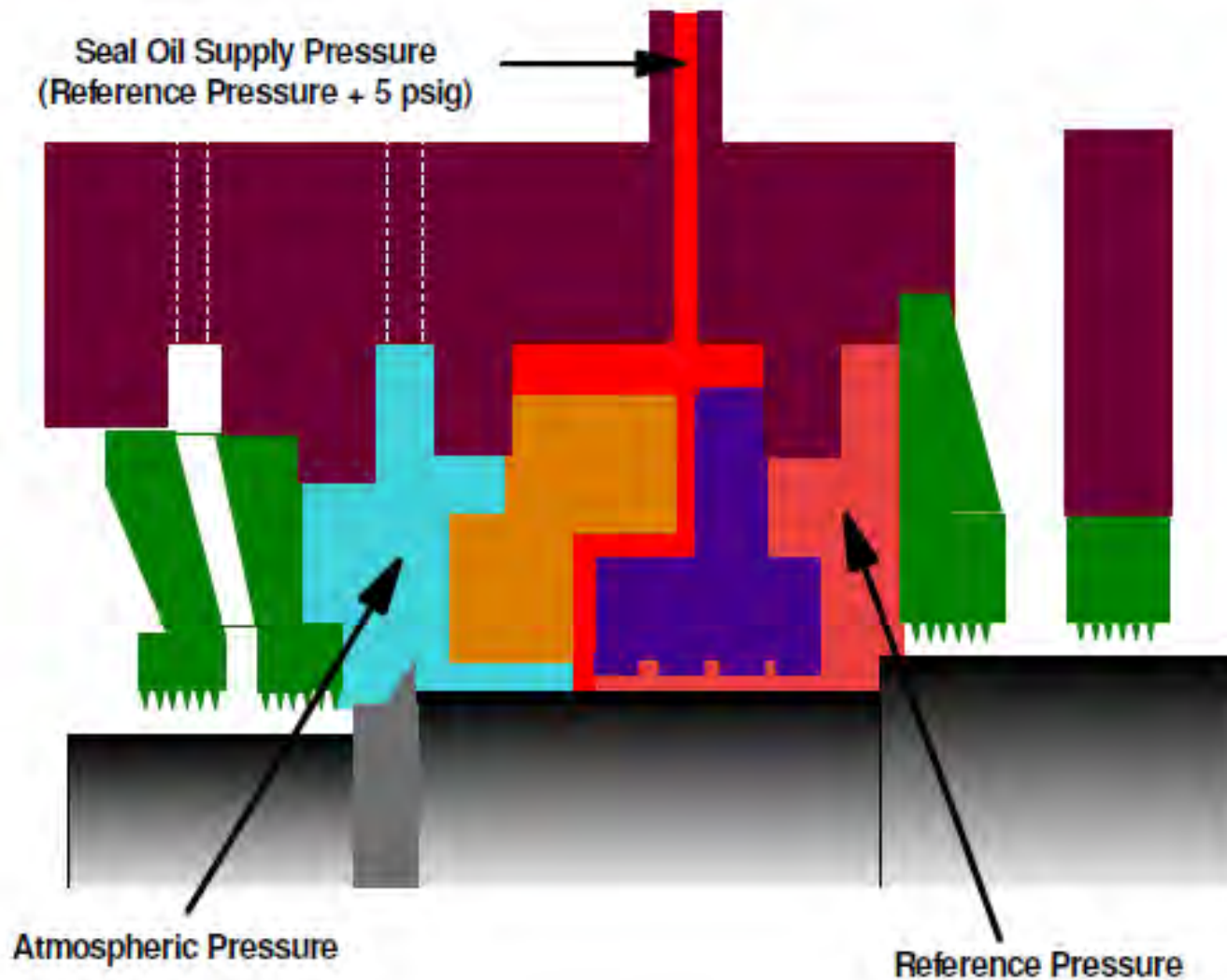
0.0025" / 0.0035"

0.0035" / 0.0045"

0.0050" / 0.0060"

0.0080" / 0.0090"

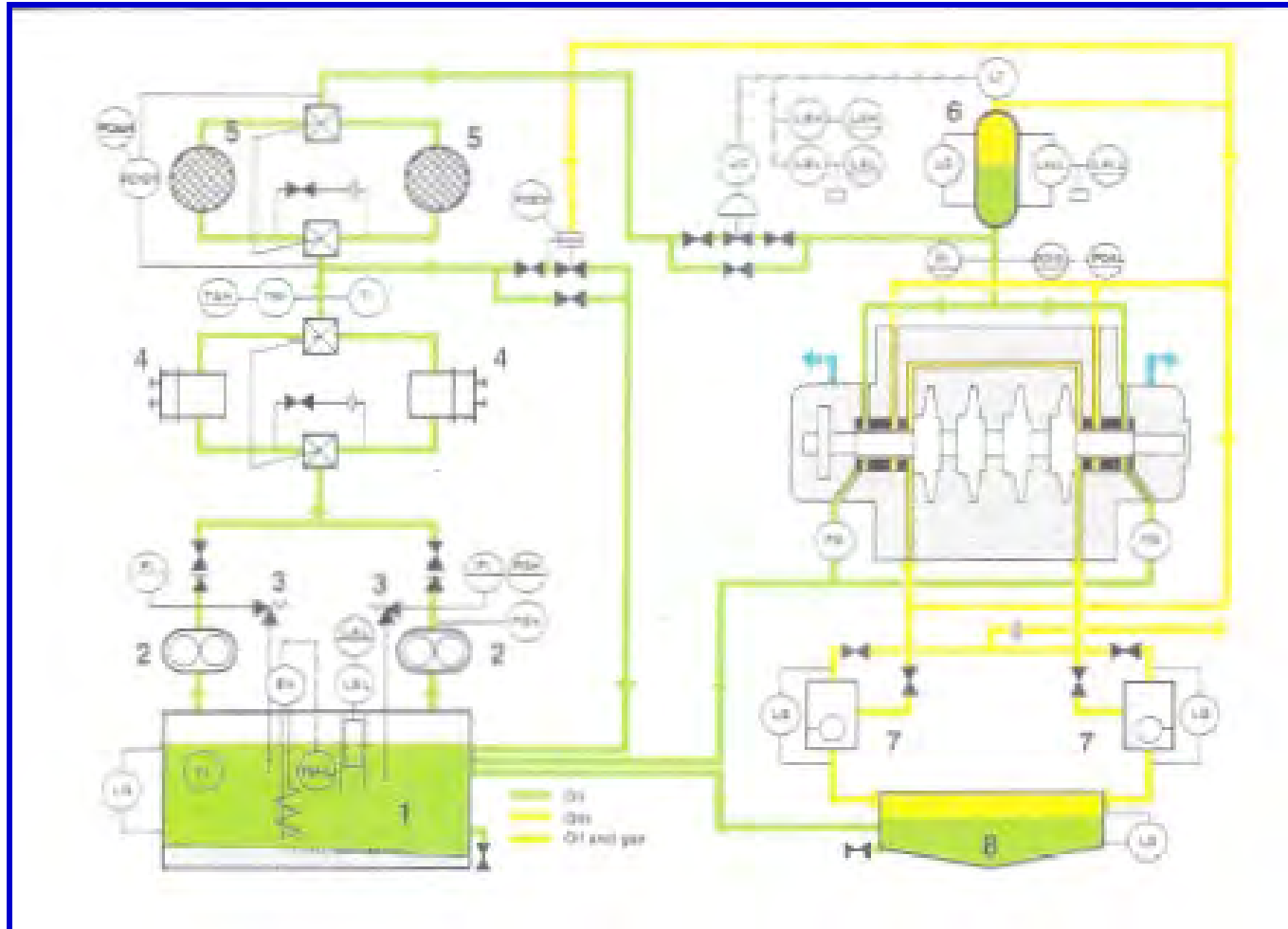
Pressure Differential



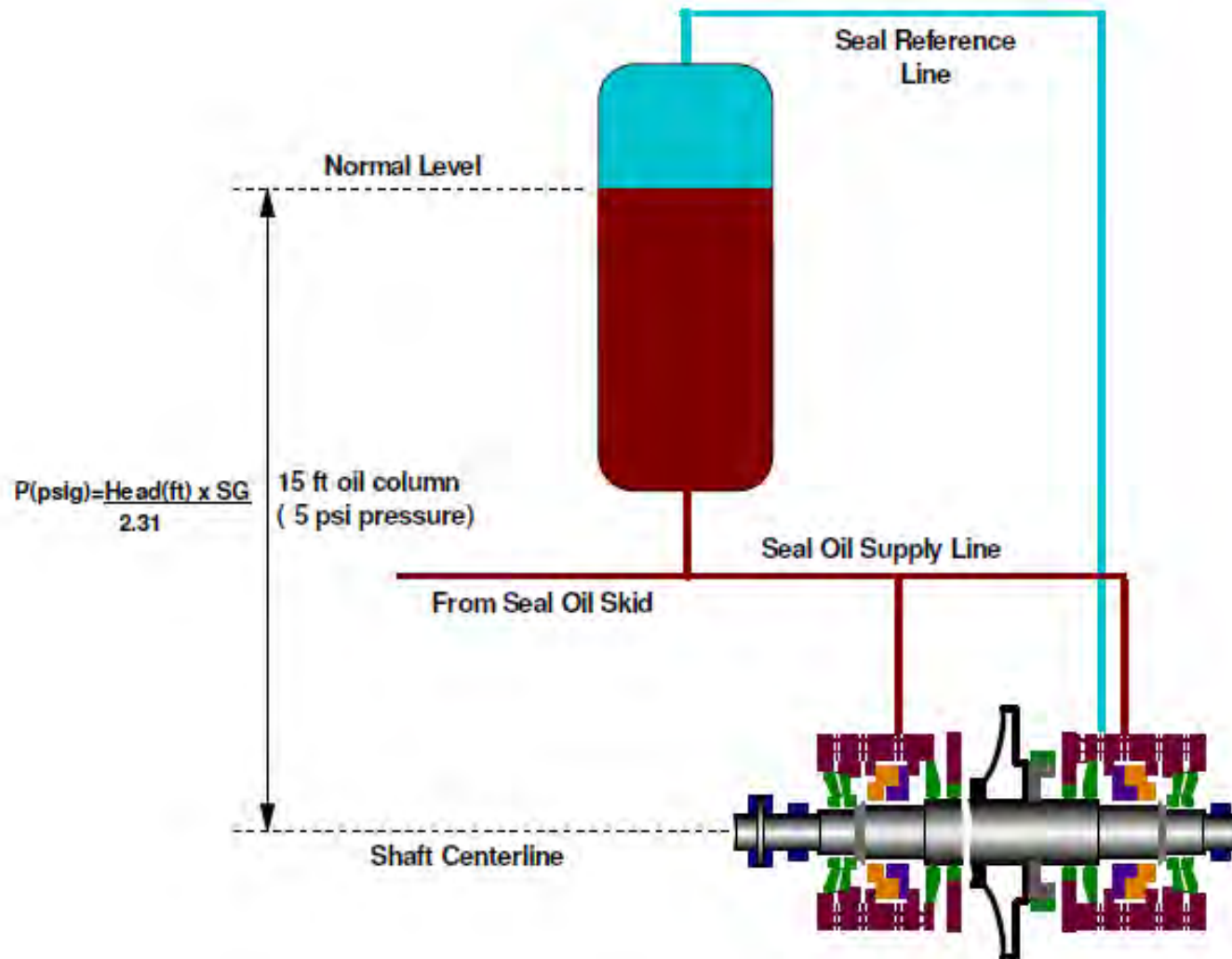
Seal Oil leakage comparison

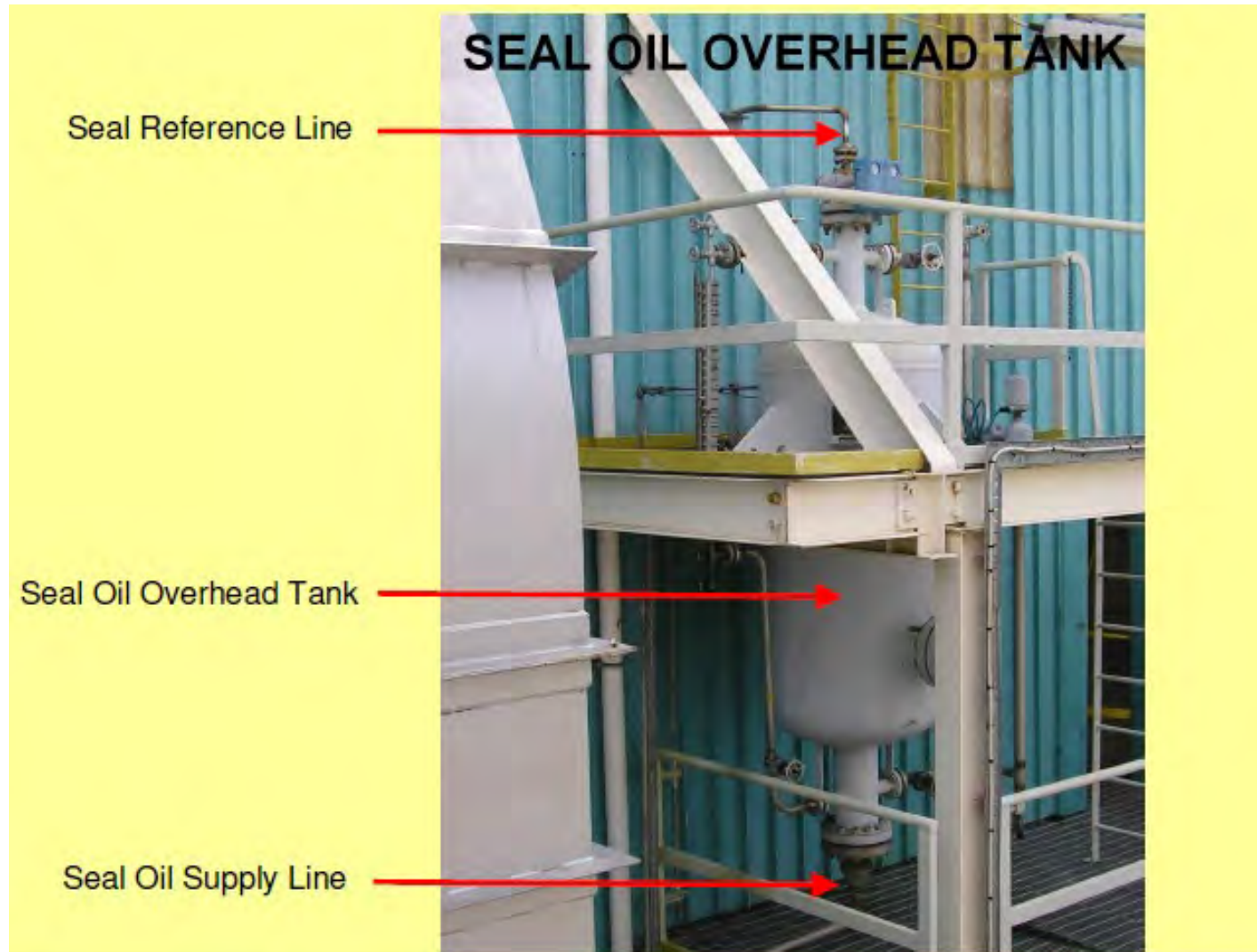
Dry Type	Oil Leak Index
Straight laby	100
Staggered laby	56
Honey comb	40
Carbon ring	20
Mechanical Contact (Dry)	2
Oil Type (Oil Loss)	
Mechanical contact	0.03 gal/Hr
Liquid Film or Bushing	1.75 gal/Hr

Seal oil System



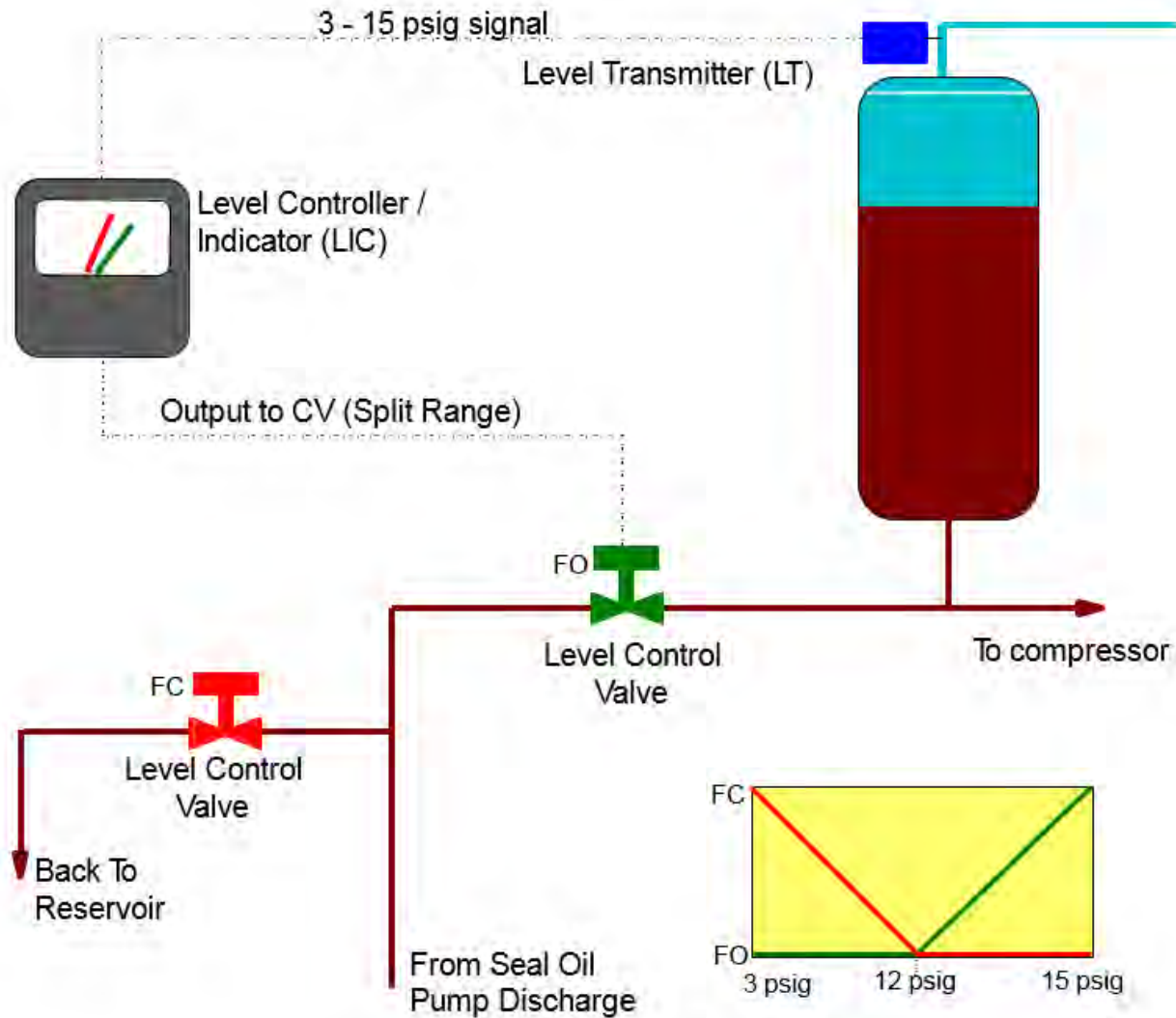
- Seal Oil Supplied shall be always 5 psig above the pressure to be sealed (reference pressure).
- How to design a system that will ensure that it will always be 5 psig at different compressor casing pressure ?

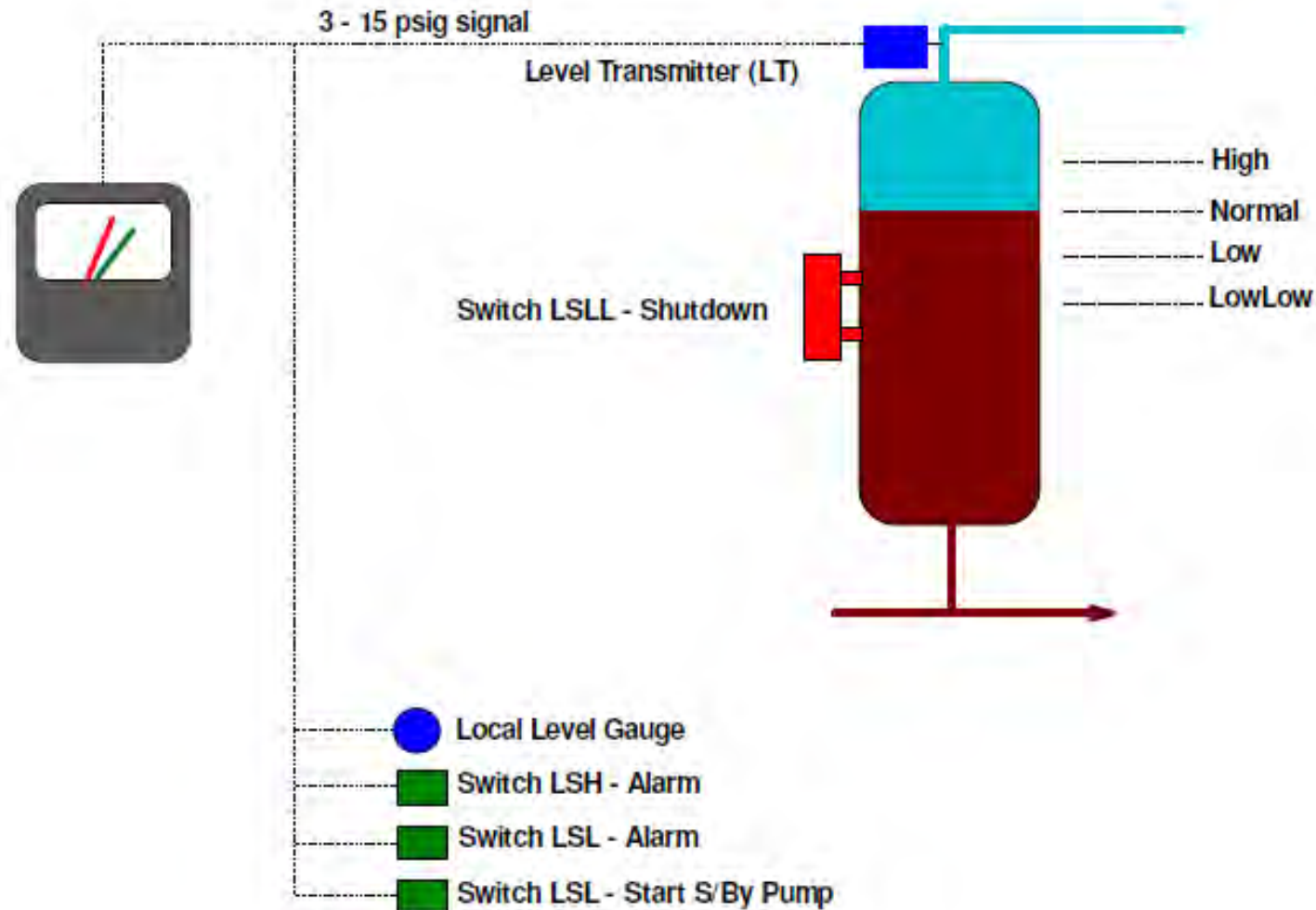


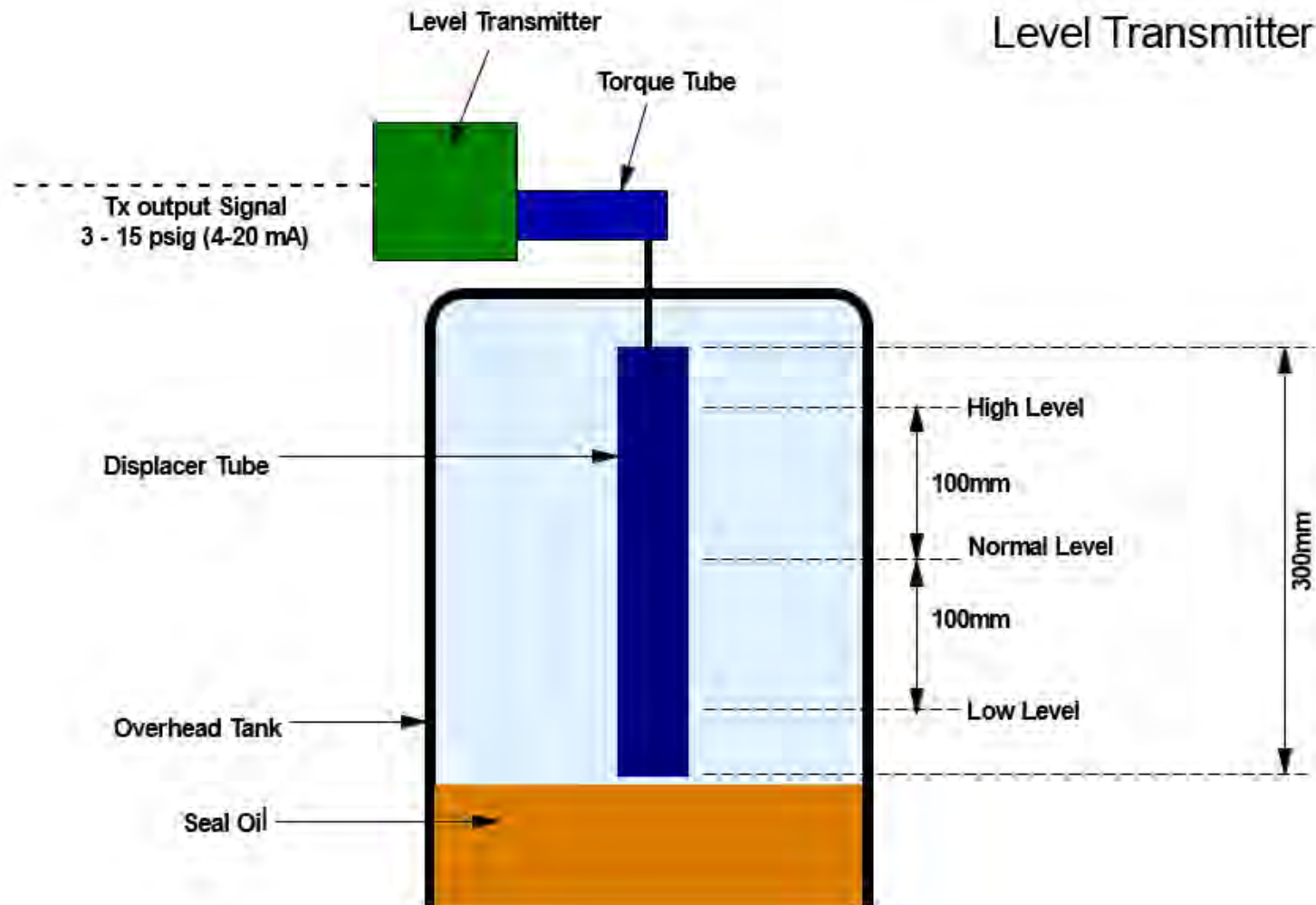


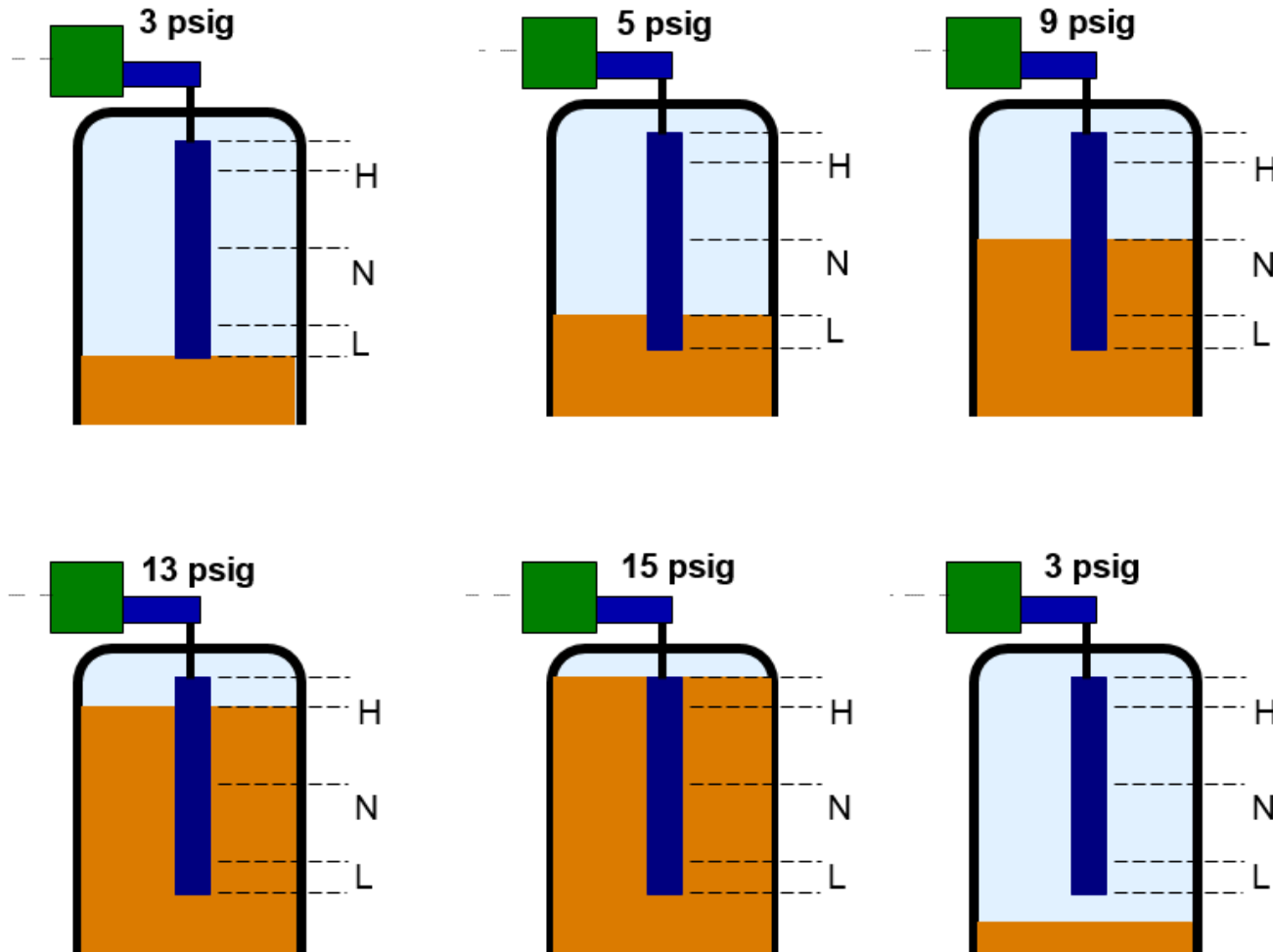
How to maintain the 15 feet of oil column inside the Seal Oil Overhead Tank ?

- Level Controller





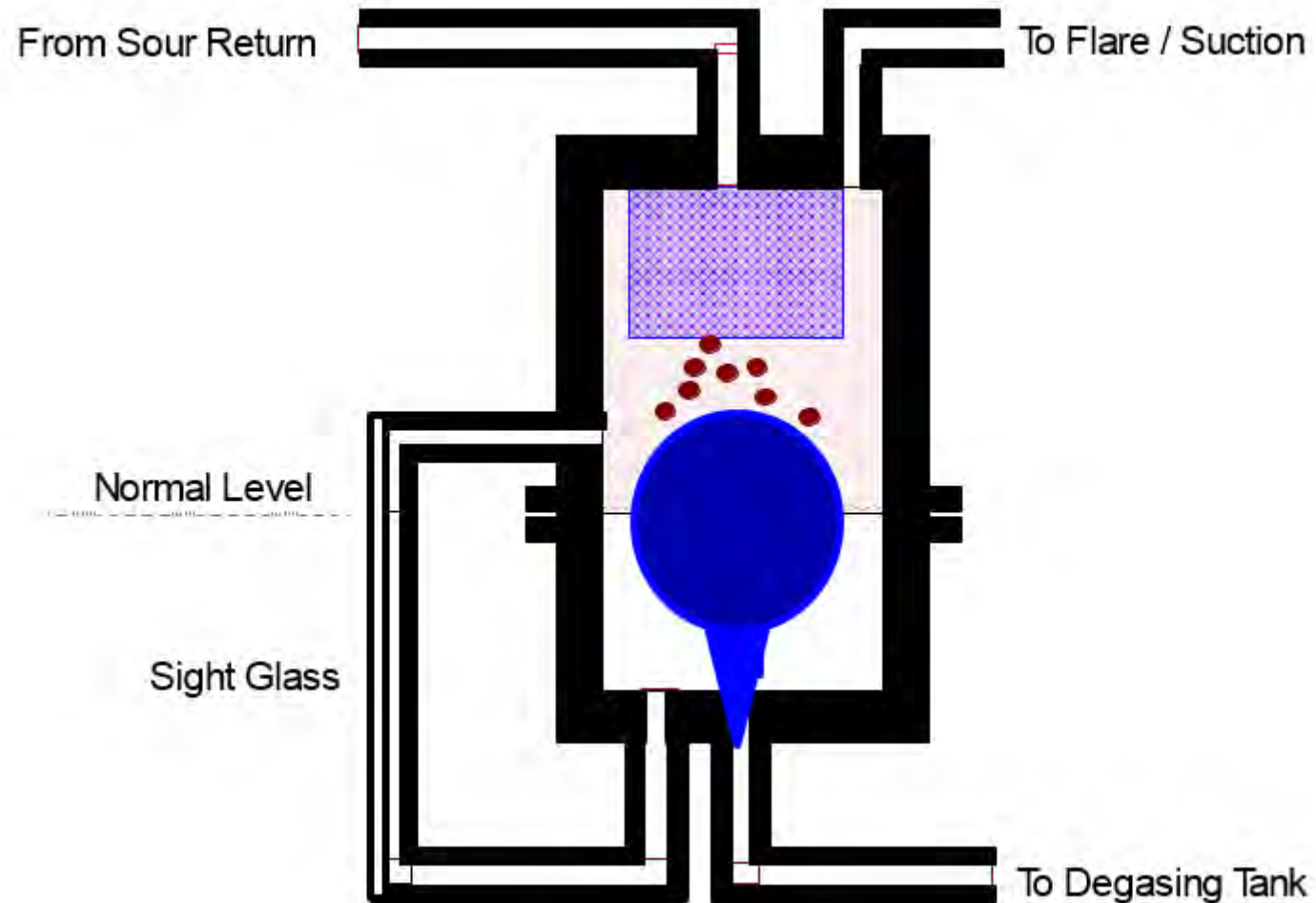




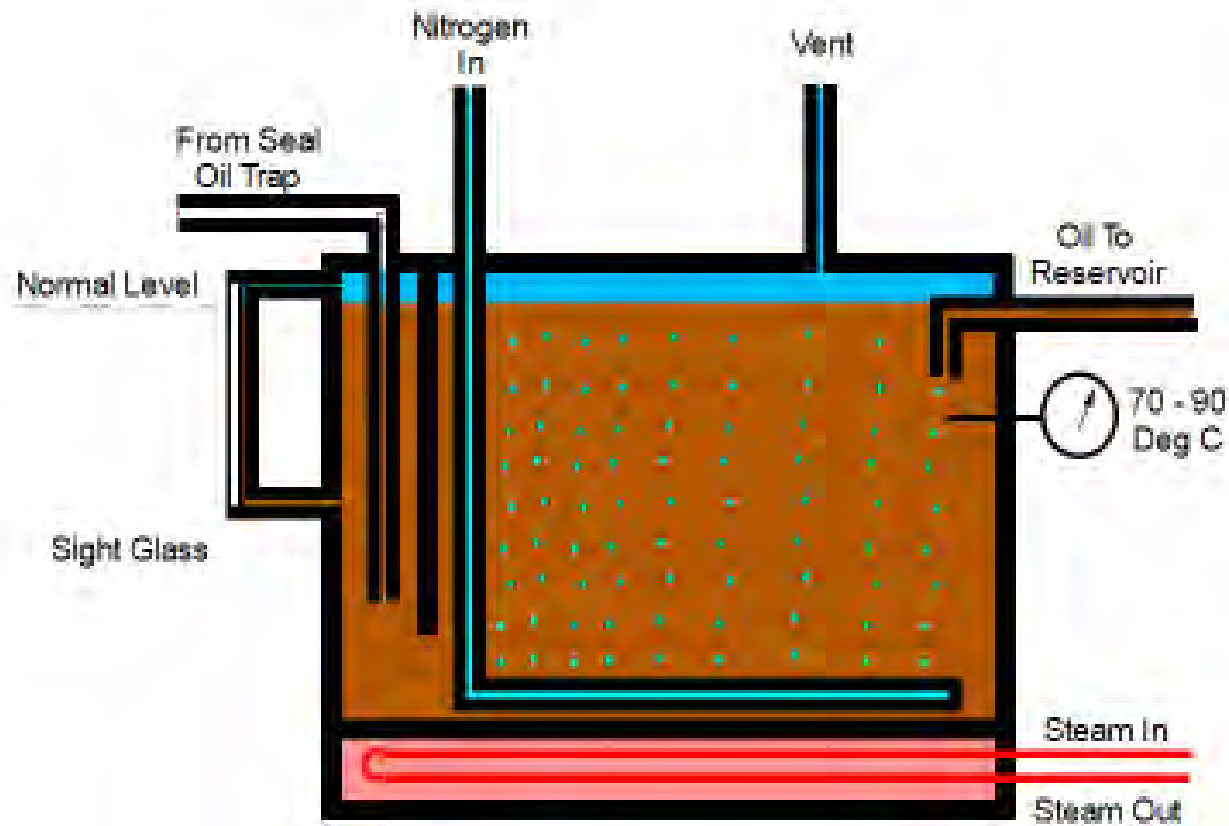
What happen to Sour Oil Return ?

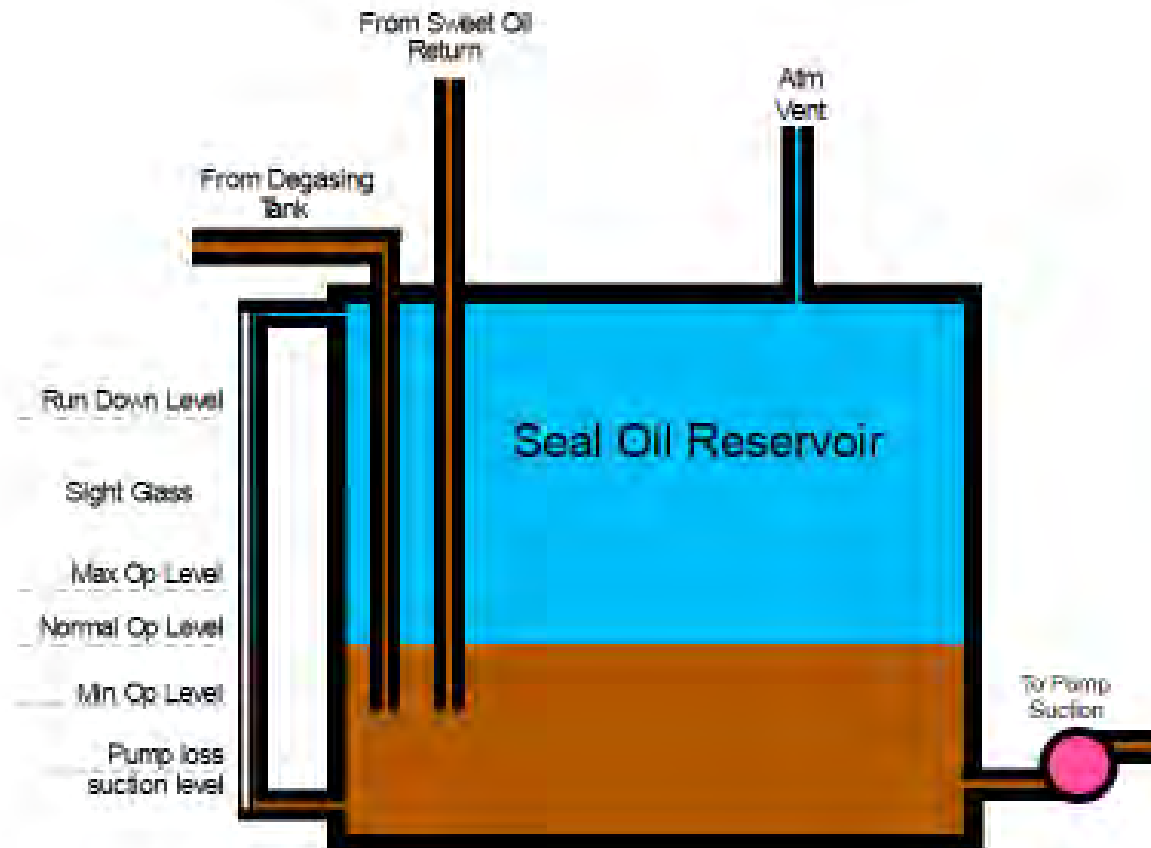
- It will flow into Seal Oil Trap
- Trapped gas will flash-off from the oil and vented to flare / reclamation point

Seal Oil Trap



Degasing Tank







What Type Of Oil Used ?

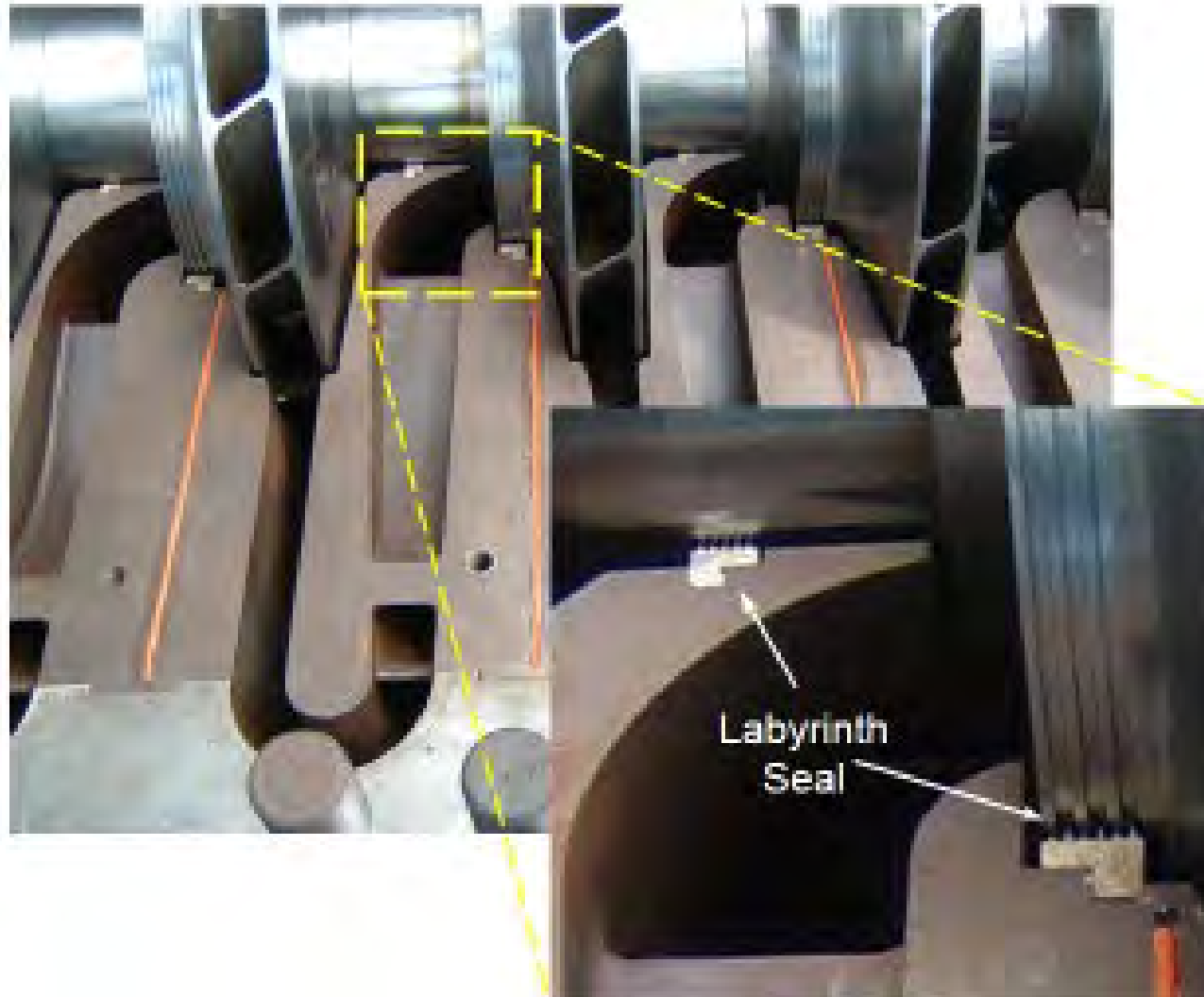
Mineral Oil Viscosity Grade 46 (Jenteram 46)

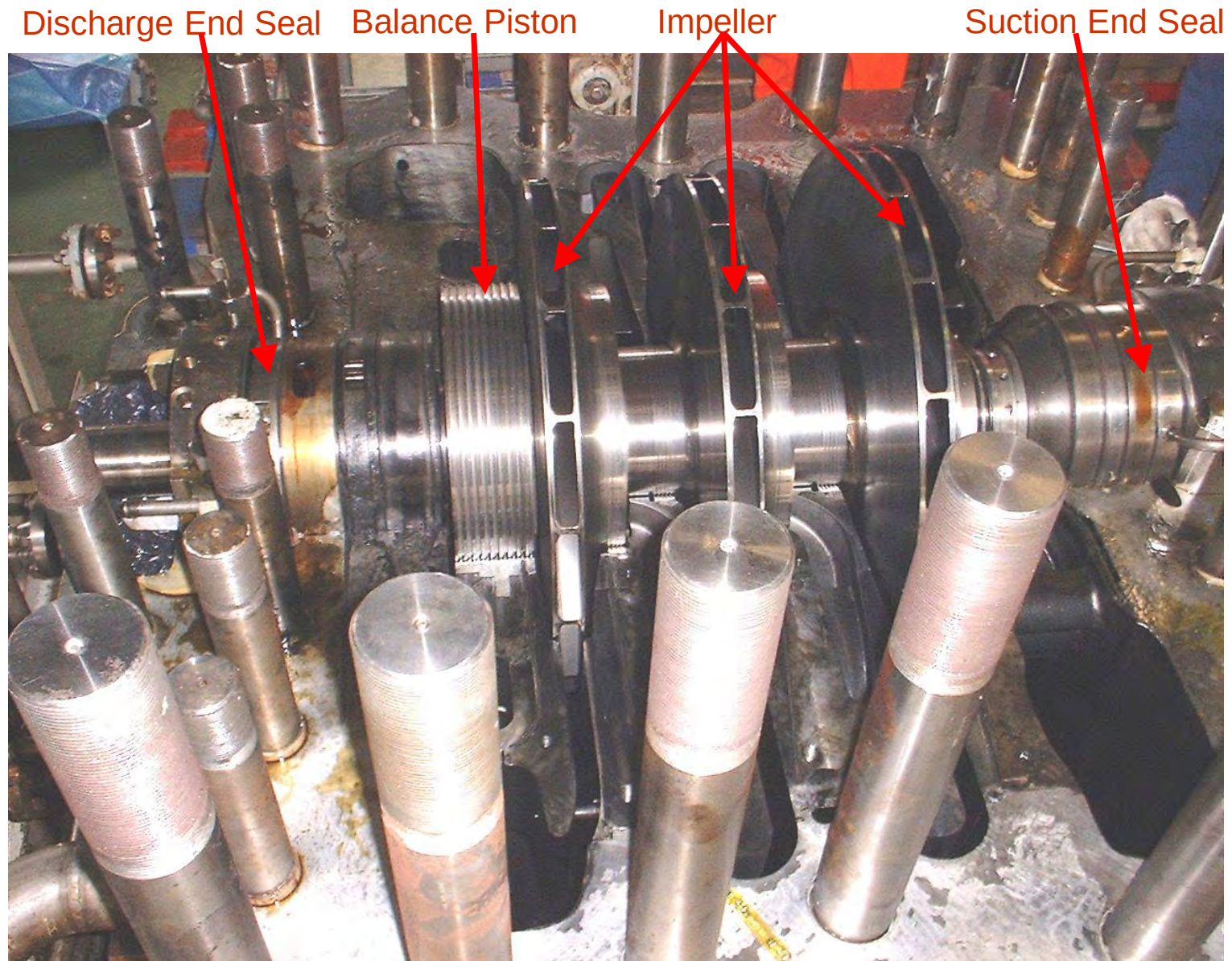
- 40 Deg C : 46 Cst
- 50 Deg C : 30 Cst
- 99 Deg C : 7.1 Cst

Why not Oil with VG 32 ?

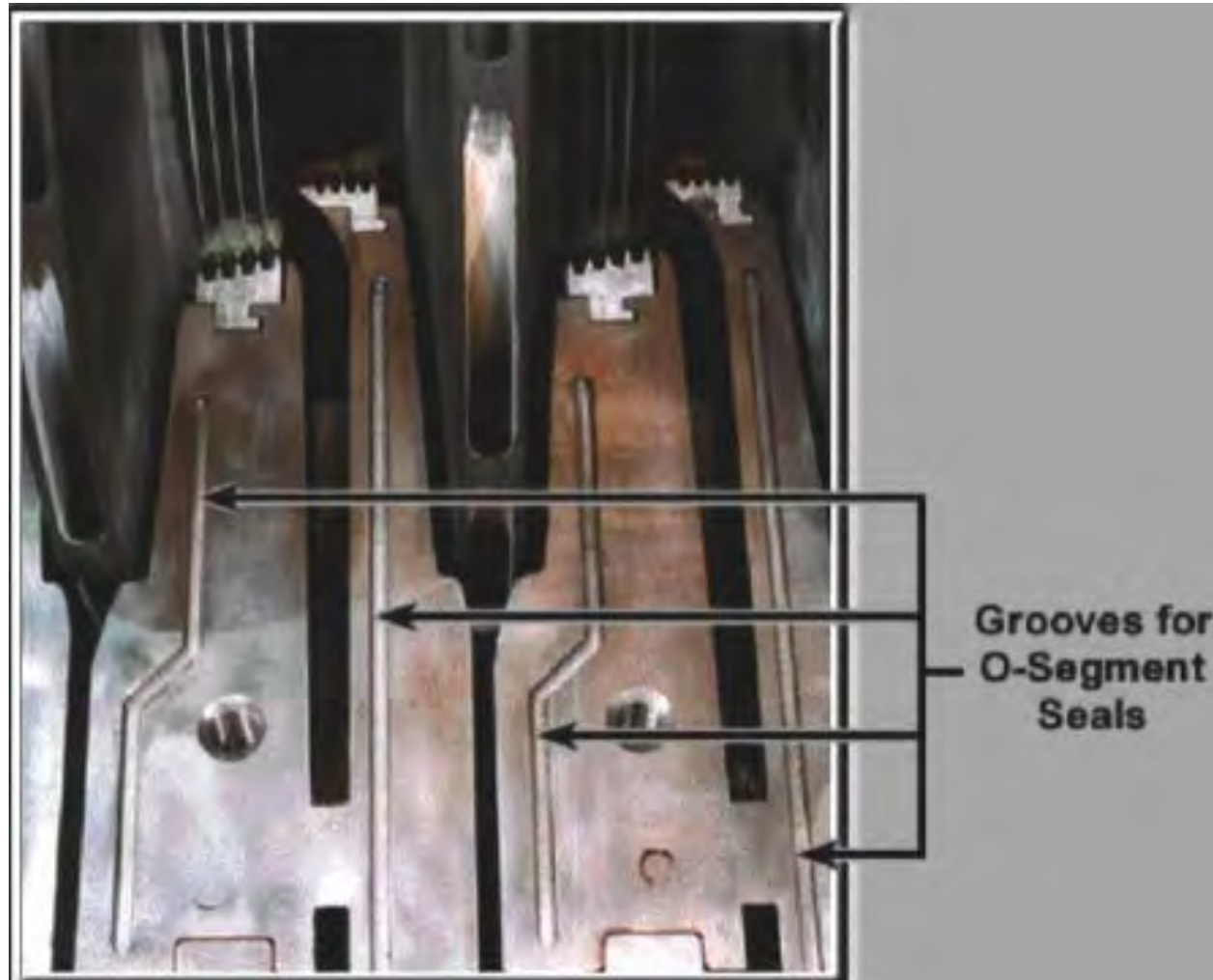
Oil Cooler Limitation. With tropical ambient emperature,normal cooler discharge temperature will be arround 55 Deg C. Using lower VG oil will make the oil suppliedto the seal has very low viscosity.

- Are devices used between rotating and stationary parts to separate and minimize gas leakage between areas of unequal pressure





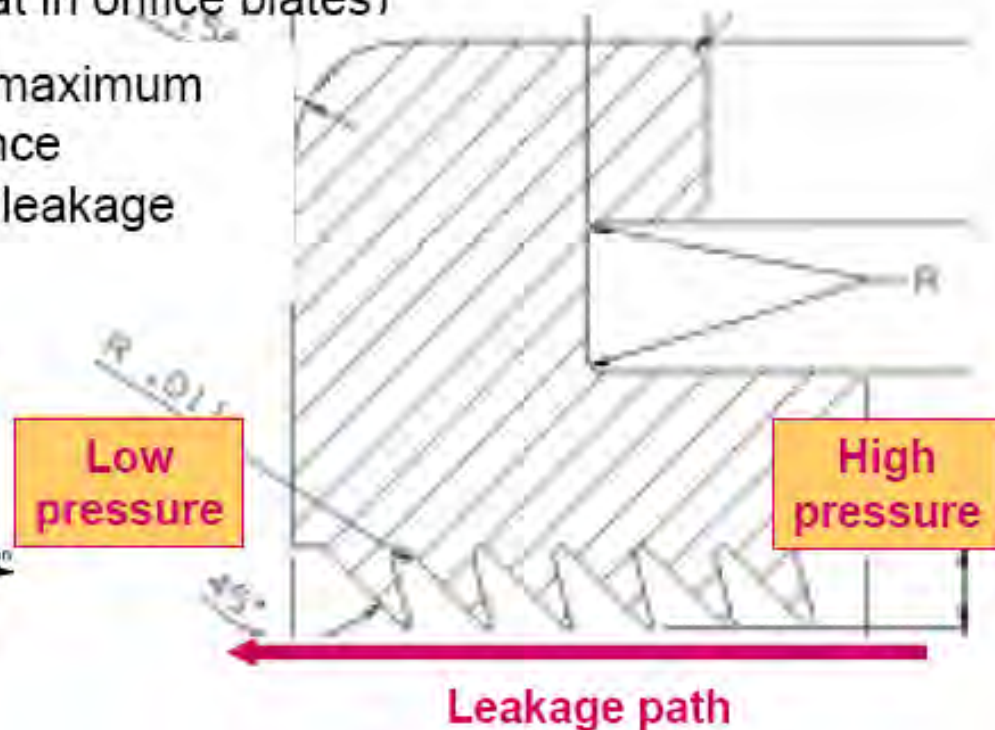
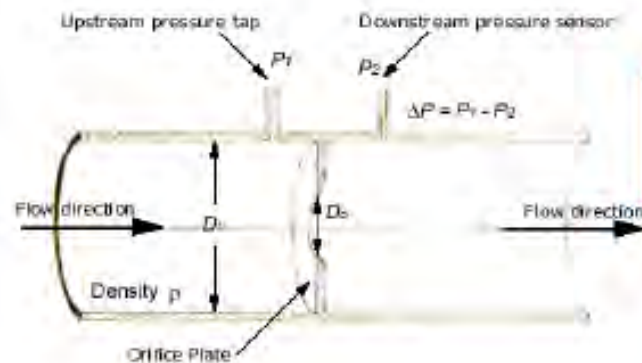
Inter-stage Labyrinth



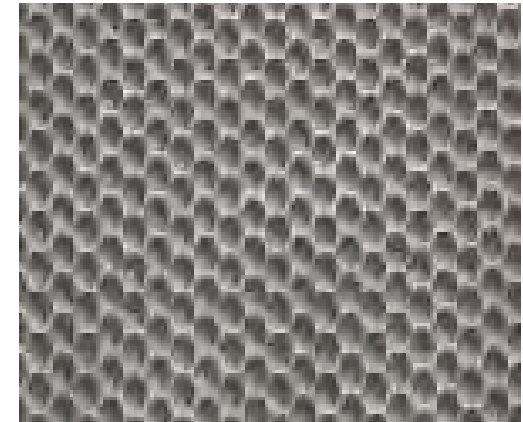
The leakage is approximately inversely proportional to the square root of the number of the labyrinth lands

To obtain labyrinth maximum sealing efficiency, the labyrinth lands must maintain sharp edges in the direction of the flow (this requirement is similar to that in orifice plates)

A sharp edge provides for maximum “*vena contracta*” effect, hence maximum restriction of the leakage loss



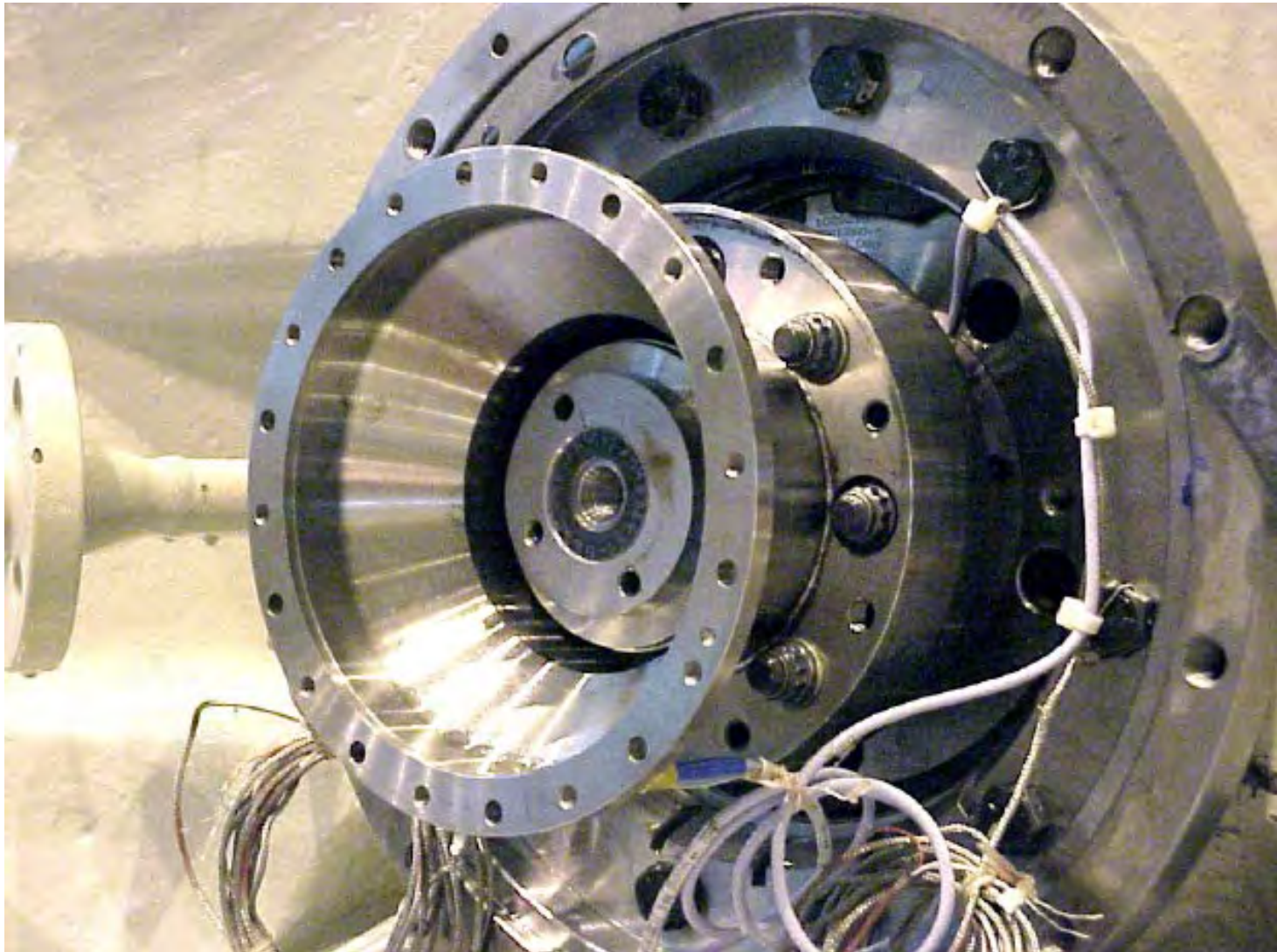
Inter-stage Honeycomb Seal



Coupling



Coupling Hub

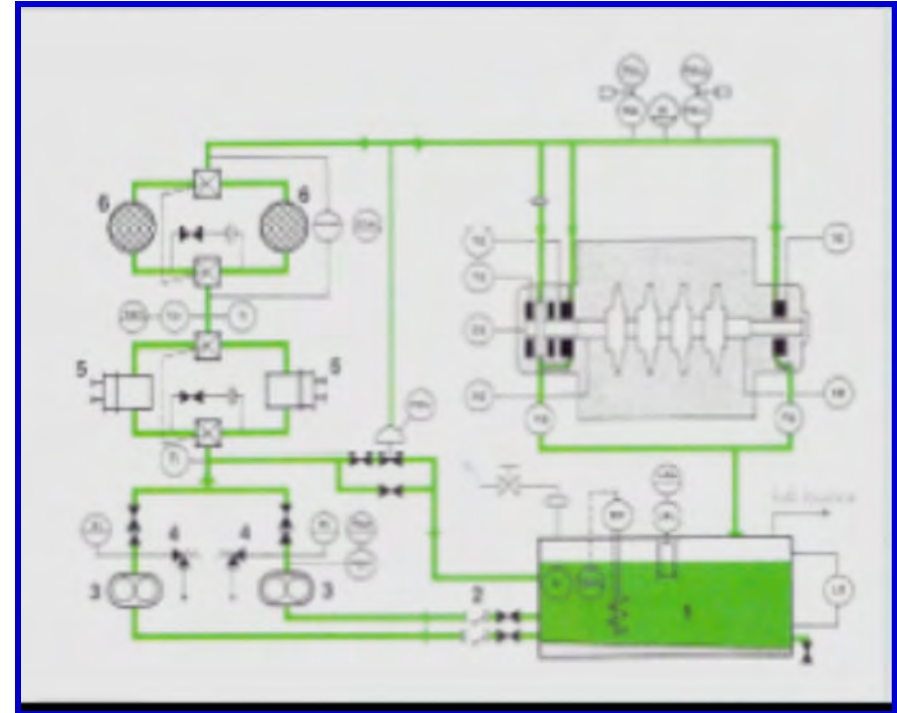


3.1 Centrifugal

- Auxiliaries (suggestion)
 - Lubrication system
 - Sealing System

Lube oil system

- Lube oil system designed as per API-614.
- Key features :
 - Enhanced reliability
 - Adequate redundancy of key components.
 - Easy maintenance & trouble shooting.
 - Adequate protection for equipment.
- ◆ Common trouble:
Equipment tripping during oil-pump auto changeover.



Lubrications Functions

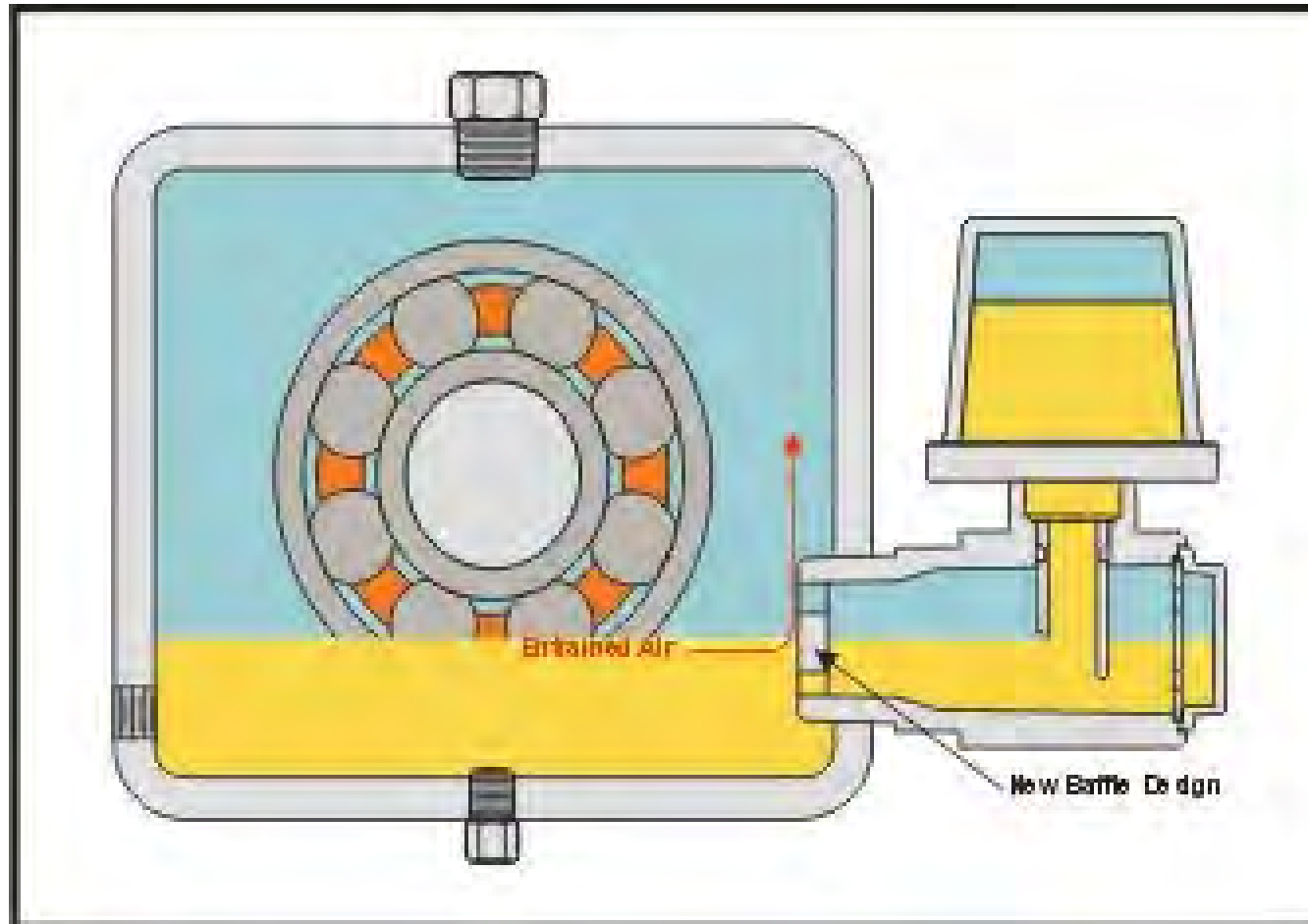
- Reduce frictions and wear
- Remove heat
- Reduce vibrations
- Transfer energy (hydraulic)
- Protection
- Contamination control

Type of lubrications system (Major)

- Grease
- Oiler
- Oil ring / thrower
- Forced lubrication systems

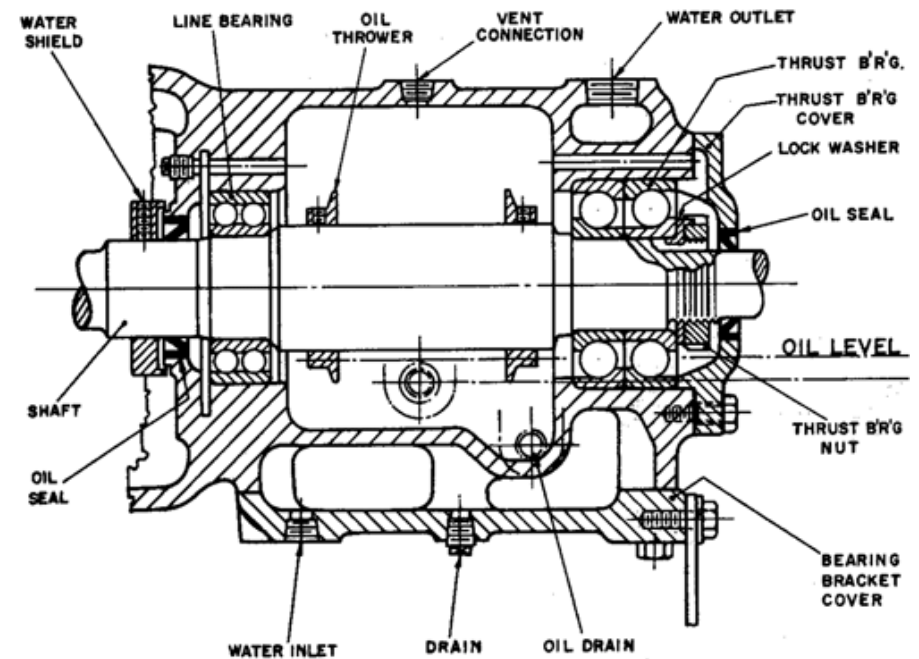
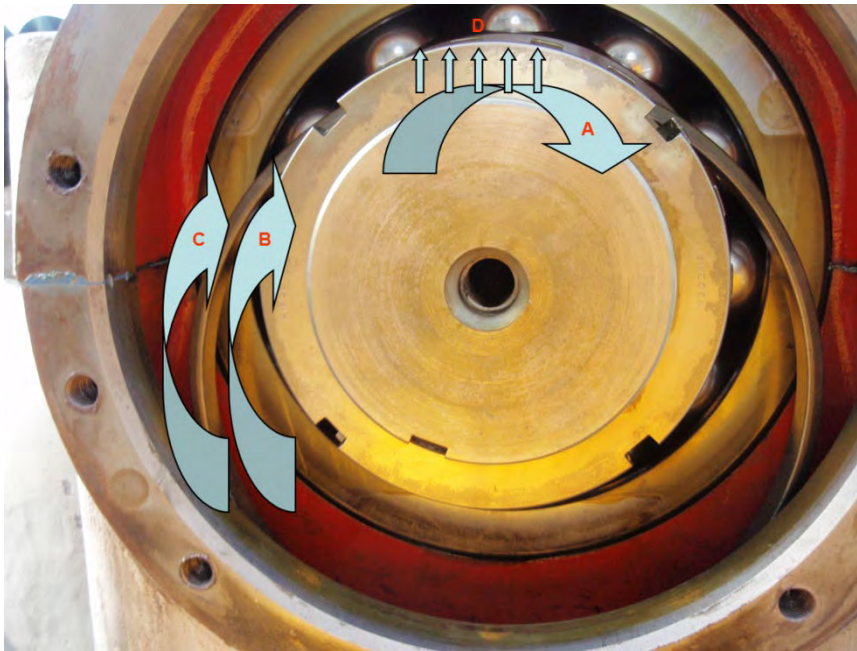
Type of lubrications system

- Oiler



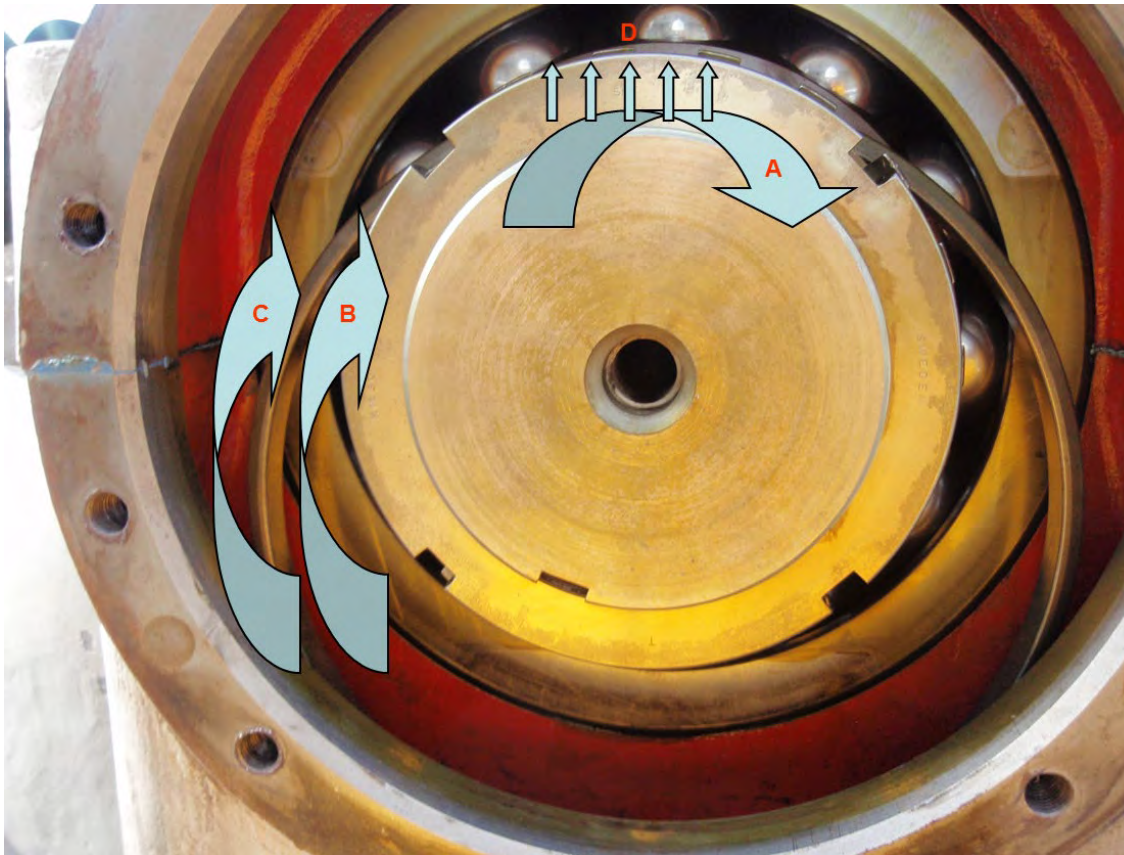
Type of lubrications system

- Oil ring / thrower



Type of lubrications system

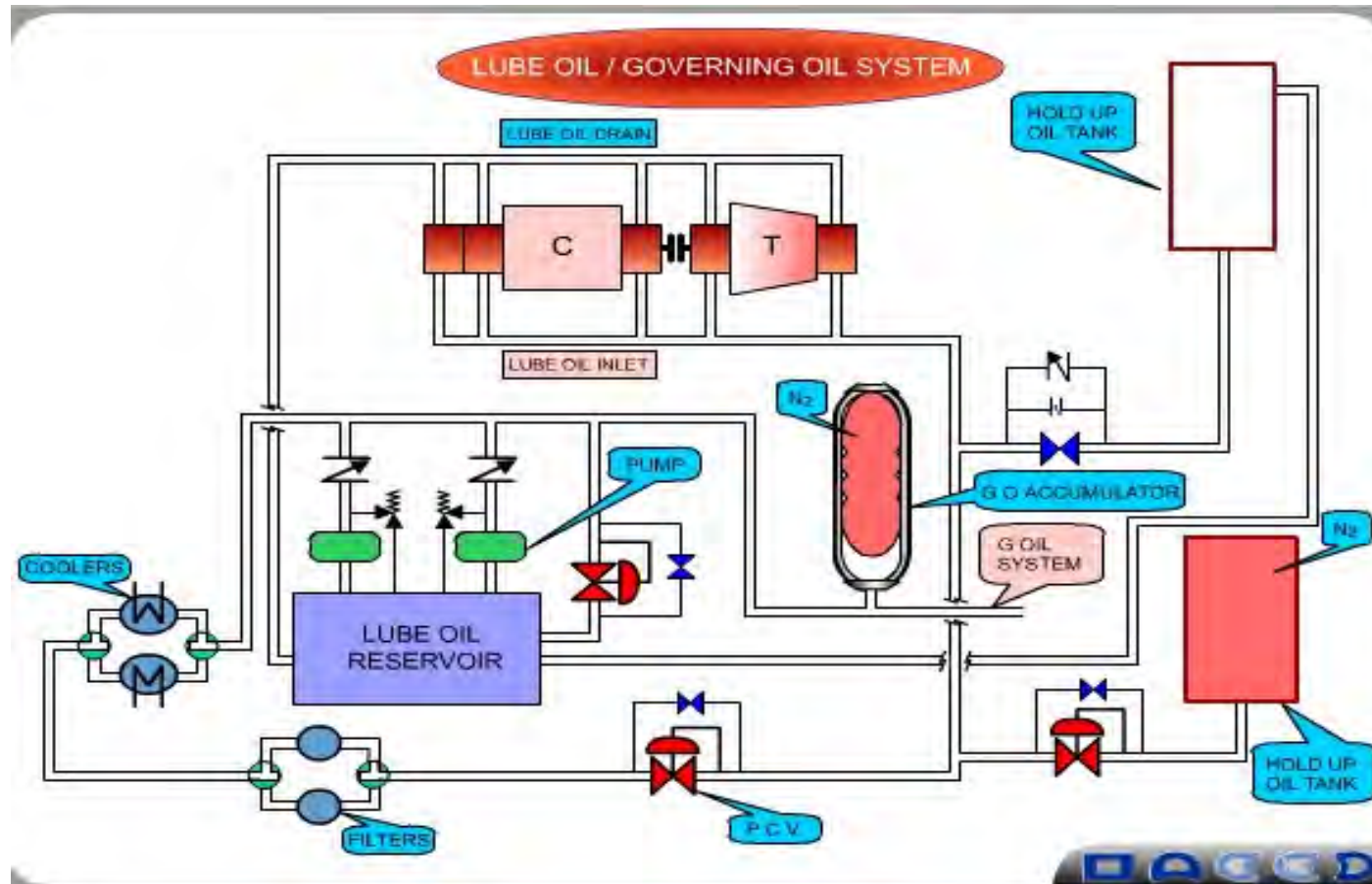
- Oil ring



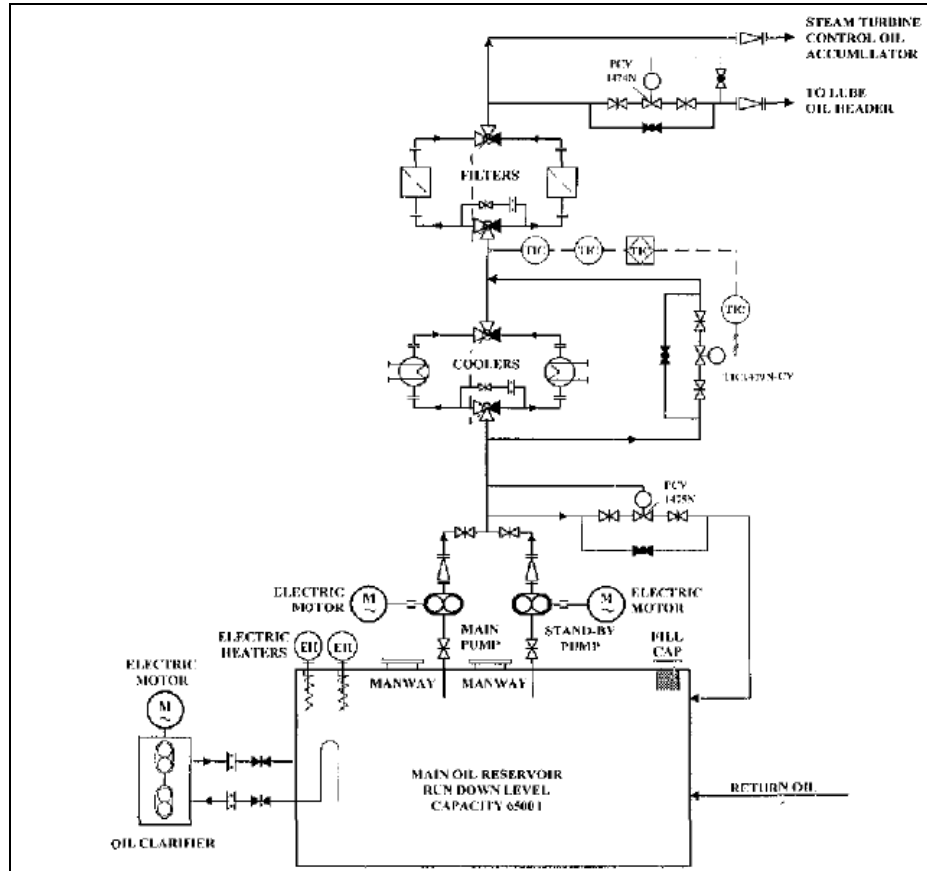
Oil ring working principle

- shaft rotate
- oil ring rotate relative to shaft rotation
- lube oil will follow oil ring
- lube oil will go into the bearings through the shaft

- Standard – API 614
- General Components:
 - MOP
 - AOP
 - Oil Filters
 - Oil Coolers
- Sometimes, connected with hydraulic and sealing systems



Pressurised closed loop lubrication system

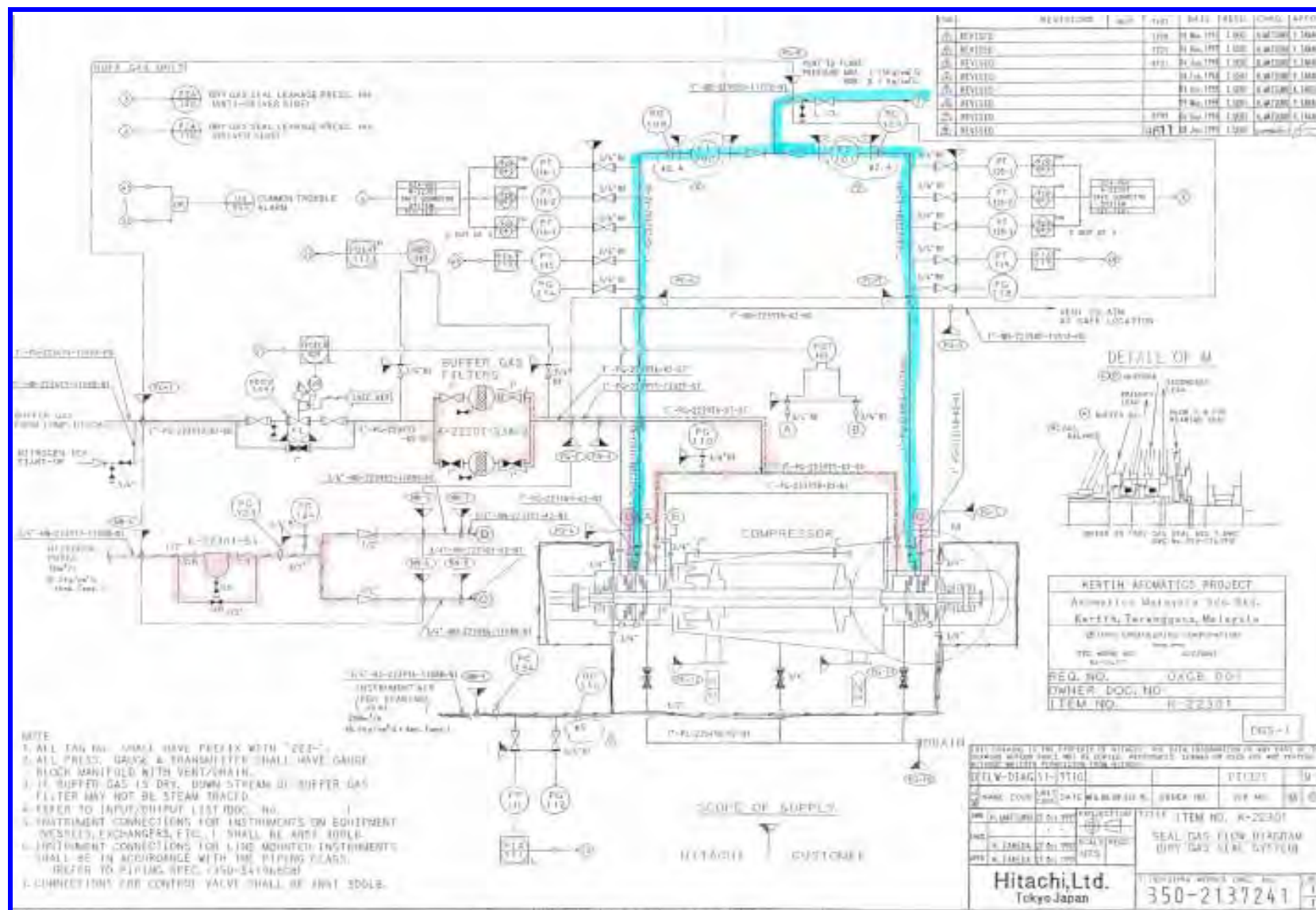


Pressurised closed loop lubrication system

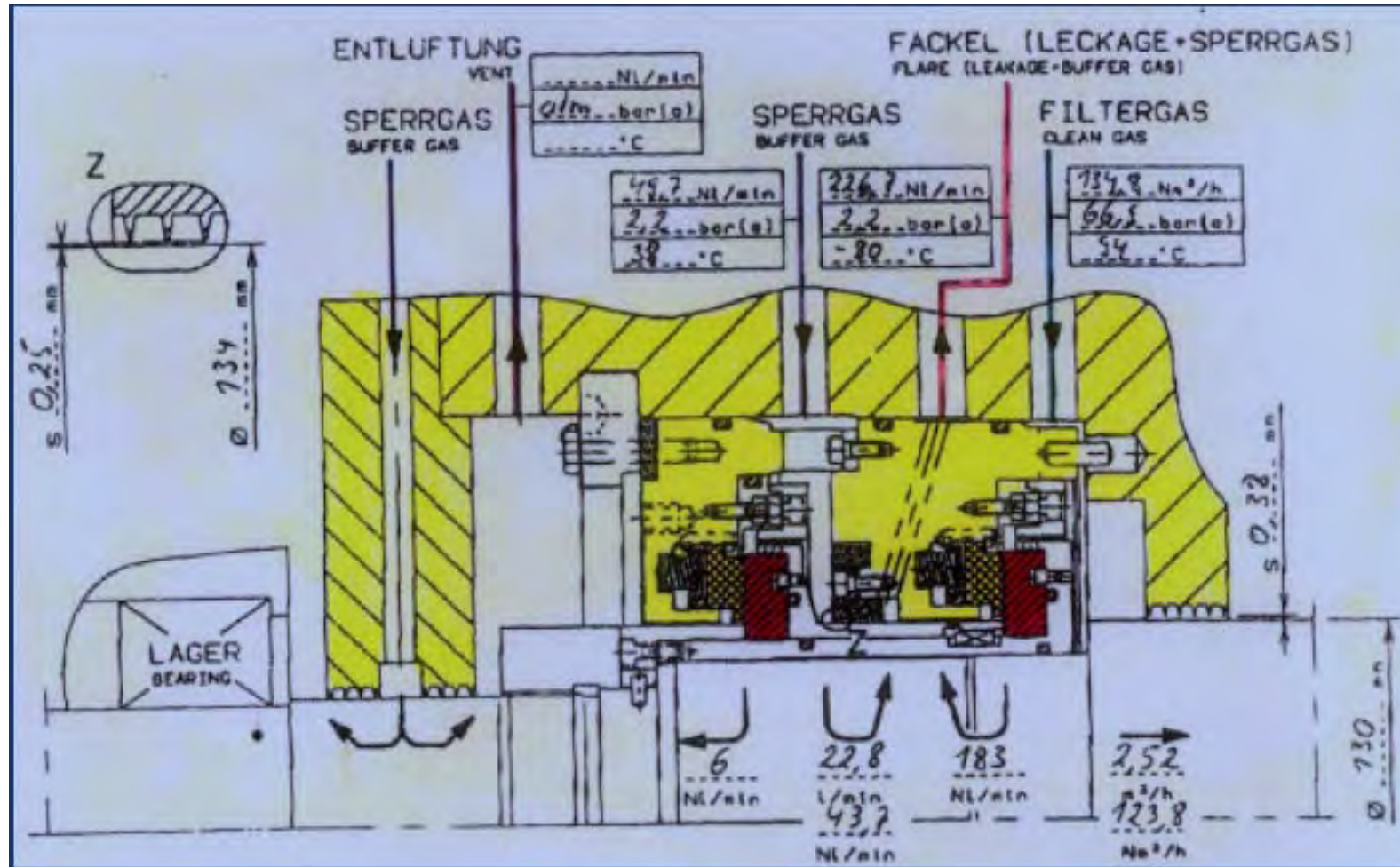
- Oil pumps
- Oil filters
- Oil coolers
- PSV
- PCV
- Main oil tank

- Gas seal control & monitoring system
 - Provides three main functions
 - filtration of buffer & purge gas.
 - Regulation of buffer & purge gas.
 - Monitoring of seal performance.
 - Offers local indication of filter and seal performance.
 - Design is simple to operate, troubleshoot & maintain.

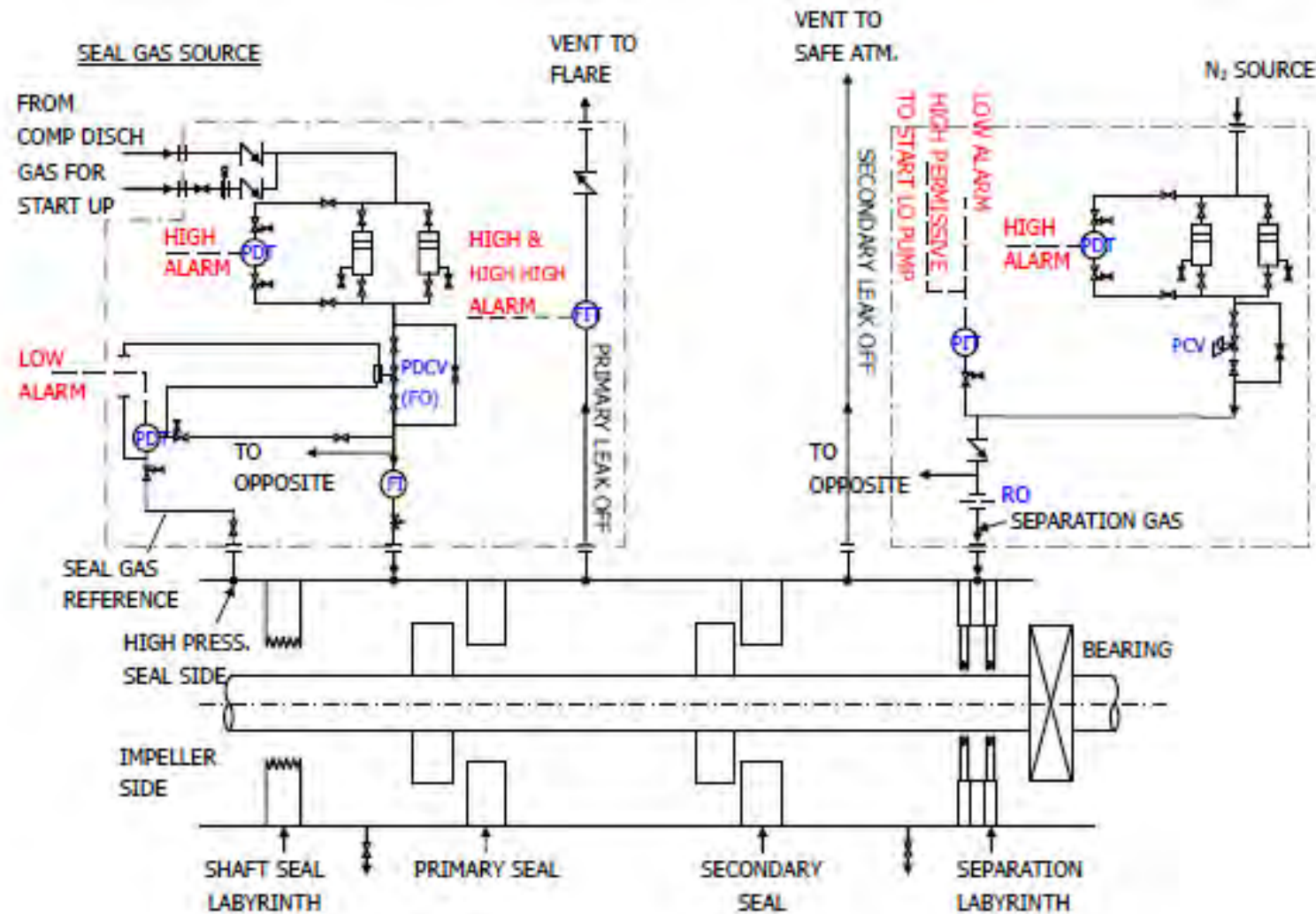
Gas seal system



Gas seal system



Gas seal System



[]: Located in gas seal panel

- Control sealing environment
 - Ensure cleanliness of process gas & seal piping.
 - Ensure adequate maintenance of upstream liquid separation systems
(Draining of knock-out drums, scrubbers)
 - Seal gas filter unit
(Clean/Change filters regularly)
 - Seal gas coalescing filter units
(regularly check for drainage)

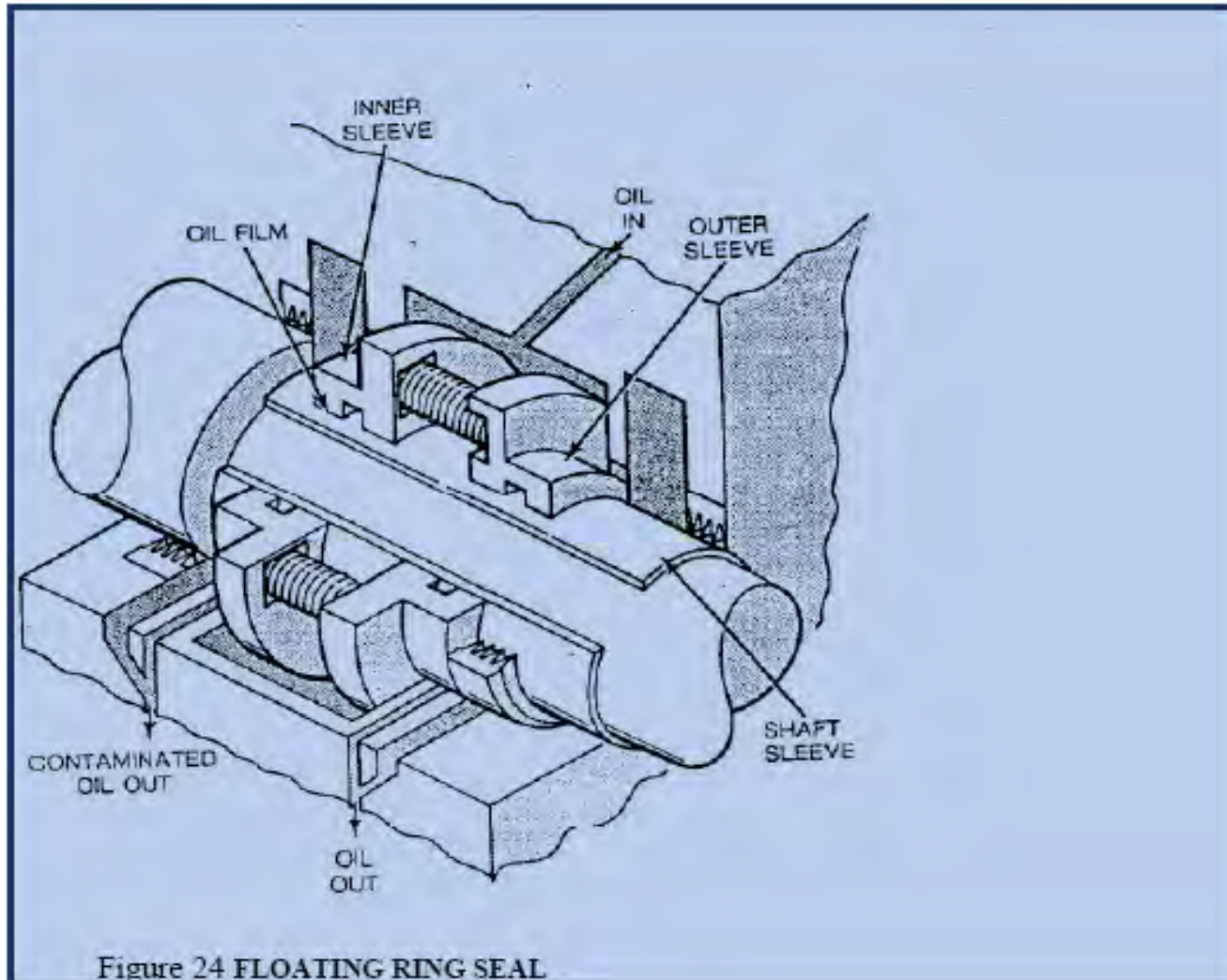
- Avoid seal reverse pressurization
 - Will lead to major seal damage.
 - Will lead to process contamination.
- Safe depressurization rates.
- Continuous Monitoring
- Avoid lube oil carry over, select best barrier seals & permissive start.

- Monitor & trend leakage/ Pressure/Flow.
- Record vibration.
- Condition Indicators
 - High Leakage
 - Damaged seal face/seat
 - Damaged polymer seal
 - Increased gap between seat & face.
 - Low leakage
 - Reduced gap between seat & face
 - Damaged outboard stage
 - High vibration
 - Damaged rotor components

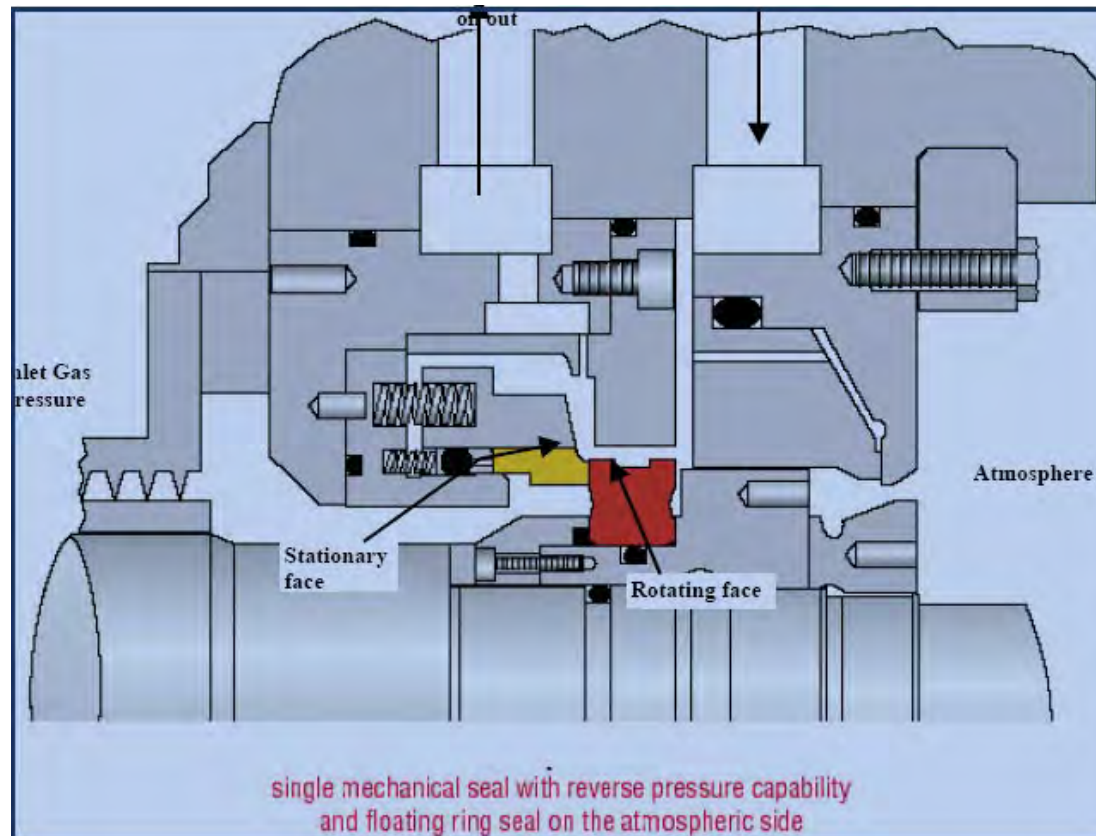
Dry gas Seal Storage

- Always store as a complete cartridge.
- Setting plates must be fitted.
- Store in original packaging.
- Grease or oil impregnated papers must not be used.
- When compressor is down for more than a week , continue supply of buffer gas.
- During initial shipping if compressor is supplied with seal installed, shaft movement must be restrained.
 - Maximum storage & running time is 10 years.
 - Maximum running time is 6 years.

Liquid Film or Floating seal

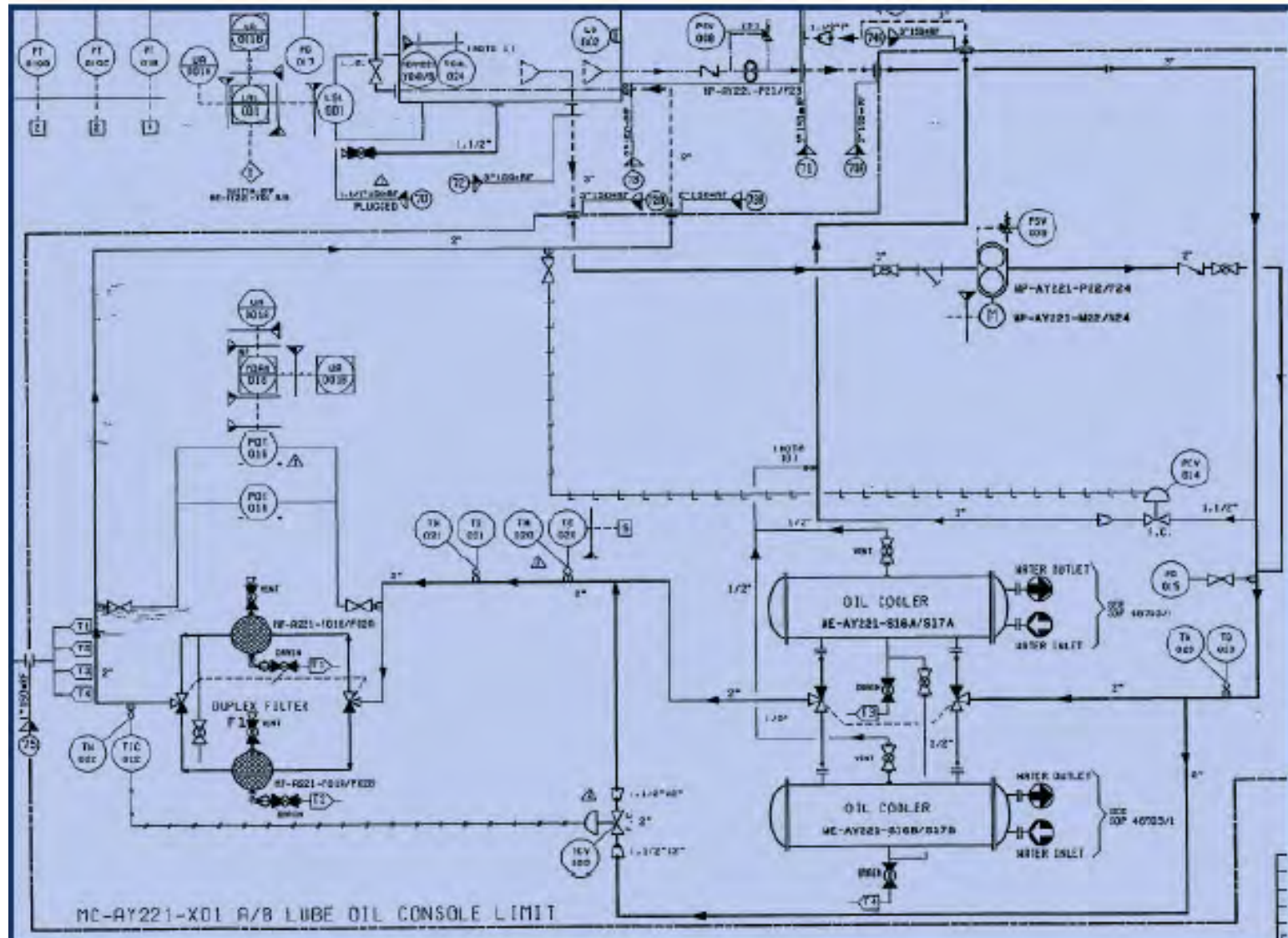


Mechanical contact seal



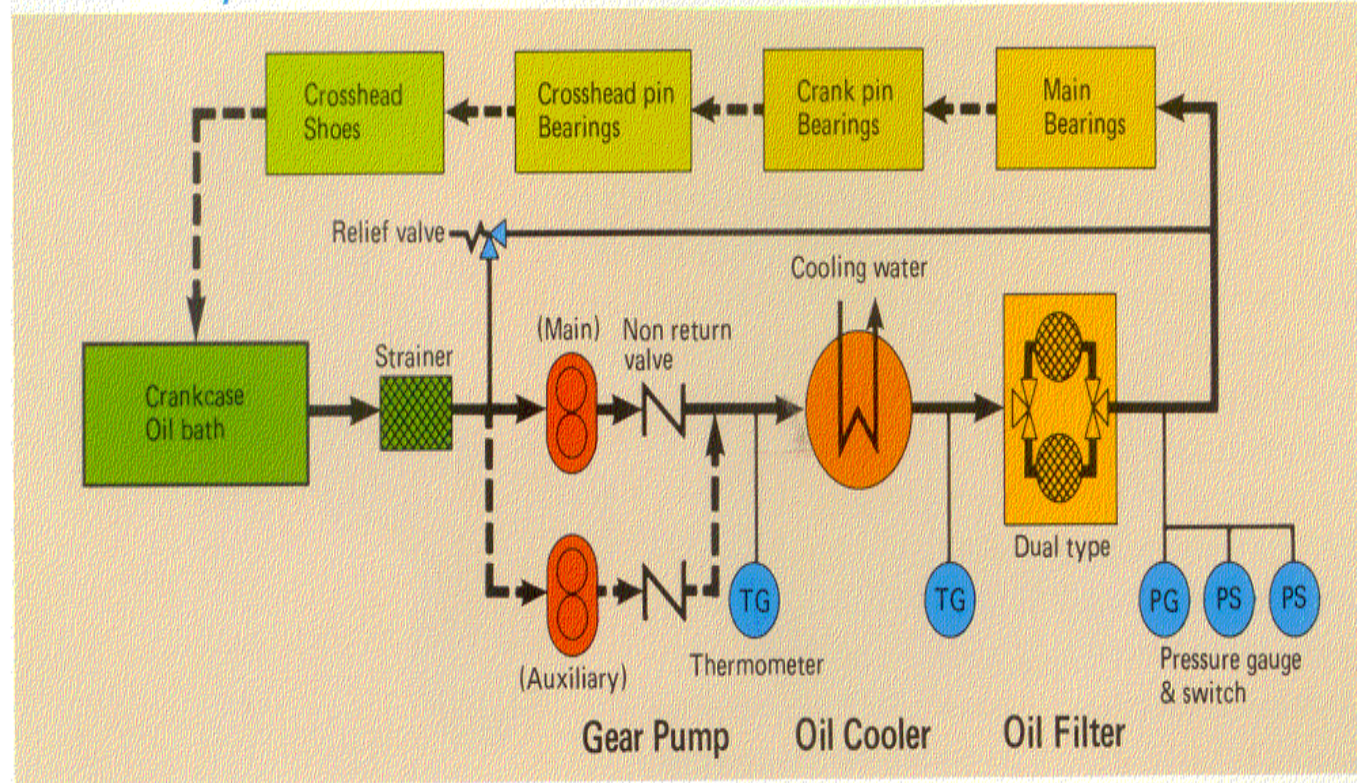
- Lubrication systems:
 - Frame lubrication
 - Provides lubrication to main bearings, connecting rod bearings & crosshead.
 - Splash lubrication for lower power size
 - Connecting rod & crank web extension dip into oil & throws oil to other parts.
 - Forced lubrication
 - Main elements are Oil pumps, filters, coolers , control valves , monitoring & protection instruments.
 - Redundant critical elements, adequate sizing of pumps for increased clearances.
 - Ease of fault findings & adequate protection.

Auxiliary System



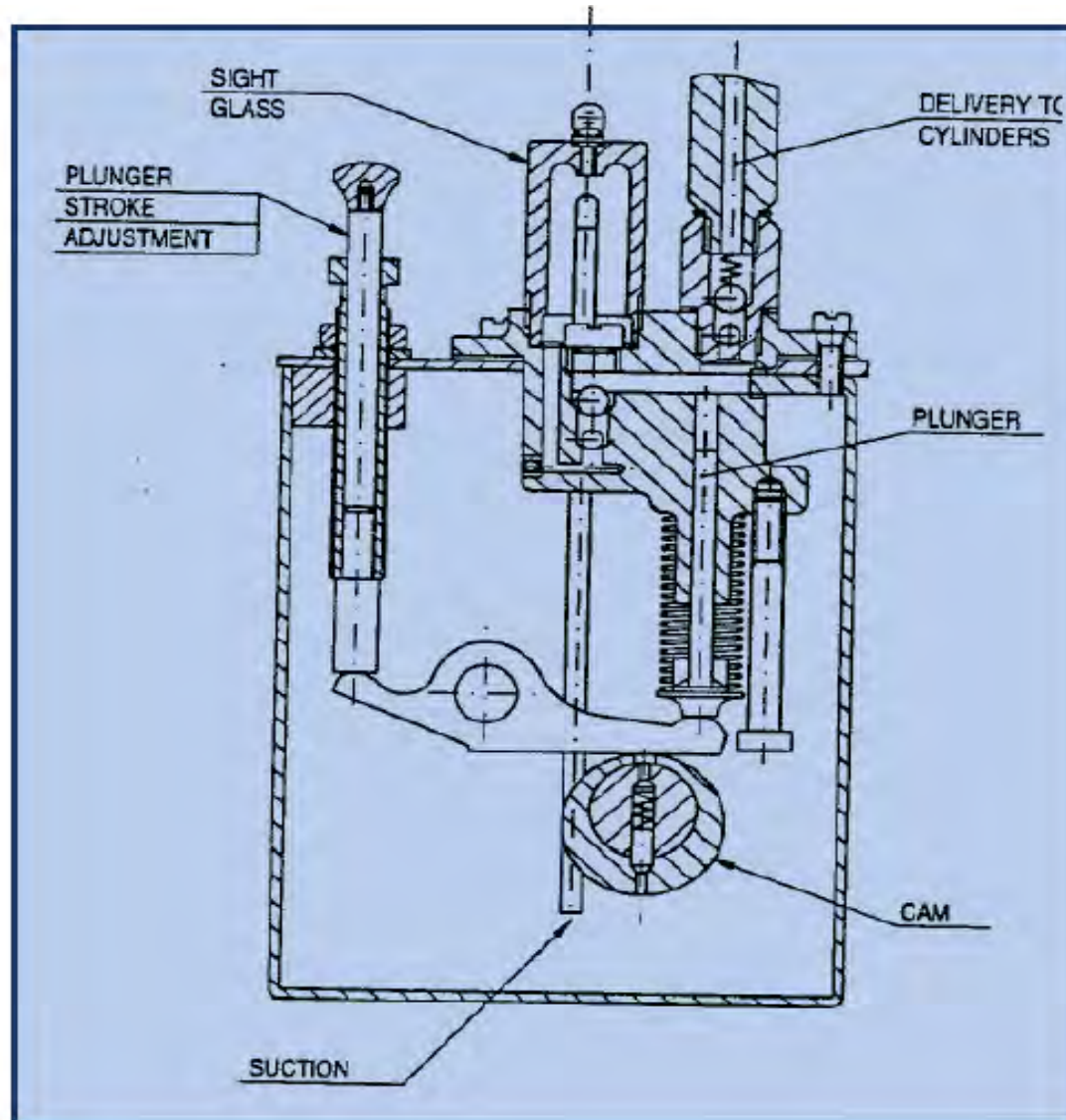
Auxiliary system

Lubrication System



- Frame Lubrication
 - For injecting metered quantity of lube oil to cylinder & stuffing box packing of lubricated compressors.
 - Main components are plunger type lubricator, sump, pre-lube facility & alarm device.
 - Mainly two types
 - Single plunger for control & delivery
 - Two separate plunger for delivery & control
 - Stroke of plunger is varied to vary oil flow control.
 - Oil Drop/Min/cylinder
- $$= \{ \text{Cylinder Dia. (In.)} \times \text{Stroke (In.)} \times \text{RPM} \} / 13800$$
 - For initial run-in period adjust 50% higher.

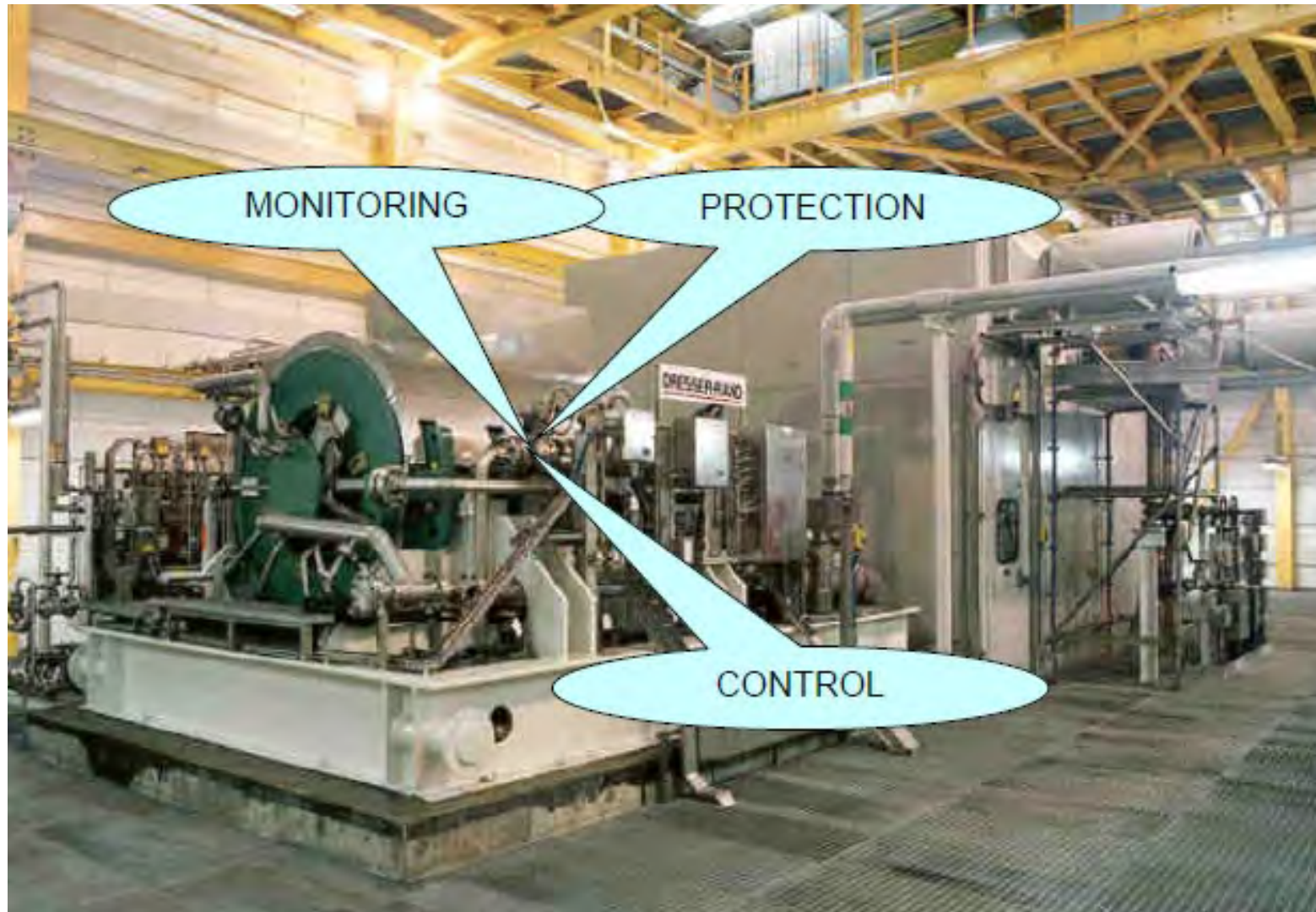
Auxiliary System



3.1 Centrifugal

- Instrumentation and control
 - Capacity Control
 - Anti Surge Control
 - Dry Gas Seal Control
 - Wet Seal Control

Main Function



- A compressor is controlled by means of a certain number of devices which are acted upon:
 1. For start-up and shut down
 2. To settle or vary operating conditions while the compressor is running, in order to obtain the required performance.
- The use of the control devices is closely tied to checking a series of significant parameters (of the operating conditions) and of their variations.
- In general, these parameters are:
 1. Pressure and temperature of the gas at suction, inter-stage and delivery;
 2. Lubricating oil pressure;
 3. Temperatures and flows of the cooling fluid;
 4. Critical variables of the reducer:
 - Lubricating oil pressure;
 - Bearing temperature
 5. Critical variables of the driver

- The machine's control devices that allow to control the parameters can be mechanical or electrical and less often hydraulic or pneumatic.
- They can be manually operated, locally or remotely, or they can be linked to automatic control systems.
- Mechanical devices are mainly cut-off and control valves.
- Among cut-off valves:
 - Cut-off gas upstream and downstream of the compressor.
 - Cut-off of cooling fluid, general plant-wide and individual on the various user applications
 - Vent or flare off
 - By-pass for unloading
 - Inlet of plant purging gas
 - Separator drain
 - Various drains and vents
 - Cut-off of control and instrument air
 - Cut-off steam for heating (e.g oil) pipe tracing
 - Other minor loops.

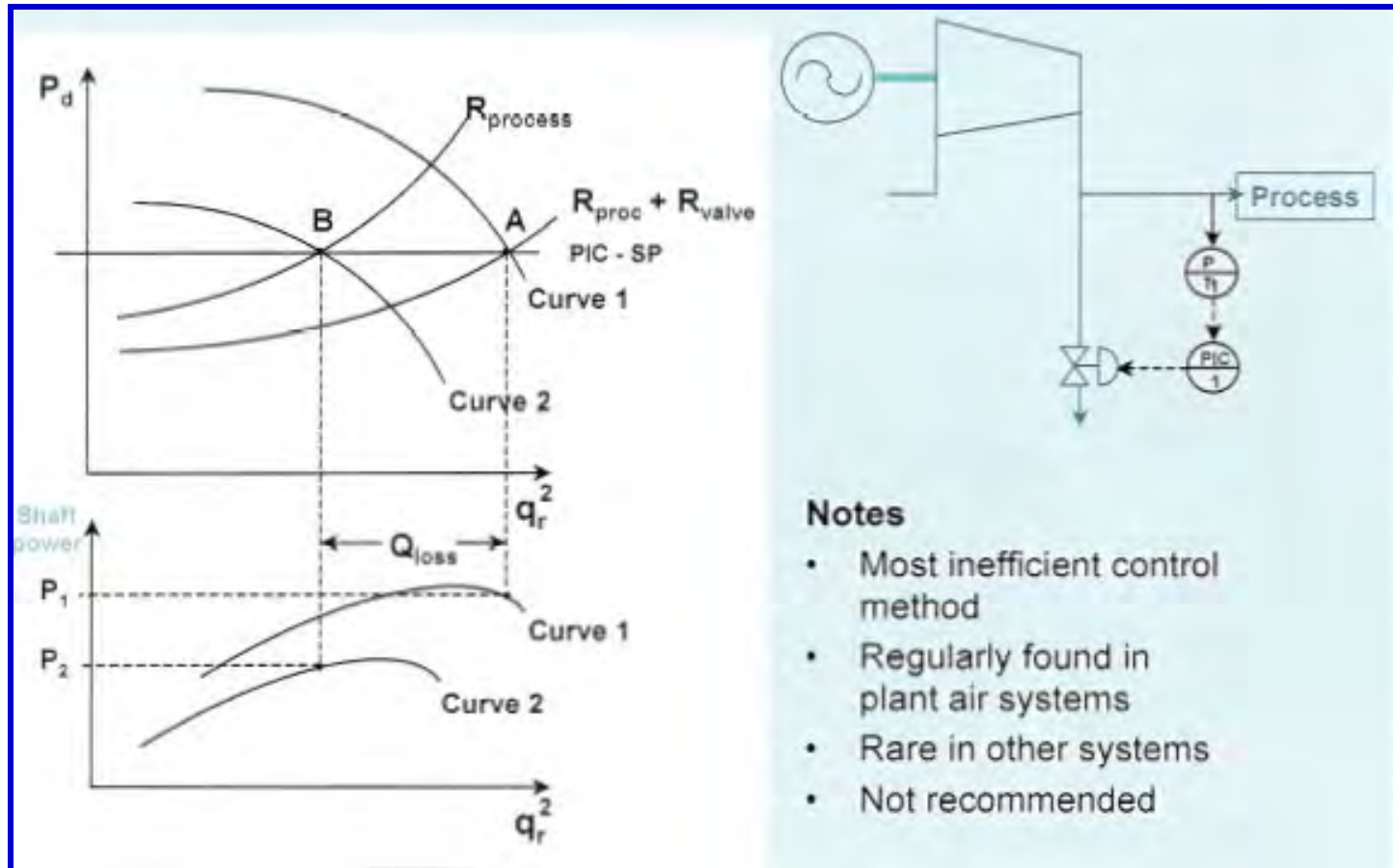
- As control valves, they may be:
 - Recycle for capacity control
 - Control of cooling fluid flows
 - Control of lubricating oil pressure.
- Electrical (or pneumatic) devices are those to start and to shut down:
 - Main driver,
 - Oil and cooling fluids pumps
 - Radiator fans
 - Crankgear and cylinder oil heaters,
 - Acoustic hood fans

- The compressor's operation and maintenance manual contains the operating instructions for the control devices, as well as the execution times various phases of the machine's operation, such as:
 - Unloading for start-up
 - Starting the auxiliaries
 - Starting the compressor
 - Loading the compressor
 - Capacity control
 - Unloading for shut-down
 - Main driver shut-down shutting down auxiliaries
- The operating sequence of the devices, i.e the logic, is determined as a function of the service the machine has to provide. The sequence can be done either manually, partial automation or automatically.

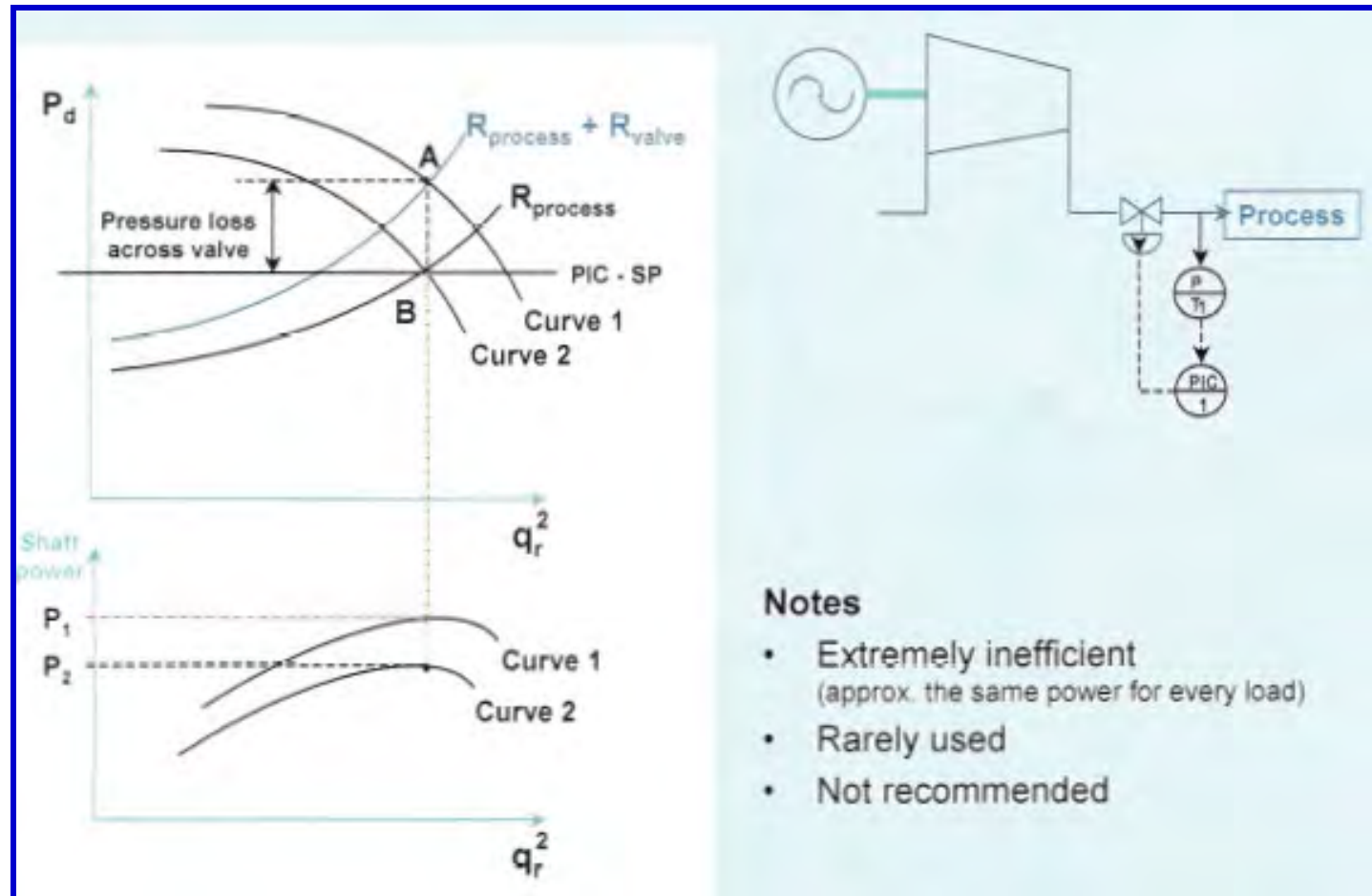
- The functions of monitoring the various sequences and parameters functions are performed through control panels, whose system includes:
 - Sequence control logic (with PLC)
 - Shut down circuits (protection)
 - Logic to signal the causes of alarm and shut down
 - Lighted indicators of the status of compressor and auxiliaries
 - Various control push buttons
- In a more complex configuration, the following may also be installed on the control panel:
 - Gauges measuring process and machine variables,
 - Regulators of the process variables
 - Recorders
 - Monitoring systems

- Matches the compressor throughput to the load.
- Can be based on controlling
 - Net flow to the user (recycle)
 - Discharge pressure
 - Suction pressure
 - Variable inlet guide vanes
 - Speed

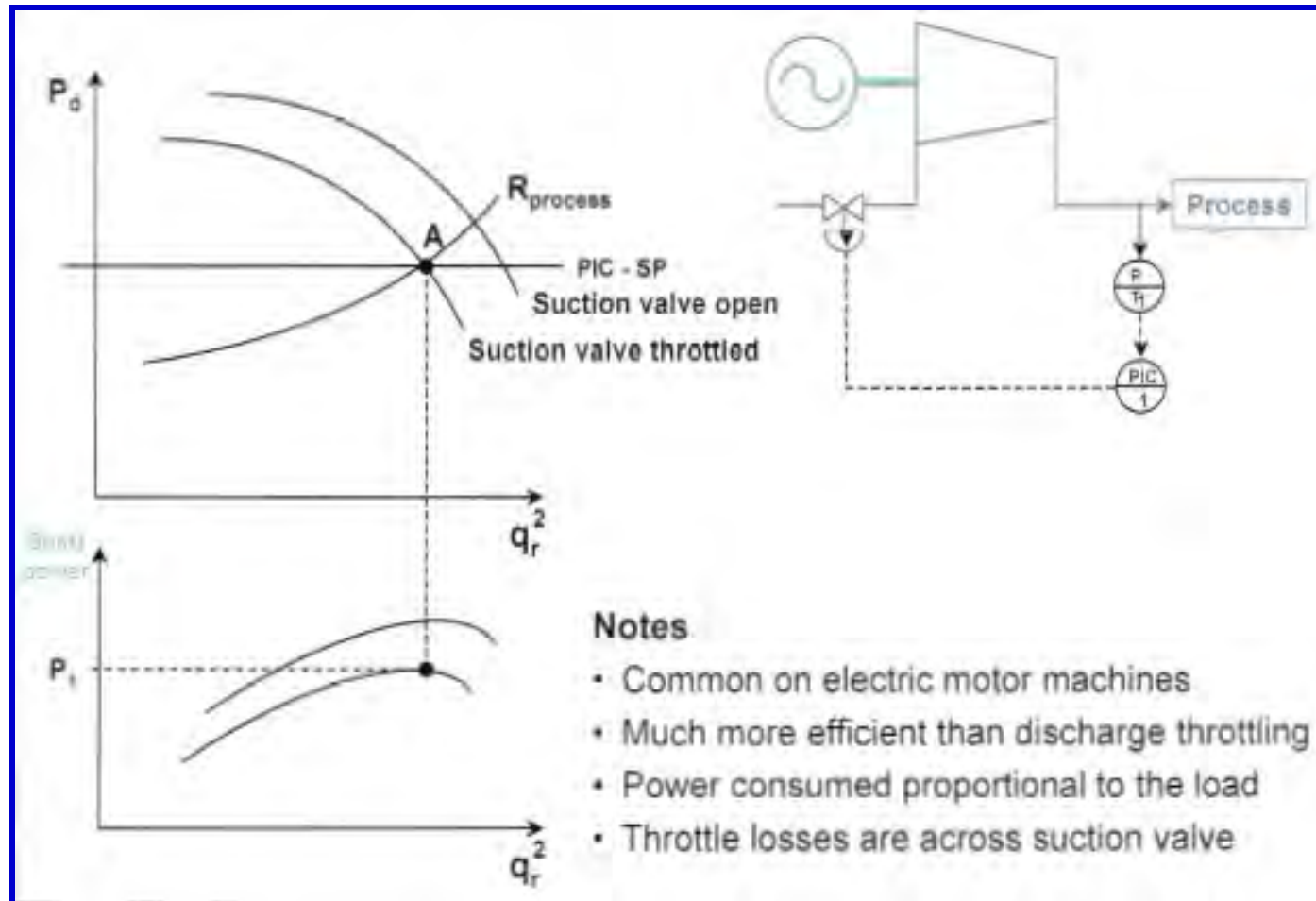
Performance control by blow-off control



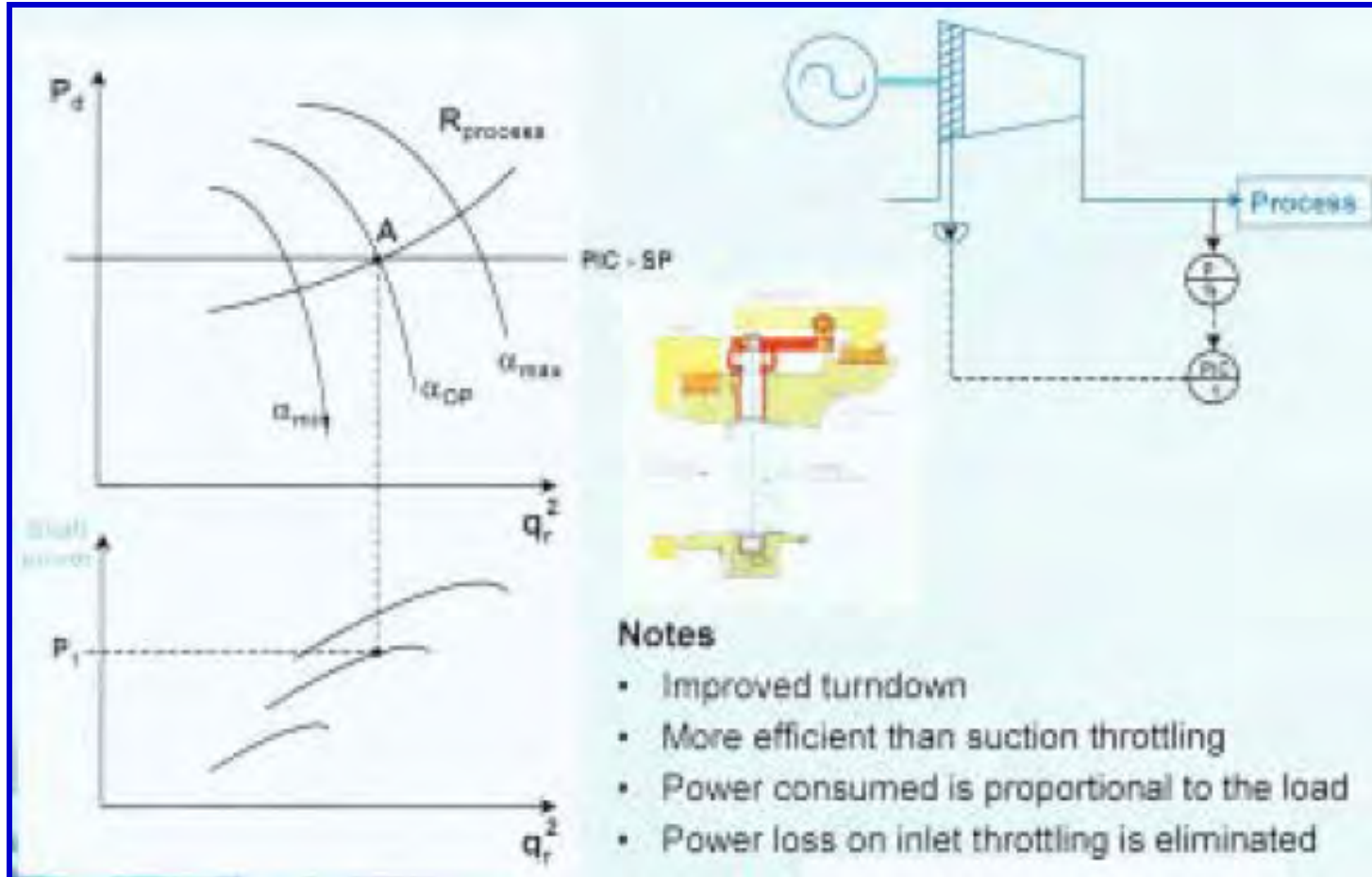
Performance control by discharge throttling



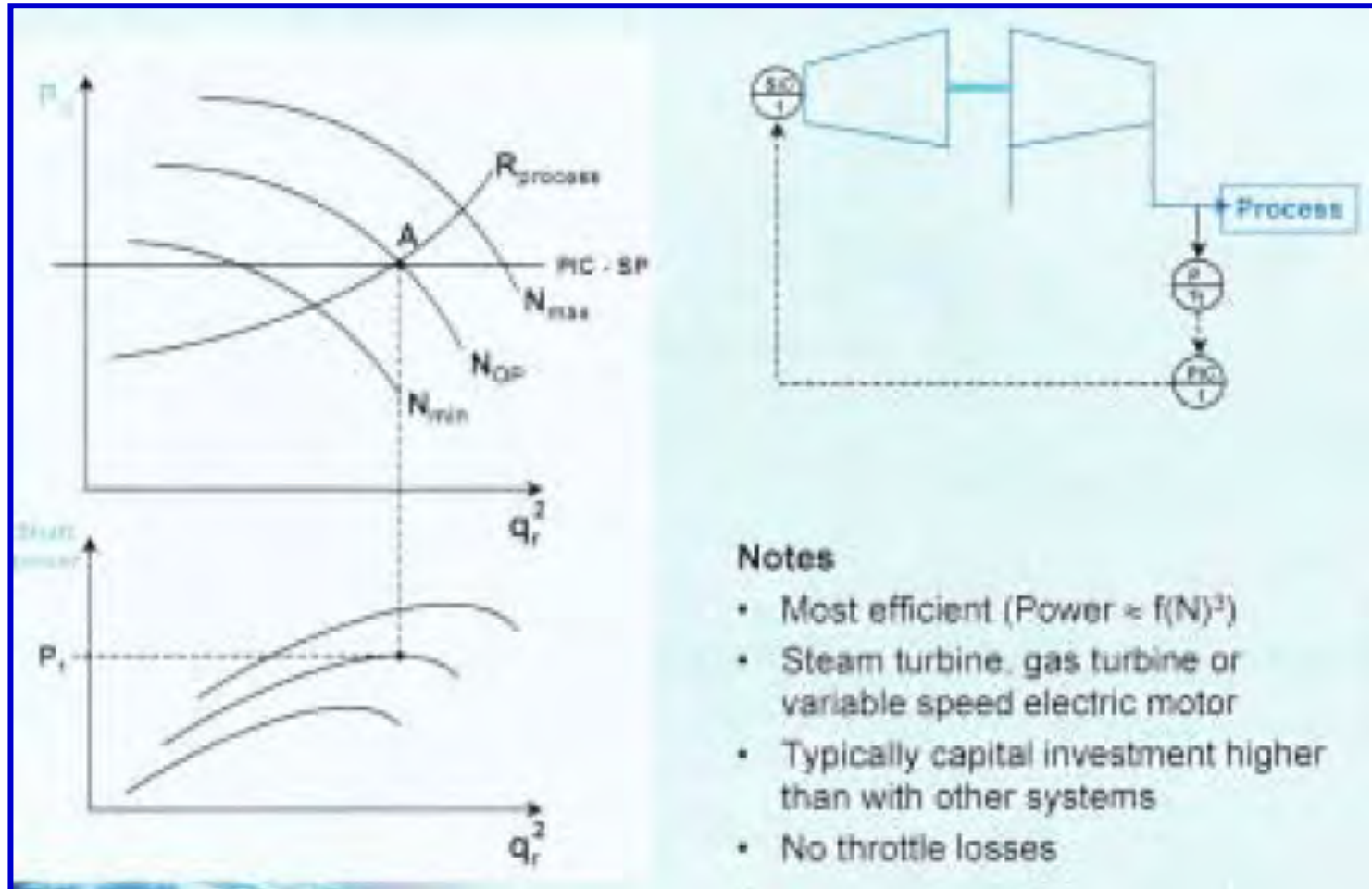
Performance Control by suction throttling

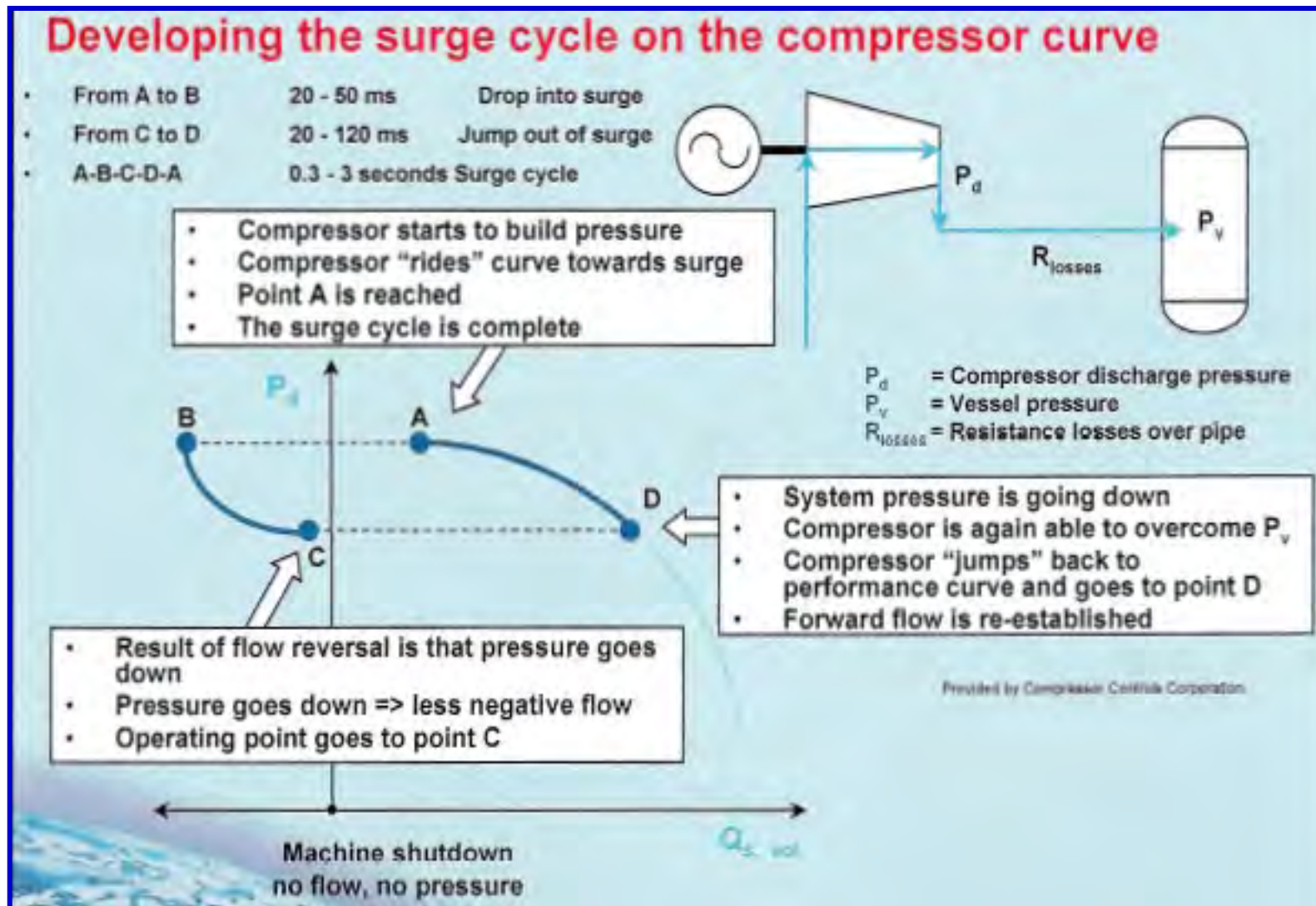


Performance control by adjustable inlet guide vanes

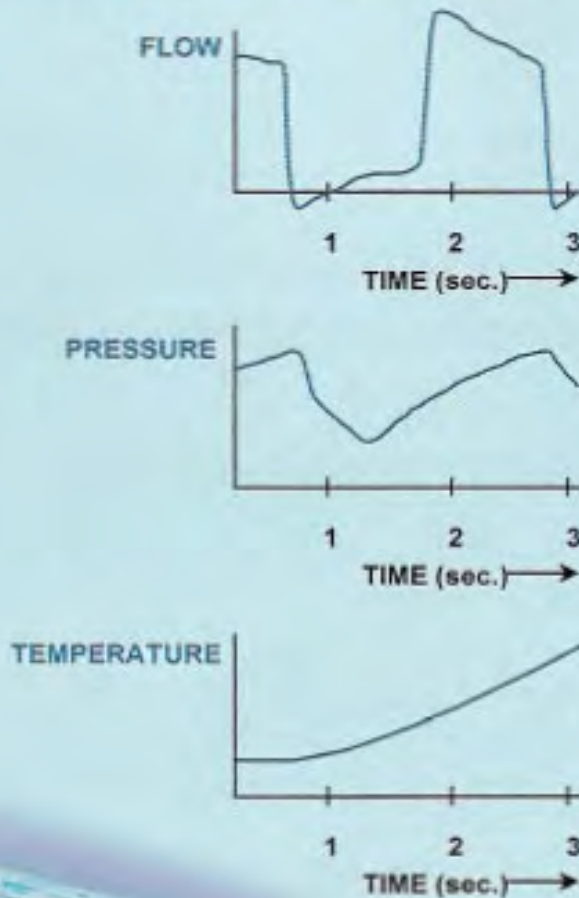


Performance control by speed variation





Process parameters during surge



- Rapid flow oscillations
 - Thrust reversals
 - Potential damage
-
- Rapid pressure oscillations with process instability
-
- Rising temperatures inside compressor

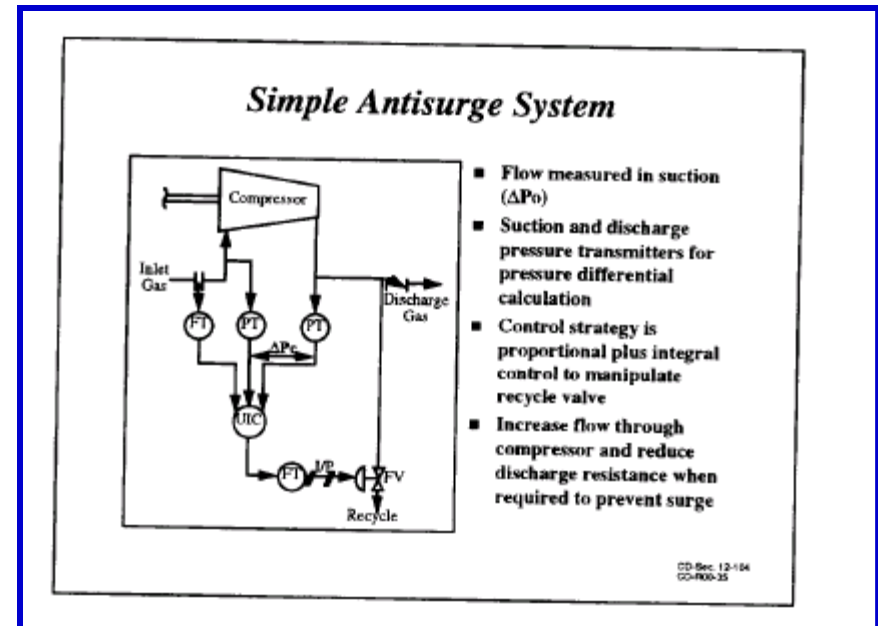
Provided by Compressor Controls Corporation.

- Surge Description
 - Flow reversals in 20 to 50 milliseconds.
 - Surge cycles at a rate of 0.3 to 3s per cycle.
 - Compressor vibrations
 - Temperature rise
 - Noise
 - Axial displacement
- Consequences of surge
 - No flow and pressure
 - Damage in sequence with increasing severity to seals, bearings, impellers & shaft.
 - Reduced compressor life.
 - Catastrophic damage
 - Heavy fluctuations of load on driver.

- Internal
 - Reduction in performance.
 - Change in gas composition.
- External
 - Fouled downstream cooler
 - Downstream upsets.
 - Fouled strainer
 - Power limits due to driver problems.
 - Human errors

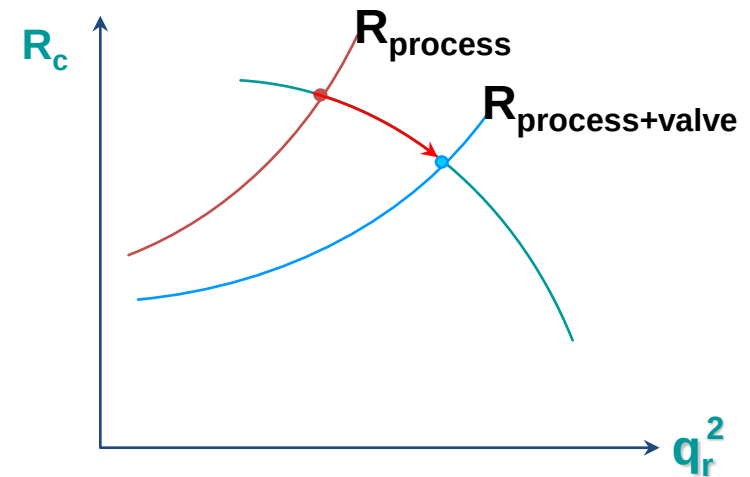
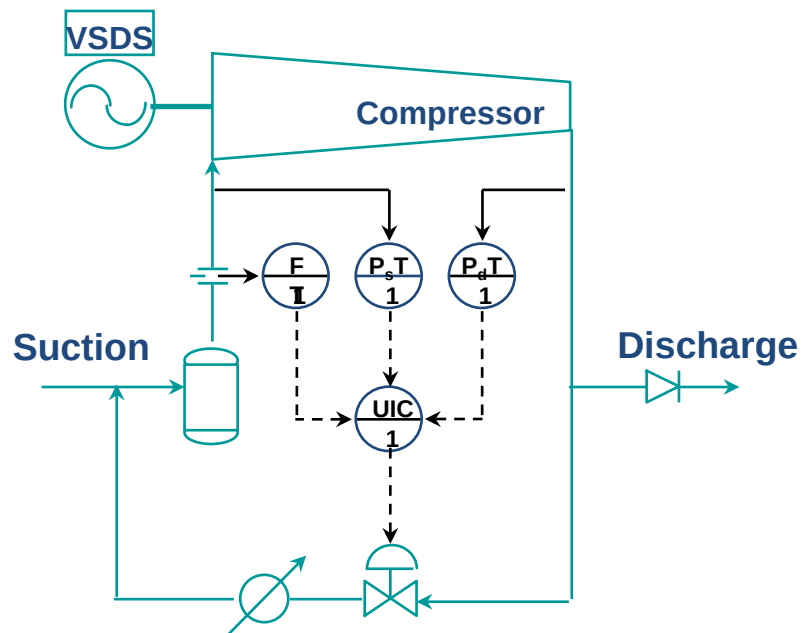
Anti-surge control system

- Simple anti-surge protections like low flow lock & based on simple algorithm.
- Sophisticated anti-surge control system required for variable speed & variable gas services. (CCC).



Basic anti-surge control system

- The anti-surge controller UIC-1 protects the compressor against surge by opening the recycle valve
- Opening of the recycle valve lowers the resistance felt by the compressor
- This takes the compressor away from surge



- Surge parameter based on invariant coordinates R_c & q_r
 - Flow measured in suction (ΔP_o)
 - P_s and P_d transmitters used to calculate R_c

- Surge line moves closer to operating point
 - Molecular weight of gas reduces.
 - Suction temperature increases.
 - Suction pressure reduces
 - Impeller becomes fouled.

- Instrument installed on the centrifugal compressor is essential for **checking their performance** and **continuous surveillance over their operation**.
- It's normally relates to:
 1. Vibrations and axial displacement of the rotor
 2. Oil pressure to the bearings
 3. Bearing temperature
 4. Axial thrust of the rotor
 5. Temperature of oil discharge from the seals
 6. ΔP oil/gas on the seals
 7. ΔP oil/gas on the outer labyrinth

- All vibration and temperature signals come to Vibration System
- These signals are hardwired from field to I/O Module
- Monitor itself has ability to provide hardwired alarms (ALERT) and trips (DANGER) per input to TCP PLC.

Vibration System (BN 3500 and BN 1701)



3

Wireless



1701

**Monitors for Balance
of Plant Equipment**

Bently-Nevada 1701 Vibration Monitor

Part of AB System

Connects via Network Adapter

Integrated I/S Isolator (optional)

Meet minimum requirement



1701

Vibration System



Vibration System BN 3500

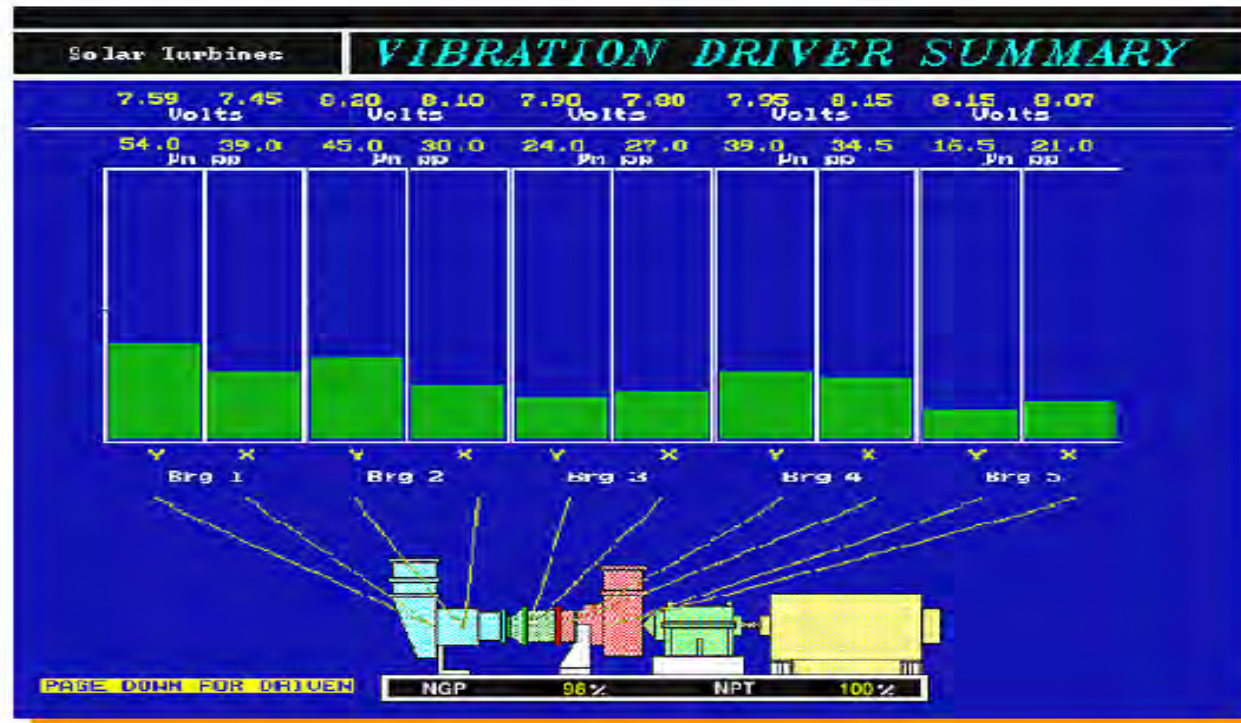
Redundant Power Supplies



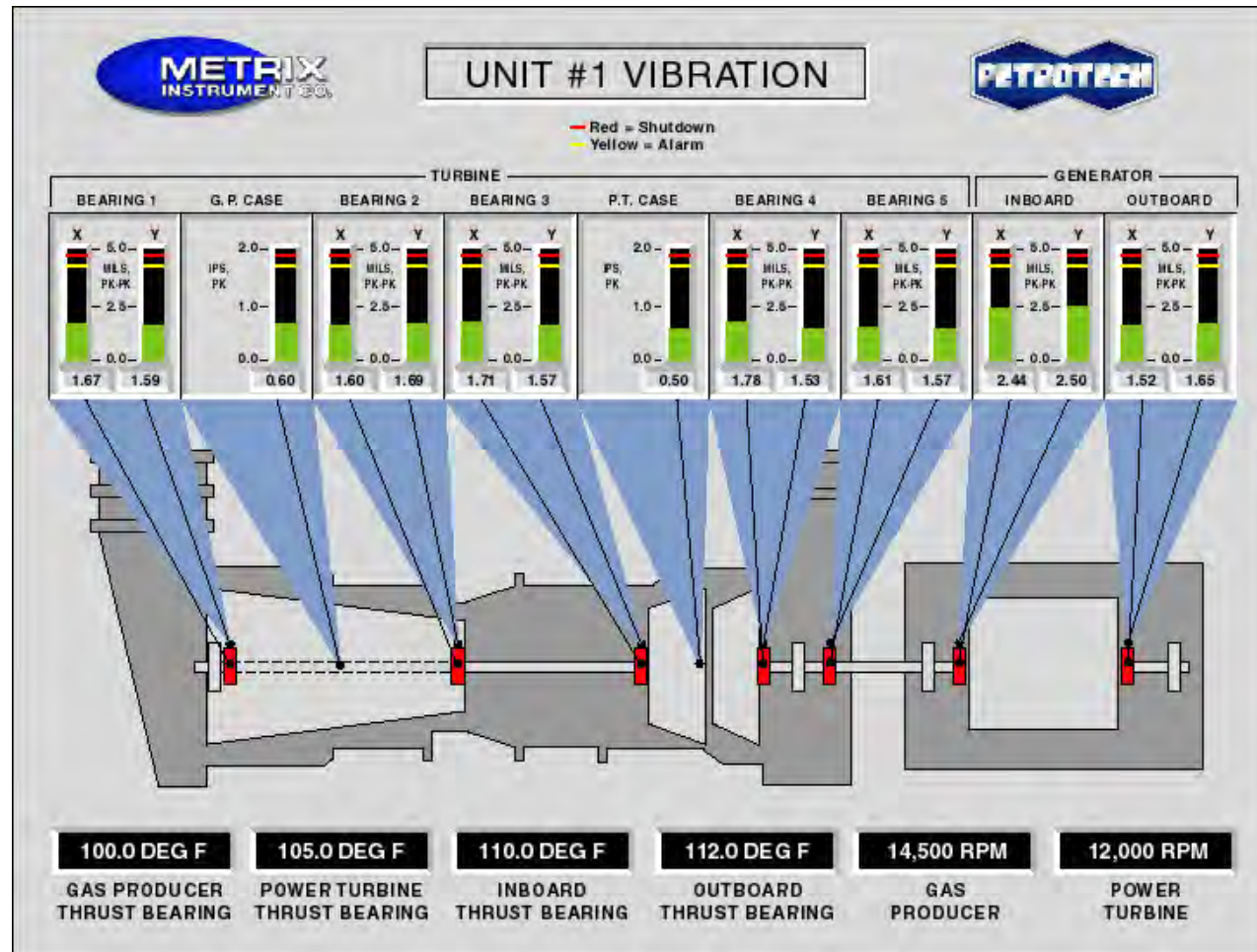
“Hot Swappable” Monitors



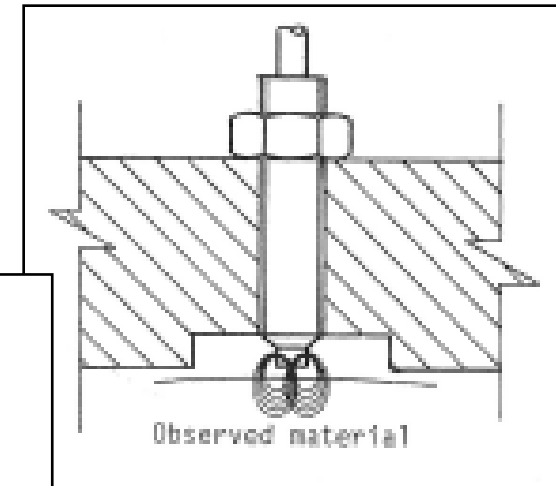
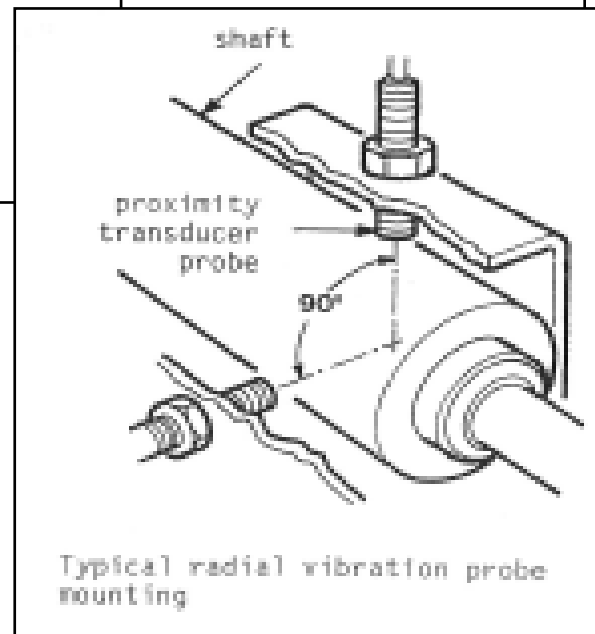
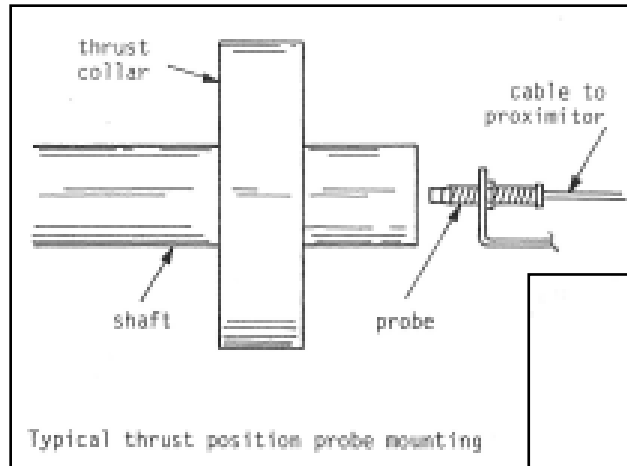
Standard Monitoring System



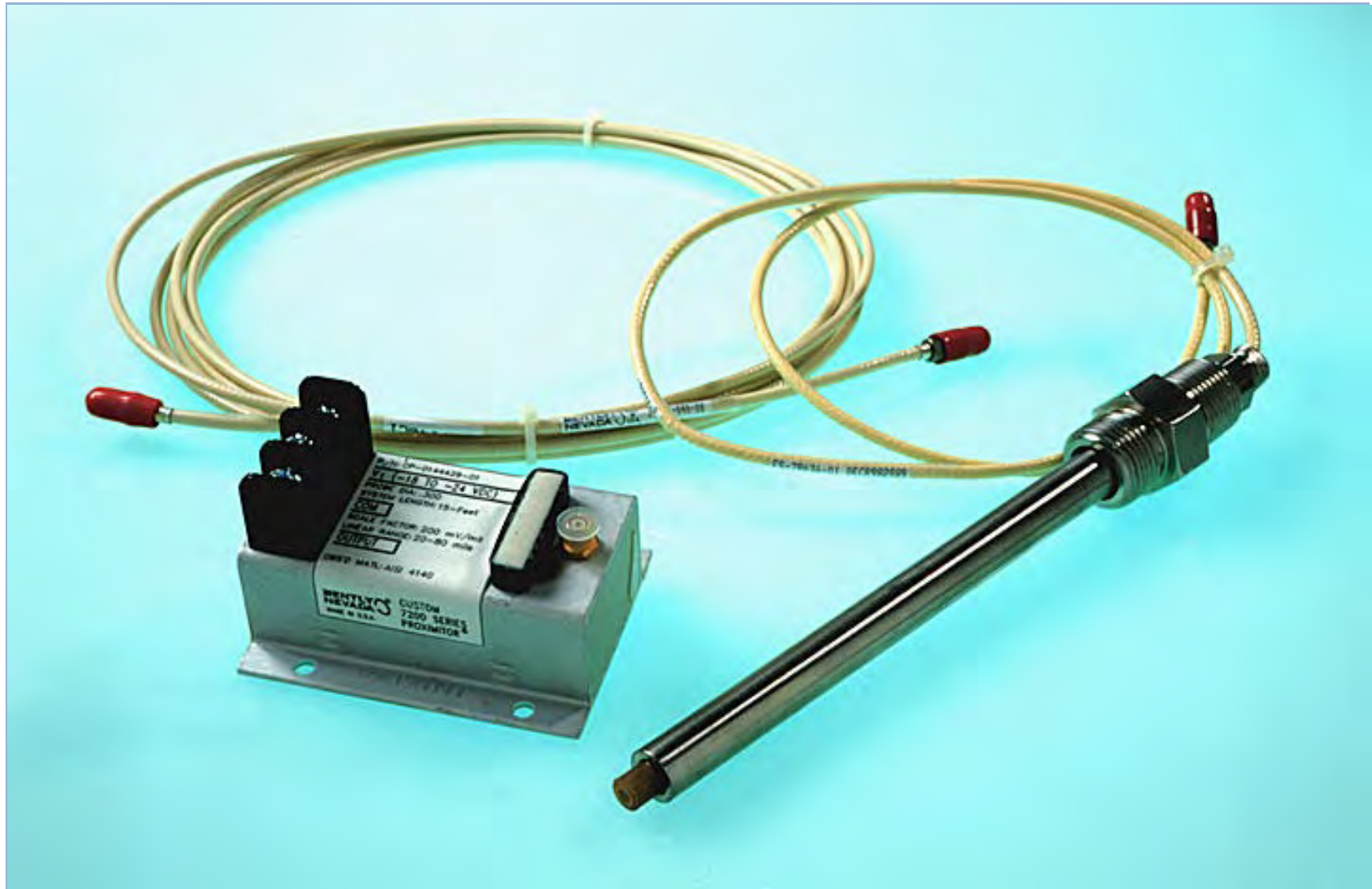
Vibration Summary



- **Proximity probes** are normally used to detect vibration and axial displacement of the rotor.
- They must be positioned about 1.5mm from the shaft, after being firmly secured to the structure. (As shown in the figure X)
- Operating principle is based on the variation of a magnetic field generated between probe and shaft, due to variation in the shaft/probe distance.
- The reading is taken on a scale graduated in thousandth of a millimeter (microns) or in thousands of an inch (1Mil = 0.0254 mm)

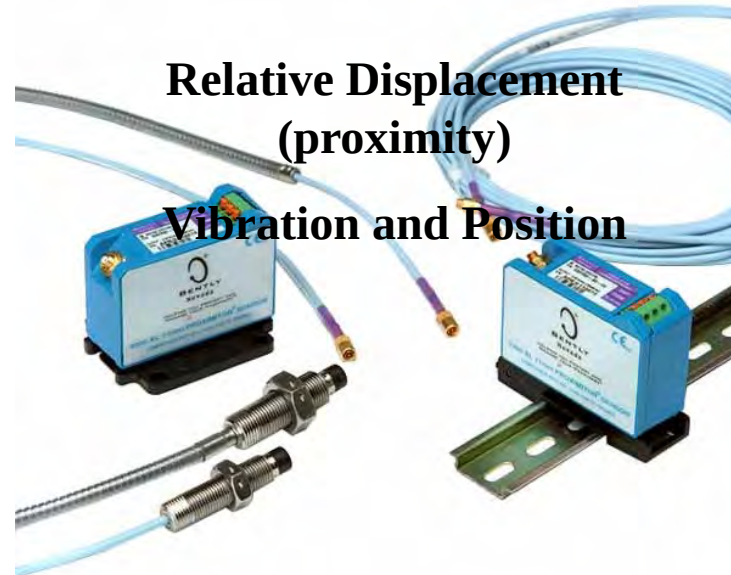


Vibration System





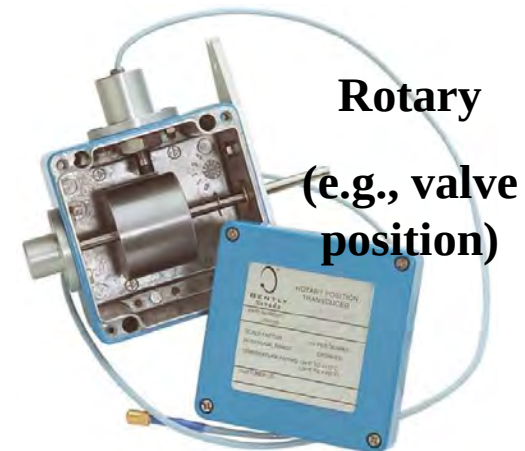
Combustion Pressure



**Relative Displacement
(proximity)
Vibration and Position**



Seal Pressure



**Rotary
(e.g., valve
position)**

Relative Displacement (proximity) Vibration and Position



BN 3300 XL Proximity Transducer



Electrical Length
(usually 9m)

1. Signal Output
2. Signal Common
3. Transducer Power



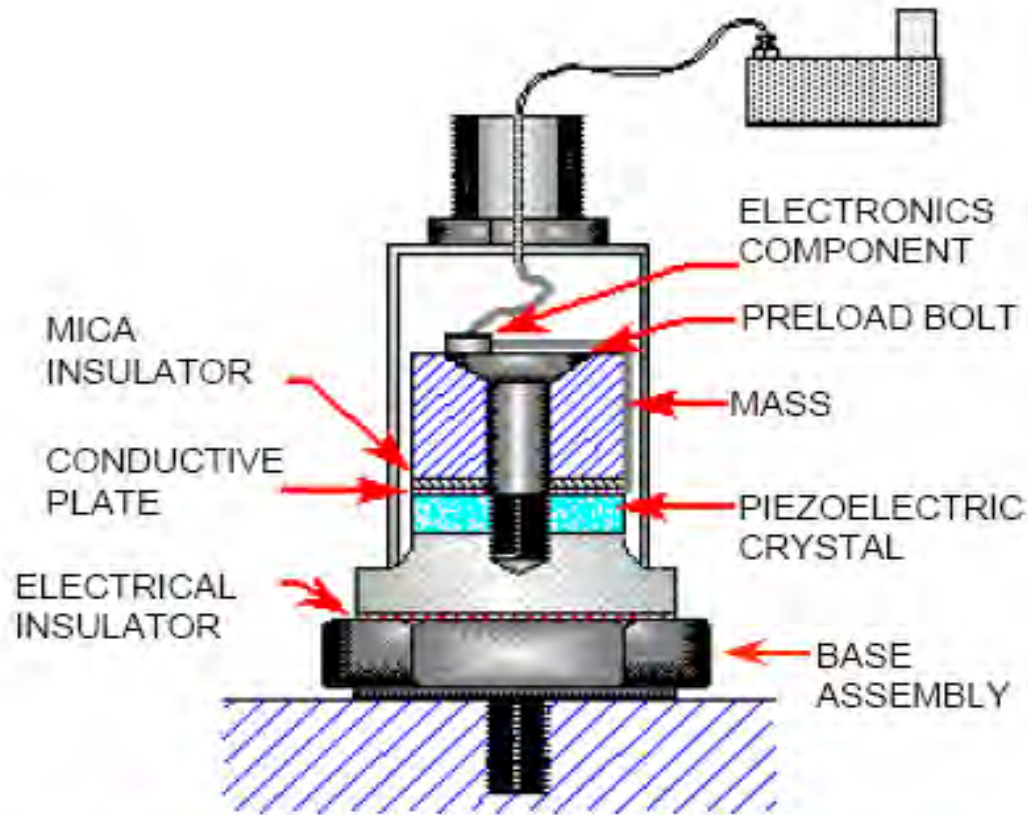
Velocity Transducer



Accelerometer Transducer



Vibration Probe



Cut away view of Accelerometer

Accelerometer



Velocity meter

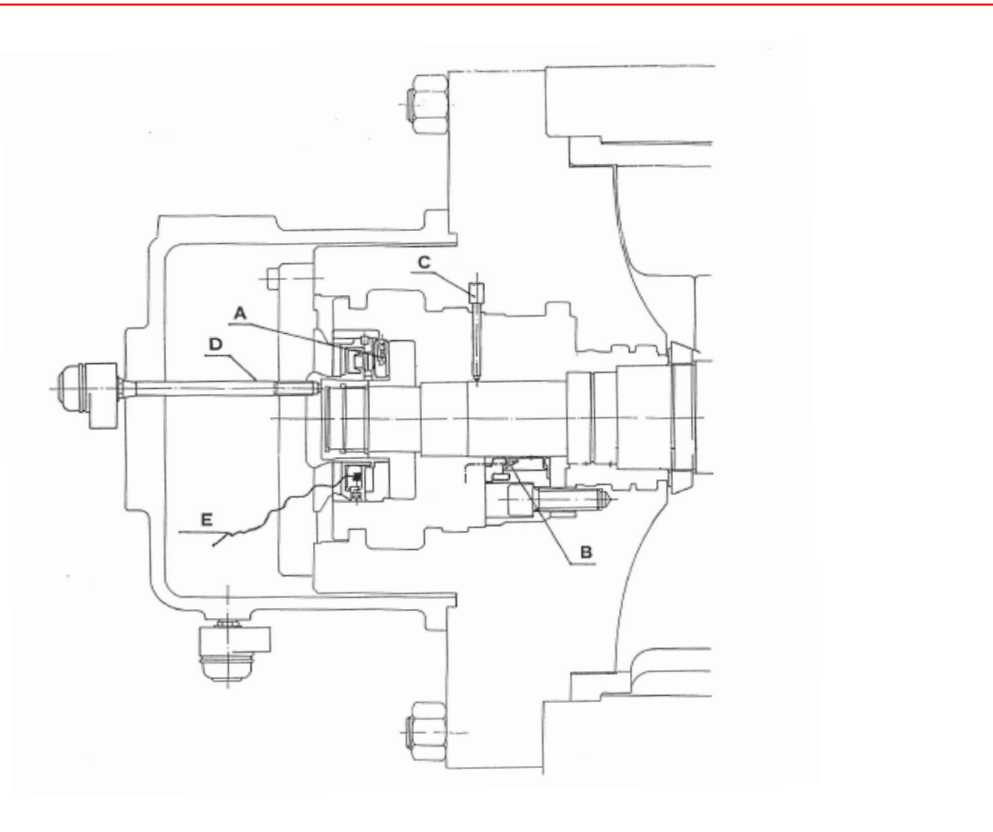


Vibration Probe

Speed Magnetic Pickup



- Typical Instrument layout installed on a centrifugal compressor:



A = Thermocouple
on thrust bearing

B = Thermocouple
on journal bearing

C = Radial vibration
sensor

D = Axial
Movement sensor

E = Load Cell

End of presentation

THANK YOU

SKG 12 Mechanical Engineering

DOMI of Compressor and Blower (4 of 7)

-Construction of Compressor (Reciprocating Compressor)

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4.0 Reciprocating

How it works

- Thermodynamic
 - Volumetric Efficiency

Reciprocating Compressor

Various compressors are found in almost every industrial facility. Types of gases compressed include the following:

- Air for compressed tool and instrument air systems
- Hydrogen, oxygen, etc. for chemical processing
- Light hydrocarbon fractions in refining
- Various gases for storage or transmission



Reciprocating Compressor



Natural gas reciprocating compressor

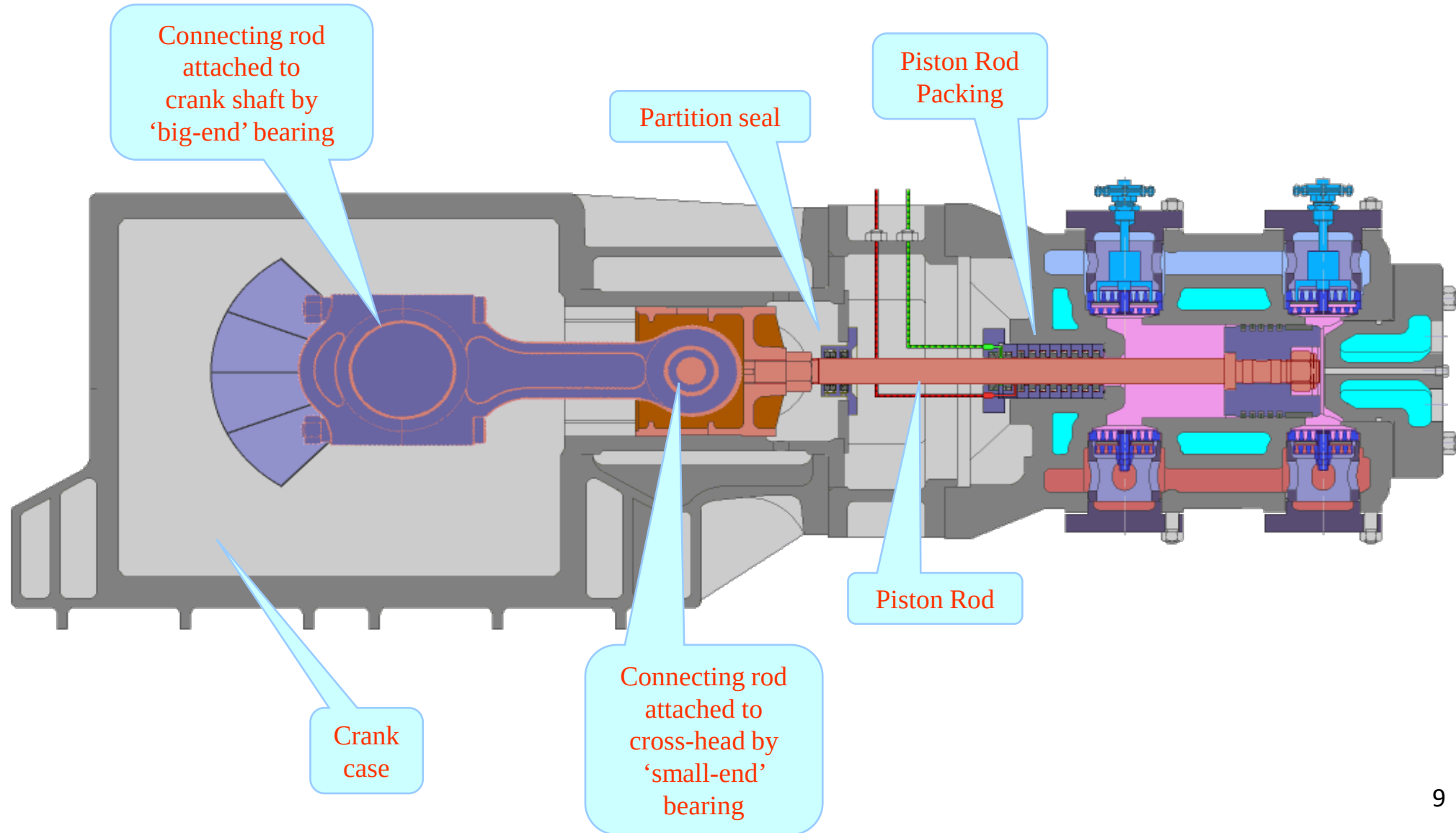


- Reciprocating compressors are the best known & most widely used among the positive displacement category.
- Advantages & Disadvantages over other types
 - Advantages
 - High Compression ratios
 - Variable pressure at constant volume
 - Molecular weight flexibility
 - Basic technology
 - High efficiency

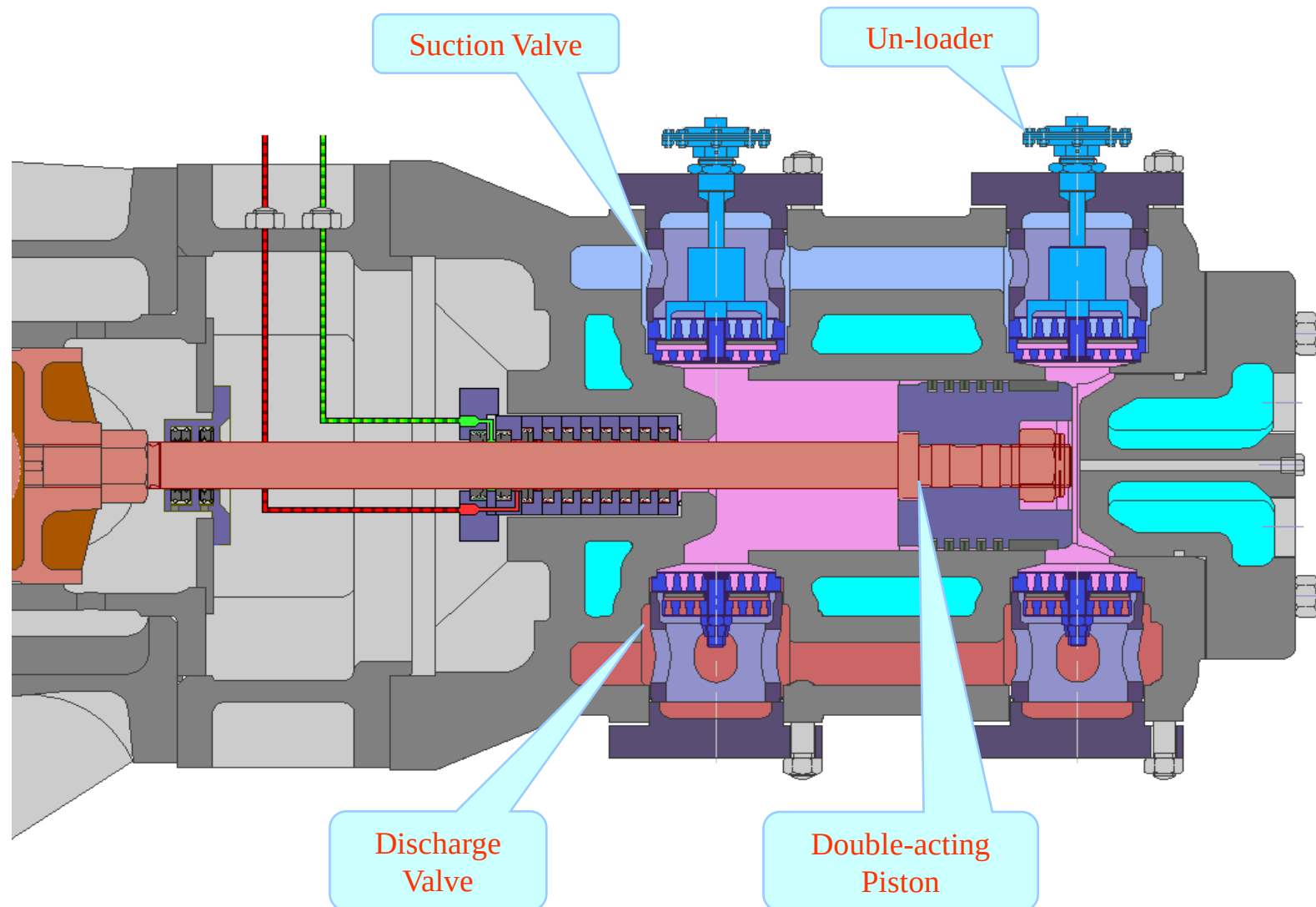
- Disadvantages
 - Separate & Heavy foundations.
 - Pulsating flow.
 - Vulnerable to dirt & liquid.
 - Low reliability.
 - Bigger Plot area.
 - Torsional implications.

- Reciprocating compressors are classified in many ways
 - Type of cylinder lubrication
 - Lubricated
 - Non-lubricated or Dry running
 - Ring-less or Labyrinth type
 - Cylinder Loading
 - Single acting
 - Double acting
 - Cooling
 - Air cooled
 - Water cooled
 - Cylinder arrangement
 - Horizontal opposed balanced
 - Vertical inline or v typed.

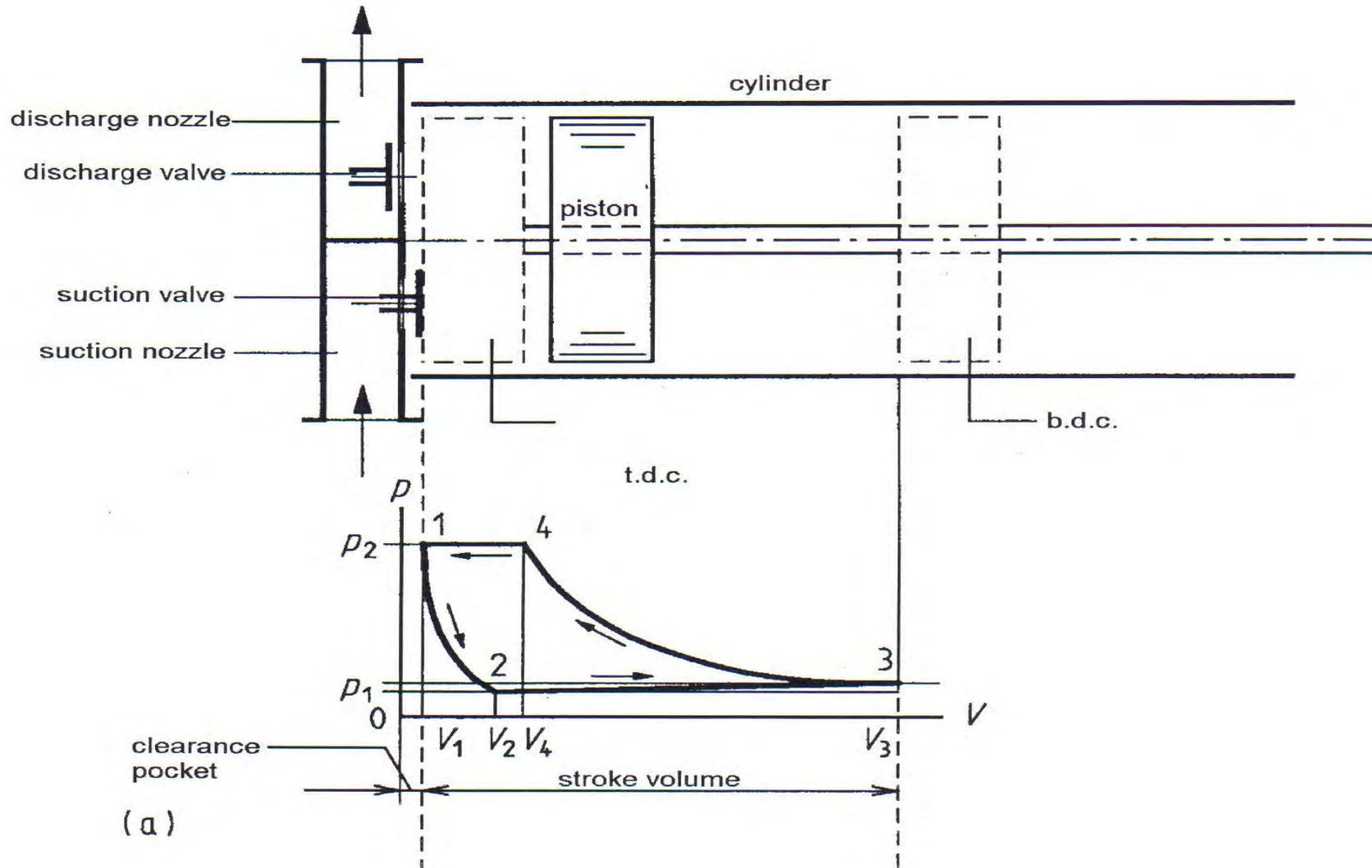
Cross-section Of Compressor



Cross-section Of Compressor

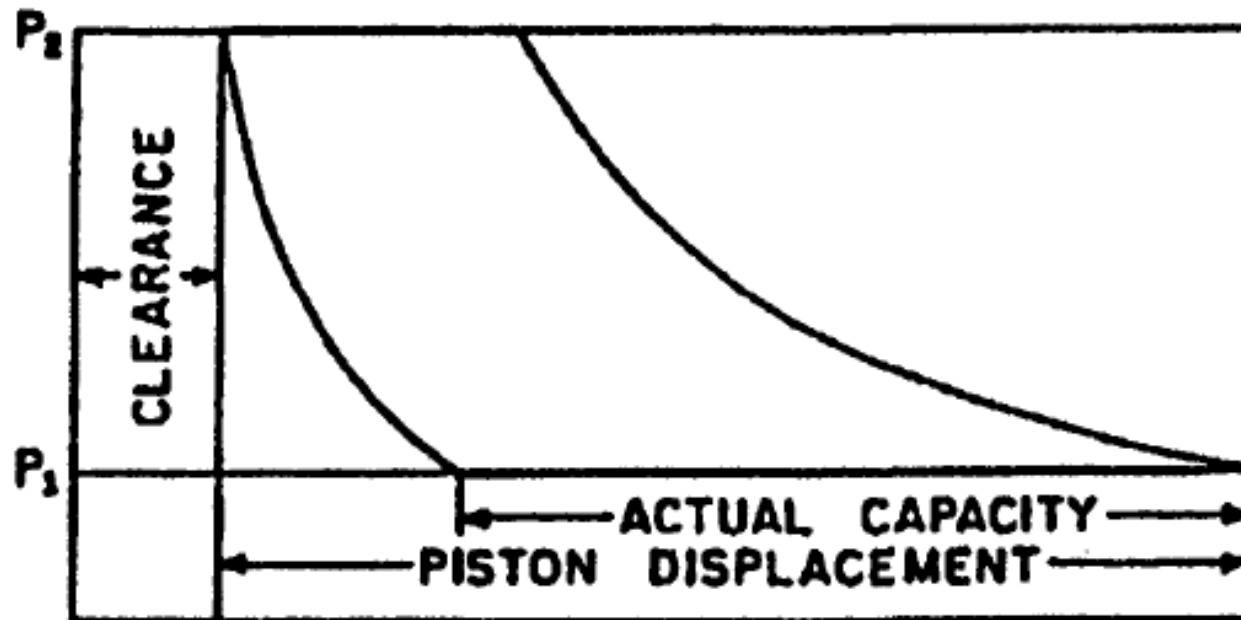


How Reciprocating Compressor Works



Volumetric Efficiency

The flow through a cylinder is a function of the volumetric efficiency and the cylinder displacement. The volumetric efficiency is the percentage of the swept volume that effectively moves the gas. The clearance volume determines how much of the swept volume contributes to the flow as the graph shows.



$$\text{Volumetric Efficiency (VE)} = \frac{\text{Actual Capacity}}{\text{Piston Displacement}}$$

The VE is calculated from this equation which takes into account the clearance volume c , compressibility factors and a correction factor L which is an indicated of leakage between the headend and crankend and other losses.

$$VE = (100 - L) - c \left[\frac{Z_{suct}}{Z_{disch}} r^{1/k} - 1 \right]$$

The value of $100-L$ may range from around 88% to 98% and will vary with different cylinders and manufacturers. The actual value used by a manufacturer is often proprietary and difficult to obtain. For rough approximation, the following values can be used:

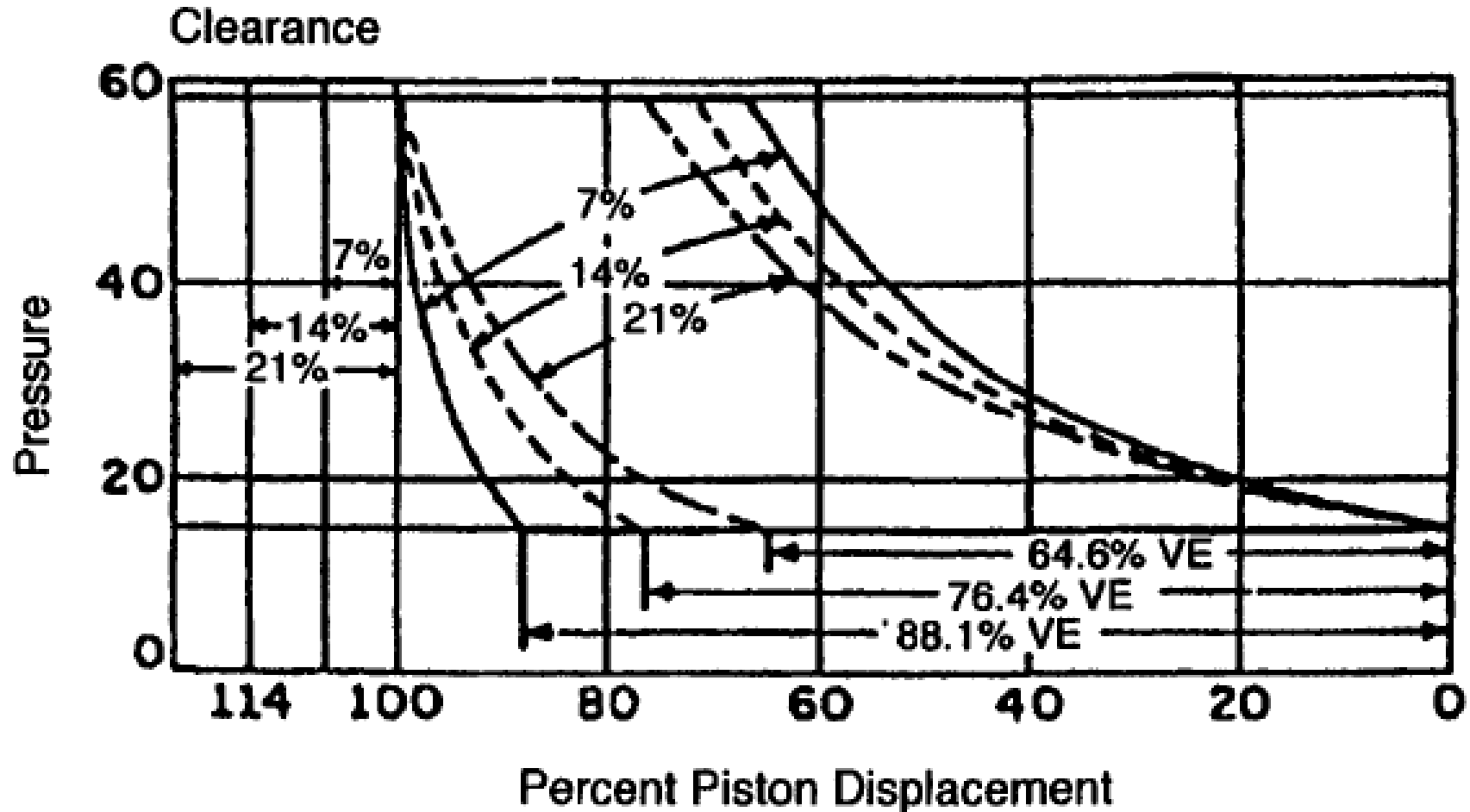
- large slow speed cylinders with low ratios (<1.5): $100-L$ may have values around 97
- slow speed process cylinders with higher ratios (>1.5): 96-r
- high speed cylinders: 96-2r

The actual flow (actual volume at the real pressure and temperature) is then found by these equations. In order to determine the standard flow it has to be corrected back to standard pressure and temperature conditions.

$$PD = \left[\begin{array}{c} \text{piston area,} \\ \text{head end} \end{array} + \begin{array}{c} \text{piston area,} \\ \text{crank end} \end{array} \right] \times \text{stroke} \times \text{rpm}$$

$$Q_a = PD \times \frac{VE}{100}$$

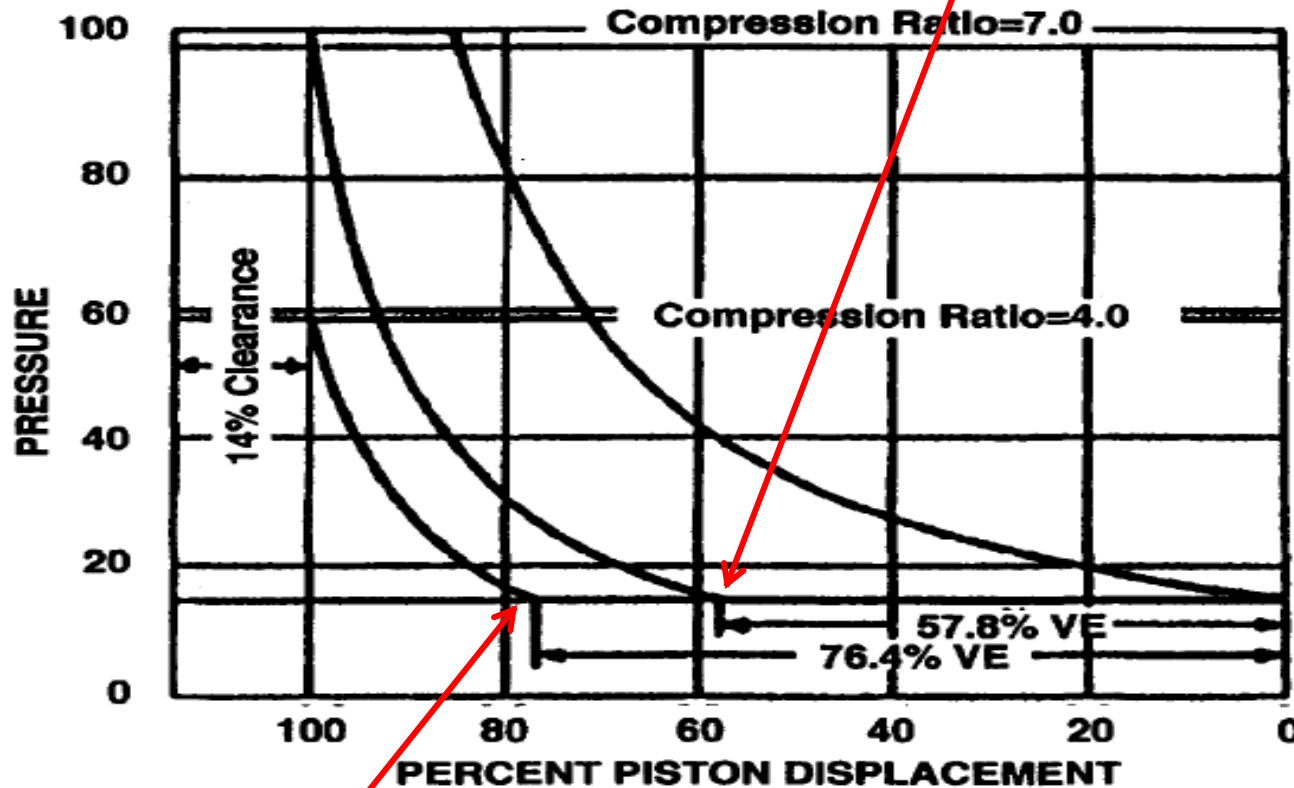
Effect Of Clearance Pocket To Volumetric Efficiency



Volumetric Efficiency :

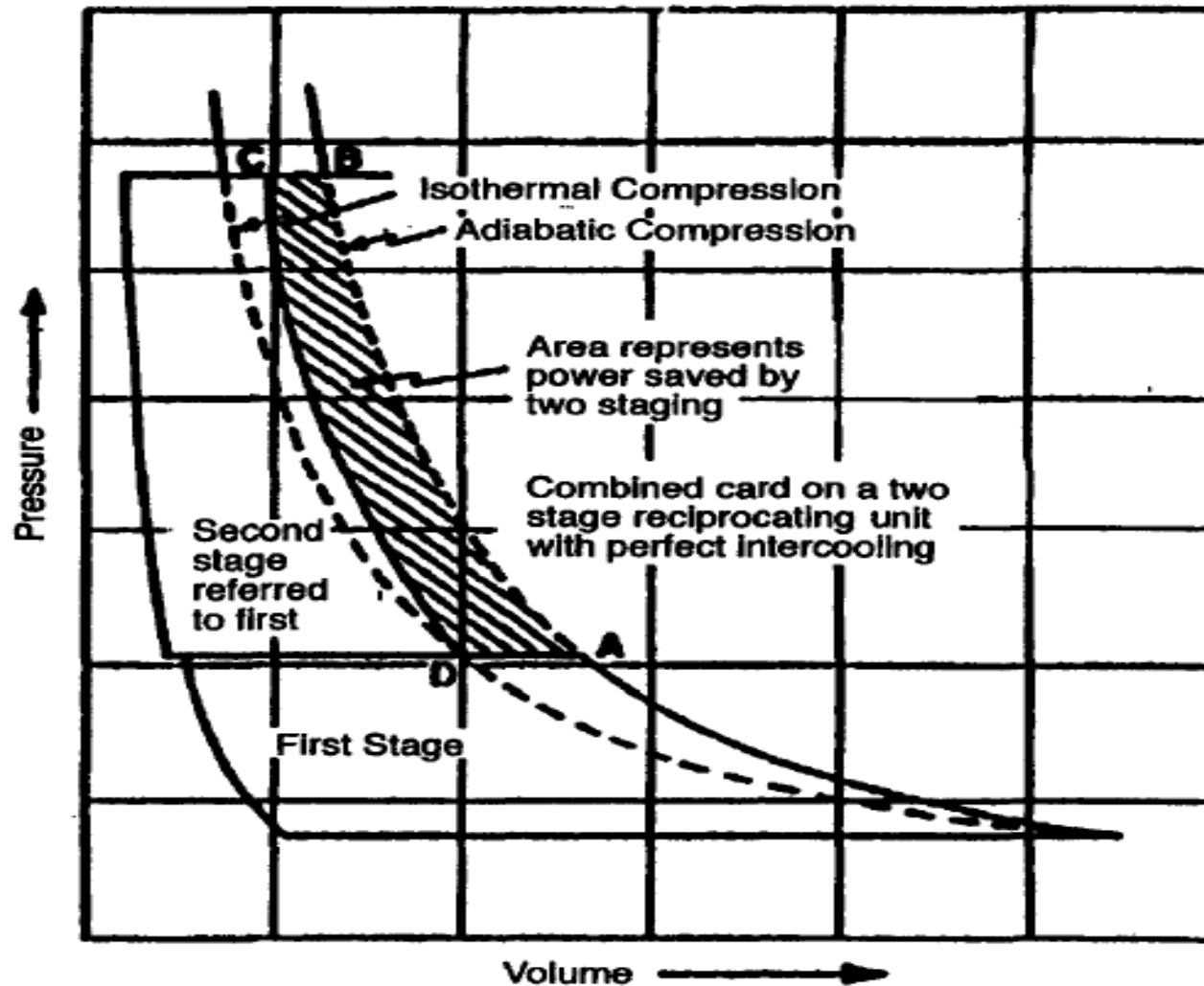
Effect Of Compression Ratio To Volumetric Efficiency

At the higher compression ratio (7.0), suction starts at 55% because the piston distance needed to reach the supply (suction) pressure is more.



At the lower compression ratio (4.0), suction starts at 75% because the piston distance needed to reach the supply (suction) pressure is less.

Compressor Staging



- The number of stages is governed by following:
 - Allowable discharge temperature.

API618

2.3.1 Unless otherwise specified and agreed, the maximum predicted discharge temperature shall not exceed 150°C (300°F). This limit applies to all specified operating and load conditions. The vendor shall provide the purchaser with both the predicted and adiabatic discharge temperature rise.

Special consideration shall be given to services (such as those with high-pressure hydrogen or applications requiring nonlubricated cylinders) where temperature limitations should be lower. Predicted discharge temperatures shall not exceed 135°C (275°F) for hydrogen-rich services (those with molecular weight of 12 or less).

PTS
121132

2.3.1 The maximum predicted discharge temperature shall not exceed 140 degree Celsius. The maximum predicted discharge temperature for non-lubricated service shall not exceed 130 degree Celsius. These temperature limits are not applicable to labyrinth compressors and also to the relief valve setting conditions.

- The number of stages is governed by following:
 - Combined Rod Loading.

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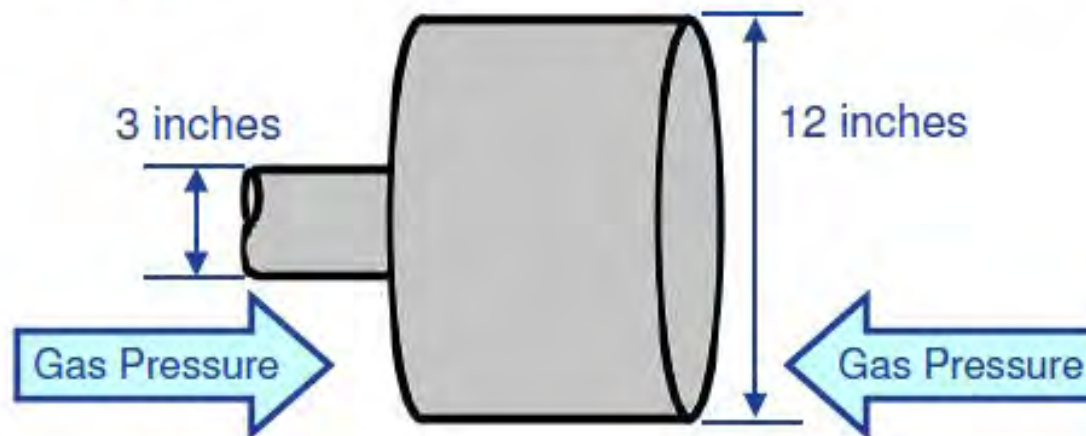
2.4 ROD AND GAS LOADS

2.4.1 These combined rod loads shall be calculated on the basis of the setpoint pressure of the discharge relief valve (inclusive of all the necessary over-pressure) of each stage and of the lowest specified suction pressure corresponding to each load step.

- Allowable of working pressure of available cylinders.

Gas Load

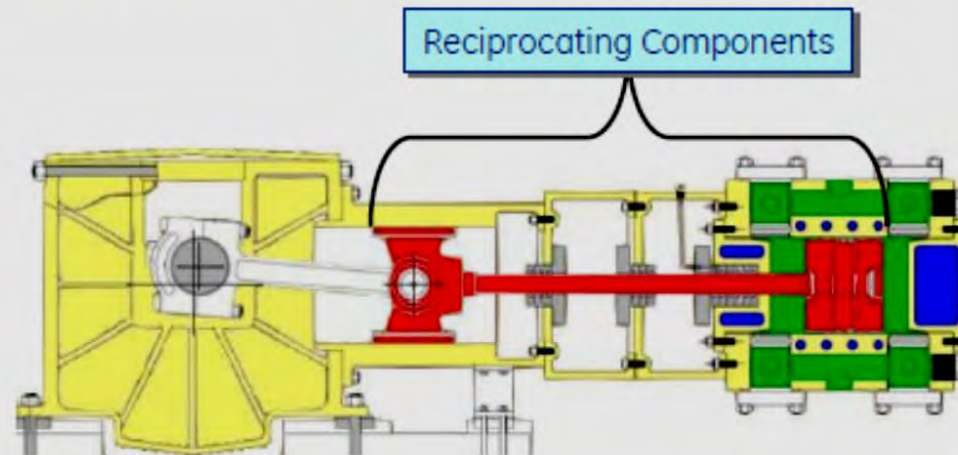
- Cyclic force caused only by pressure of the process gas on the heads of the piston:



Gas load forces are in line with the piston rod and the axis of the cylinder and crosshead guide bores.

Inertial Load

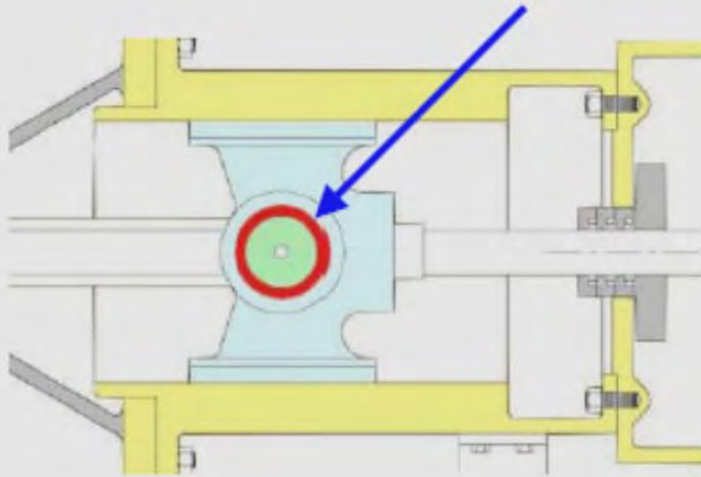
- Cyclic force caused only by acceleration of reciprocating components (piston, piston rod, and crosshead):



Inertial load forces are in line with the piston rod and the axis of the cylinder and crosshead guide bores.

Rod Load Reversal

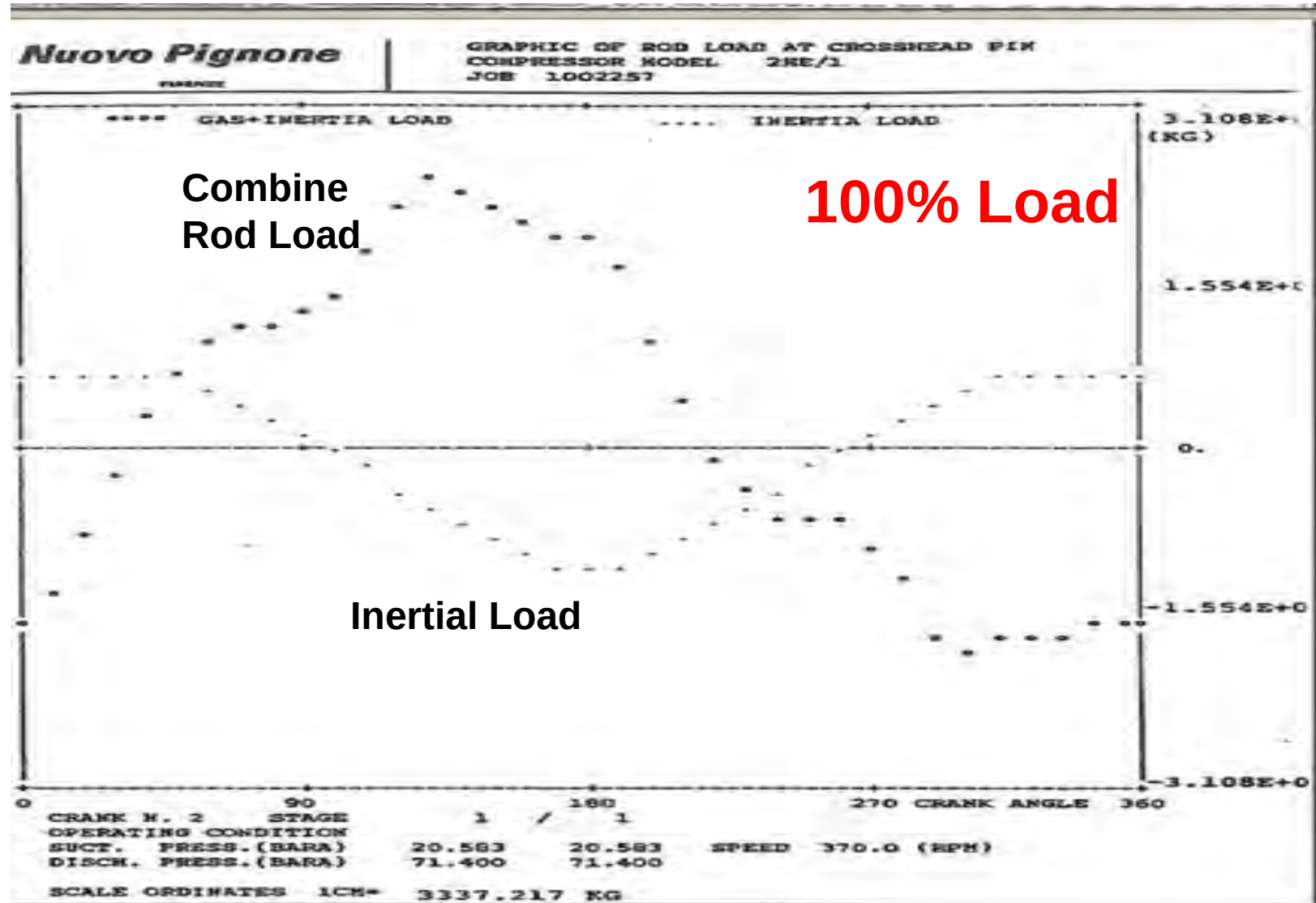
- Rod load reversal is the periodic shifting of piston rod load between tension and compression during a complete stroke.
 - Important for adequate crosshead pin & bushing lubrication.



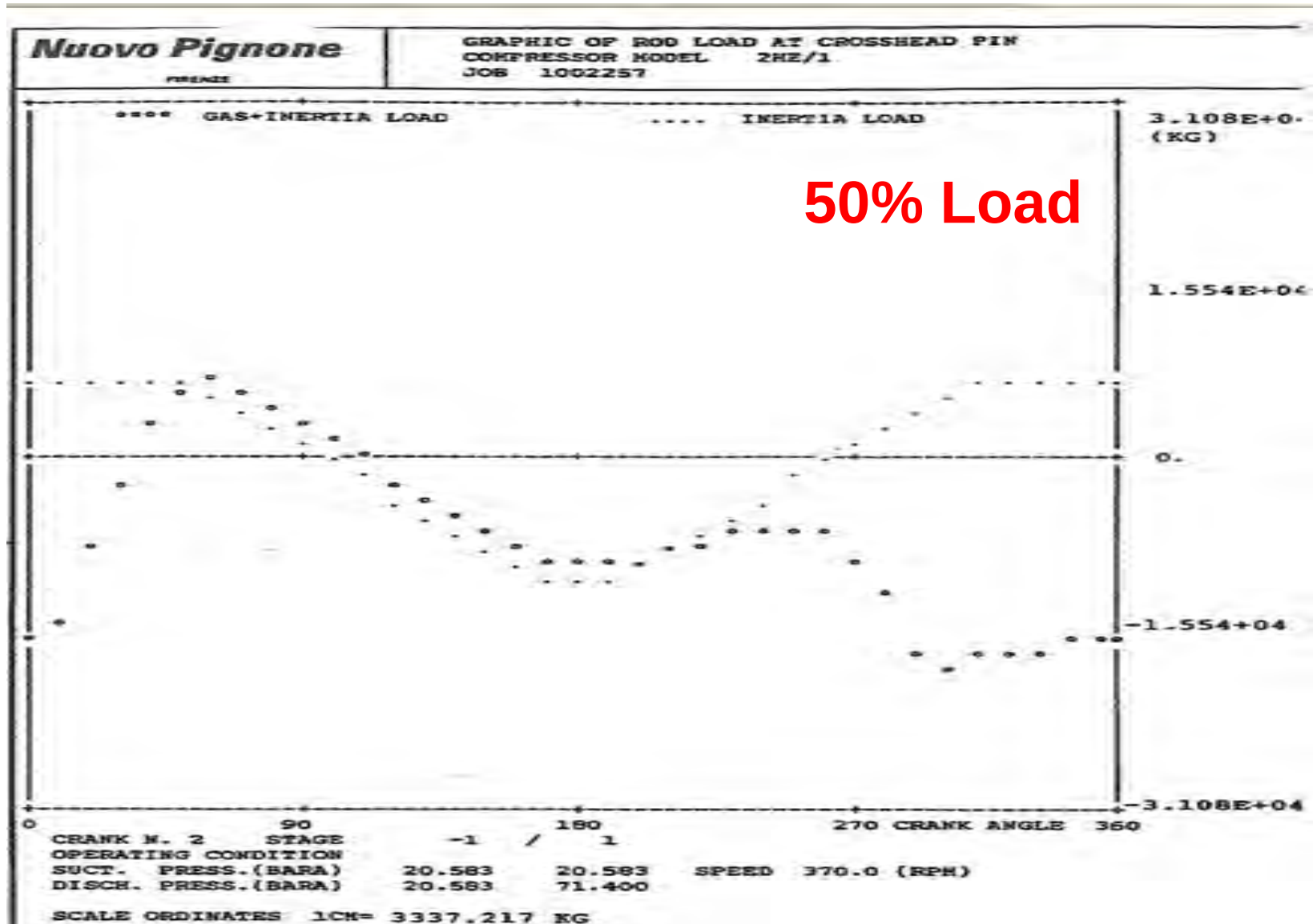
Animation: Reversal.avi

API 618 Guideline: At least 15 degrees of reversal, with peak magnitude at least 3 percent of the actual combined load value in the opposite direction.

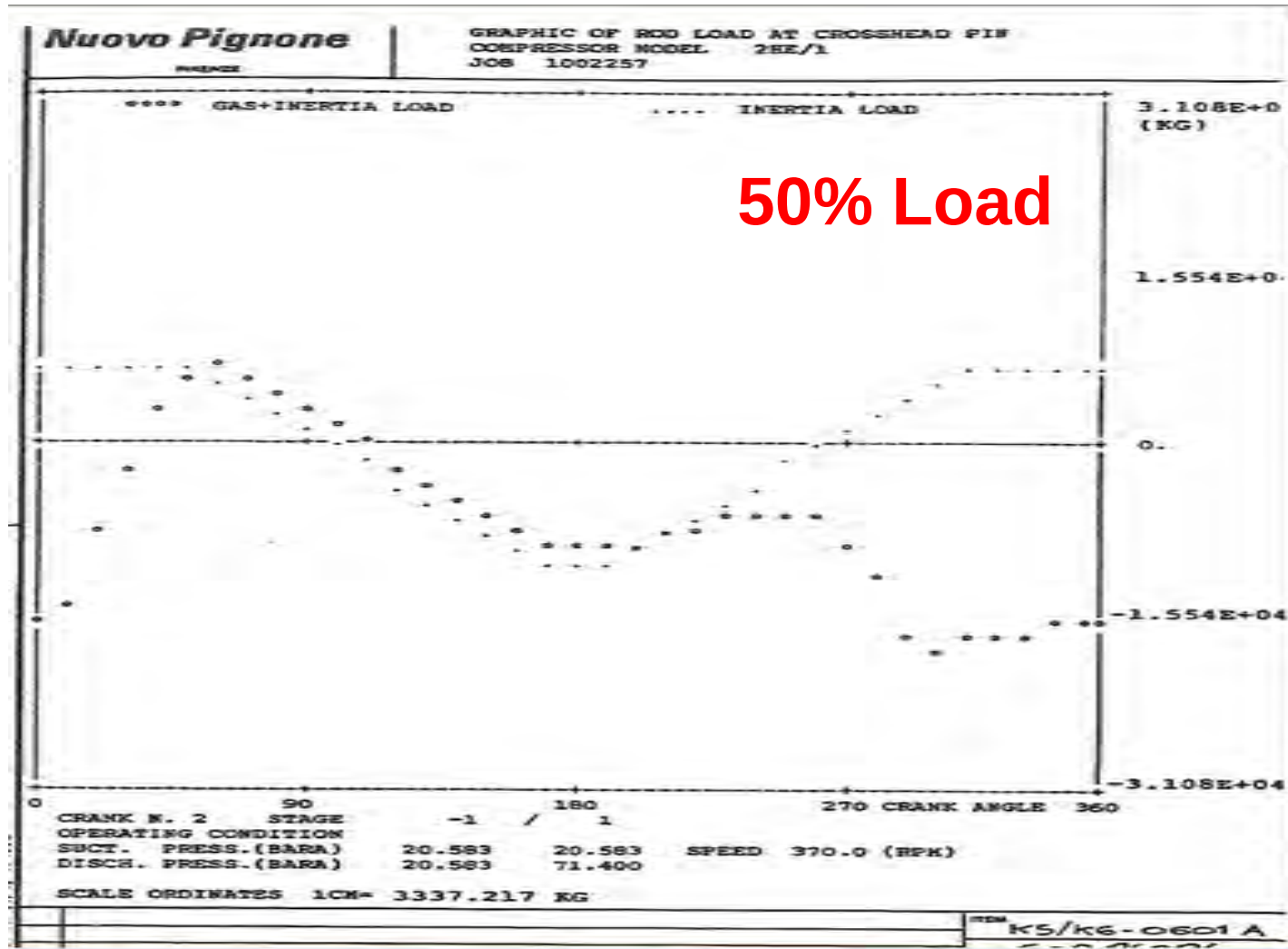
Example of Combine Rod Load



Example of Combine Rod Load



Example of Combine Rod Load



PTS 121132

2.2 ALLOWABLE SPEED

2.2.1 The limits for the speed of the compressor and the corresponding average piston speed shall be as stated values below:

Power (Installed) in Kilowatt (kW)	Compressor Speed (Maximum) in rev/minute (rpm)	Average Piston Speed (Maximum) in metre/second (m/s)
150 or less	500	5
150 to 250	420	5
250 and above	375	4

Table 2: Limits for the speed of the compressor and its corresponding average piston speed

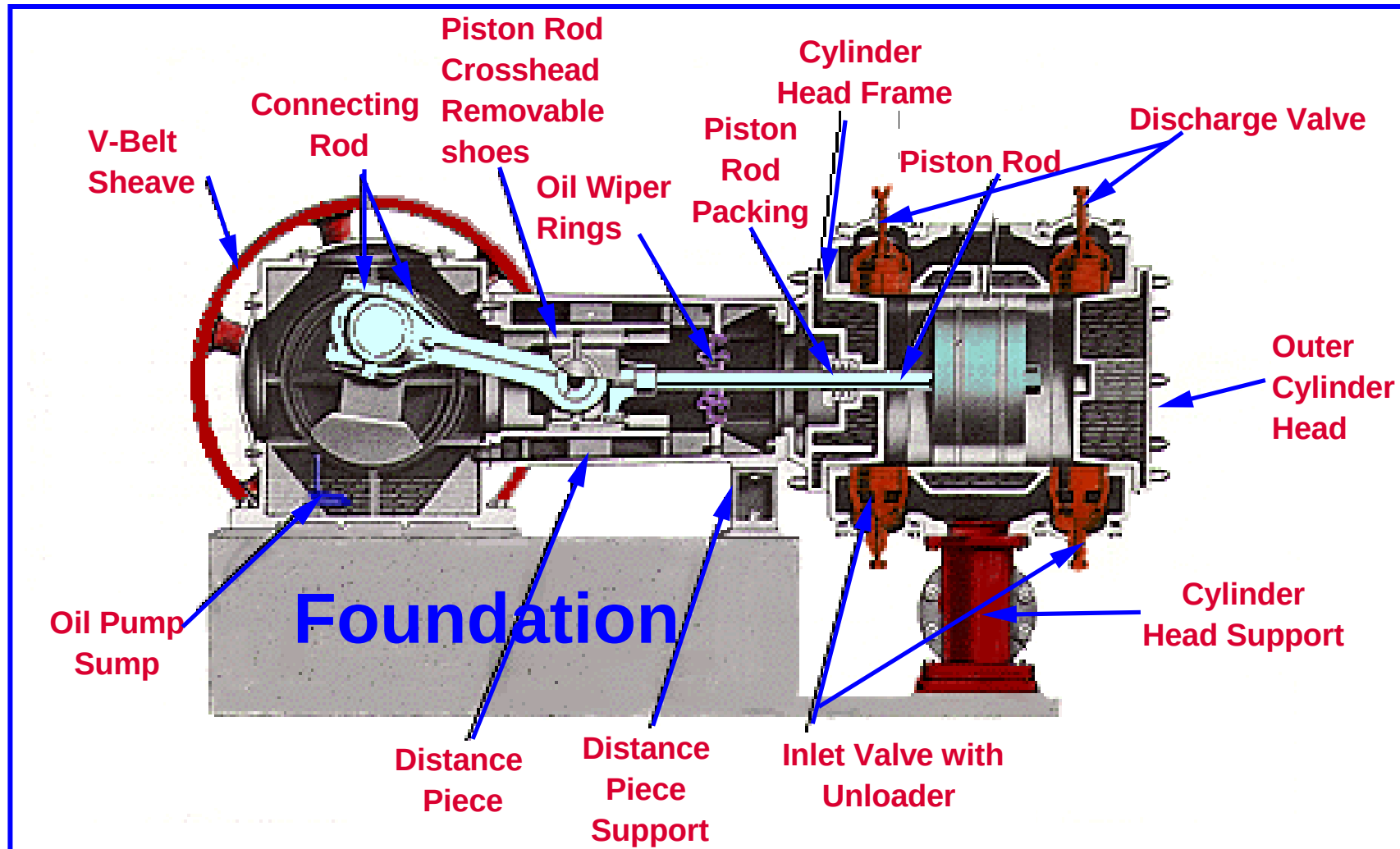
2.2.2 For compressors with non-lubricated cylinders and handling saturated hydrocarbon gasses, the average piston speed shall not exceeding 3.5 meter per seconds.

Note : High speed compressor based on API 11P are subjected to Principal approval and can be considered for non-critical application.

Compressor Major Component

- Main components of reciprocating compressor
 - Frame
 - Crankshaft
 - Crosshead & guide
 - Connecting rod
 - Piston rod
 - Cylinder
 - Piston assembly- Piston, Rider rings & Piston rings.
 - Bearings
 - Valves
 - Stuffing box packing
 - Distance Piece
 - Oil wiper packing

Compressor Major Component



- To provide rigidity & hold all moving & stationary parts in accurate alignment.
- Frame encloses crank shaft & supporting bearings.
- Designed with suitable supports & ribbings.
- The base of frame serves as oil reservoir.
- Normally made of high-grade cast iron castings.
- Normally no failures except fatigue cracks over very long period.
- During routine maintenance check mounting bolt tightness, frame level & rib stud tightness.
- Includes inspection covers on top & anti-explosion valve.

- The main function is transmit power from driver & provide oscillatory motion to connecting rod.
- Crankshaft is supported in frame on main bearings & located axially by end bearing.
- Connecting rods are attached to each crankpin.
- Crankshaft has cross drilled holes for conveying lube oil to big end bearing from main bearing.
- Counter weights are attached to the crank-web to counter balance weight of connecting rod.
- Normally made of carbon steel forgings.
- Requires dynamic balancing when operating speeds are more than 900 rpm.

- Failure & maintenance.
 - Classified as long lasting part, failures are very rare.
 - Normal failure is wear of journal area at main bearing & big end bearing.
 - Routine maintenance checks:
 - Crankshaft alignment with driver. (web deflection)
 - Axial positioning.
 - Counter weight bolt tightness.
 - Journal & Crank pin area NDT inspection.

Crankshaft



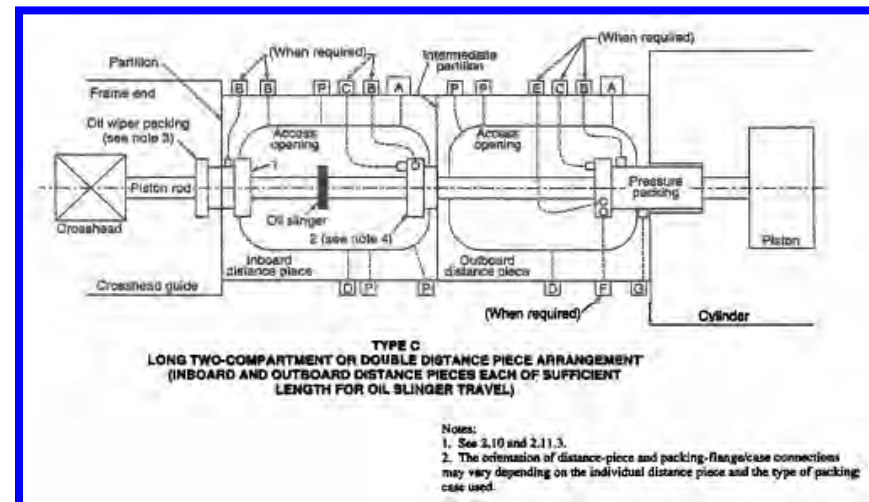
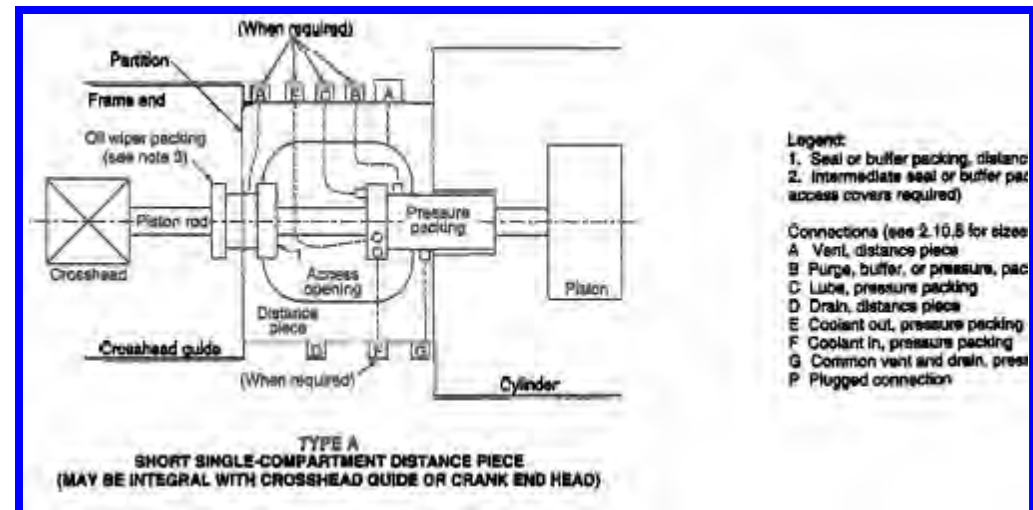
- Distance piece separates the cylinder from the cross head guide.
- It provides housing for both wiper packing and pressure packing.
- It prevents entry of compressed gases into crank case & access for maintenance of packing & rod.
- Various arrangements are defined in API-618 to suit various process gases.
- During routine maintenance , distance piece attachment bolts & base bolts shall be checked for tightness.

- Distance piece types:
 - Type A: Short single distance piece used only for lubricated services where oil carry over is not objectionable.
 - Not recommended for flammable & toxic gases.
 - Type B : Long single compartment distance piece used for non-lubricated service or when specified.
 - Compartment shall be of sufficient length so any portion of piston rod at crosshead do not enter packing areas.
 - Type C : Long/Long two compartment designed to contain flammable or toxic gases.
 - Only used for extremely hazardous or flammable gases.
 - Piston rod length will be longer.

Distance Piece

- Distance piece types:
 - Type D: Long / Short two distance piece used to contain flammable, hazardous or toxic gases.
 - No part of the piston rod shall alternately enter the wiper packing and the intermediate seal packing.

API 618



PTS 121132

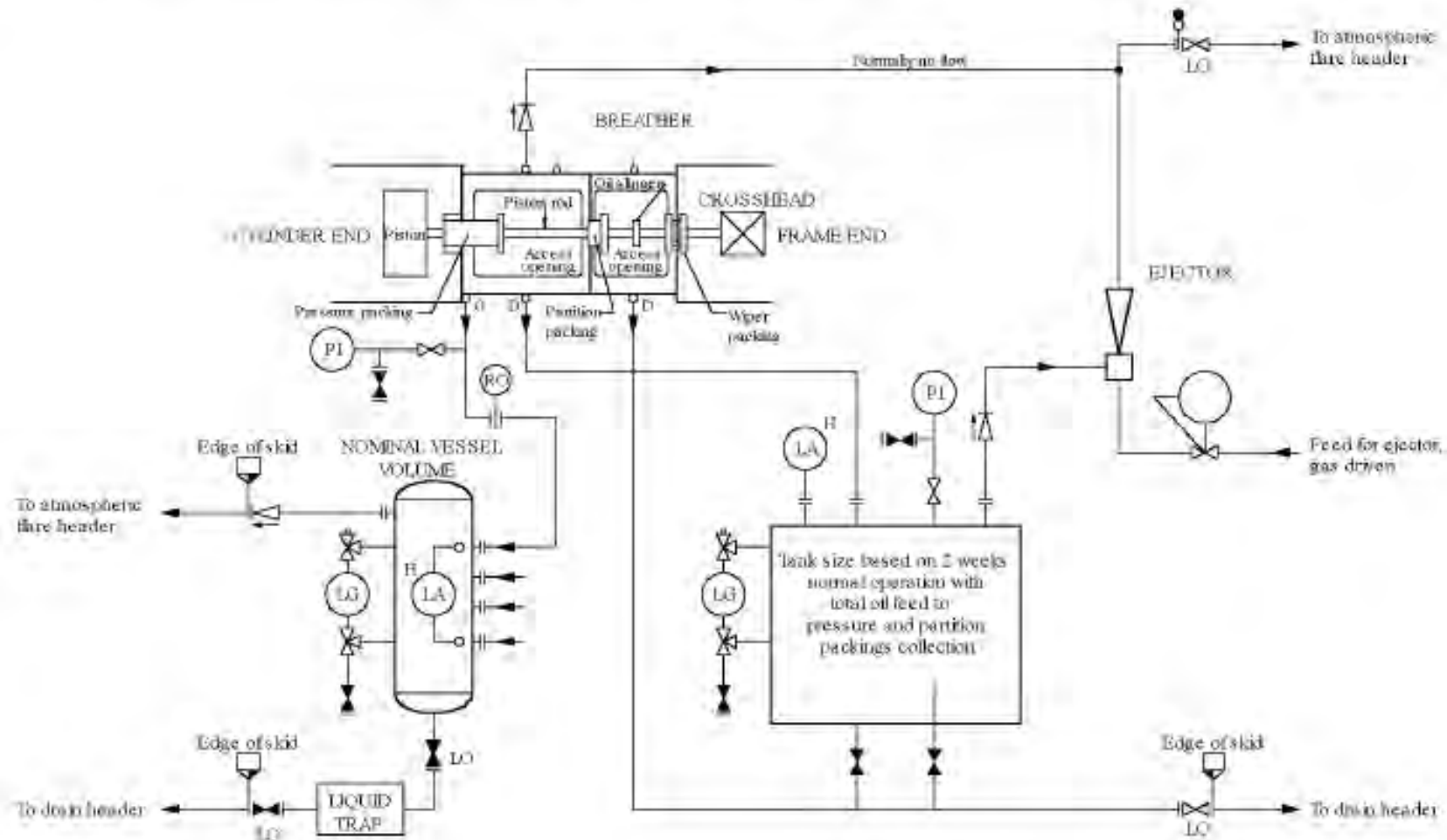
2.13 DISTANCE PIECES

2.13.1 Unless differently stated in the engineering data requisition sheet, the required distance piece type is as follows;

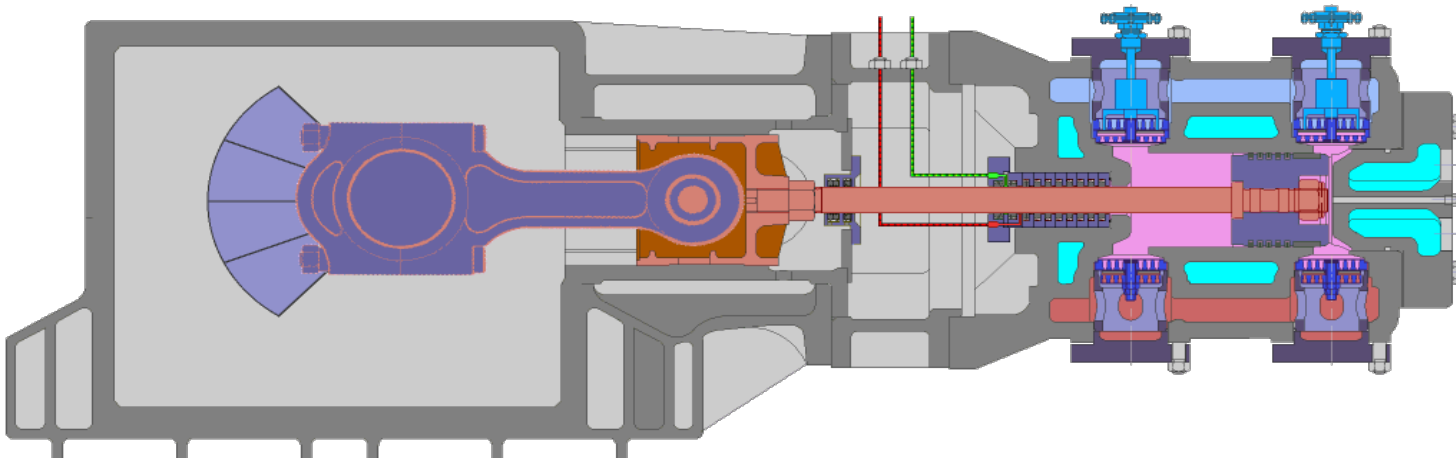
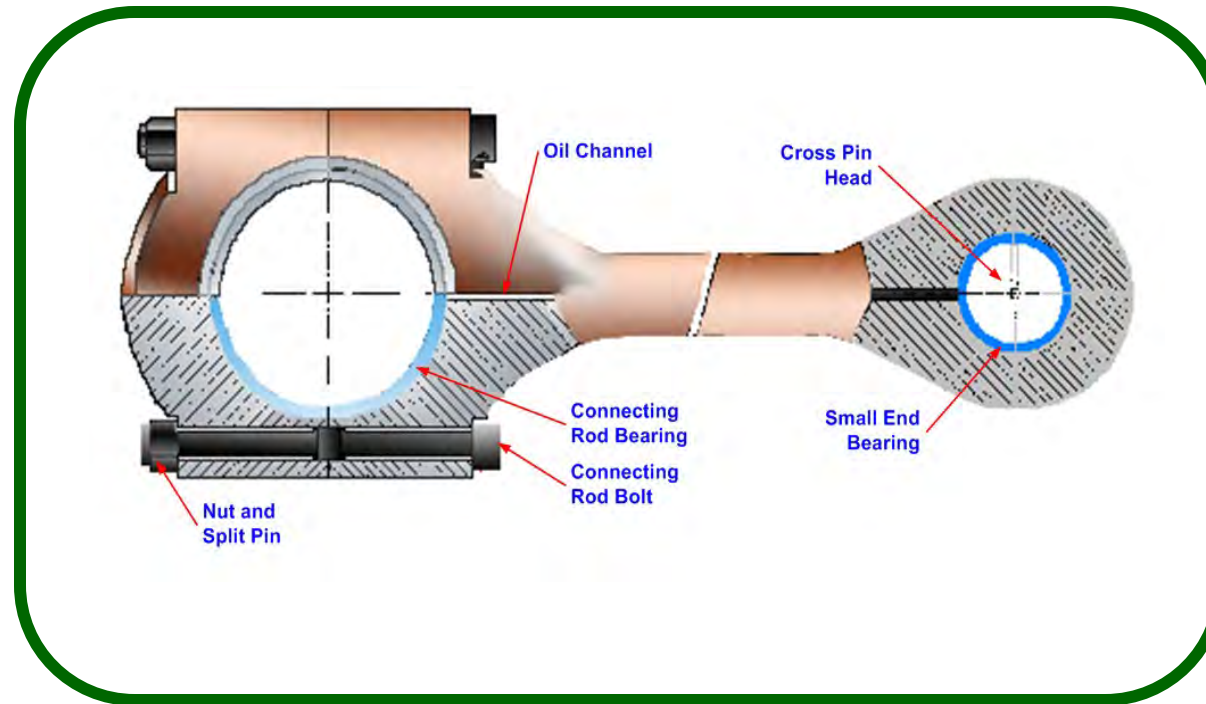
- i. For air or pure nitrogen: Type B
- ii. For all other gases in lubricated and non-lubricated services: Type C or D.

PTS 121132

APPENDIX 1 DISTANCE PIECE VENTING AND DRAINING FOR ARRANGEMENTS C AND D



Connecting rod



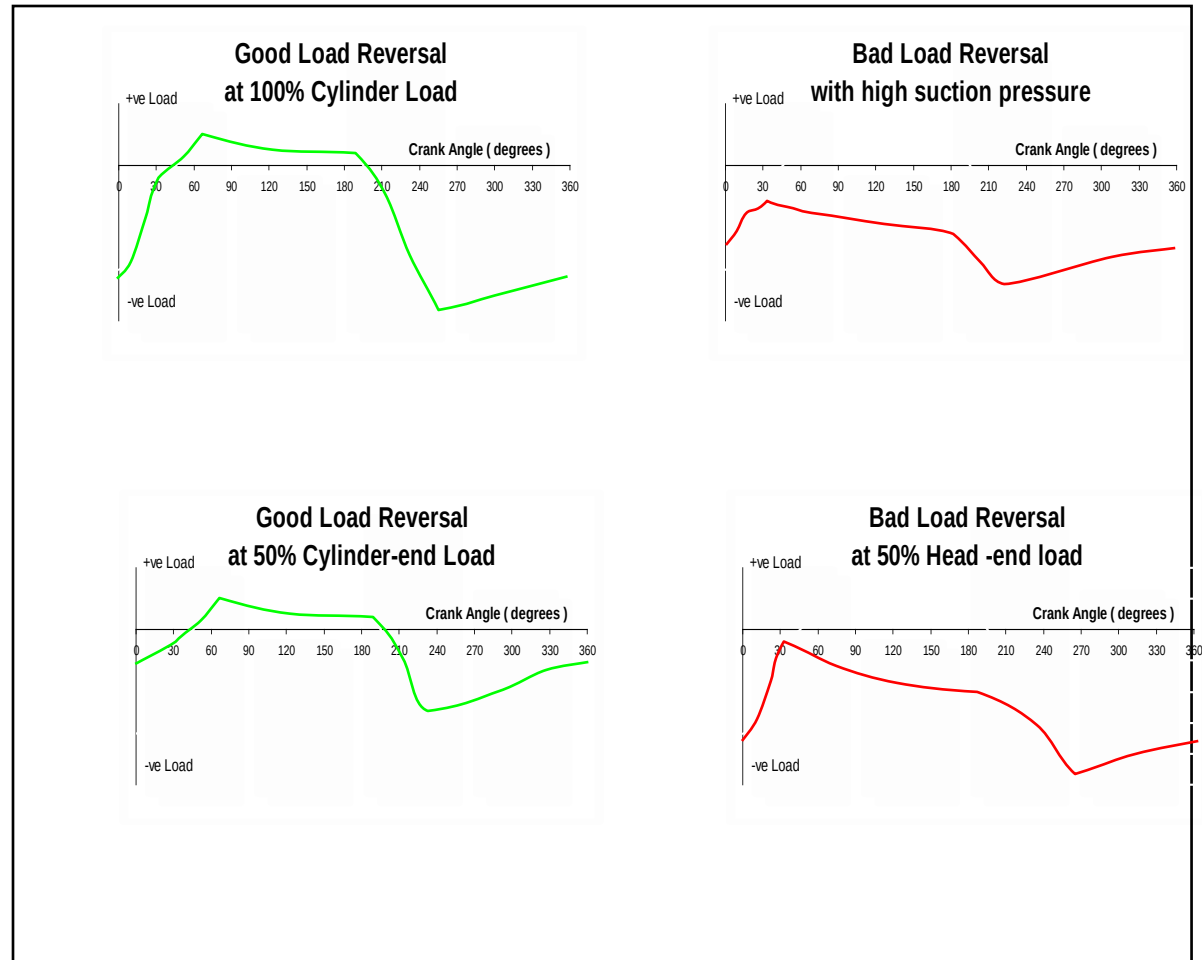
- To convert rotary motion of crank shaft into linear motion at crosshead.
- Combined load.
- Big end
 - Big-end bearing is similar to the main bearing it receives lubricating oil from drilled hole connecting main crankpin & bigend crankpin.
- Small end.
- At small end motion is oscillatory, so condition are not favorable for hydrodynamic lubrication, hence bush is provided with grooves.

- Connecting rod :
 - Connecting rods are made from drop forged steel.
- Failure & Maintenance:
 - Common failures are of small end & big end bearing
 - The main causes are attributed to lubrication problems.
 - On small end bearing improper design for load reversal also results in inadequate lubrication.
 - During routine maintenance
 - Lube oil passage.
 - NDE of Rod bolts & rod.
 - Physical check of bearing condition.
 - Clearance check.

Small End Bearing Load Reversals

Combined rod load = (gas + inertia + valve losses)

Combined rod load will determine the load on the crosshead pin bearings and will define reversal at the crosshead pin for proper lubrication. Reversal is the change from a compression load to a tension load at the crosshead pin, thus allowing oil to flow into the loaded side of the bearings.



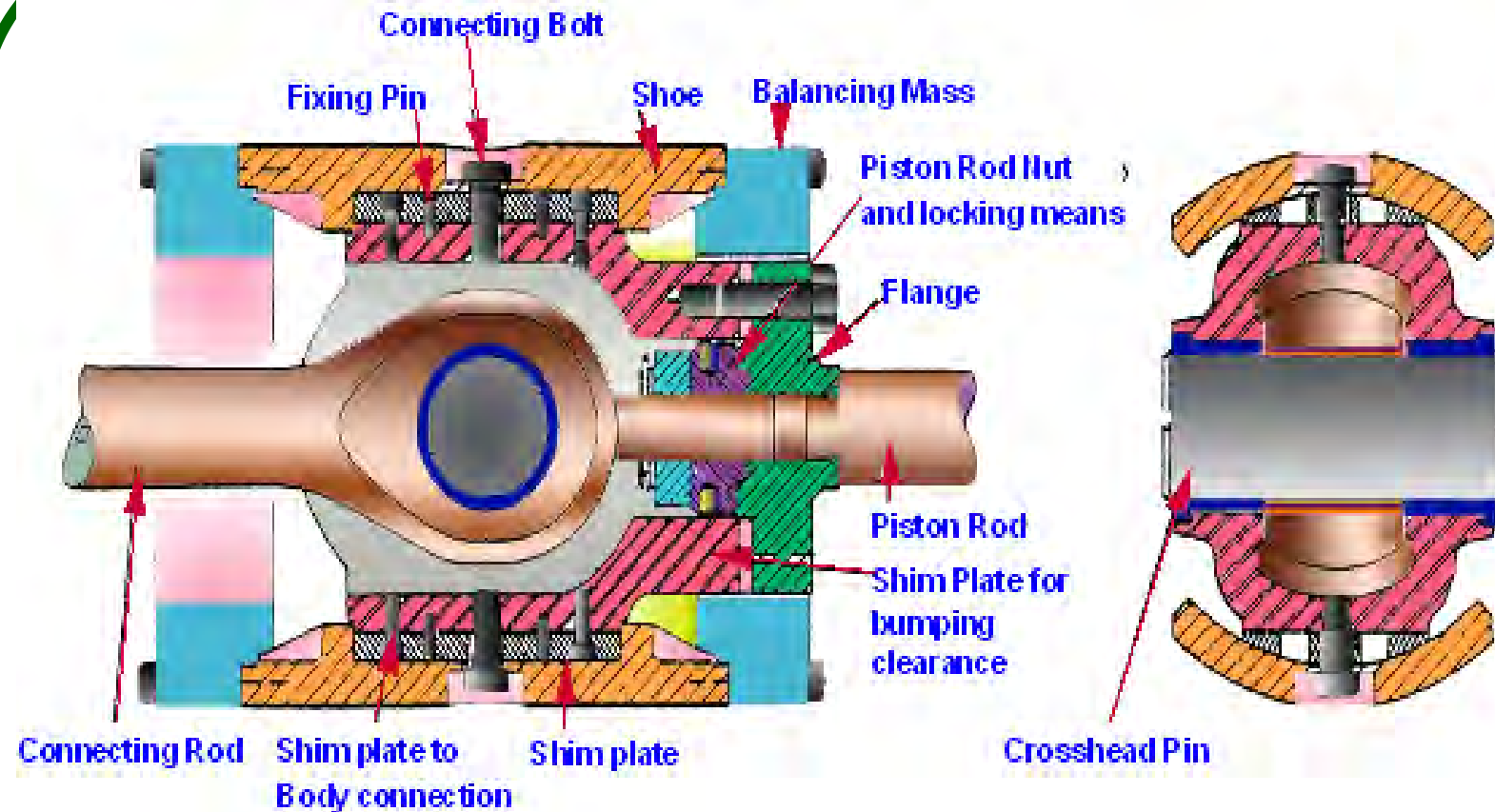
- It maintains reciprocating motion of piston & rod in horizontal plane.
- It bears the forces produced by motion of connecting rod connection with it.
- Cross head are equipped with separable shoes on which it slides in cross-head guide.
- Shims or shoes are adjustable for clearance adjustment.
- Cross head are normally made of cast grey iron or nodular iron or cast steel.
- Cross head to piston rod connection is quite critical for medium to heavy machines.

Cross head

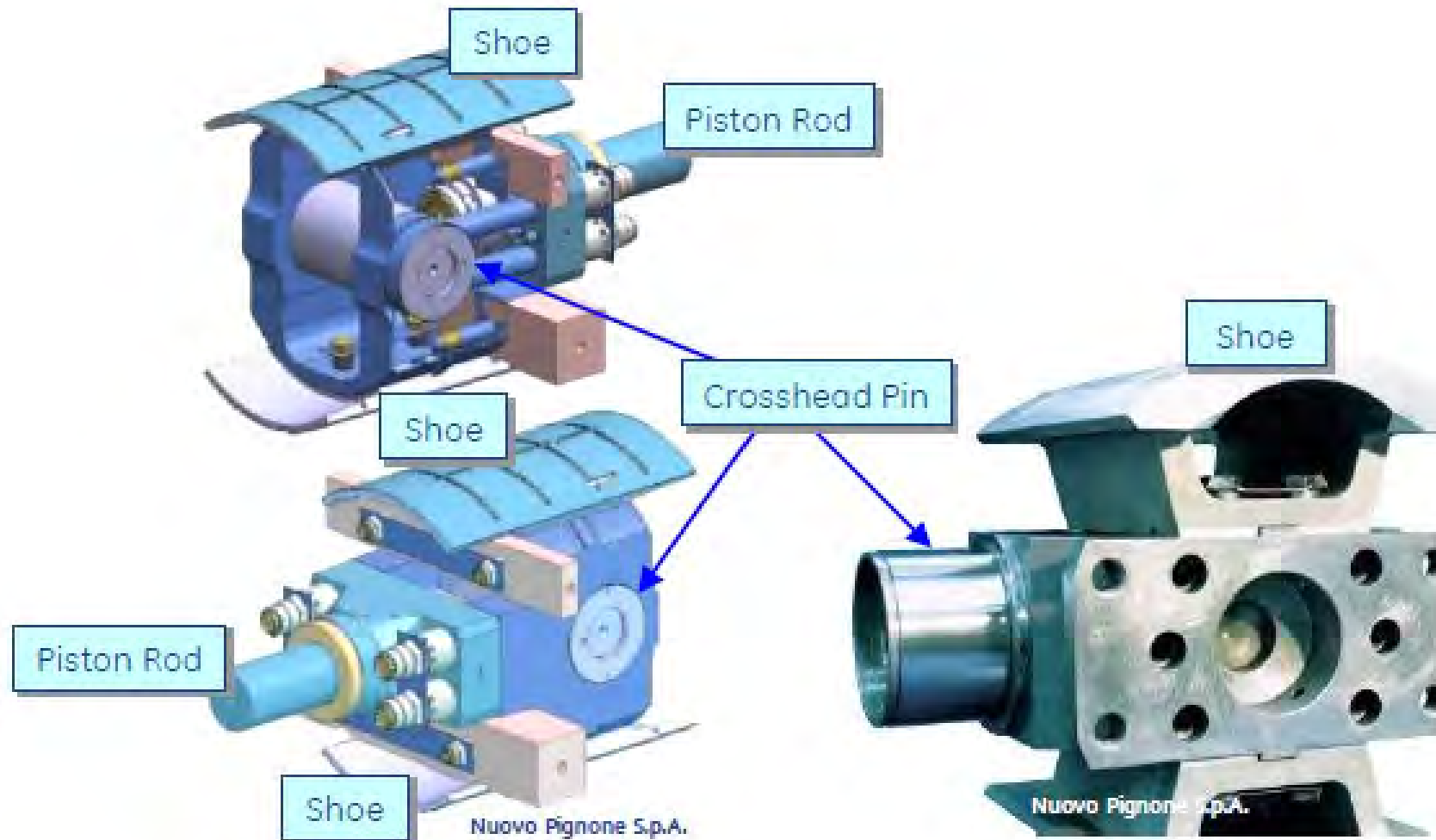
- Common failure on cross heads are worn out shoes.
- Damage at piston rod connections.
- Worn-out cross head pin bushes.
- During normal maintenance check
 - Cross pin dimensions.
 - Cross head alignment & top clearance.
 - Physical condition of guide & shoes.



Cross head



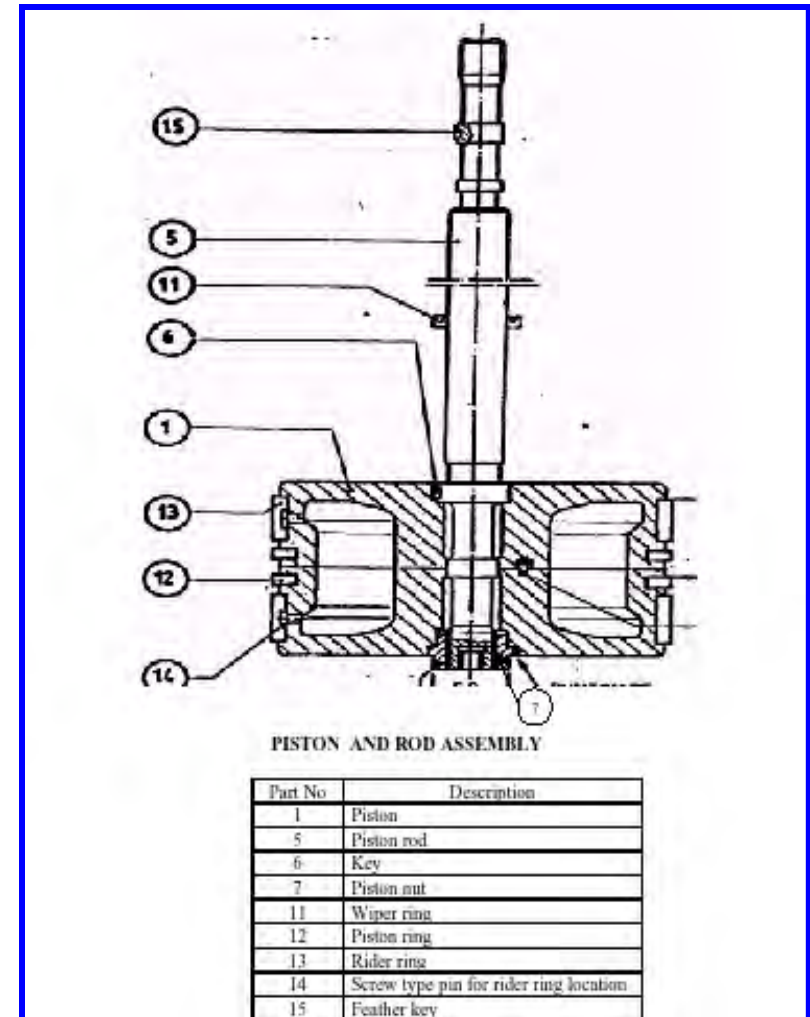
Cross head



- Transmits reciprocating motion from cross head to piston.
- Subjected to combined bending and compressive stresses.
- Critical joints are cross head to rod & rod to piston due to heavy cyclic stresses.
- Other critical aspect is surface at rod packing.
- The common material for rod is AISI 4140 with maximum Rc40 at core & minimum of Rc 50 at surface.
- If stress corrosion is a factor, material is annealed to a hardness of 22C maximum (core) & 50 minimum at surface.
- 17-4 Ph steels are used for corrosive services.
- Rolled threads are preferred over cut threads.

- Piston
 - Compresses or reduces trapped gas volume & increase pressure.
 - Piston is guided on rider ring during reciprocating motion in cylinder.
 - Pistons are made in one piece, two piece or three pieces
 - One piece construction for smaller diameters & higher differential pressures.
 - Two piece designs are used for large diameters & lower differential pressures.
 - Three piece design are used for large diameters coupled with high differential pressures.
 - Easy for rider ring installation.

- Factors affecting piston design
 - Cylinder bore diameter
 - Discharge pressure
 - Compressor speed
 - Compressor stroke
 - Required piston weight.
- Factors affecting material selection
 - Weight
 - Strength for differential pressure
 - Corrosion resistance
 - Outside diameter & wear resistance
- Common materials
 - Cast iron & Steel
 - Aluminum
 - SS



- Cylinder
 - Cylinder diameter defines the basis of displacement volume along with stroke & speed.
 - For heavy duty double acting compressor, cylinder are fitted with liner as replaceable wearing part.

Material	MAWP PSIG API 618	MAWP,PSIG Oil Field < 8 in Dia.	MAWP, PSIG Oil Field > 8 In Dia.
Gray cast iron ASTM A 278	1000	1600	1000
Cast Nodular iron ASTM A395	1000*	2500	1500
Cast steel, ASTM A 216	2500	2500	2500
Forged Steel	>2500	7500	-

- Common failures for piston rods are
 - Wear out at packing area.
 - Damage at thread & locating collar.
 - In worst case damage to piston rod attachment at cross head.
 - Routine maintenance check
 - Piston rod dimensions at critical areas.
(Taper limit :0.0005 in/ in of stroke)
 - Surface condition specifically at packing areas.
 - Rod run-out (Limits 0.00015 in./in. stroke).
 - Thread conditions & gauging.
 - Crack check of all critical areas by DP.
 - Torque tight or hydraulic joint

- Valves:
 - Controls or allows one-way entry & exit of gases.
 - Operates on differential pressures across it.
 - Main contributor for poor compressor reliability most of the time.
 - Main types
 - Plate type or slotted disc type
 - Concentric Ring type
 - Channel/ Feather valve
 - Poppet valves

CYLINDER VALVES

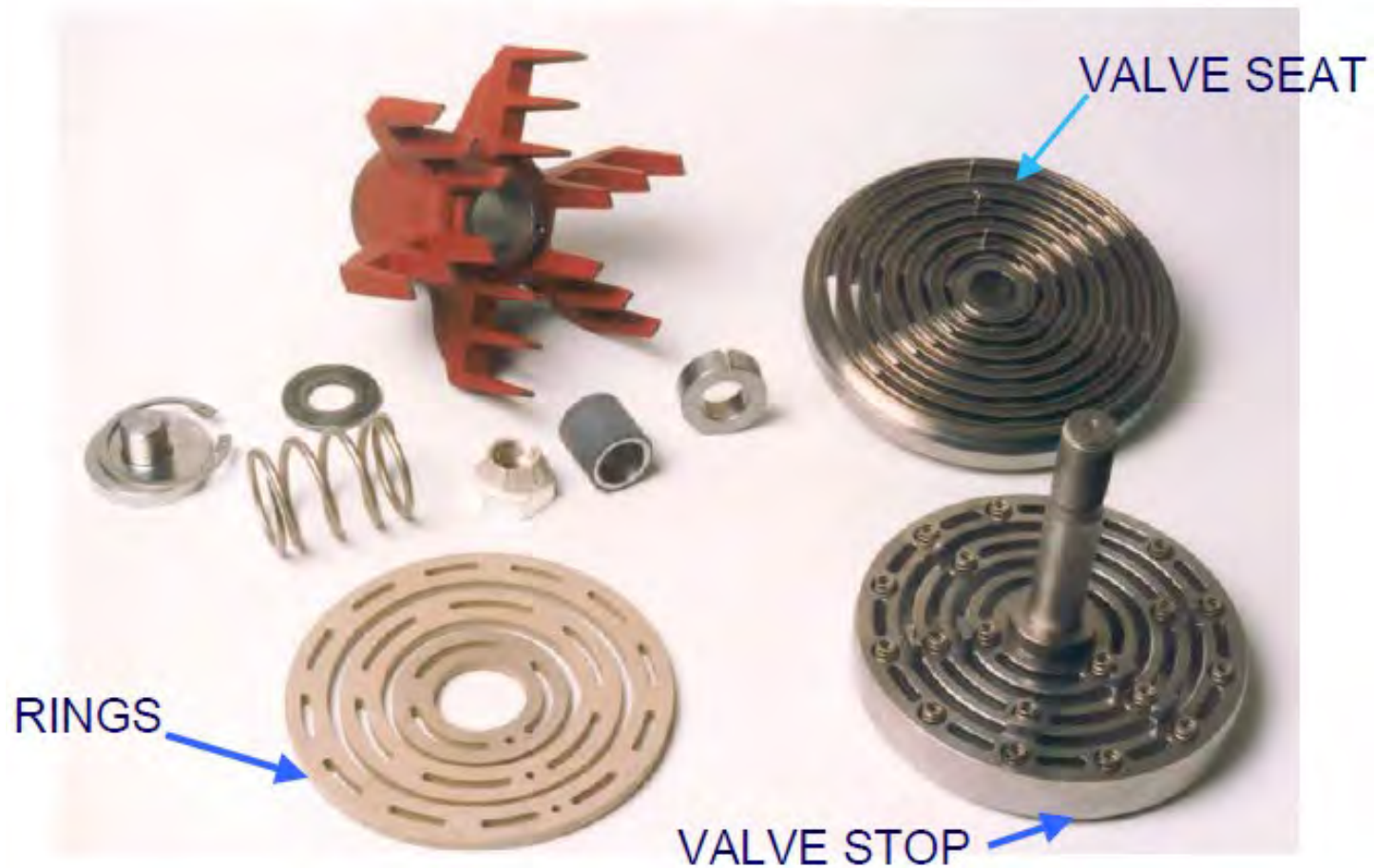


Suction Valve



Discharge Valve

SUCTION VALVE



DISCHARGE VALVE



Poppet Valve

- Can use 50% more valve lift area than ring or plate type valve.
- Normally used for pressure above 1000 bar.
- For low ratio, high lift operations like Gas Transport, Poppet valves are the cheaper solution
- For high ratio, high lift operations like Gas Lifting, Gas Gathering, Ethylene Process etc., Poppet valves are not to be recommended
- For low ratio hydrogen applications and eventually gas storage the use of poppet valves may be considered

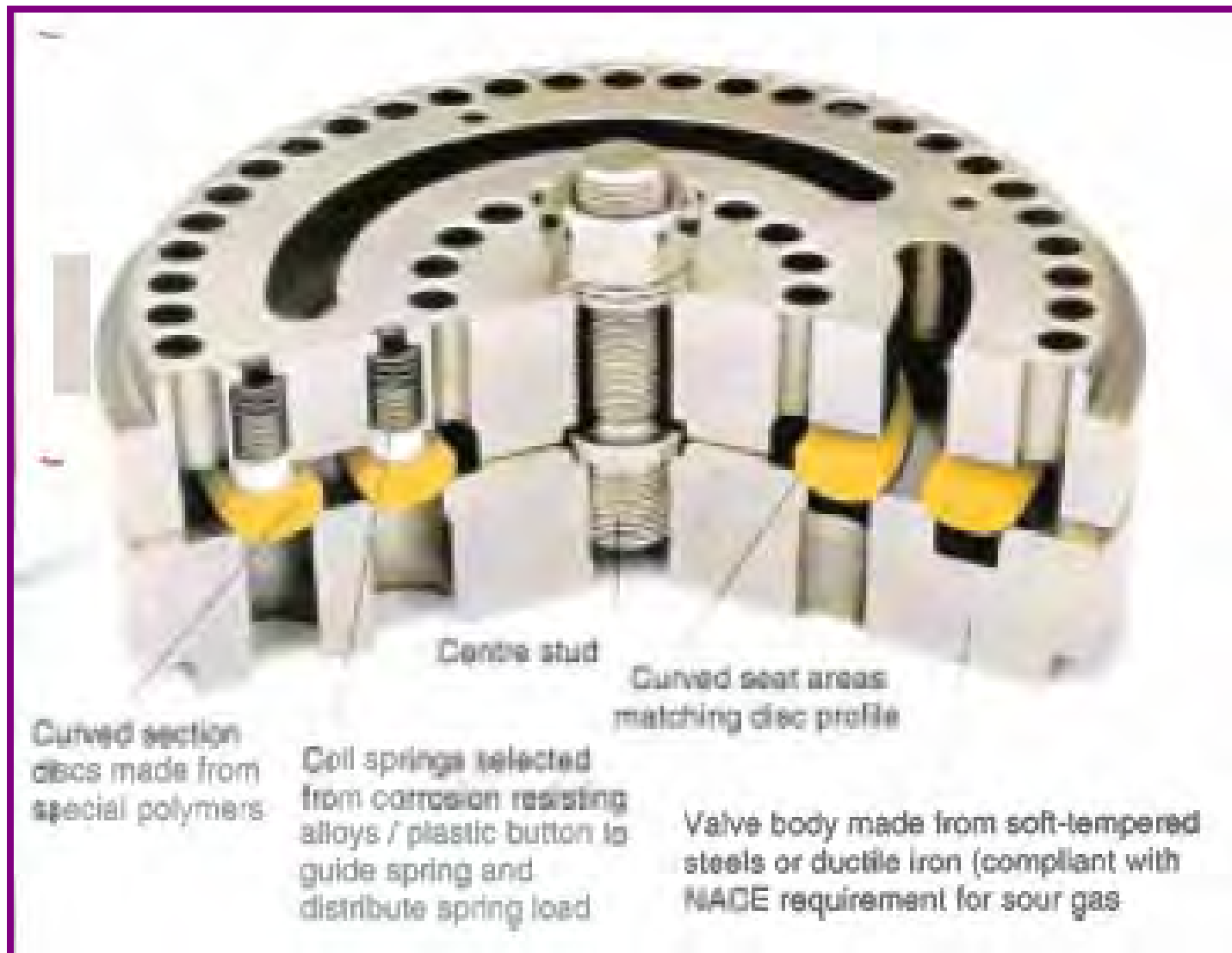


Ring Type Valves

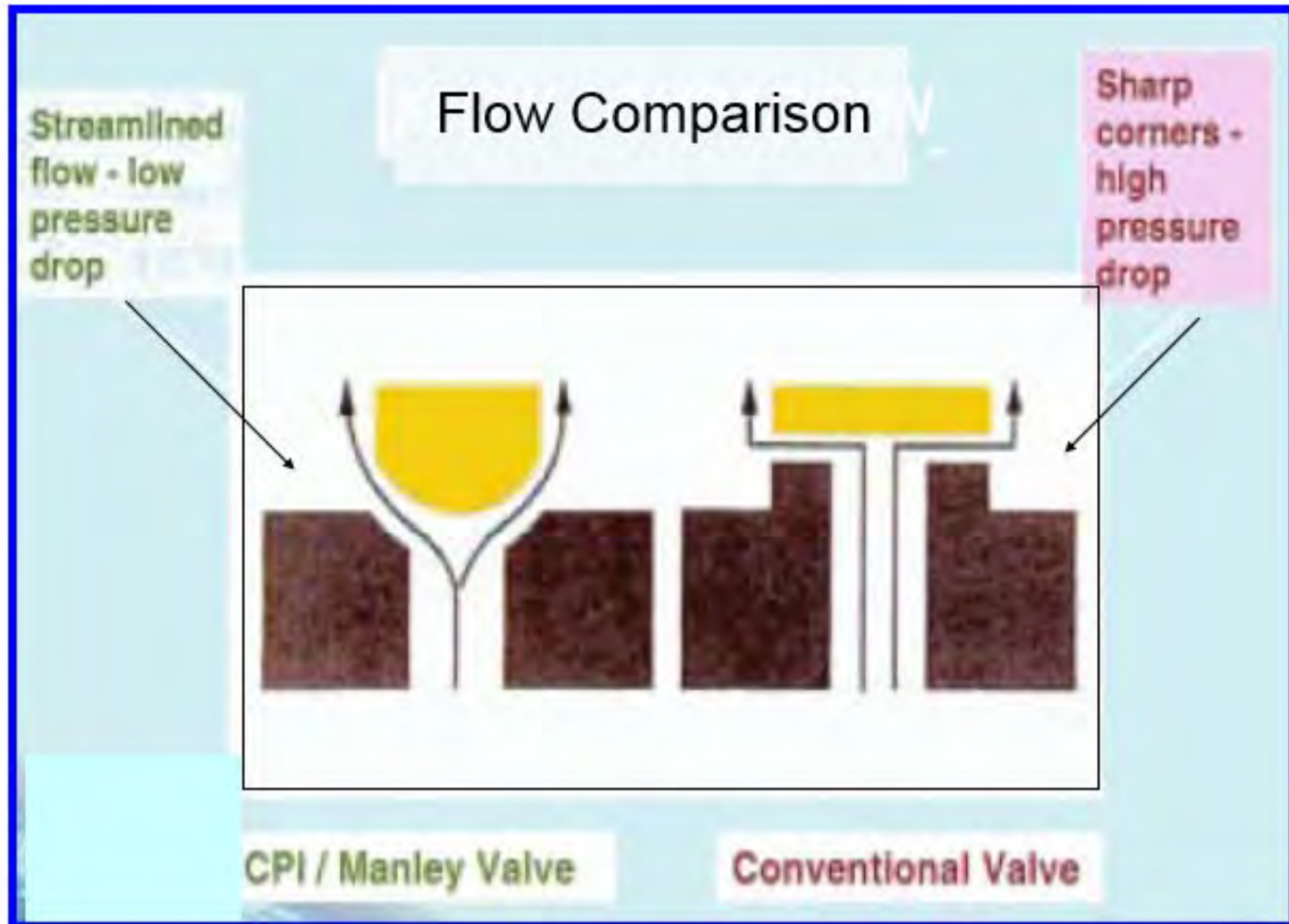
- Are used for non-metallic ring materials.
- Used for pressures between 200-1000 bar.
- Used in majority of applications for refinery & petrochemical plants.
- With improved flow design, it has become more popular choice for wide variety of applications.



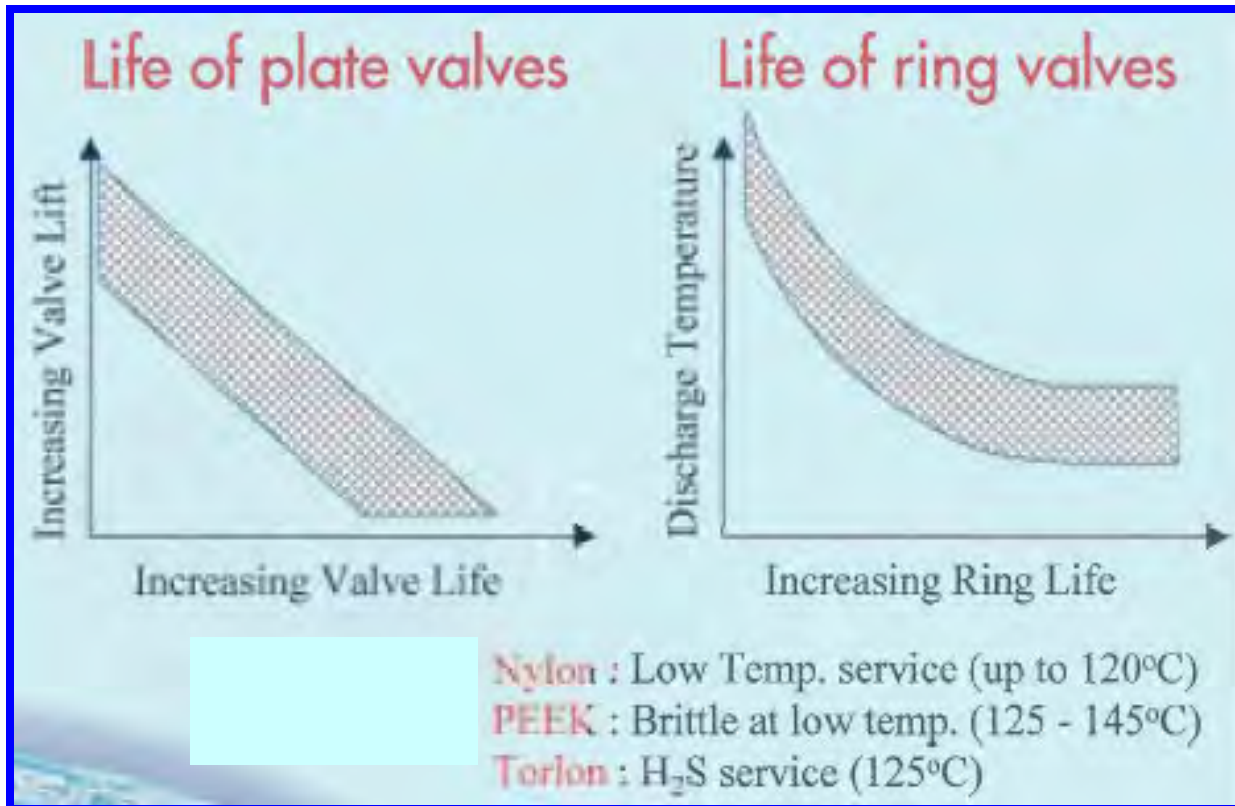
Ring Type Valves



Compressor Valves



Compressor Valves



Typical Valve Materials

Materials	Applications
1141	Light Duty Non-Corrosive
Heat treated 1141	Light Duty Non-Corrosive
Ductile Iron	Light to Medium Service – Resistance to some chemical attack.
4140	Medium to High Strength- Resistance to some chemical attack.
400 Series SS	Corrosive service
300 Series SS & 17-4 Ph Steel	Extreme corrosive service

Valve guard & Seat materials

Typical Valve materials

Material	Applications
Glass Filled Nylon Thermoplastics	Good Impact and corrosion Resistance. 270 Deg. F Limit
PEEK- Polyetherether Ketone	High Strength -High Temp. (375 Deg.F)
Linen Based Phenolic Laminate	Clean gas low compression ratio .225 Deg. F limit.
Laminated cloth Based Phenolic	High Temp. up to 400 Deg. F, available only for ported plates
410 Series steel	Moderate corrosion resistance
17-7 Ph Steel	Moderate corrosion resistance, Good impact resistance

Valve - inspections

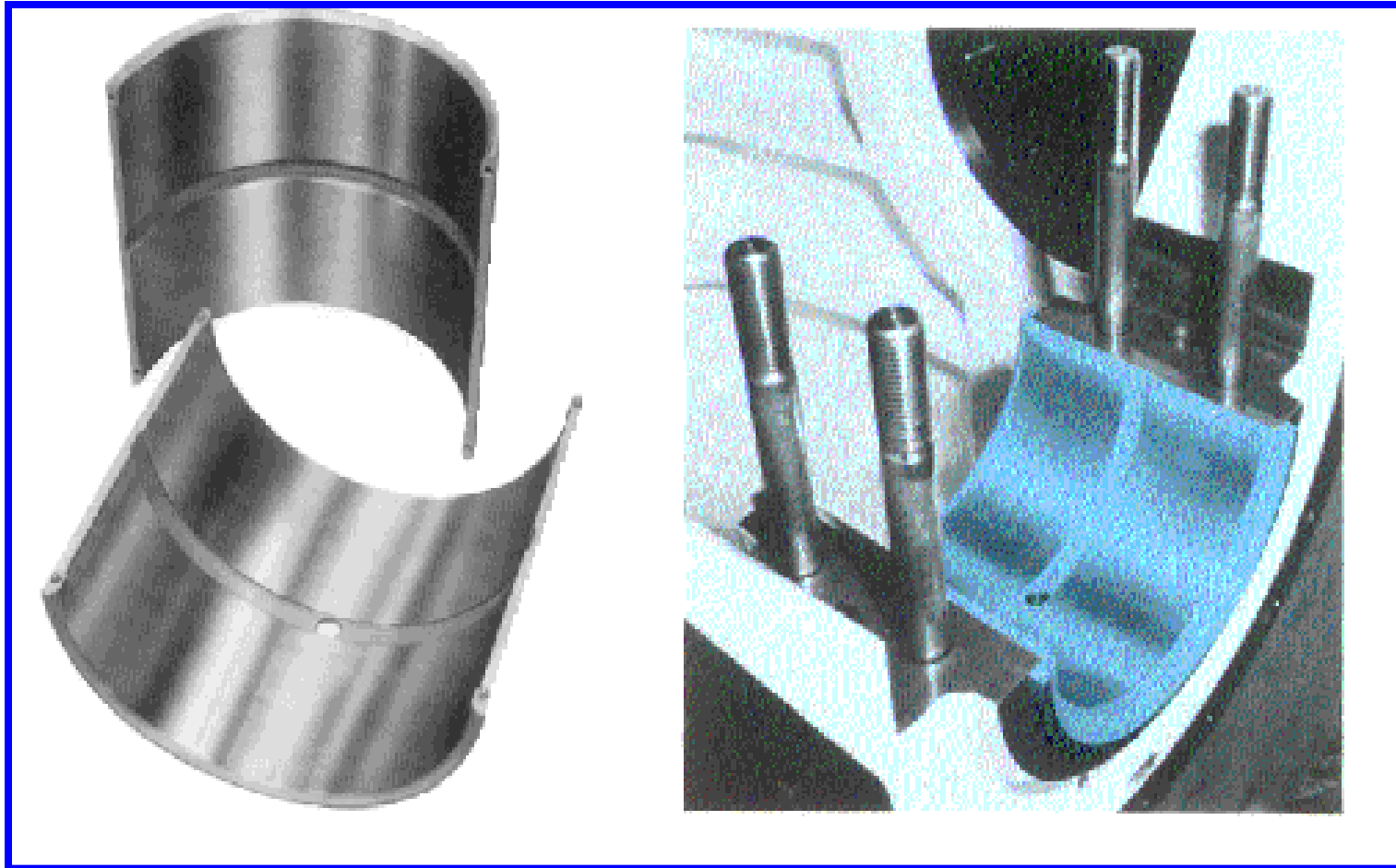
- Valve failure Mechanisms:
 - Plate stiction
 - Due to excessive lube oil or liquid in gas.
 - Plate fracture
 - Due to bending caused by dirt under plate or sticking.
 - Spring fracture
 - Due to fatigue or corrosion.
 - Improper design
 - Higher valve lifts or high impact velocities

Allowable valve Plate
Impact Velocities (In/Sec.)

Valve Design	At Valve Guard	At seat
Std. Plate	150	75
Damped Plate	200	100

- Valve maintenance Requirements
 - Visual inspection for scratch, nicks or cuts.
 - Clean with solvent.
 - Non destructive testing to see major cracks etc.
 - Grind/ Lap seats
 - Plate replace if required.
 - Replace springs.
 - Liquid leak test or Pneumatic test

Bearings



- Main Bearings
 - Main function of the bearing is to support the crank under all operating conditions.
 - Bearings for bigger machine are normally in two piece & lubricated by hydrodynamic lubrication.
 - Bearing are normally made of steel shell with white metal lining.
 - Common failure modes & causes are
 - Unexpected or excessive wear –
 - Inadequate lubrication.
 - Improper alignment.
 - Wrong lube oil
 - Dirt in oil
 - Foreign material.
 - Immature fatigue failure
 - Wrong material
 - Manufacturing defect.
 - Excessive loading etc.

- Main bearing checks
 - During periodic maintenance
 - Check physical condition for any abnormal wear, scratches etc.
 - Check dimension & clearance.
 - Carryout DP for integrity.
 - Check blue match for proper positioning.
 - Check bearing crush .

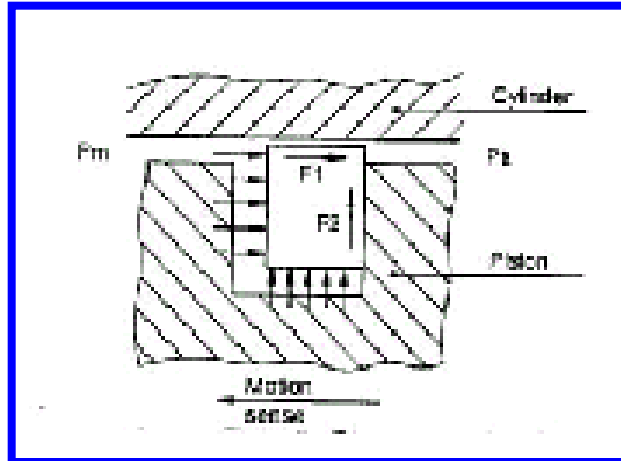
Guide Line for Common materials

Service	Operating pressure	Lube or Non-Lube	Rod Material	Minimum Rod Hardness (Rc)	Surface Finish (RMS-Mi)
Non-Corrosive	Up to 6000	L, NL	Low Alloy steel. Th or Case hardened	50	10-20
	Above 6000	L		55	10-20
	Above 6000	NL	Hardened or Coated	60	6-8
Corrosive	Through 1000	L,NL	17-4 Ph	50	10-20
	Above 1000	L		55	10-20
	Above 1000	NL	Plated or coated	60	6-8

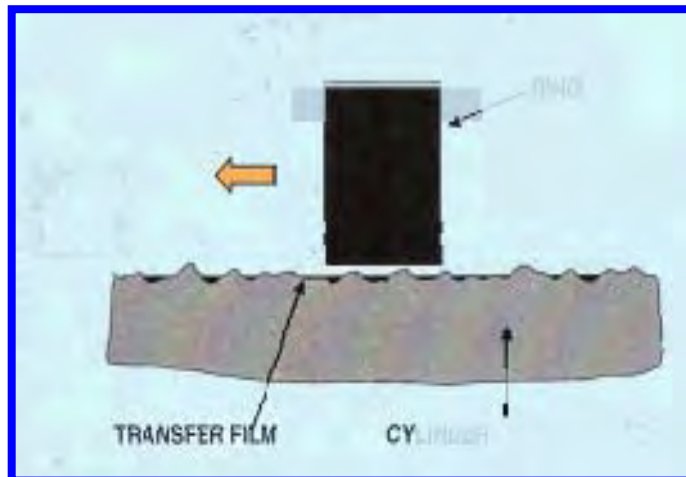
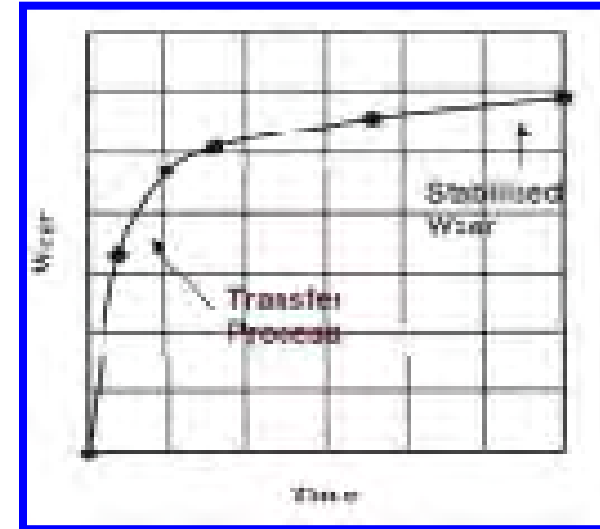
- Piston checking during maintenance:
 - Ring groove depth & width checking.
 - Rod hole & nut area checking for cracks, damage etc.
 - Squareness check: Rod to piston surface HE /CE.

- Rider ring & Piston rings
 - Rider rings are provided on the piston for supporting the weight of piston & rod assembly.
 - The location & no. are dictated by design diameter, length & weight of piston.
 - As a guide 5 Psi /Ring is the bearing limit for non-lube & 10 Psi/Ring is the limit for lube service.
 - The most common material in majority of service is PTFE with various fillers.
 - Cut or solid both designs are used. Cut designs are easy for installation.
 - Rider rings provided on the end shall have vent hole or groove to prevent it to work as piston ring.

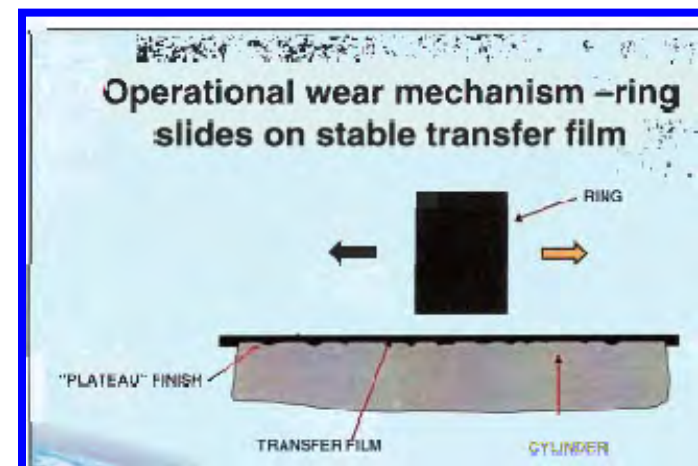
- Piston rings
 - The piston ring prevents blow-off of gases from one side to other side.
 - Piston rings are energized by gas pressure as shown in fig. & makes the seal at OD as well as on side.
 - Some designs use inner ring to energize piston ring.
 - Piston rings are also available in various design i.e. angle cut, step cut ,seal cut etc.
 - Piston rings are also made from the PTFE with various filler to suit process gas.
 - Piston rings & Rider rings are the most wearing parts & life depends on various factors like : Speed, Pressure, design, gas conditions etc.
 - Rod drop monitoring is used to measure rider ring wear.
 - During routine maintenance
 - Rider ring thickness & width shall be checked.
 - Piston rings shall be also checked for height, width & end gap.



Piston ring sealing mechanism

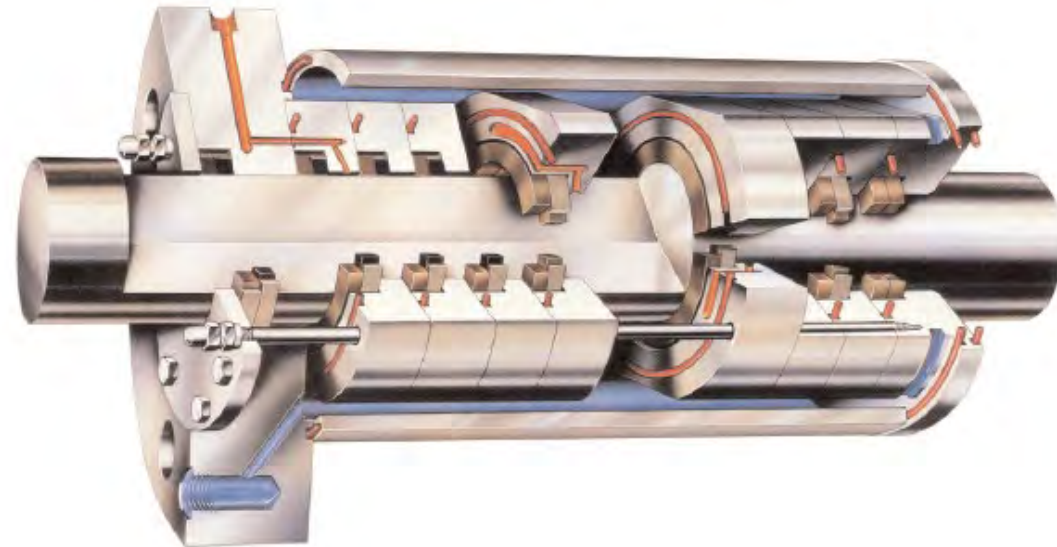


Running in wear mechanism



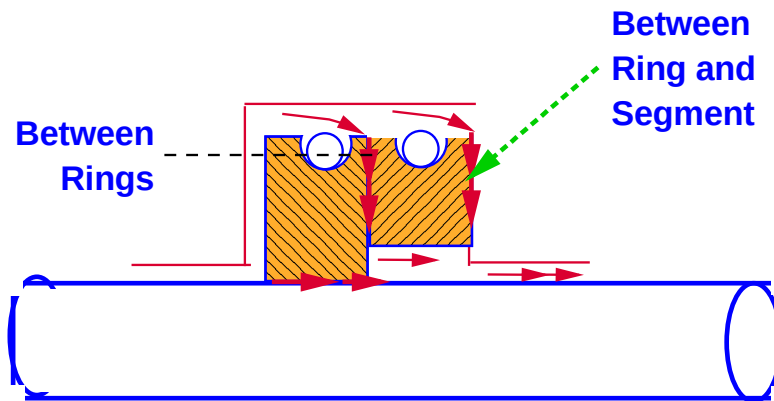
Normal operating

- Piston rod packing
 - Main function of packing ring is to seal the pressurized gas inside of cylinder & prevent it from leaking to atmosphere.
 - Packing ring consists of series of sealing elements to break-down the pressure across them & reduce leak to atmosphere.
 - Packing rings are restrictive type of seal. For hazardous gases, purged packing boxes are preferred.

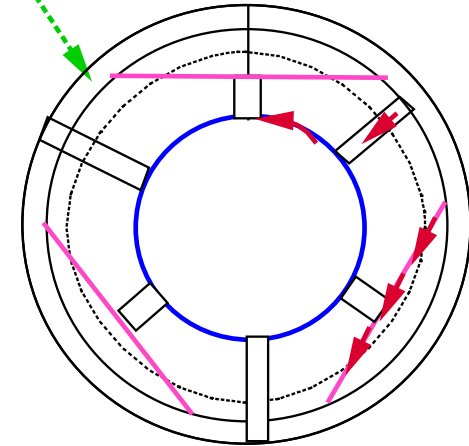


Leakage in rings

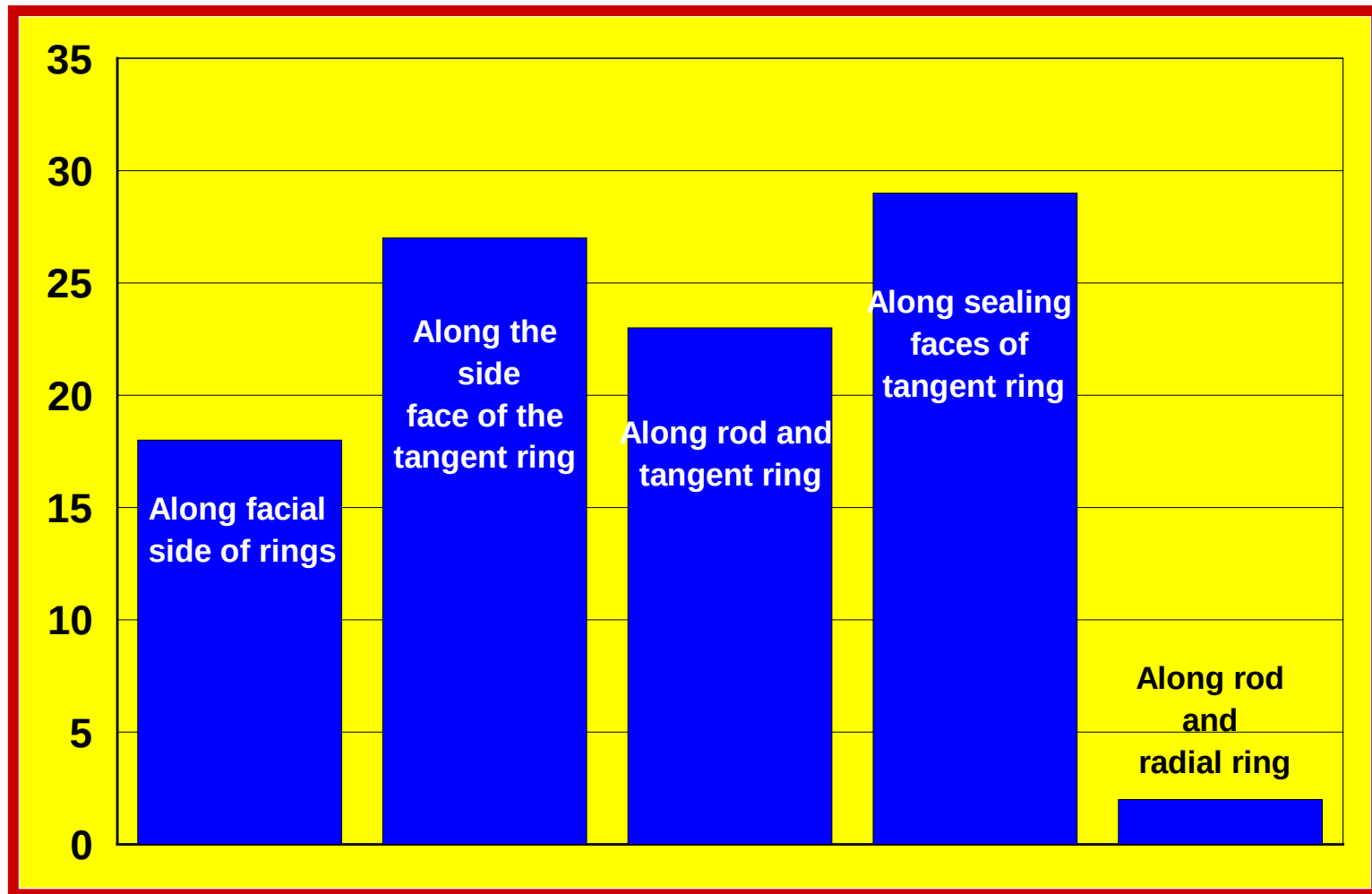
- ▶ Three ways in which leaks can occur in rings are
 - Between rings
 - Between ring and segment
 - Along Sealing surfaces



Between
Rings

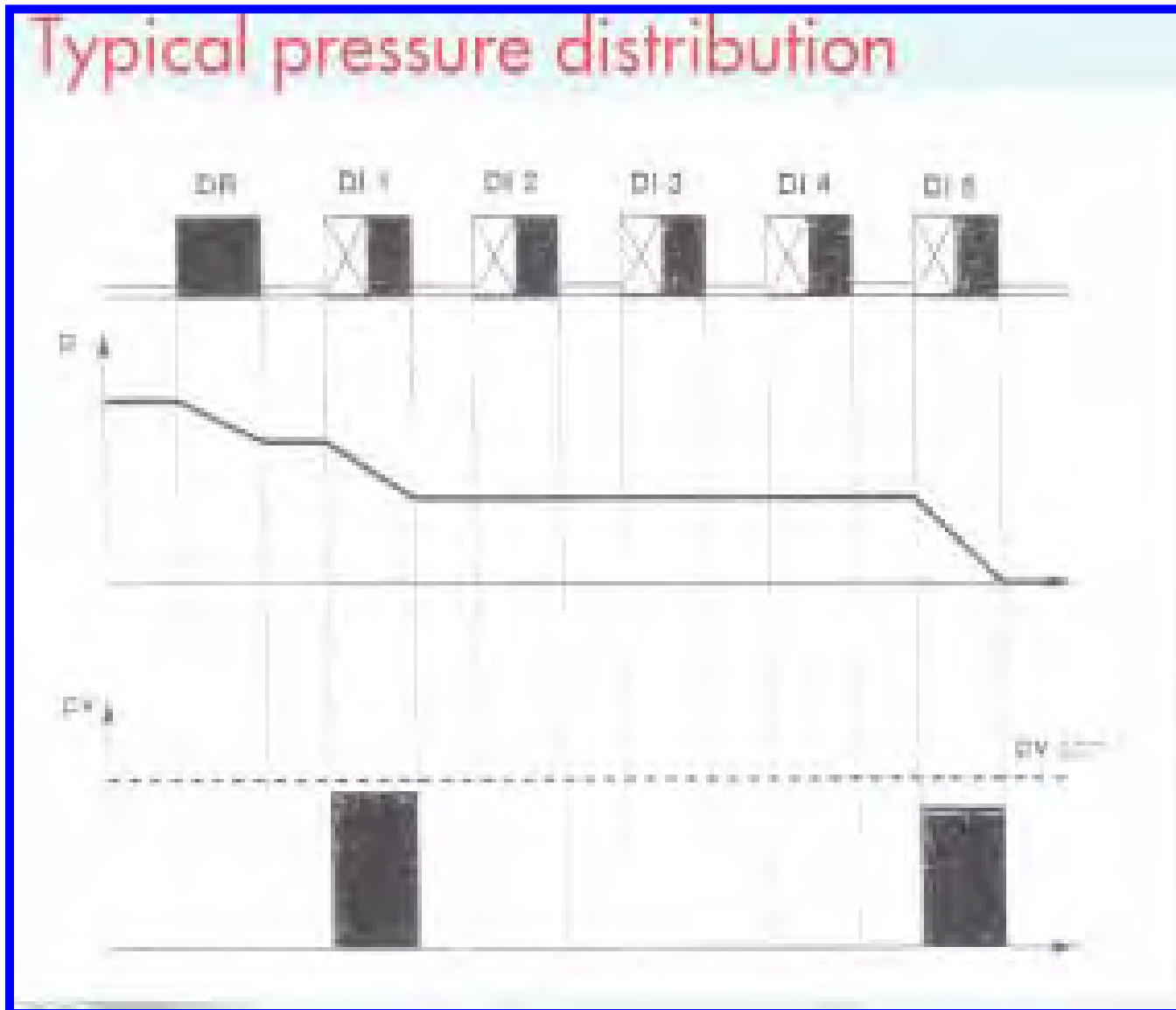


Average Leakage For Stuffing Boxes



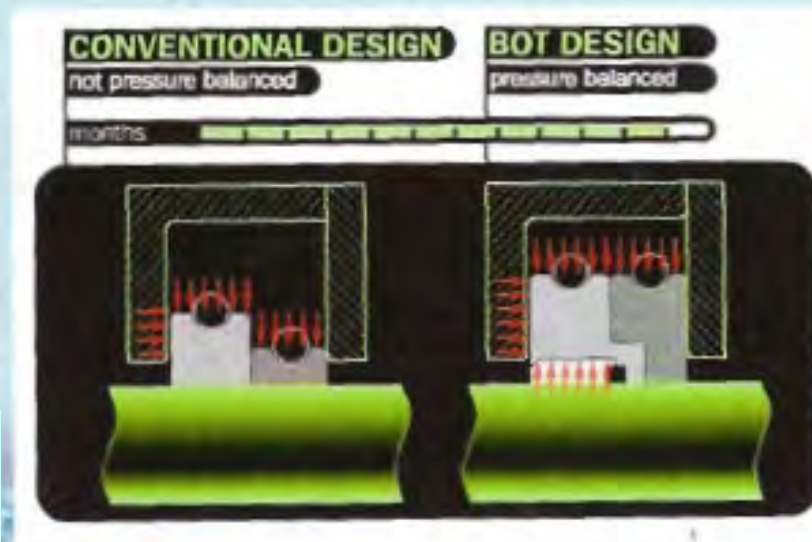
The left diagram is a cross-sectional view of a multi-layered structure. It shows a central core with a wavy line, surrounded by several layers. Labels include 'LAYER 1' at the top, 'LAYER 2' below it, and 'LAYER 3' further down. A dimension line on the right indicates a thickness of '1.5 mm'. The bottom part of the diagram shows a series of vertical lines, possibly representing a stack of components or a cross-section of a pipe.

The right diagram shows six individual components, which appear to be rings or washers, arranged in two rows of three. Each component is labeled with a number from 1 to 6. Labels 1, 2, and 3 are on the top row, and labels 4, 5, and 6 are on the bottom row. A label 'LAYER 1' is positioned near the top row, and a label 'LAYER 2' is positioned near the bottom row. A label 'LAYER 3' is positioned near the middle of the components.



BOT packing ring pair

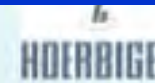
- **B**alanced **O**verlapping **T**angent pressure packing ring pair
 - Reduced wear (evenly distributed pressure drop across all the ring pairs)
 - Significantly increased life
 - Reduced rod temperature



- Piston rod packing : Failure & maintenance
 - Wear Radial & Tangential rings.
 - Cup face wear.
 - Sludging
 - Extrusion
 - Rotational wear
 - Spring failure.

- Causes
 - Improper application
 - Dirty gases
 - Inadequate cooling
 - Excessive rod run-out
 - Lubrication failure
 - Excessive lubrication

Thermosleeve pressure packing

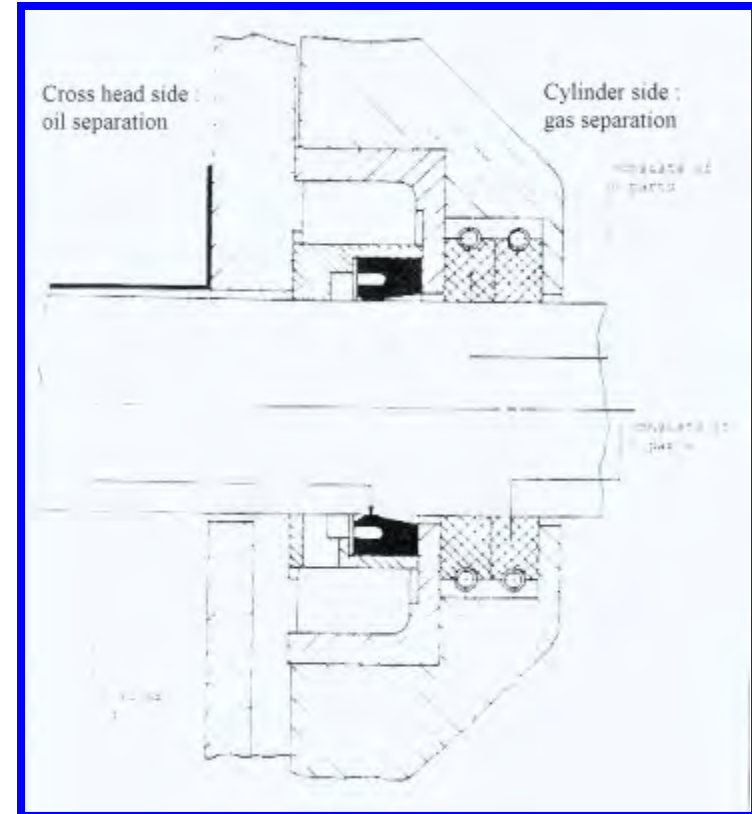
 HOERBIGER

- Lower operating temperature of packing rings
 - Direct heat removal through intensive heat conduction into the compressor housing
 - Suitable for both water-cooled and air-cooled compressors
- No water cooling of the pressure packing
 - No extra piping
 - No danger of coolant contamination
 - No clogging of narrow cooling channels



Packing and sealing

- Main function of combination seal on cross head side is to prevent lube oil carryover to cylinder side.
- To prevent leakage of cylinder gases to crank case side.
- Combined of oil wiper seal & gas seal.



OT oil wiper packing

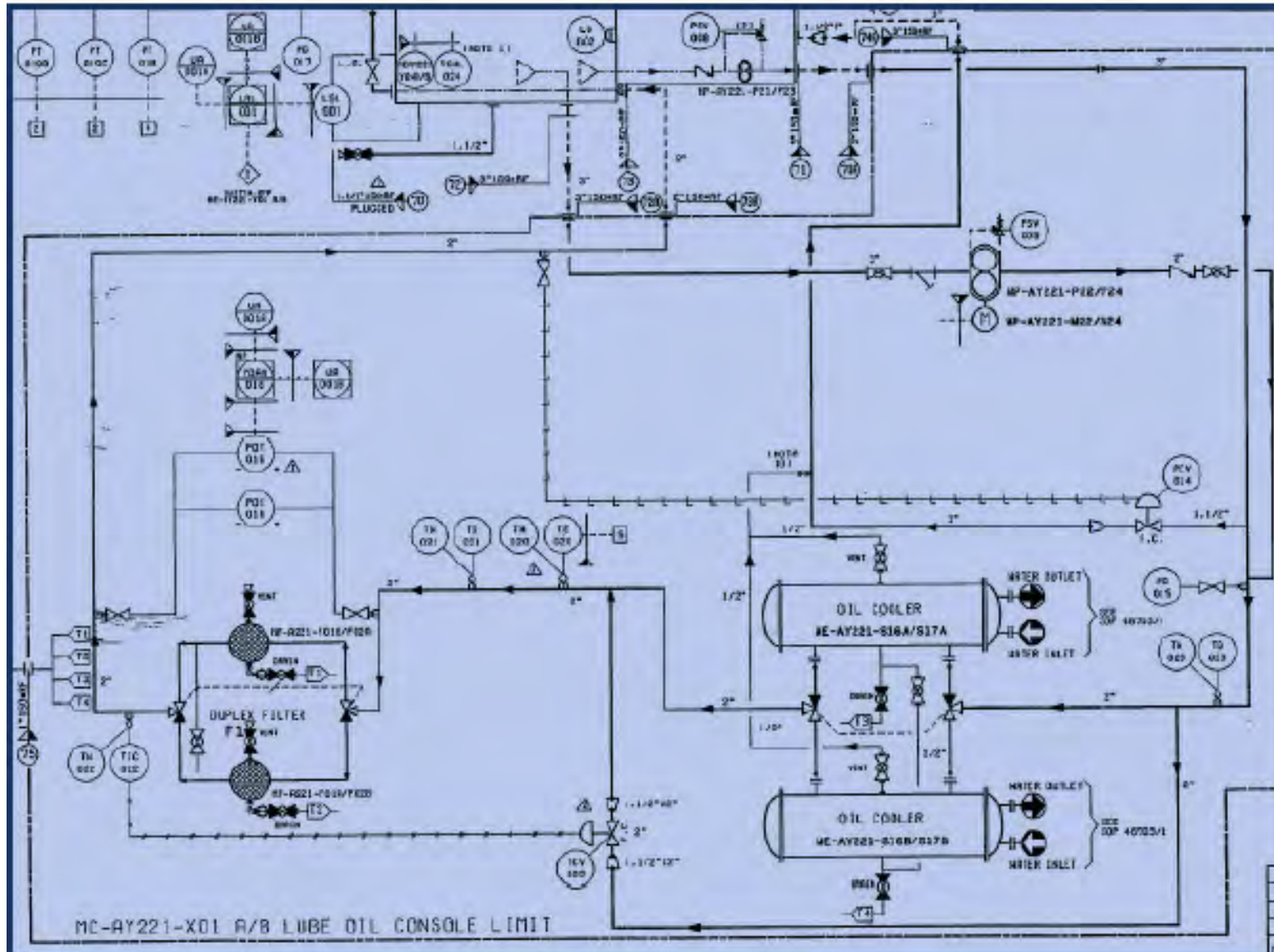
- Optimal oil wiping is a “must”
 - Environmental laws and regulations
 - Protection of critical gases from contamination (oxygen, technical gases)
 - Longer life of materials in non-lubricated pressure packings
- OT wiper packing
 - Features new designed rings for optimum efficiency of the wiper packing



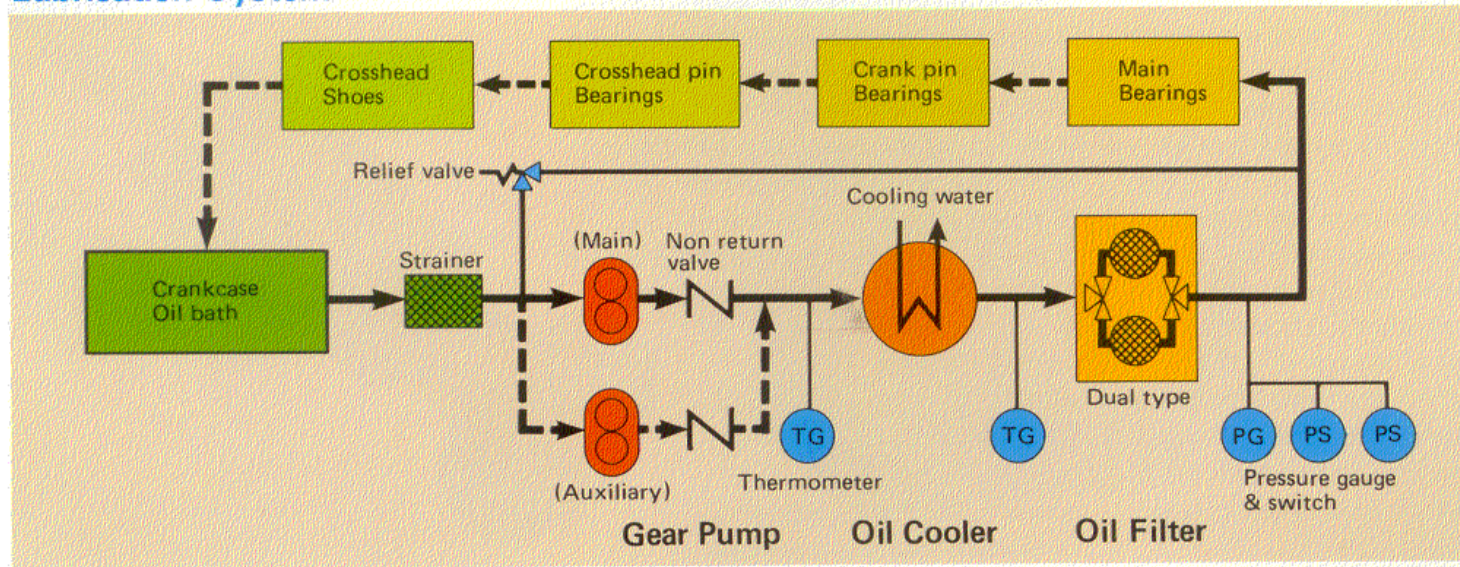
Auxiliary System

- Lubrication systems:
 - Frame lubrication
 - Provides lubrication to main bearings, connecting rod bearings & crosshead.
 - Splash lubrication for lower power size
 - Connecting rod & crank web extension dip into oil & throws oil to other parts.
 - Forced lubrication
 - Main elements are Oil pumps, filters, coolers , control valves , monitoring & protection instruments.
 - Redundant critical elements, adequate sizing of pumps for increased clearances.
 - Ease of fault findings & adequate protection.

Frame Lubrication



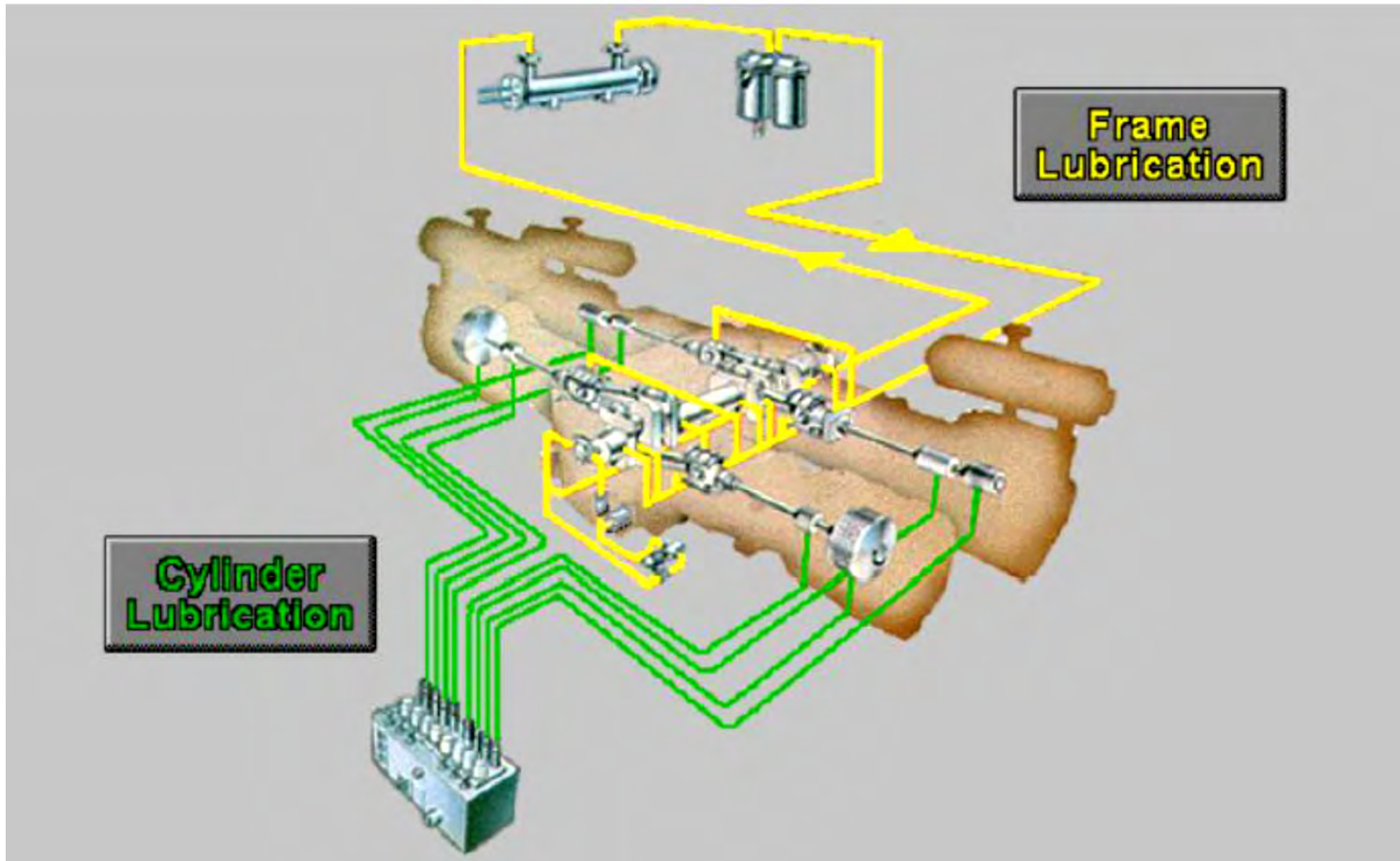
Lubrication System



- Cylinder Lubrication
 - For injecting metered quantity of lube oil to cylinder & stuffing box packing of lubricated compressors.
 - Main components are plunger type lubricator, sump, pre-lube facility & alarm device.
 - Mainly two types
 - Single plunger for control & delivery
 - Two separate plunger for delivery & control
 - Stroke of plunger is varied to vary oil flow control.
- Oil Drop/Min/cylinder
$$= \{ \text{Cylinder Dia. (In.)} \times \text{Stroke (In.)} \times \text{RPM} \} / 13800$$
- For initial run-in period adjust 50% higher.

- Cylinder cooling
 - Cylinder cooling is required to dissipate heat generated due to friction of rider ring ,piston ring with cylinder & heat of compression.
 - Air cooling
 - Generally used on smaller machines & more common portable air compressors.
 - Water cooling
 - Once through (Open system)
 - Low cost
 - Difficult to control water quality
 - Closed system
 - Easy to control water quality
 - Higher capital cost.

Lubrication System



Cylinder/Packing Lube System

Pump to point:

- Single pump unit for each point lubricated
- Visual indication of the oil sent to each point
- Lubrication Monitoring (shaft rotation and low level) through dedicated pump units.
- Very reliable and proven System
- Lubricator can be driven by compressor shaft or dedicated electric motor.
- An electric heater is installed to keep heated the oil inside lubricator sump

Progressive system:

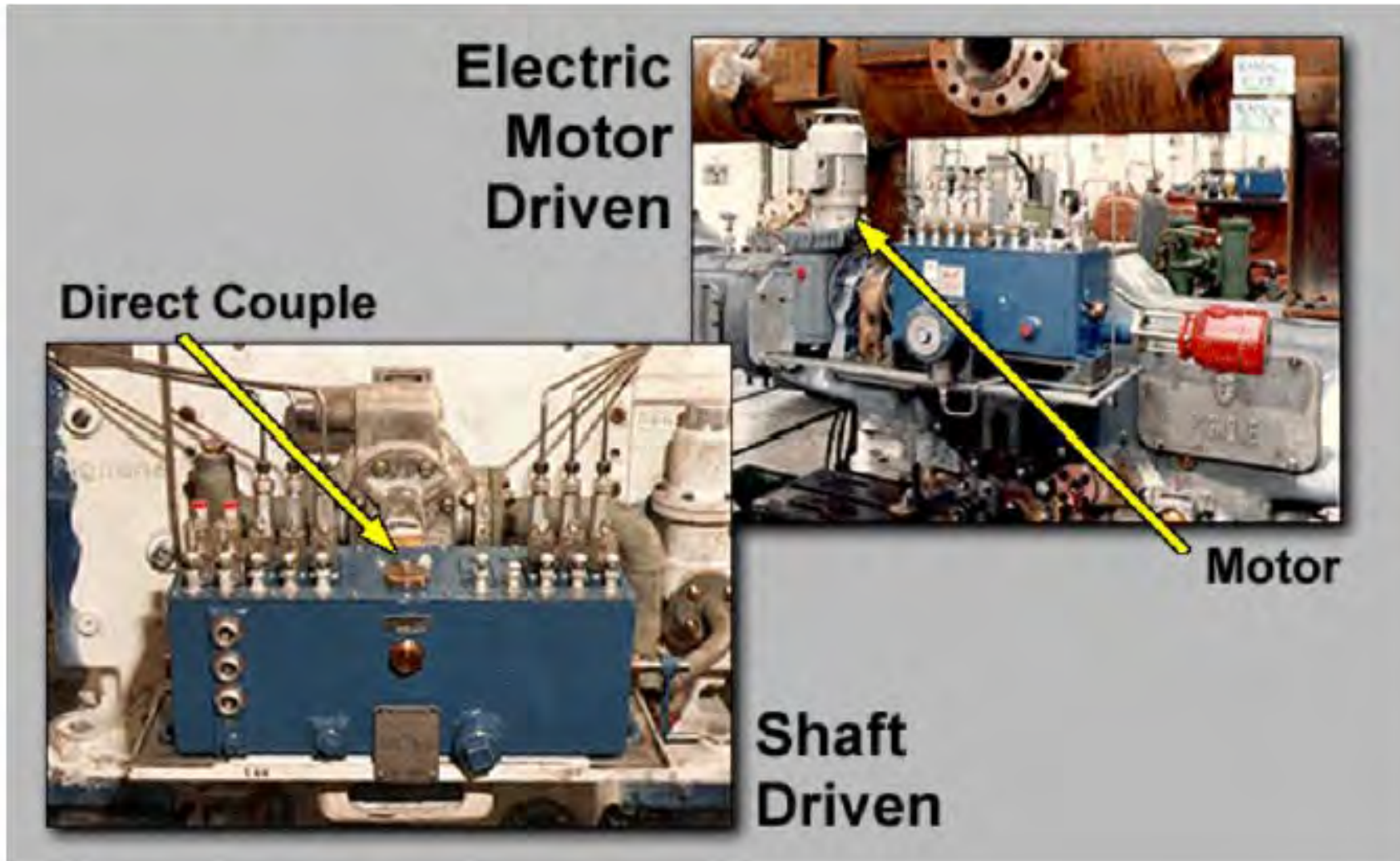
- Electric motor driven, may be equipped with a primary distributor and a secondary distributor for each cylinder.
- Each distributor may be equipped with a dedicated device to monitor the operation functioning directly from control panel or in field by a dedicated visual indicator.



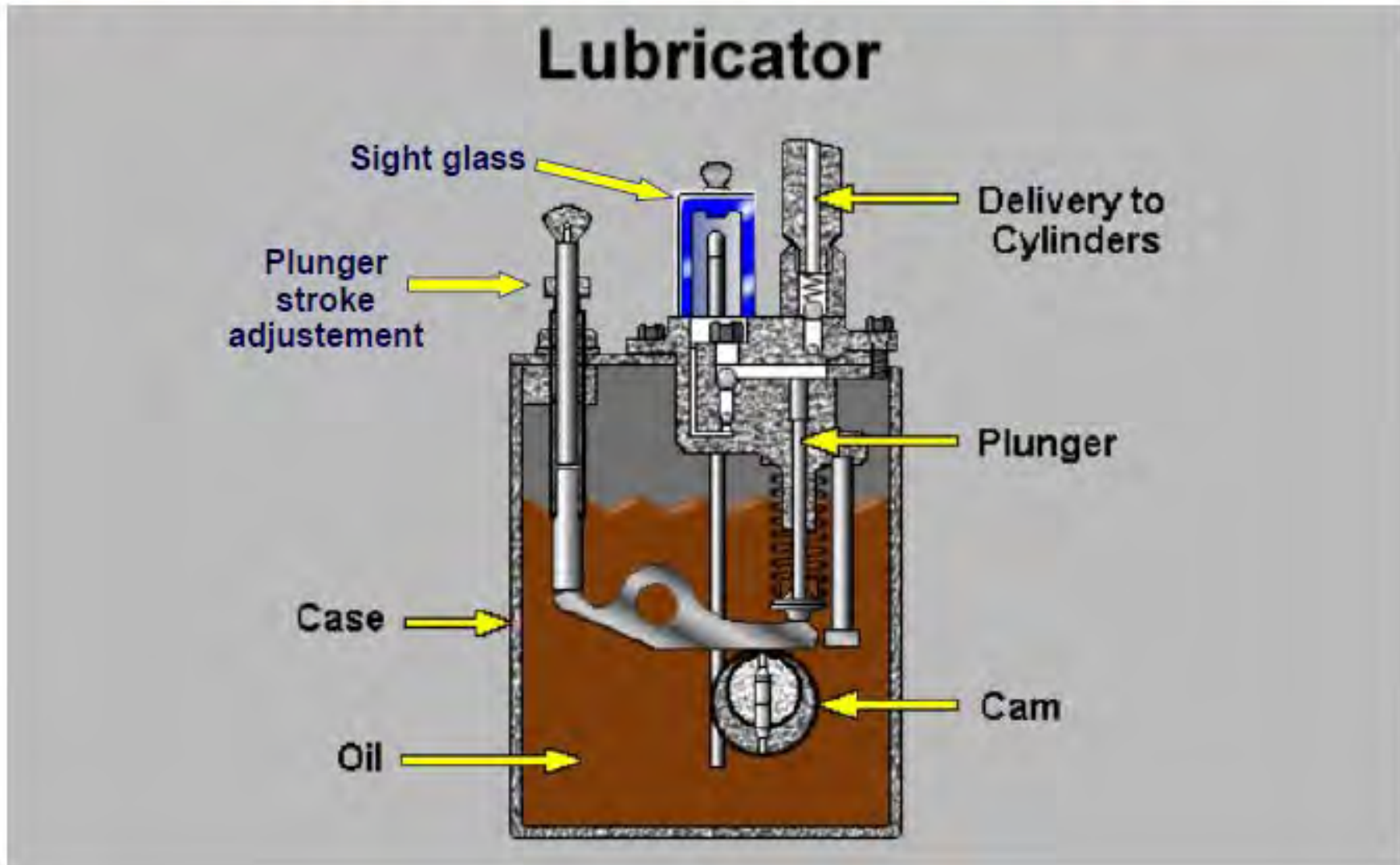
2.16 CYLINDER AND PACKING LUBRICATION

- 2.16.1 Separate electric motor shall be provided for the cylinder lubricators and both the motor and lubricator shall be suitable for outdoor installation.
- 2.16.2 Minimum of two lubricating points shall be provided for each cylinder.
- 2.16.3 Minimum of two lubricating points shall be provided for cylinders piston rod packing with 100 barg or above of discharge pressure.
- 2.16.4 Location of lubricating points shall be at the upper side of the piston rod (at each packing assembly end).
- 2.16.5 The use of divider blocks is not permitted.

Lubrication Systems (Pump to Point)



Lubrication Systems (Pump to Point)



PTS 121132

2.8 CYLINDER COOLING

- 2.8.1 A pressurized thermosyphon system is not allowed to be used.
- 2.8.2 The temperature of the coolant inlets shall be maintained at the minimum of 6 K higher than the gas temperature and at the minimum of 10 K higher than the inlet gas temperature for saturated and wet gases. For 'bone dry' gases, these limits are not applied.

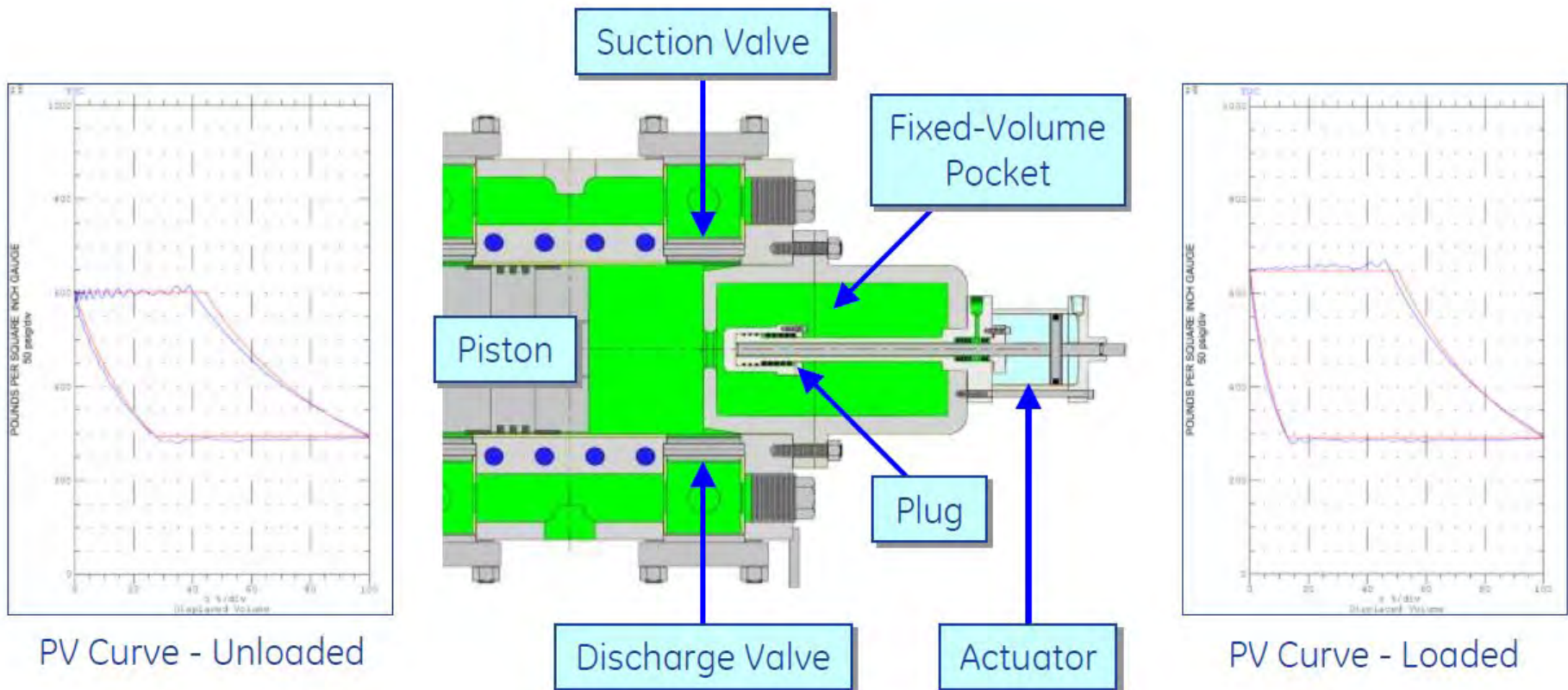
Capacity Control

Reciprocating Compressor Capacity Control

- Speed Control
- Recycle Control
- Valve Unloader
- Clearance Pocket

Reciprocating Compressor Capacity Control

Clearance Pocket



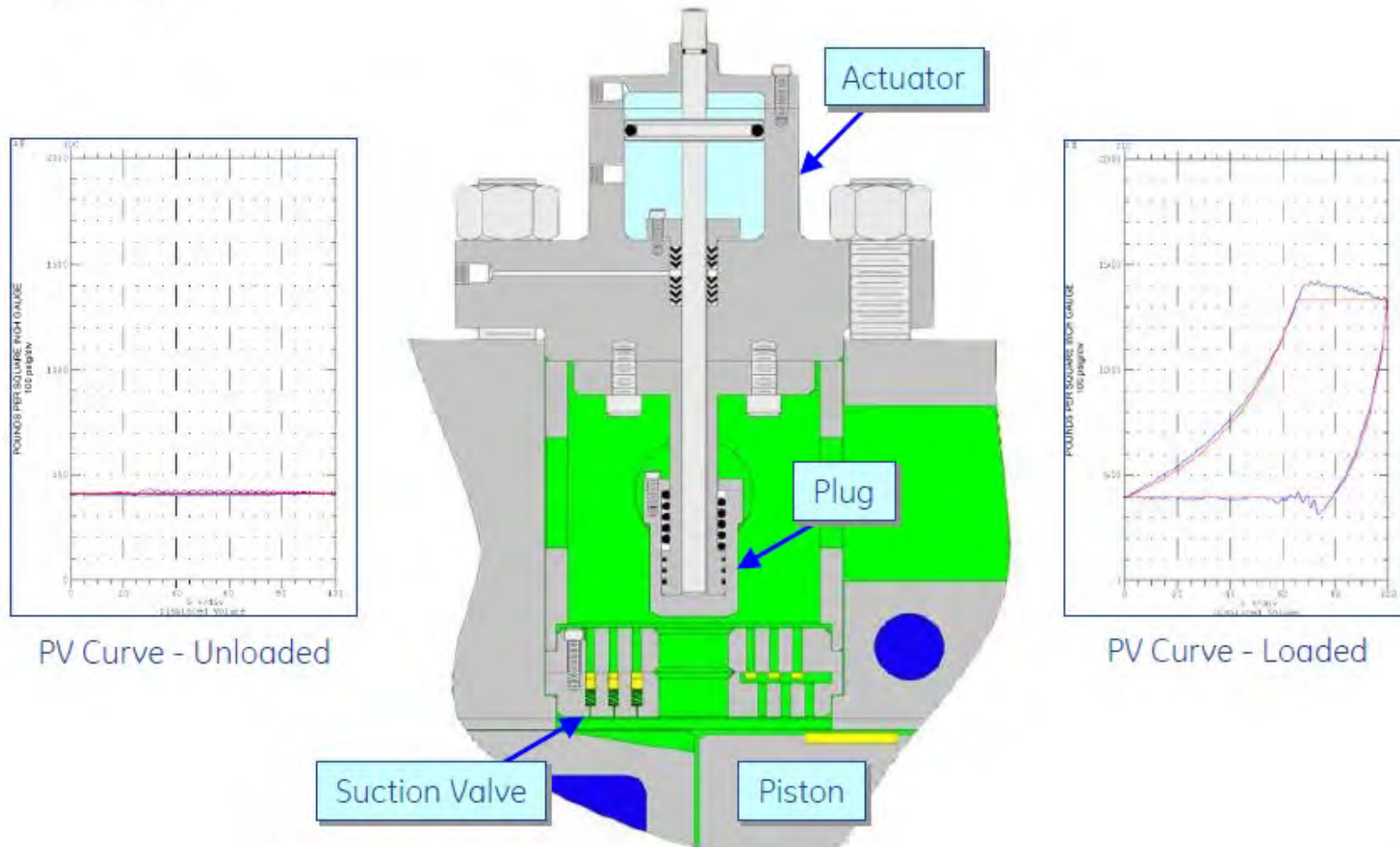
Clearance Pocket



Example: Head-end clearance pockets with hydraulic actuators.

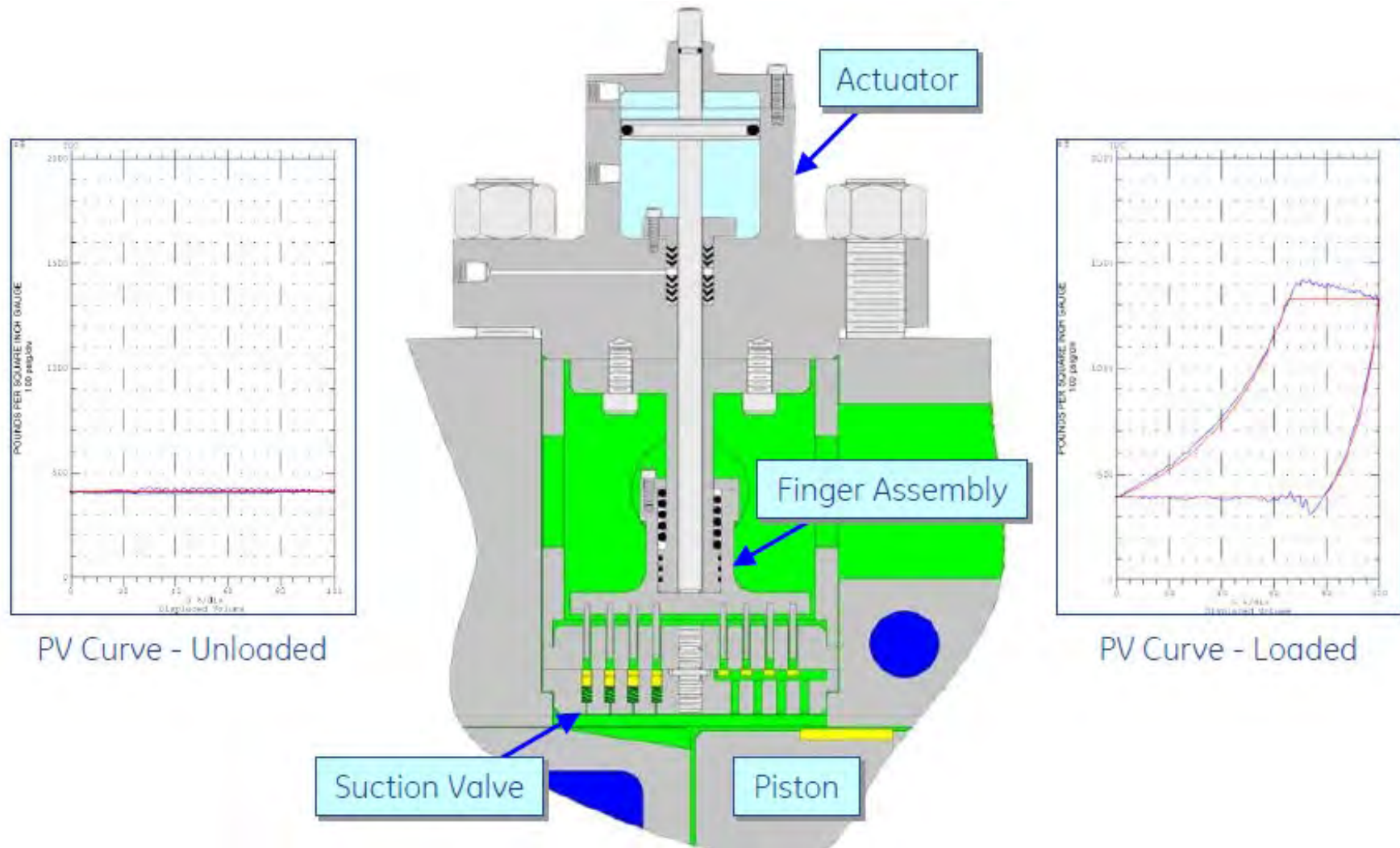
Reciprocating Compressor Capacity Control

Plug Unloader

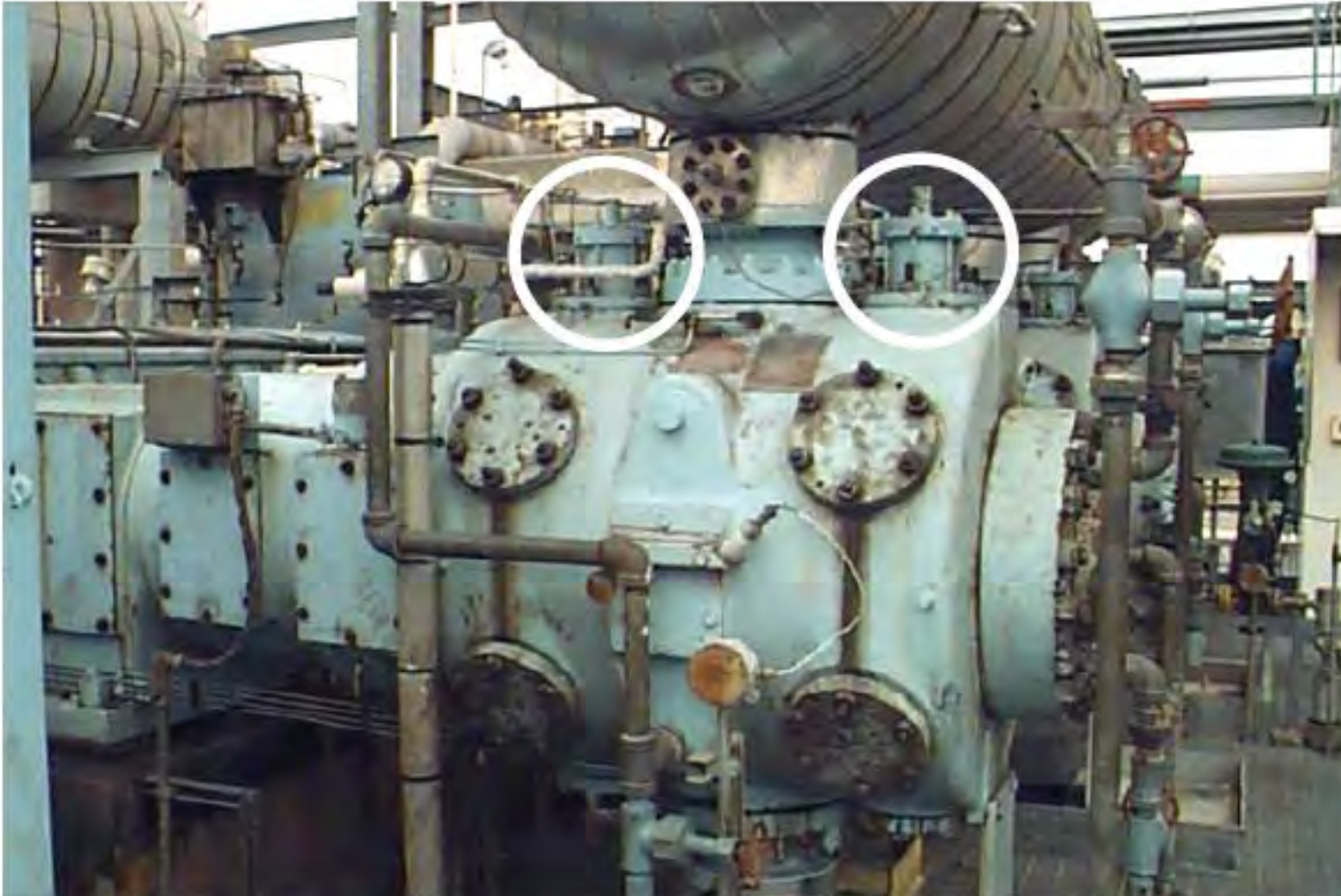


Reciprocating Compressor Capacity Control

Finger Type Valve Unloader



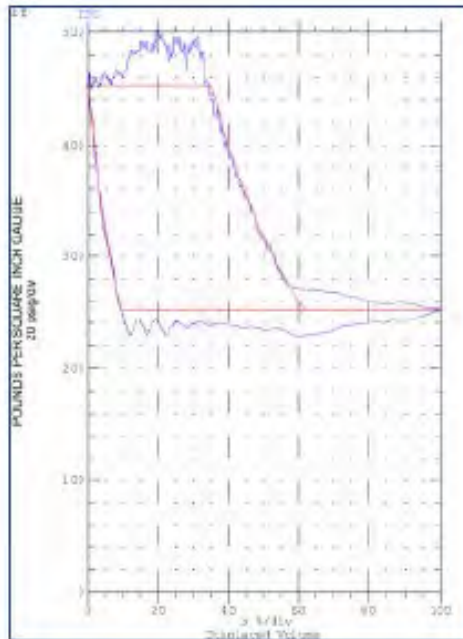
Reciprocating Compressor Capacity Control



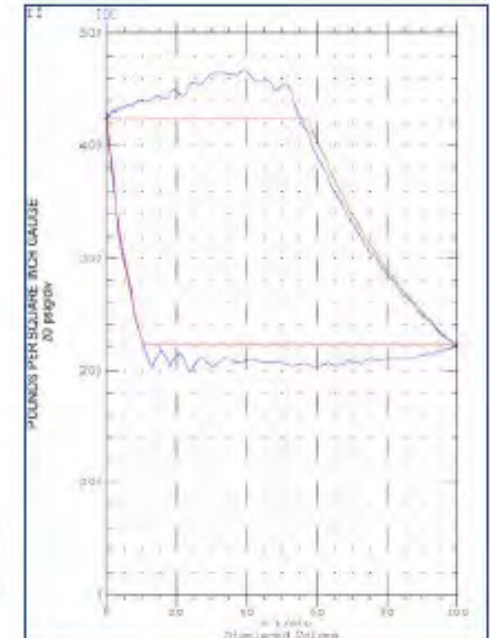
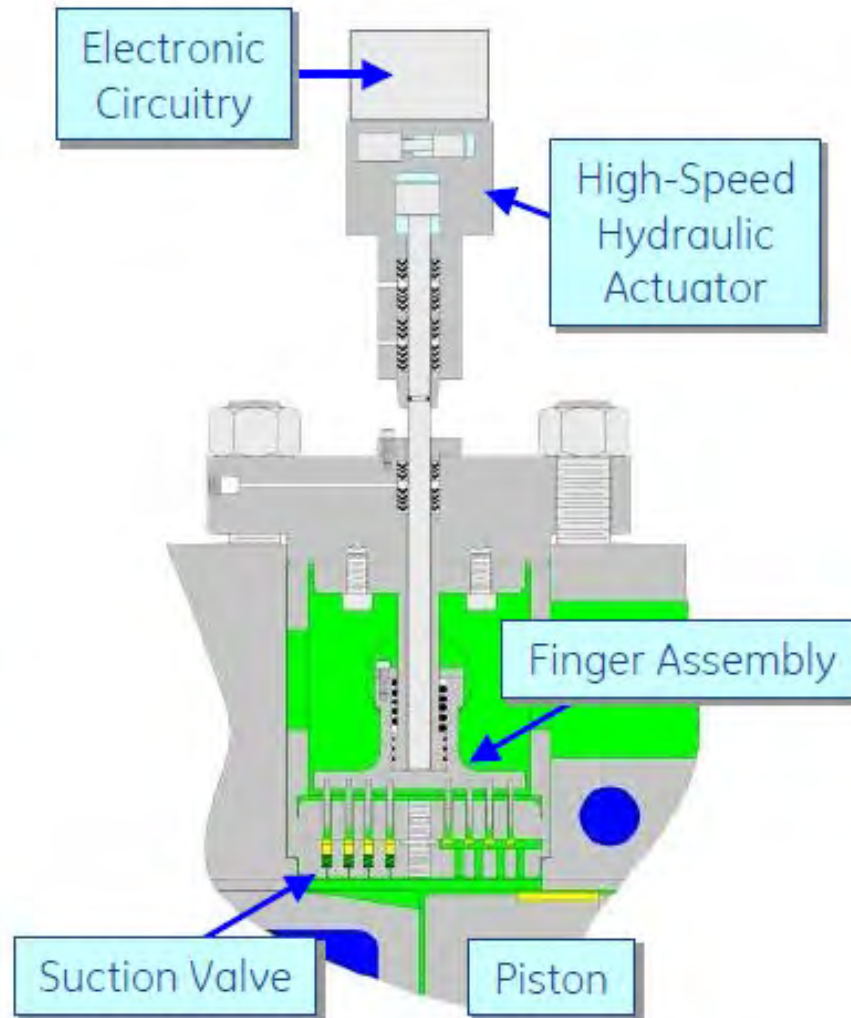
Example: Suction valve unloaders with hydraulic actuators.

Reciprocating Compressor Capacity Control

Active Valve Control



PV Curve - Unloaded



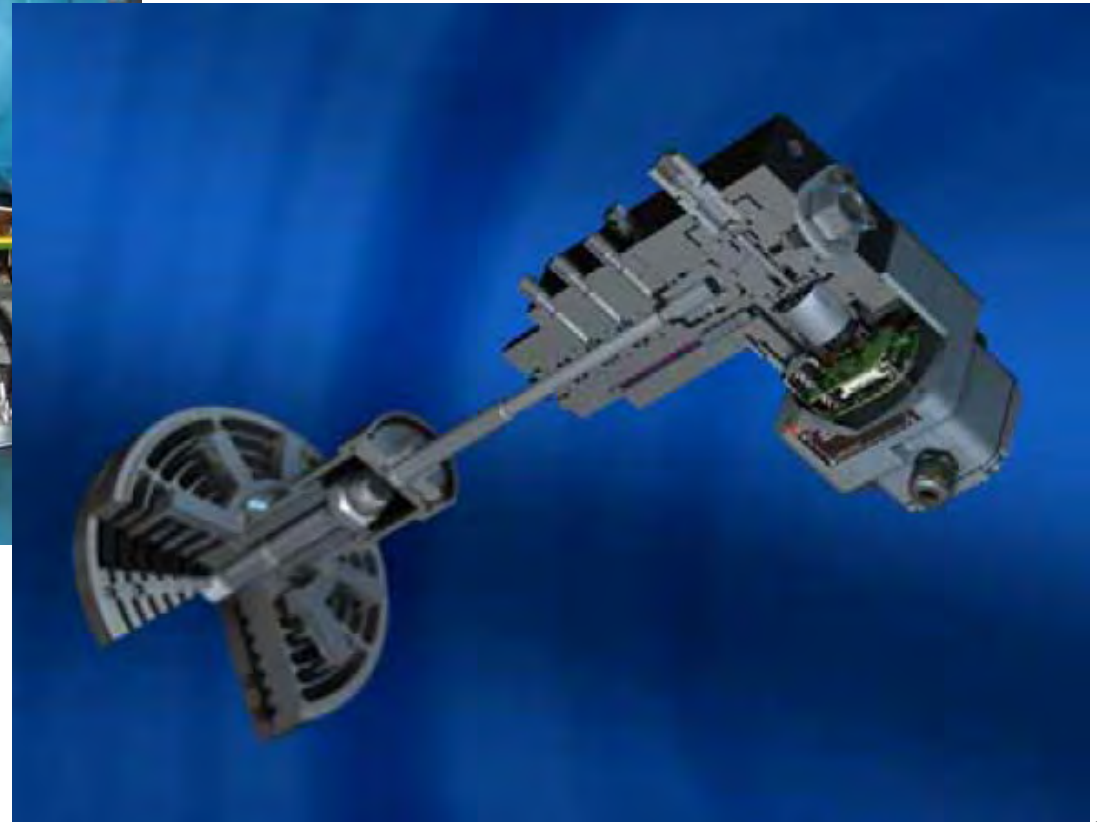
PV Curve - Loaded

Reciprocating Compressor Capacity Control

Active Valve Control



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HydroCOM



End of presentation

THANK YOU

SKG 12 Mechanical Engineering

DOMI of Compressor and Boiler (5of7)

- Installation , Testing and Commissioning

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- Adequately designed & defect free foundation is key for trouble free reciprocating compressor operation.
- Foundation attenuate vibratory forces generated by machinery & reduce transmission of these forces to surrounding plant & equipment.
- Foundation also maintains machinery in alignment.
- In reciprocating machines due to unbalanced primary & secondary forces as well as couples. Large reactions are generated at main bearings which are transmitted to foundation.
- The natural frequency of foundation shall be 30% above or below primary & secondary frequencies.
- As a good rule of thumb foundation mass shall be five times the combined weight of compressor & driver.

- Medium sized reciprocating compressor are rarely shipped as complete assembly.
- Three methods for mounting compressor on foundation
 - Direct grouting
 - ✓ Used for small machines i.e. Air compressors.
 - Mounting on sole plate
 - ✓ Preferred method for medium to heavy compressors.
 - ✓ Heat from frame can cause distortion of foundation.
 - Mounting on base plate
 - ✓ Class B units are sometimes skid mounted.

Typical steps for site mounting

- Foundation checking & fix leveling plates.
- Installation of frame & leveling .
- Installation of distance piece & cylinders.
- Leveling of cylinders.
- Web deflection check.
- Install driver & rough alignment set.
- Grouting of frame , crosshead & cylinder supports.
 - Take care not to grout complete below frame.
- Grouting of driver.
- Final alignment check.
- Piping connection (Main & Auxiliary)
- Instrument hook-up.

Foundation and Installation (continue)

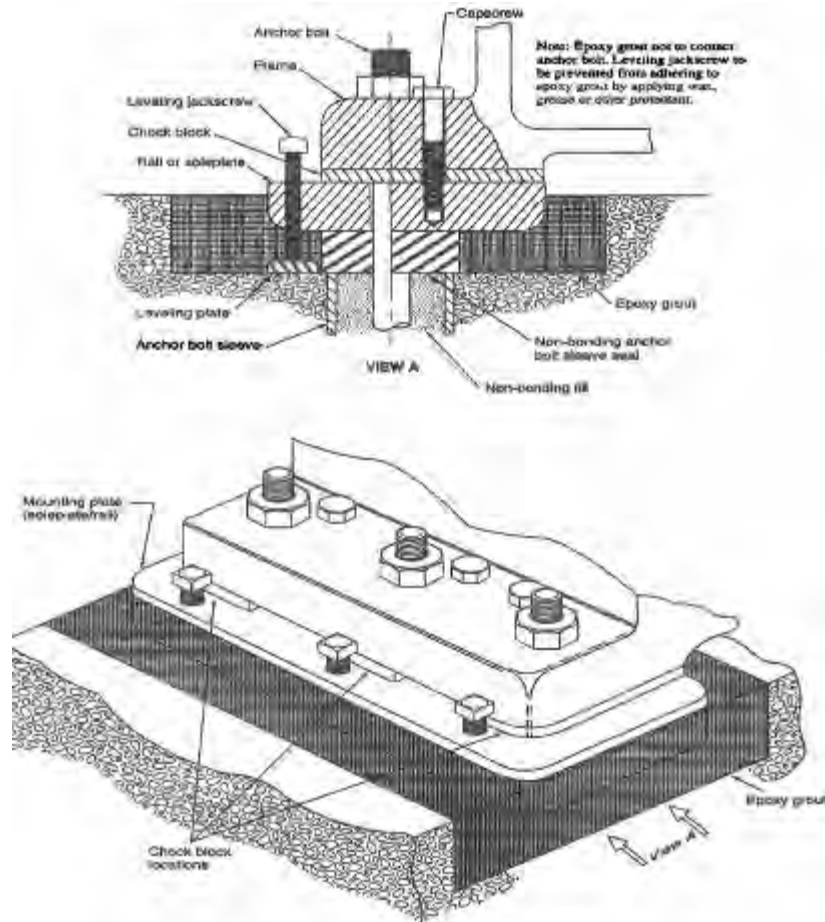
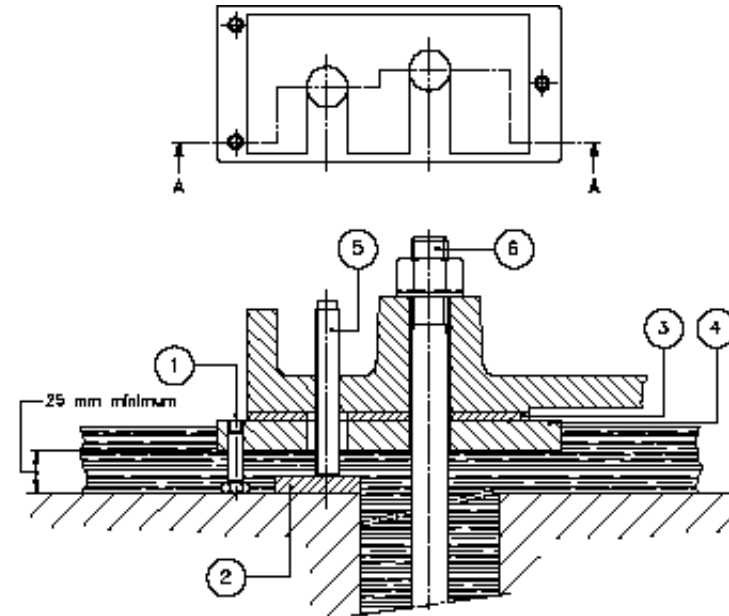


Figure L-1—Typical Mounting Plate Arrangement



1. Sole Plate positioning bolt
2. Steel Plate
3. Shims
4. Sole Plate
5. Machine leveling spindle
6. Anchor bolt

Flushing

- Commissioning
 - Process pipe cleaning & preservation.
 - ✓ Suction piping from knock-out drum to damper shall be chemically cleaned.
 - ✓ Lube oil piping shall be chemically cleaned & lube oil circulation with by-passing bearings shall be done prior to run.
 - ✓ All other auxiliary piping : Cooling water shall be mechanically cleaned.
 - Auxiliary piping cleaning & flushing.
 - ✓ Fine strainers shall be placed on inlet on pulsation damper.
 - ✓ Fine strainers shall be left in place for initial operation.
 - ✓ Cone type strainer with fine mesh (# 100) on inside is preferred.
 - Installation of temporary strainers.
 - Protective function simulation & verifications.
 - Driver no-load run.

Testing and Commissioning

- Compressor no-load run.
 - Parameters to be monitored during initial run
 - ✓ Compressor suction pressure & flow
 - ✓ Delta P across strainer
 - ✓ Compressor vibrations, bearing temp., thrust positions.
 - ✓ Start-up bode plots
 - ✓ Power drawn

Mechanical run test

- Main objective is to verify mechanical suitability of compressor internals for operating speed region.
- To verify rotor dynamics calculations i.e. Critical speed map, AF, Separation margin etc.
- To verify bearings sensitivity to change in oil pressure & temperatures.
- It is a four hour no-load test at maximum continuous speed.
- Compressors are accelerated first to 110% of rated speed to see mechanical suitability.
- Filtered & unfiltered vibrations, bearing temperatures are captured throughout the test.
- This test is mandatory for all compressors & spare rotors ordered at same time API-617.

Shop Performance testing:

- In order to verify compressors thermodynamic performance i.e. Head v/s Flow characteristic, performance test as per PTC10 is carried out.
**This test is optional.
- There are three classes of test
 - Class-I, where actual gas composition , pressure & temperature are simulated.
 - ✓ The most accurate simulation of site conditions.
 - ✓ Constraints are safety & shop cooling limitations.
 - Class-II & III, where gas composition other than actual is used to simulate site conditions within allowable aerodynamic limits of standards.
 - Class-II uses perfect gas equations for conversion of results.
 - Class-III uses real gas equation for conversion of results to specified condition.

End of presentation

THANK YOU

SKG 12 Mechanical Engineering

DOMI of Compressor and Blower (6 of 7)

-Construction of Compressor (Reciprocating Compressor)

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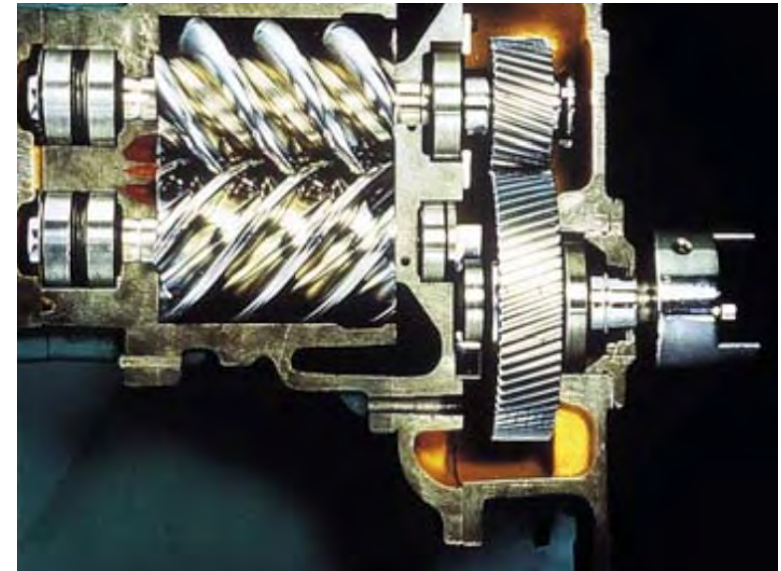
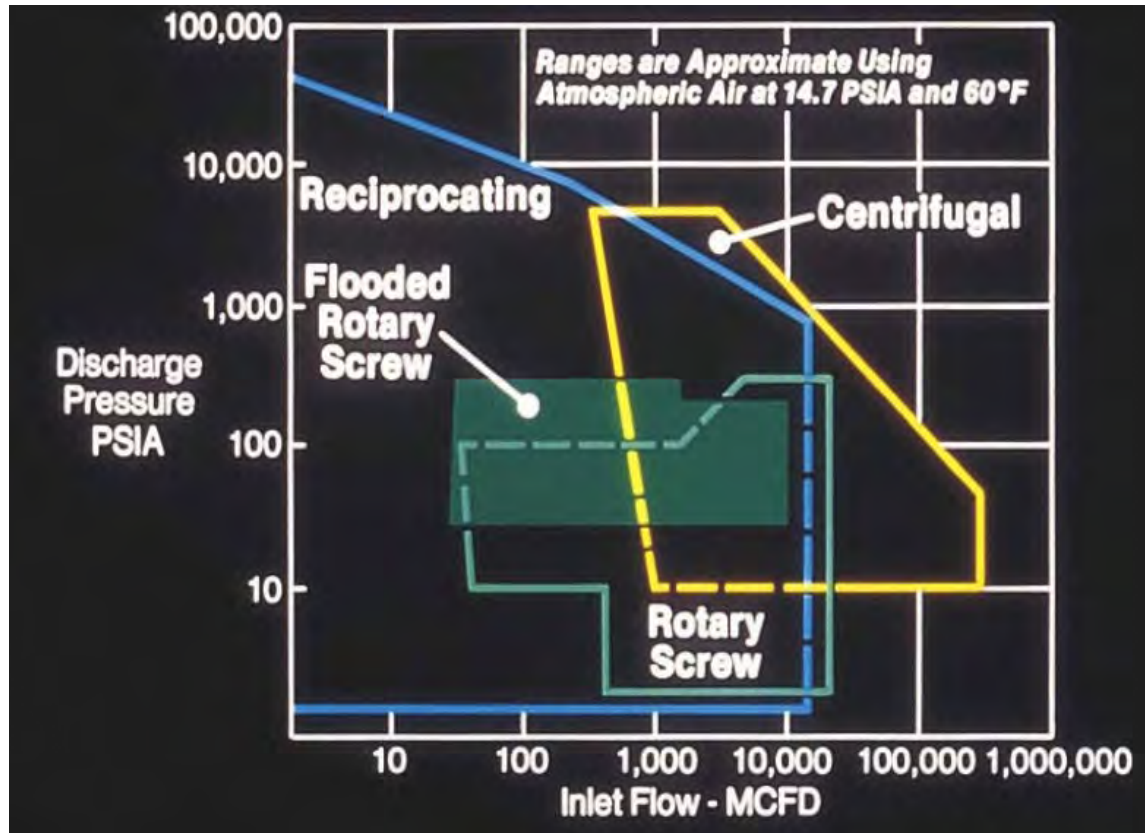
5.0 Other Type Of Compressor

Other Type Of Compressor

- Screw Compressor
- Hyper Compressor
- Iso Thermal Compressor

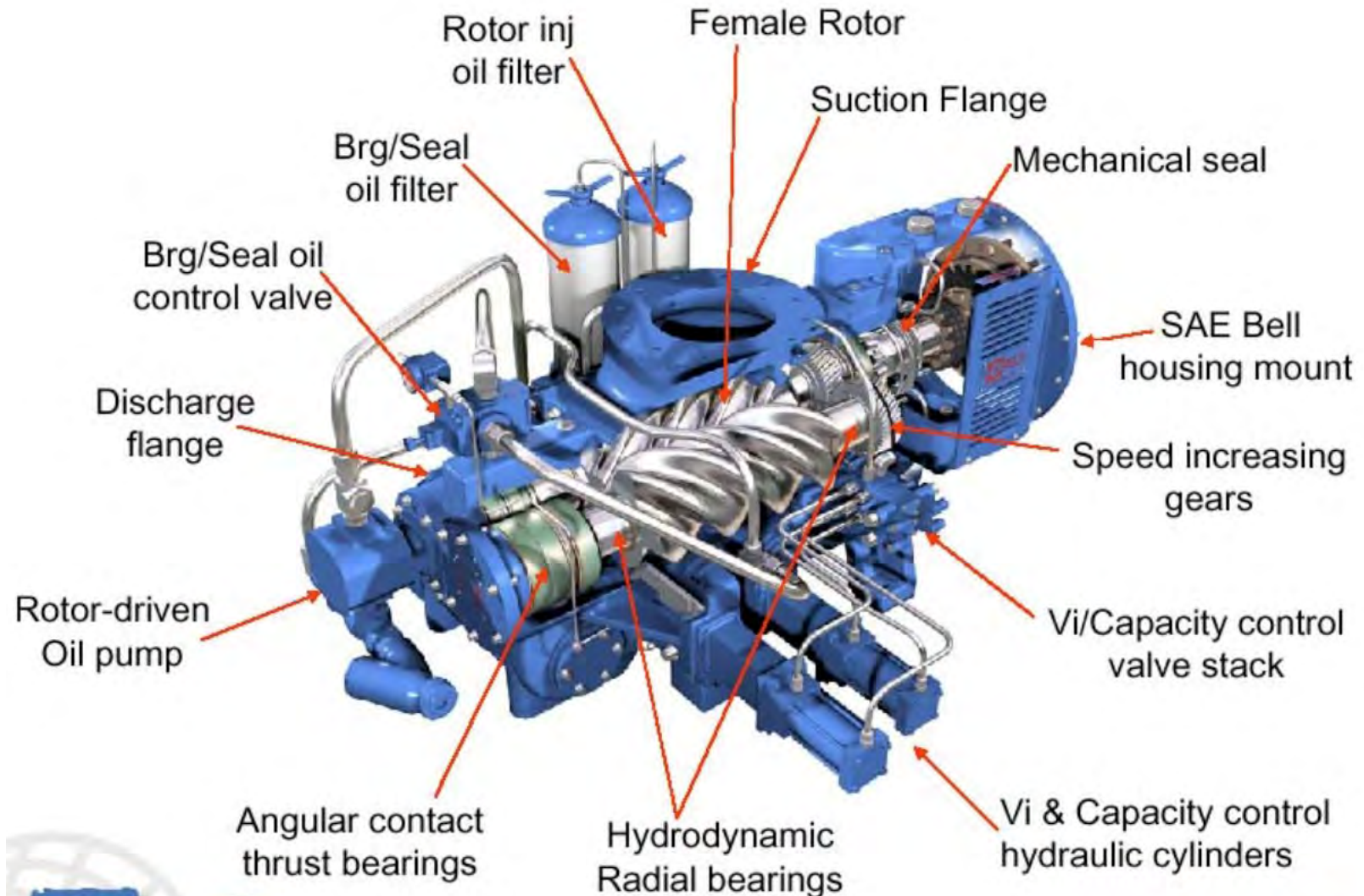
Screw Compressor

Performance Comparison

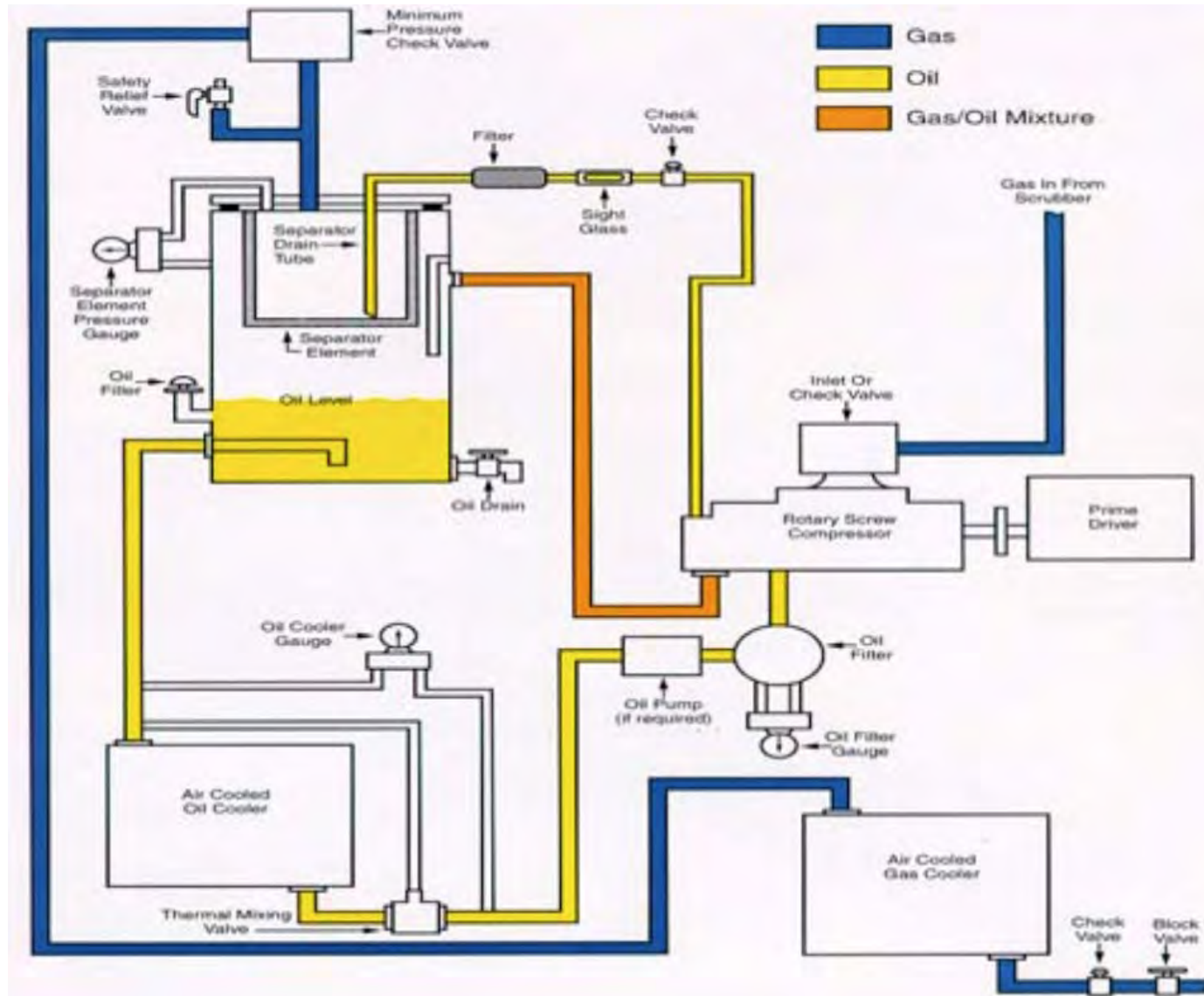


[Screw Compressor for Gas Process Applications](#)

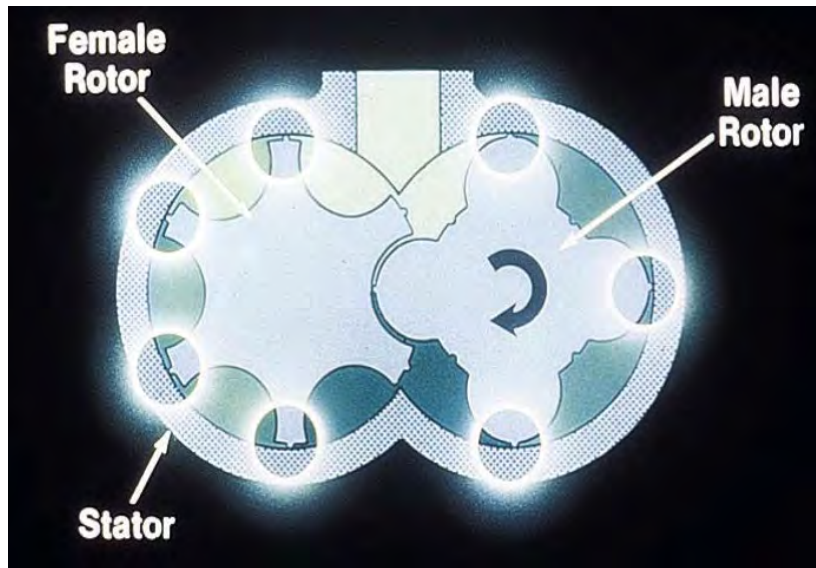
Screw Compressor Components



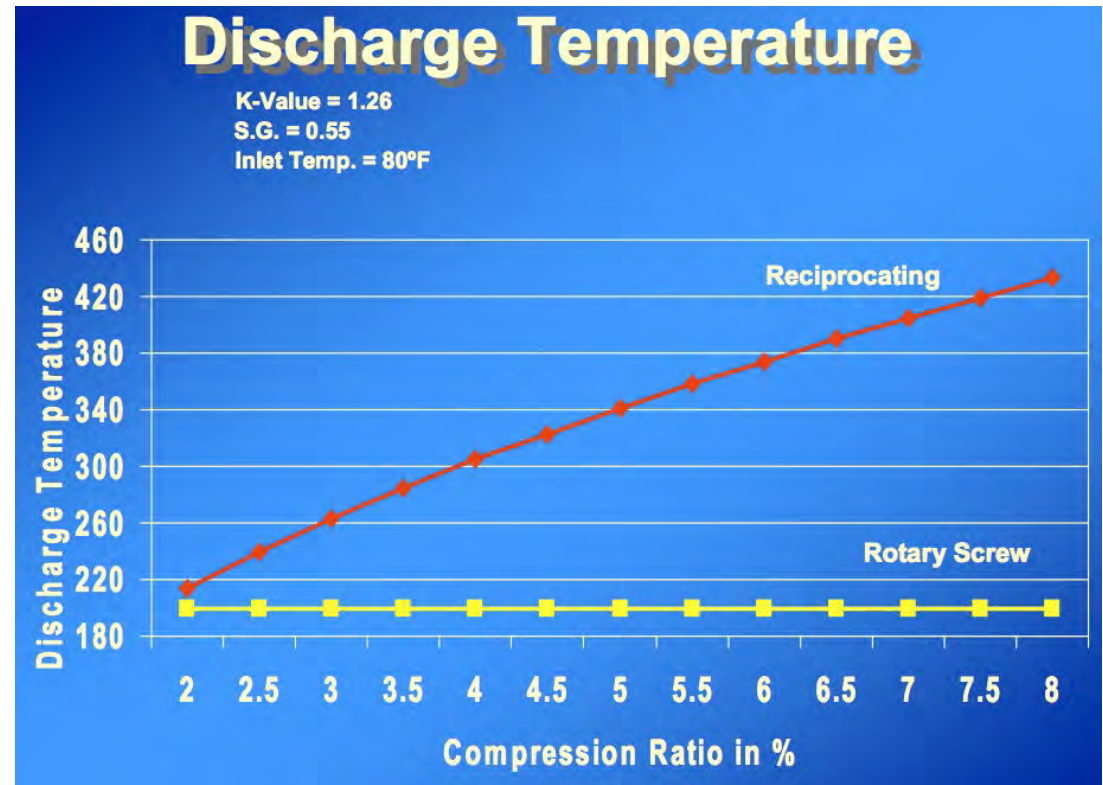
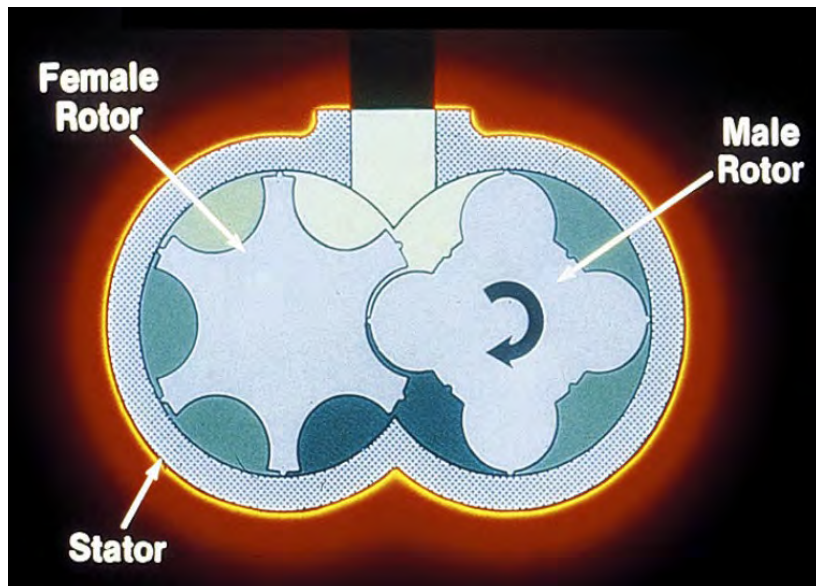
Screw Compressor System



Flooded Screw Compressor

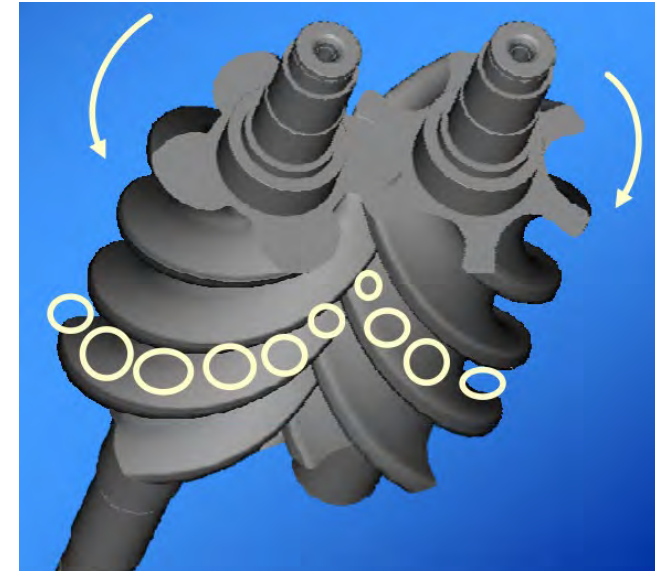
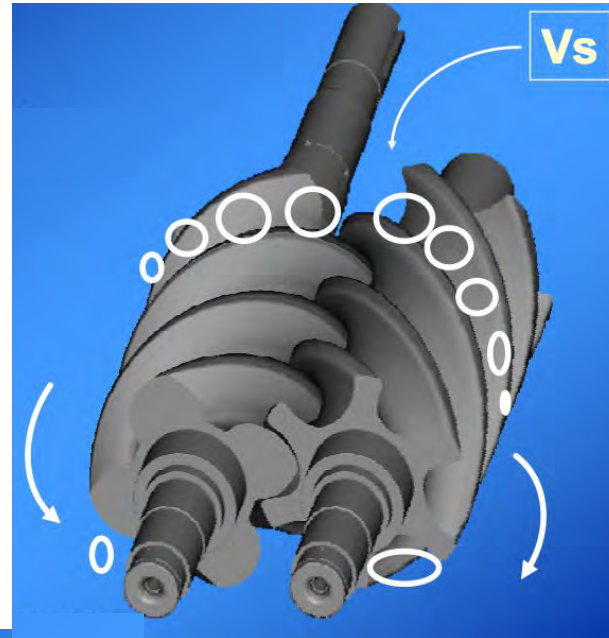
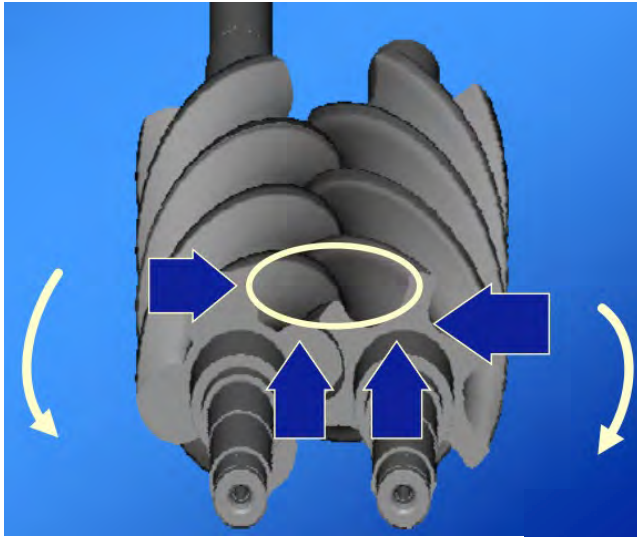


Leakage occurs around the rotors but this is minimized by oil flooding



The oil absorbs much of the heat of compression and improves the compression ratio

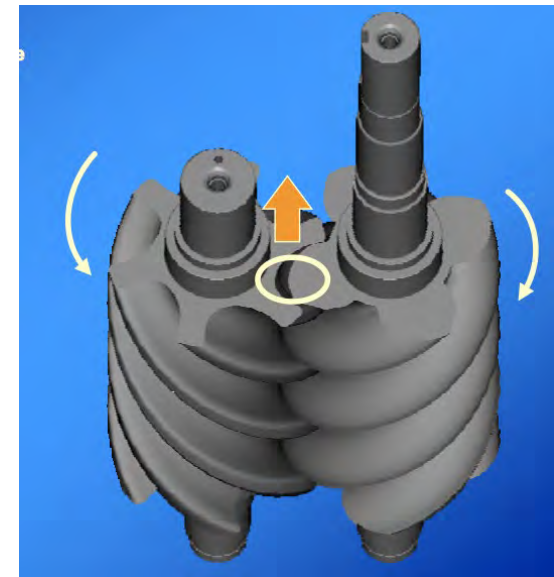
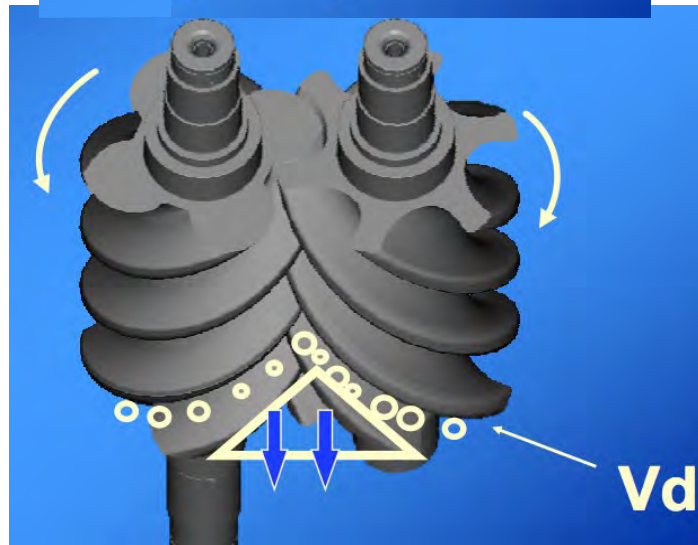
Compression Process



- Suction process starts
- Maximum suction volume trapped
- Compression takes place
- Final discharge volume
- End of discharge

Screw Compressor Basics

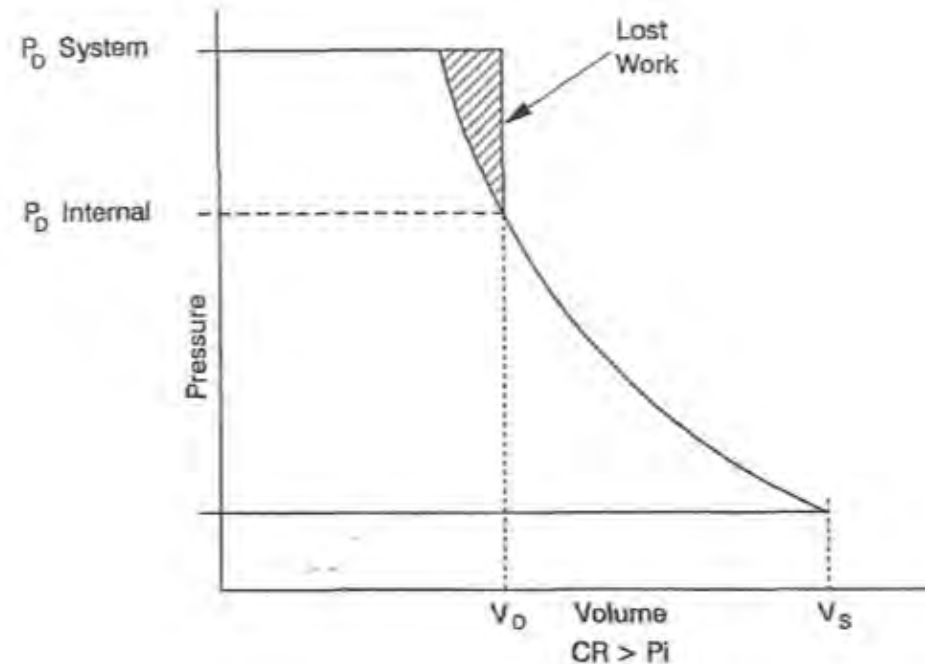
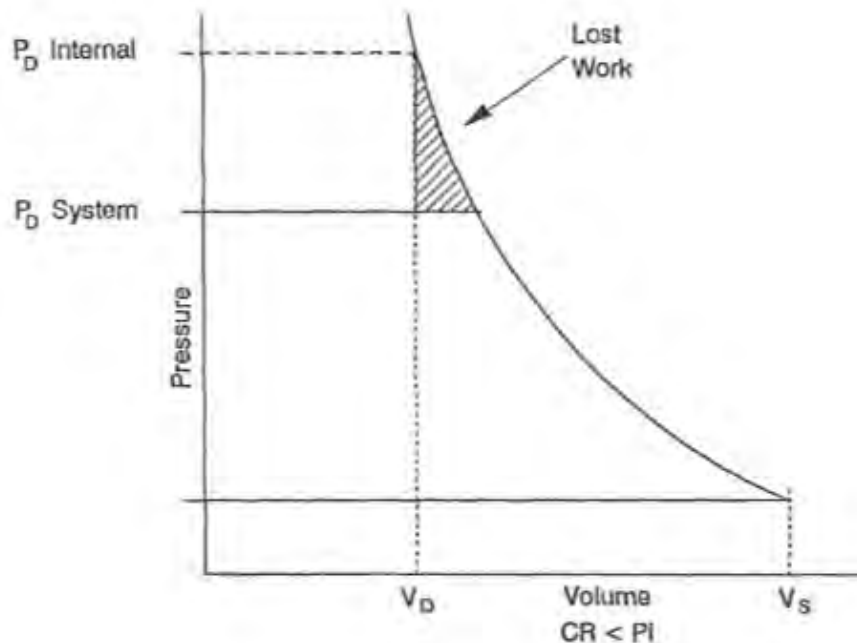
Mathematical Analysis



$$V_i = \frac{V_s}{V_d} \quad \text{Volume index}$$

$$P_i = V_i^k \quad \text{Internal pressure ratio}$$

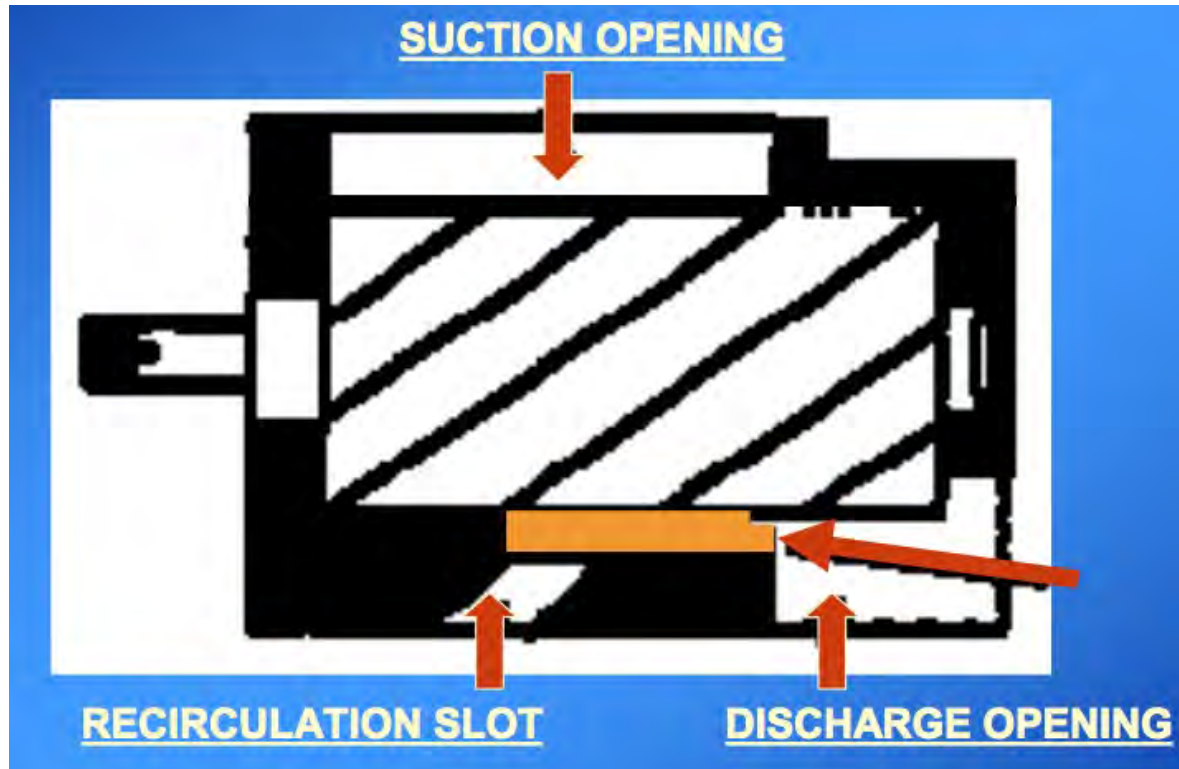
- Screw compressors are a volume reduction device and volume index is the prime performance characteristic
- The suction pressure and volume index determine the discharge pressure of the final trapped gas
- If the system discharge pressure is different, there will be over or under compression, both of which result in lost work



Types of Capacity Control

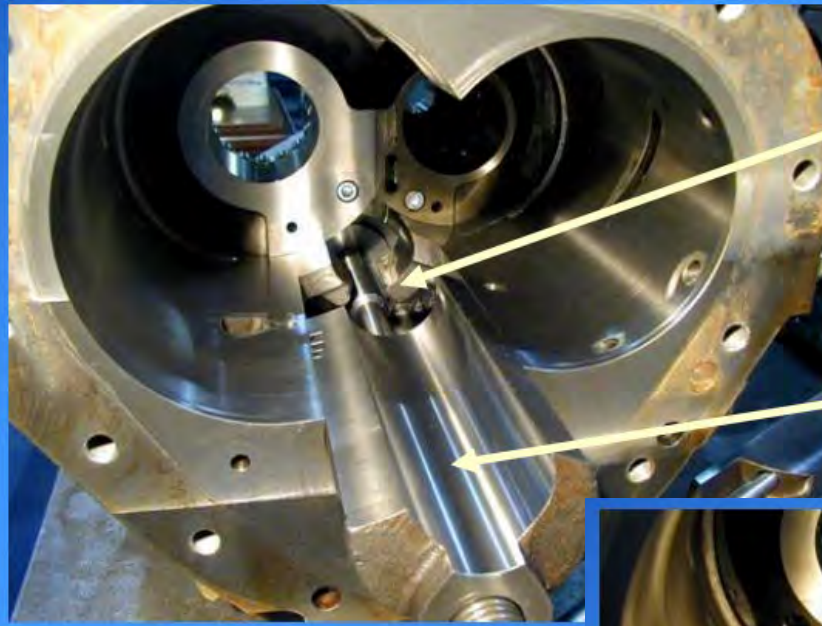
- Fixed V_i – slide valve controlling discharge port
- Variable V_i – slide valve and movable slide stop
- Variable speed

Fixed Vi Capacity Control



Some designs have a single slide control. Moving the slide opens a recirculation slot on the discharge side which increases the discharge volume and reduces the Vi

Variable Vi Capacity Control



Bypass channel to suction side opened at part load operation

Slot for Vi slide

Some designs have a dual slide control:

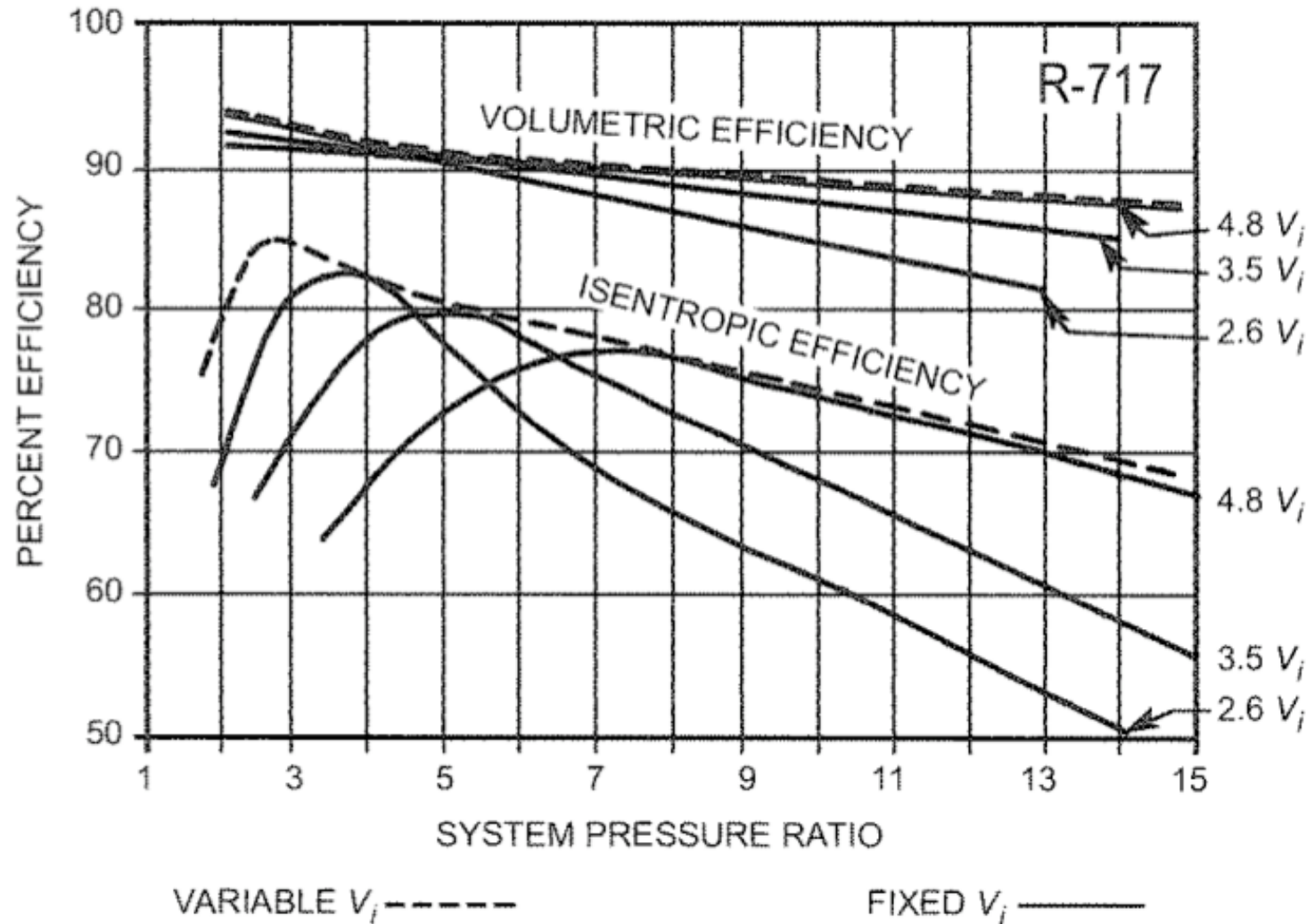
- One slide on the suction side controls capacity by recirculating suction gas
- The other slide adjusts the volume ratio by changing the position of the discharge location



Bypass channels to suction side opened at part load operation

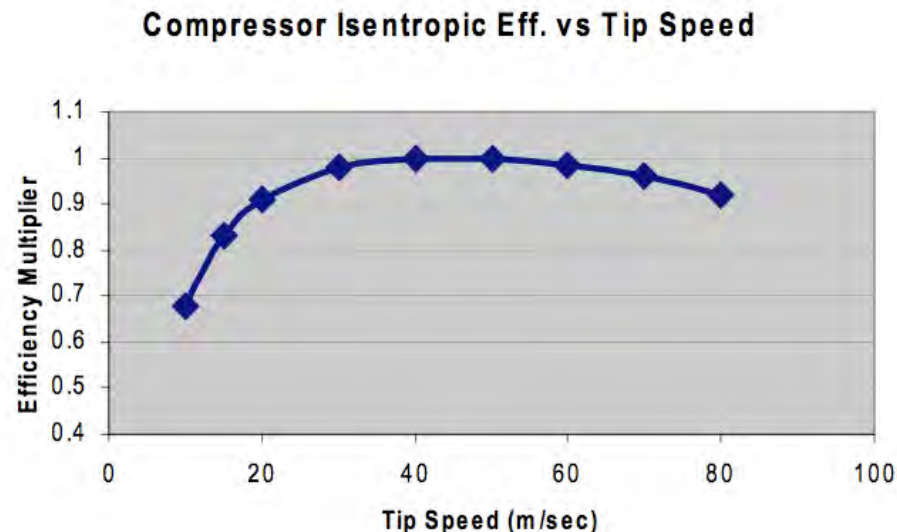
Vi slide

Performance Graph



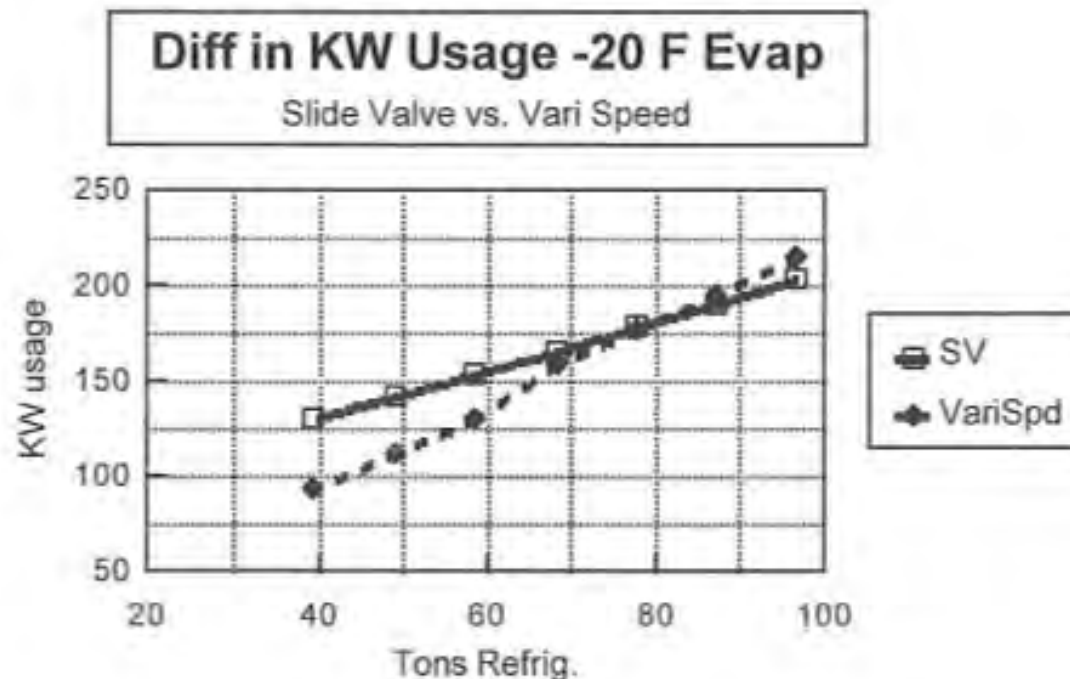
Variable Speed Capacity Control

- Allows for large variability in capacity
- May need one size bigger driver to handle VFD losses
- Cooling of the motor may not be adequate below 50% speed
- Note variation in efficiency with rotor tip speed



Variable Speed Capacity Control

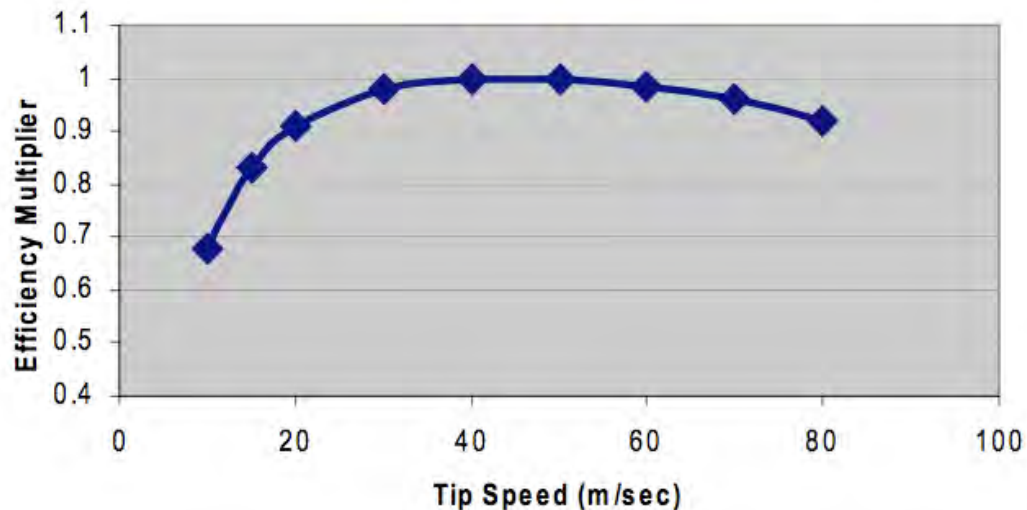
- Using VFD has a power advantage at low speeds and part load but is slightly worse at full load compared to a slide valve
- VFD efficiency varies between 98% at full speed to 94% at 50% speed



Variable Speed Control

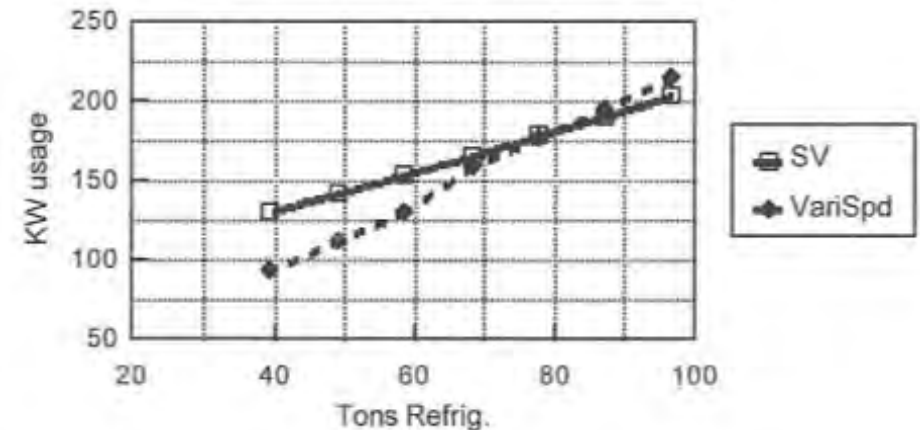
Using VFD has a power advantage at low speeds and part load but is slightly worse at full load compared to a slide valve

Compressor Isentropic Eff. vs Tip Speed



Diff in KW Usage -20 F Evap

Slide Valve vs. Vari Speed

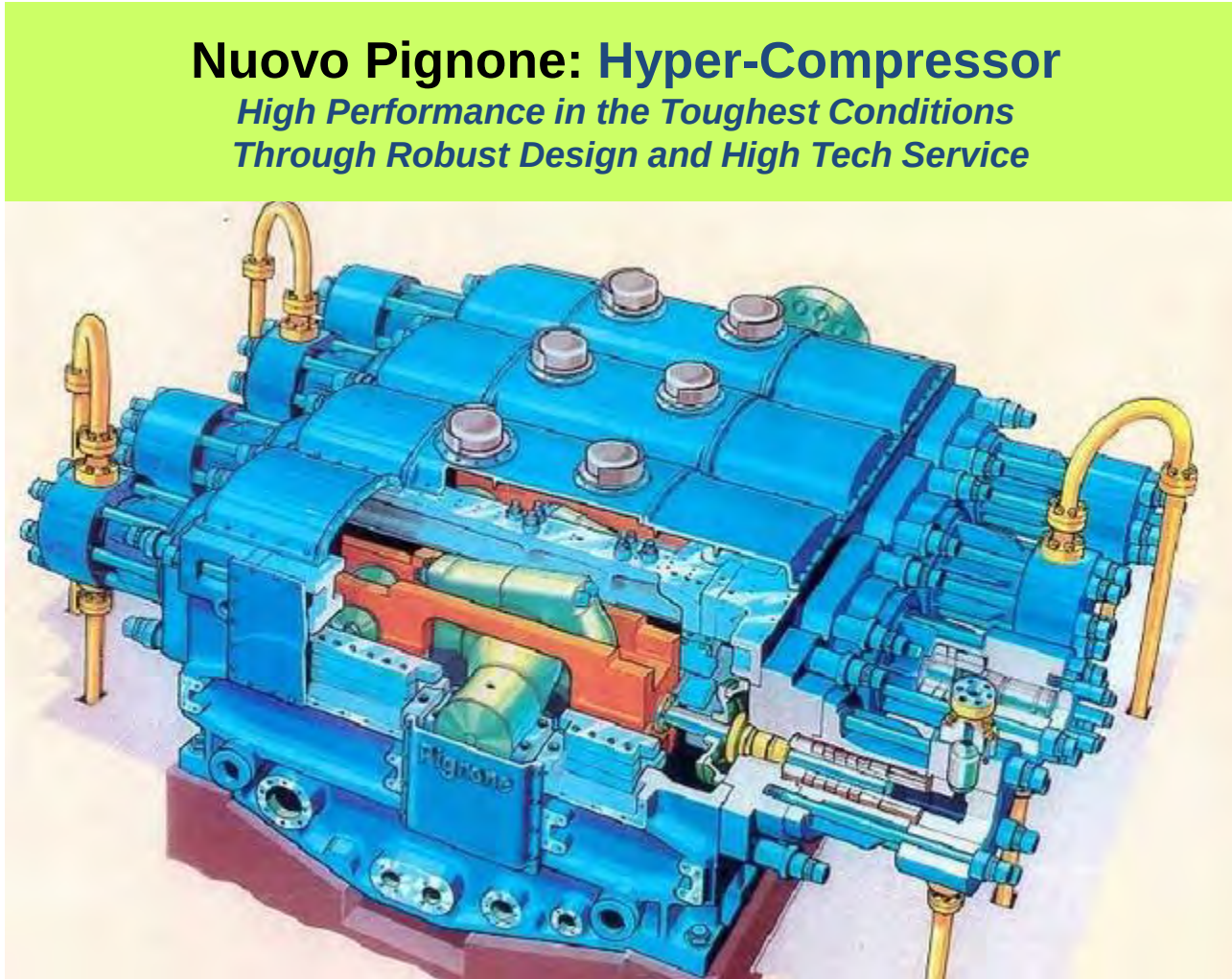


Hyper Compressor

Special Application: Nuovo Pignone Hyper-Compressor

Nuovo Pignone: Hyper-Compressor

*High Performance in the Toughest Conditions
Through Robust Design and High Tech Service*



Special Application: Nuovo Pignone Hyper-Compressor

51 units installed (@ Apr.05)

- Large size Compressors up to 25000 kW
- Highest pressures in HP Industry, up to 3500 bar (50,000 psi).
- Unique Frame Design:
 - No bending for the extremely hard, but brittle, tungsten carbide plungers.
 - Compact, easy to install
- Cylinder Components at leading edge of fatigue resistance technology
- Unique In-house capability for optimization of piping pulsations and vibrations
- Extremely long components life: 2nd stg packing up to 20.000 hrs, valves up to 40,000 hours
- Dedicated Service Team



Special Application: Nuovo Pignone Hyper-Compressor

Design & Service Approaches

Design Approaches

- DFSS, STD, CFE, FMEA
- Advanced Mathematical Models:
 - ✓ Thermodynamic of Gas Mixtures
 - ✓ FEA on critical components
- Acoustic Analysis of pressure pulsations
- Rolled threads for critical bolting
- Hydraulic tensioning

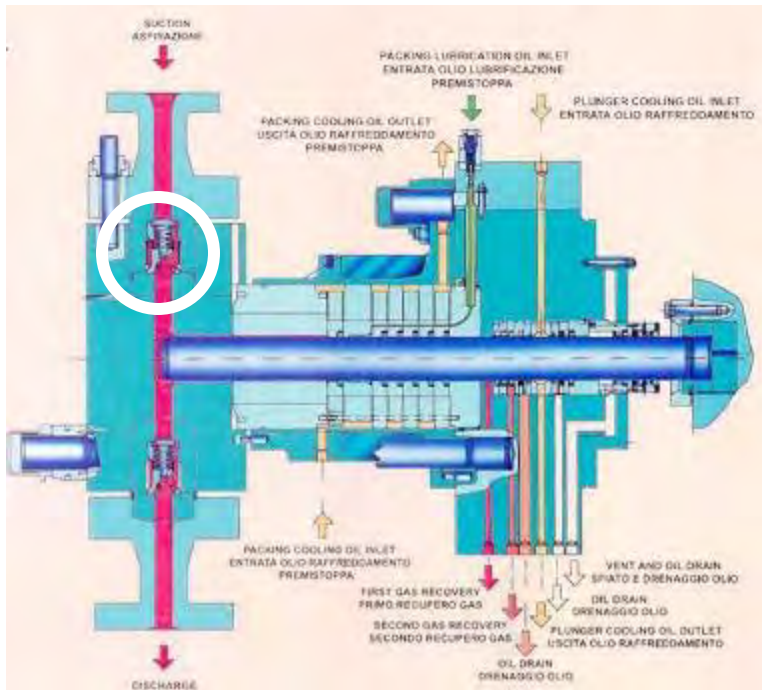
Service Approaches

- Dedicated Service Team
- Predictive Maintenance
- Performance Monitoring
- Diagnostics
- Machinery modernization

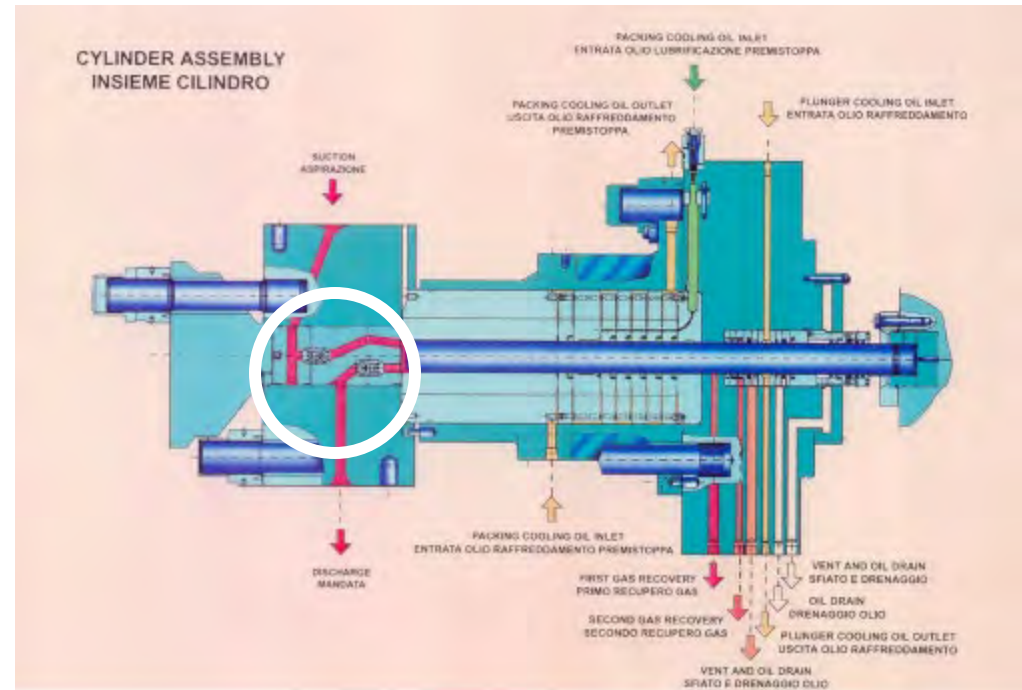
- Reliability
- Efficiency
- Maintainability
- Extended life of Wearable Parts
- Continuous Updating & Upgrading

Special Application: Nuovo Pignone Hyper-Compressor

The Cylinder and its Components



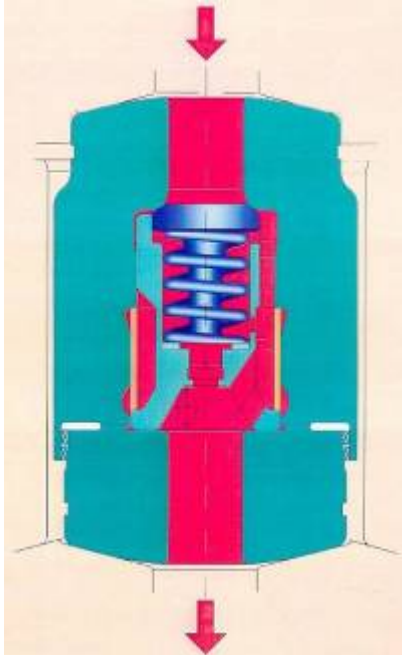
Cylinder with Radial Valves



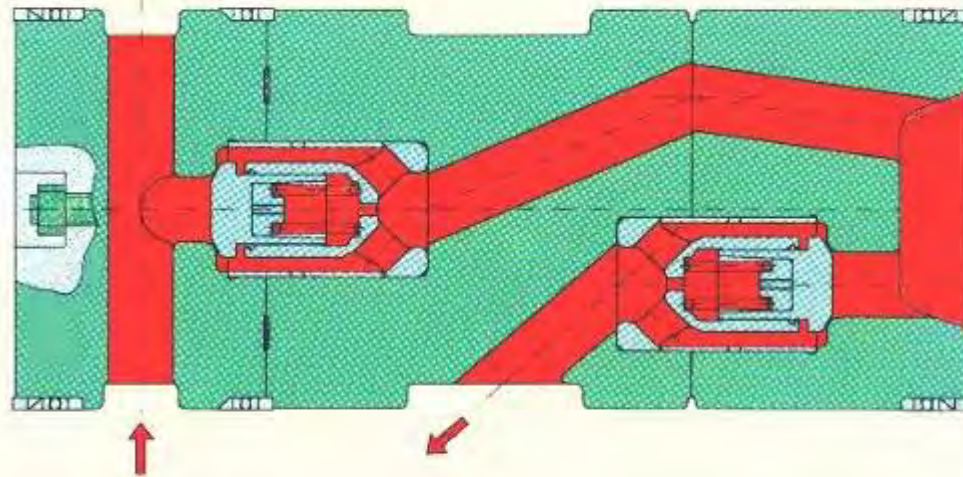
Cylinder with Axial Valve

Special Application: Nuovo Pignone Hyper-Compressor

Cylinder Valves



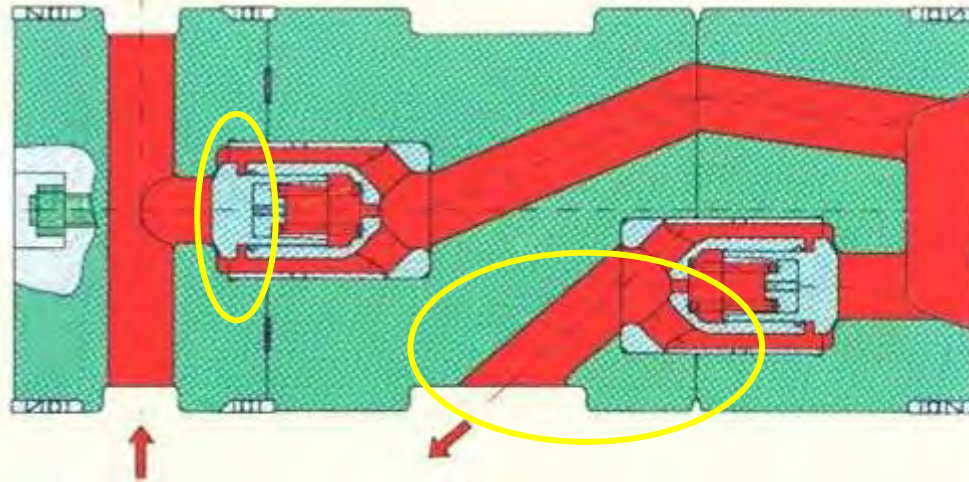
Radial Valve



Suction/Discharge Valve (Axial

Special Application: Nuovo Pignone Hyper-Compressor

Axial Cylinder Valve Improvement



1. Enlarged mating surface between poppet and seat to reduce contact pressure
2. Improved discharge valve flow-dynamic

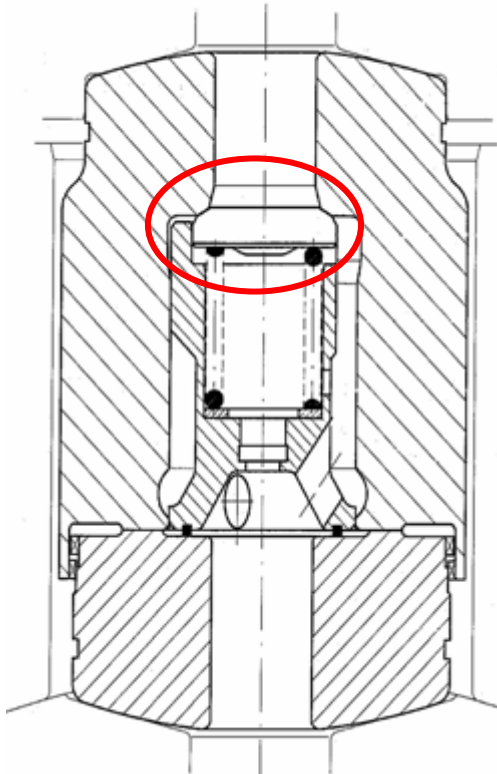


Wear on seat and poppet



Special Application: Nuovo Pignone Hyper-Compressor

Radial Cylinder Valve Improvement



The reduced mass stem-less poppet design enhances valve reliability



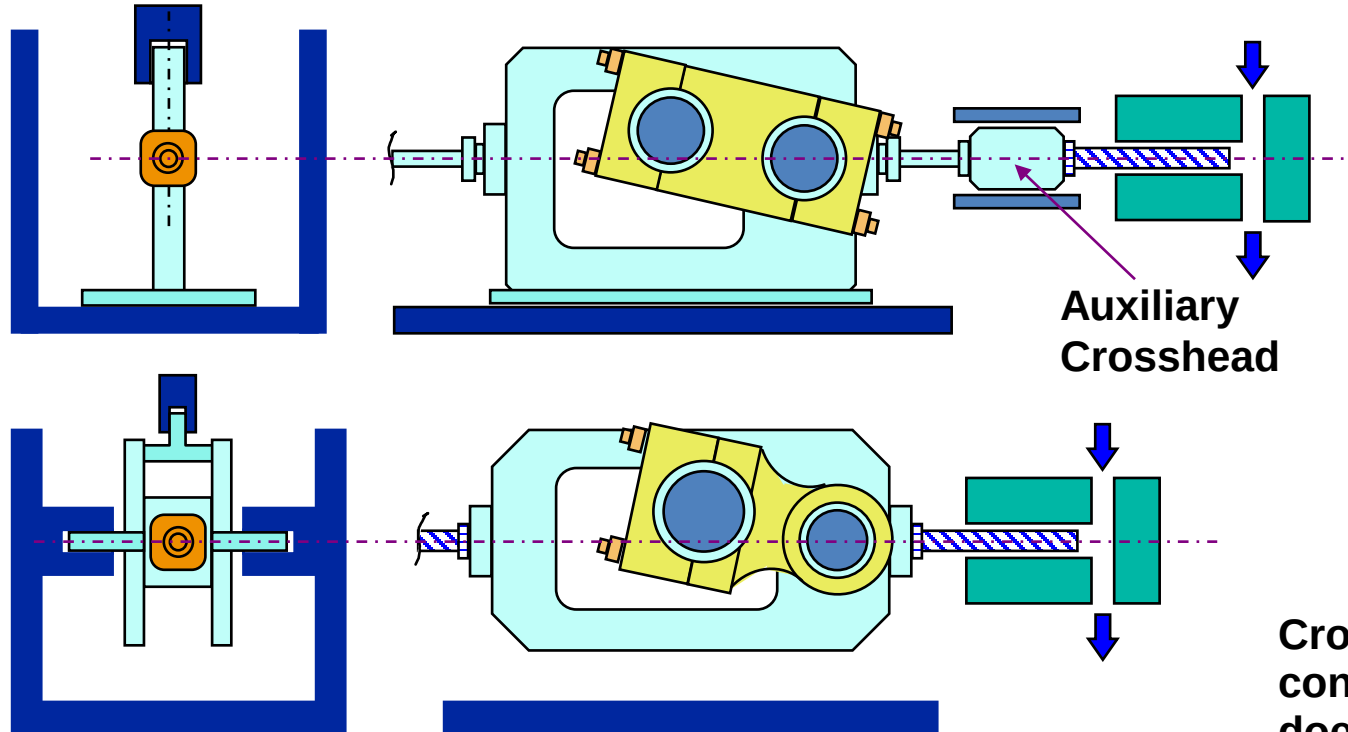
Radial valve internals with standard poppet



Operating Case:
excellent condition
of the stem-less
poppet after a 8100
hours Inspection

Special Application: Nuovo Pignone Hyper-Compressor

Crankgear Arrangements

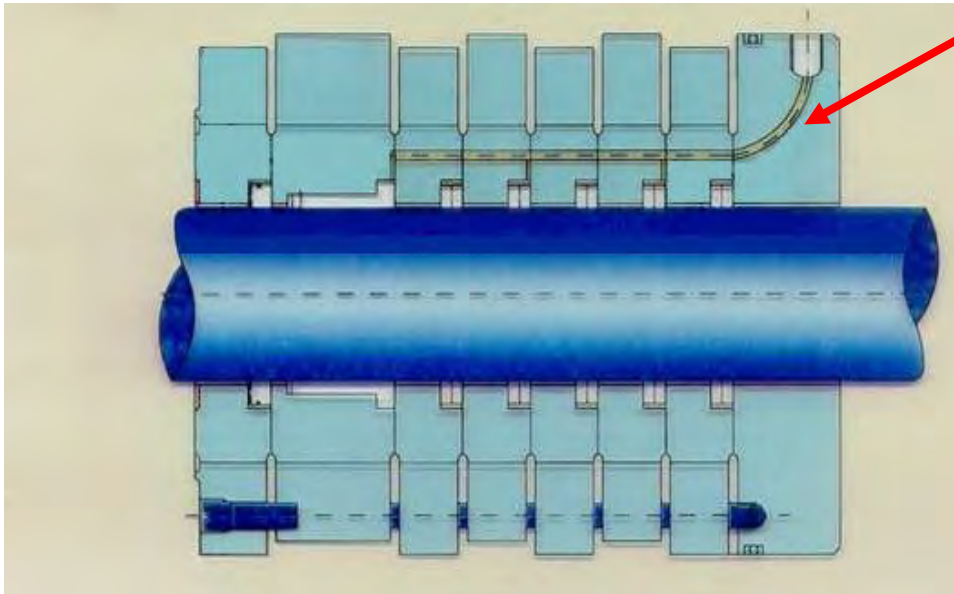
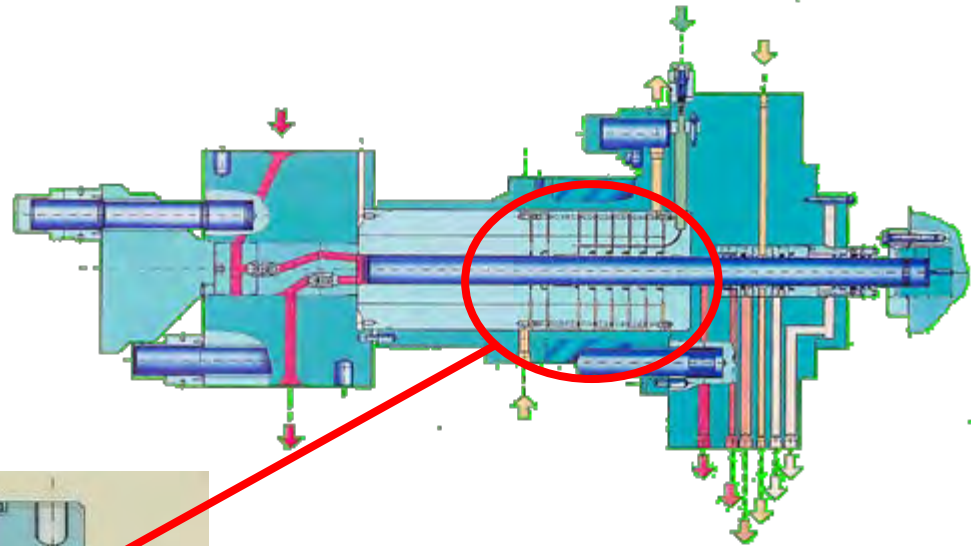


Crosshead sliding plane containing cylinder axis does not need an aux. crosshead

- Compact machine width
- Low investment cost for piping, foundations, building
- Low erection time and cost
- Short maintenance downtime
- Cylinders alignment set once forever
- Reduced number of parts, high overall reliability

Special Application: Nuovo Pignone Hyper-Compressor

Packing Stack

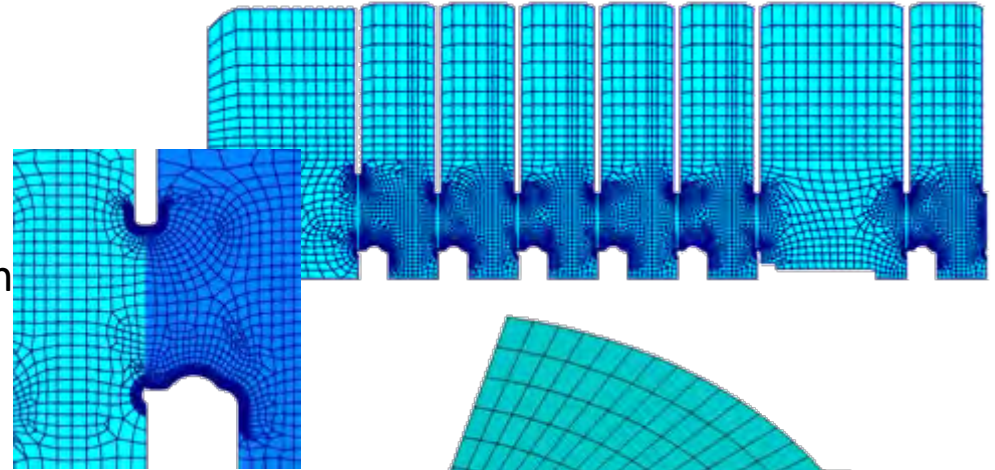


Special Application: Nuovo Pignone Hyper-Compressor

Packing Stack: Design Approach

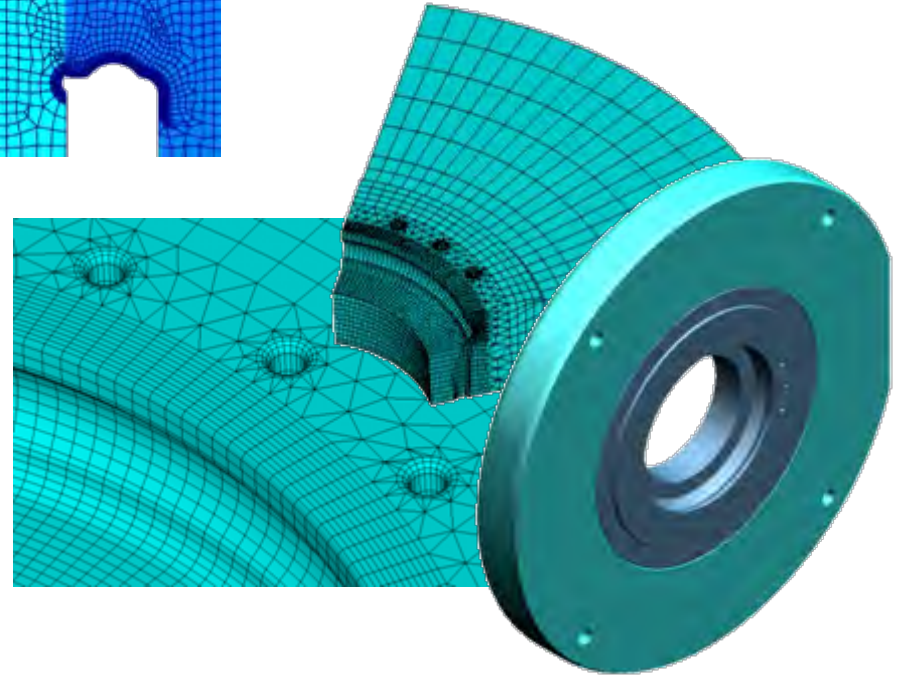
Packing Cup Assembly

- 2D FEA :
- Optimize packing cup profile
- Minimize contact pressure peaks between cups



Single Packing Cup

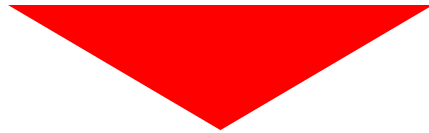
- 3D FEA :
- Optimize stress gradient close to stress concentration points (i.e. oil channel and fillets)
- Accurate assessment of autofrettage effect



Special Application: Nuovo Pignone Hyper-Compressor

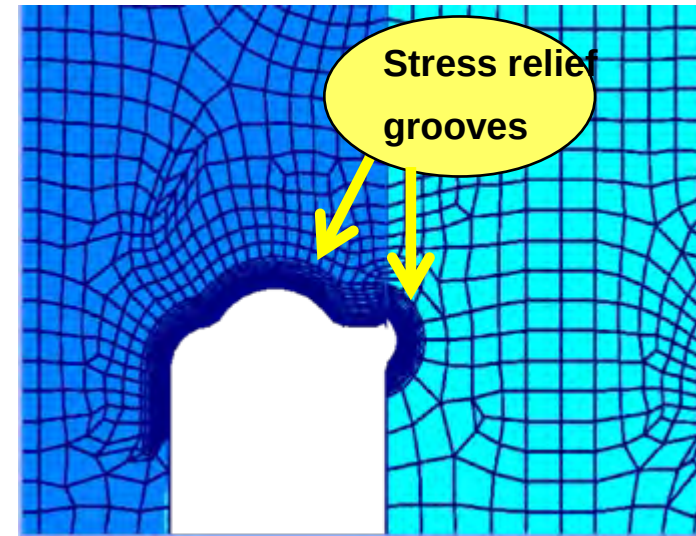
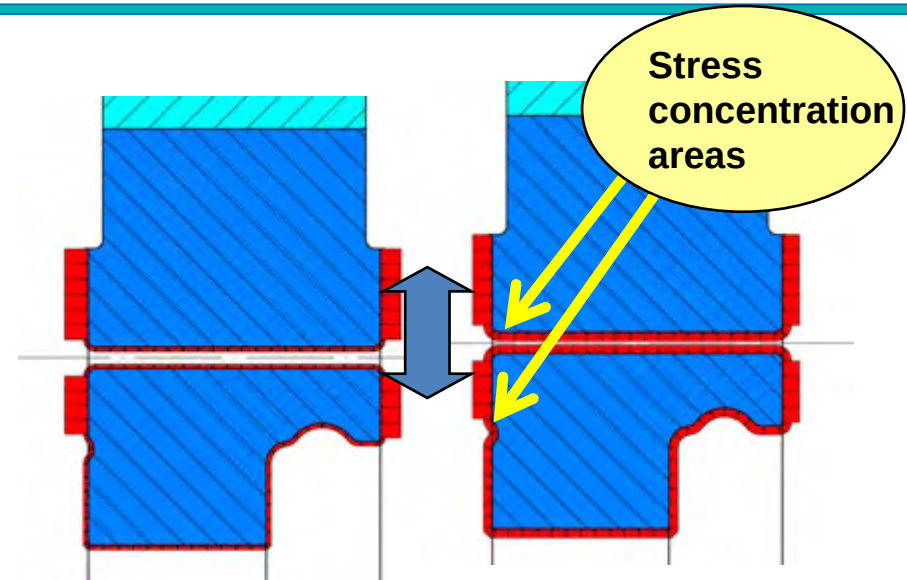
Packing Cup

Discontinuity of internal pressure of adjacent cups and heavy low-polymer presence may cause an abnormal cyclic radial movement between adjacent cups



HP cup design optimized to withstand high fatigue stress:

- The highest available material quality (by VAR, ESR)
- Special groove profile
- Autofrettage of oil holes
- Shot peening of grooves and contact
- Silver coating on the contact surfaces

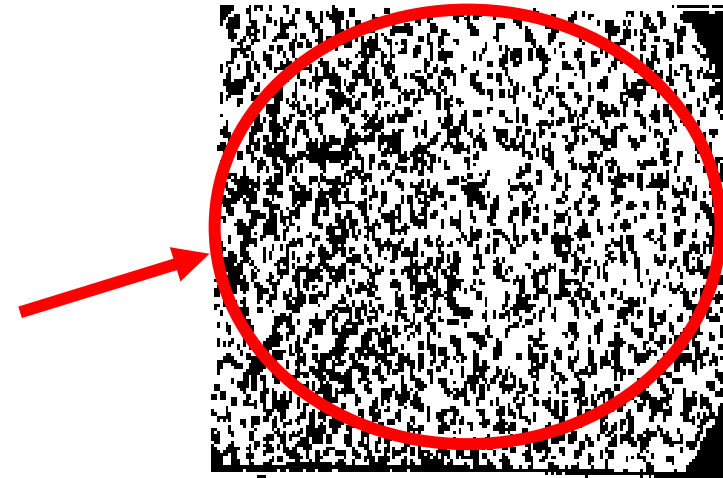


Special Application: Nuovo Pignone Hyper-Compressor

Improved Packing Ring design

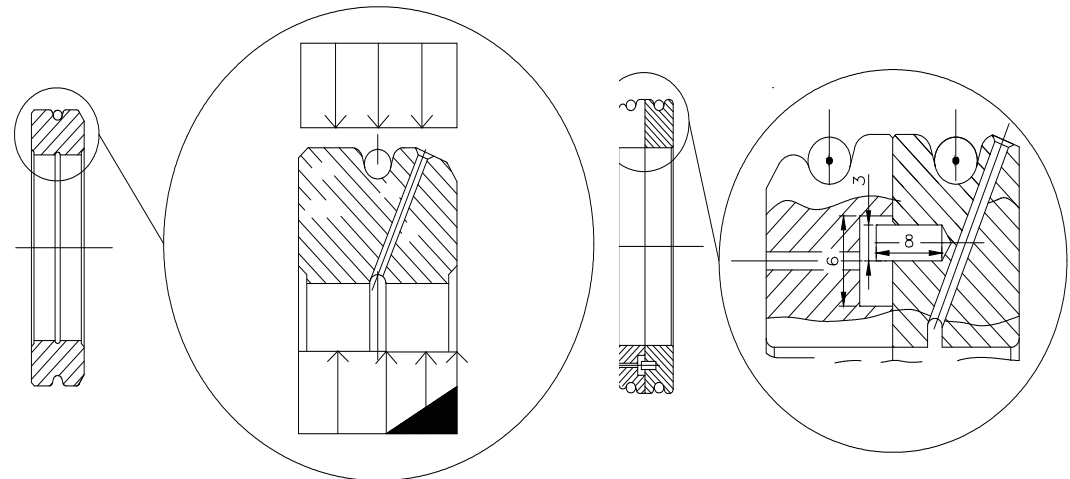
Enhanced Ring Material:

- Special alloy with limited impurity
- Lower friction coefficient
- Excellent self lubricating property by fine, evenly distributed lead particles
- Medium-Low hardness



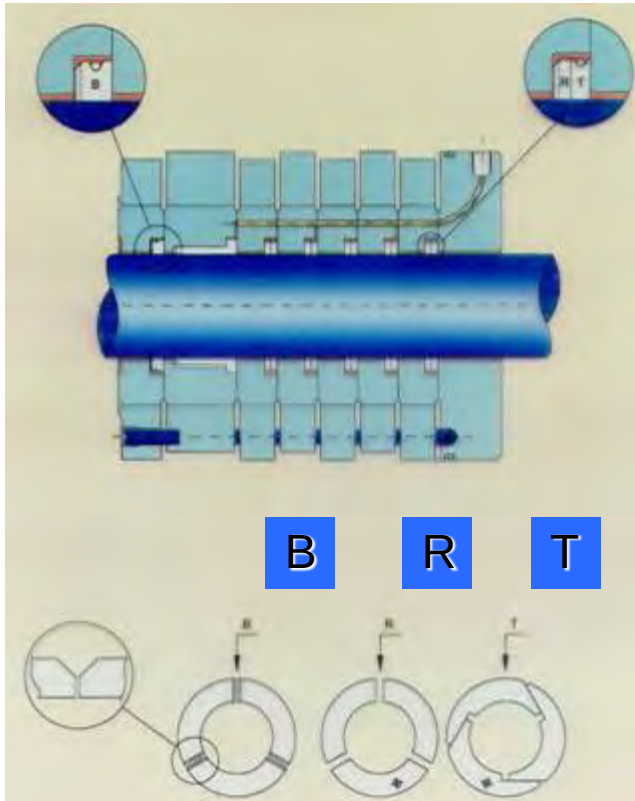
New Ring Design:

- Balancing hole to reduce the contact pressure
- Special pin housing to allow radial movement & reduce shear stress



Special Application: Nuovo Pignone Hyper-Compressor

Operation knowledge by experimental measurement

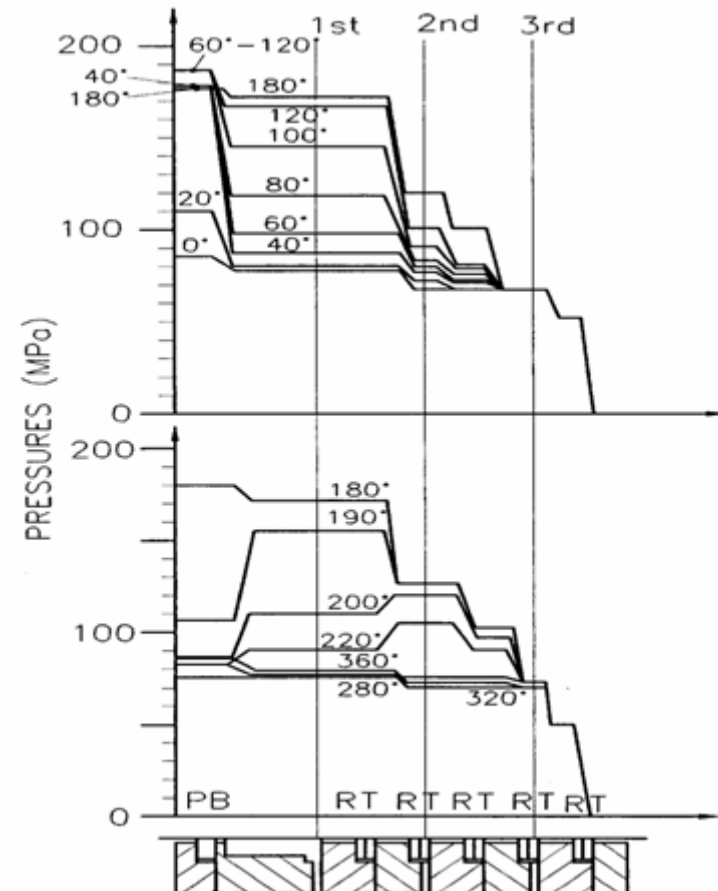


B = Pressure breaker ring

R = Radial ring

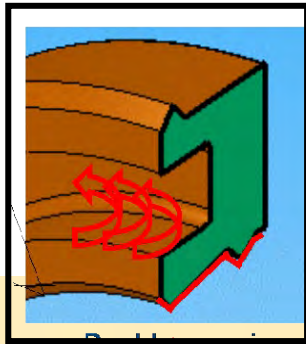
T = Tangential ring

Pressure distribution inside the packing



Special Application: Nuovo Pignone Hyper-Compressor

Partition Cover and Plunger Cooling Chamber



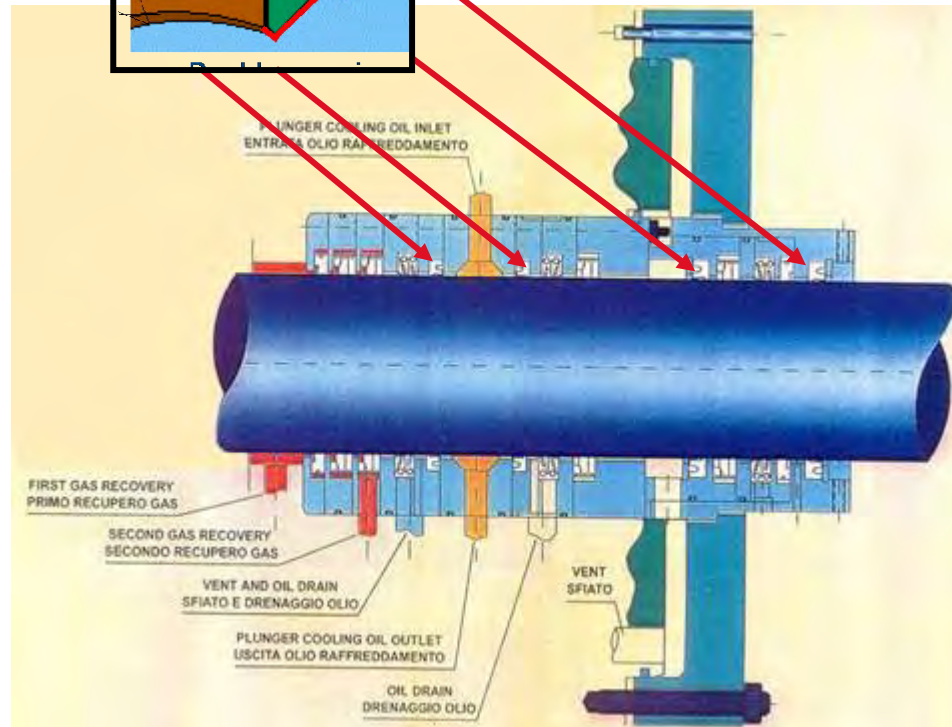
New improved
seal ring



- Double scraping profile
- Large back-flow section
- Single piece cut without any garter spring



- Enhanced scraping action
- Cooling oil consumption can be reduced up to 50% max
- Easier maintenance



Special Application: Nuovo Pignone Hyper-Compressor

New Bearings & related troubleshooting tools

Bi-metallic AlSn bearings

for greater fatigue strength and the finest bonding quality

Dynamic Simulation tool

(BERC© Proprietary Software)

I
N
P
U
T

- 3D bearing geometry (grooves, holes, etc.)
- Oil condition (viscosity, pressure, temperature)
- Load vs. Crank Angle

B
E
R
C

O
U
T
P
U
T

- Pressure distribution map
- Orbit diagram
- Oil film thickness

- Robust Design
- Sensitivity Analysis
- Work optimization and up-rating
- Troubleshooting

MAIN BEARING

BIG END BEARING

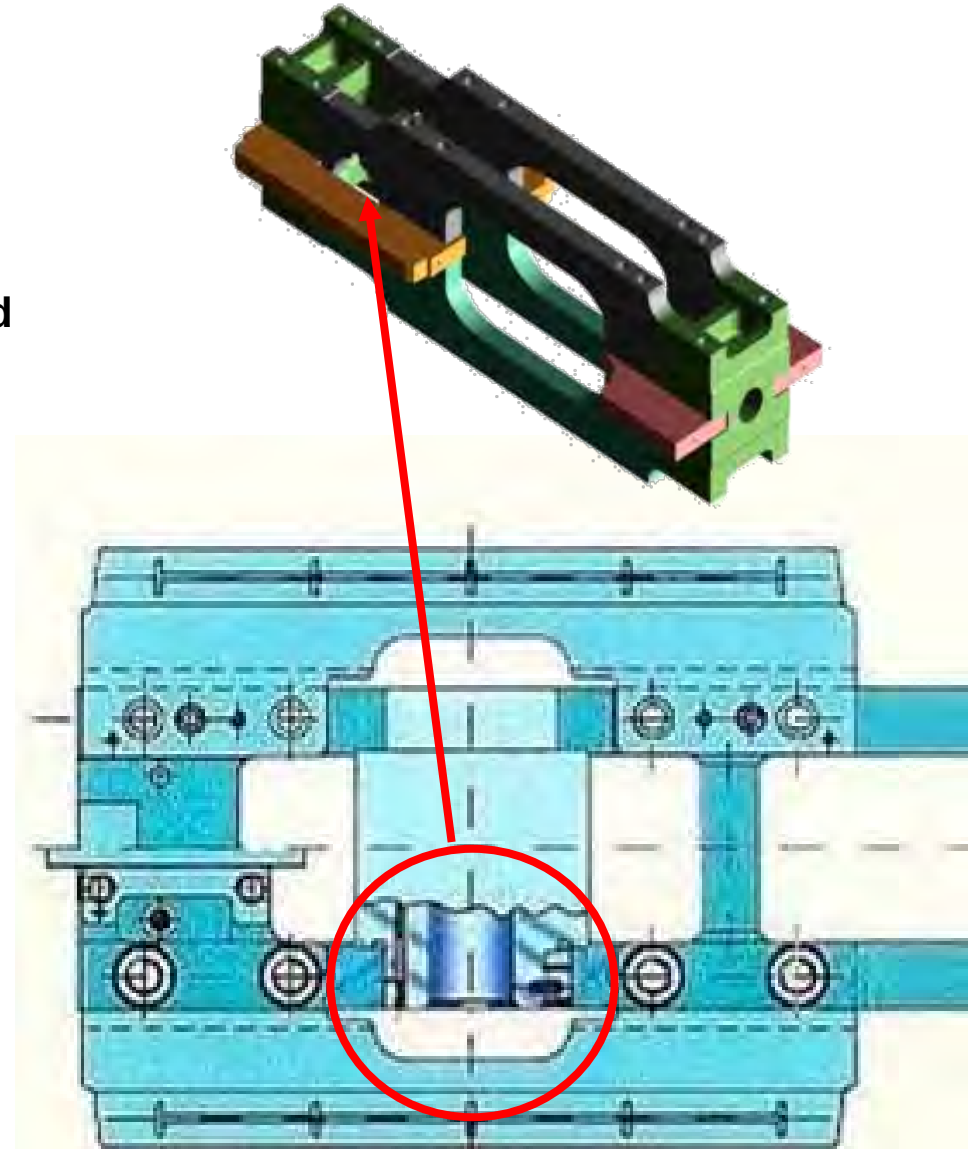
SMALL END BUSHING

Special Application: Nuovo Pignone Hyper-Compressor

Crosshead Pin Design

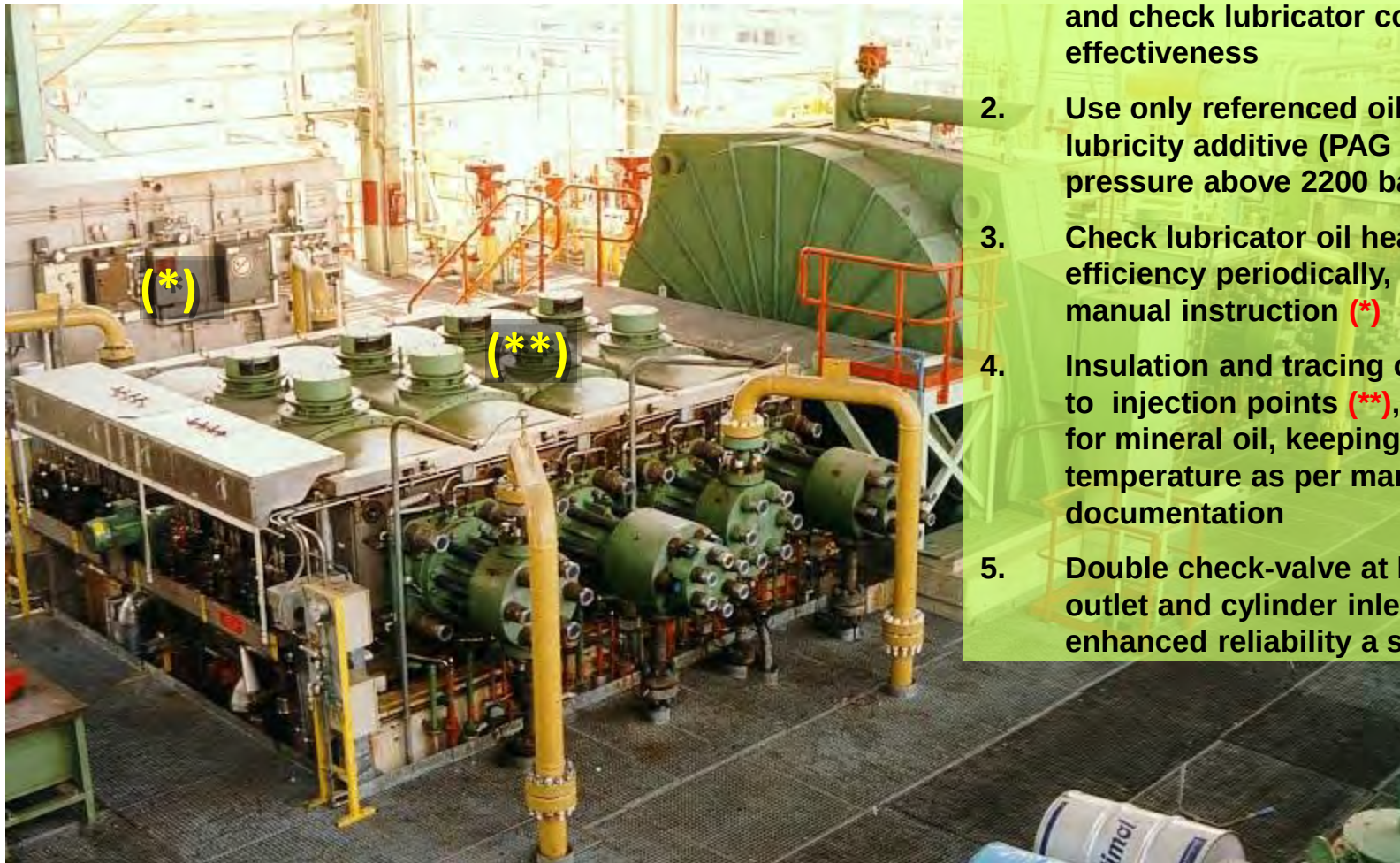
Cylindrical coupling between pin and crosshead have given a more rigid and stable connection than traditional Tapered type and permits:

- Positive tightening
- Increased reliability
- Easier assembly & maintenance
- Possibility of performance up-rating



Special Application: Nuovo Pignone Hyper-Compressor

Cylinder lubrication: component life fundamental



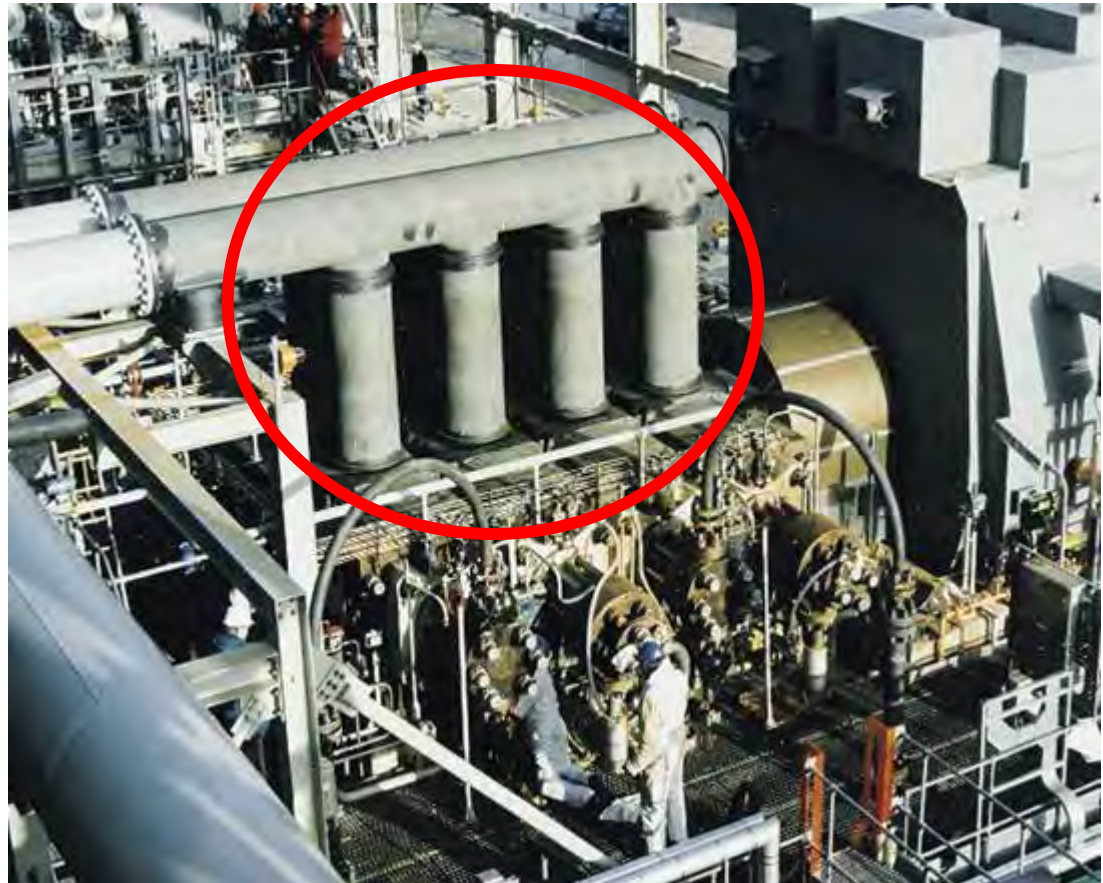
The Top 5 Rules

1. Keep recommended oil flow rates and check lubricator components effectiveness
2. Use only referenced oil with lubricity additive (PAG preferred for pressure above 2200 bar)
3. Check lubricator oil heating system efficiency periodically, as per manual instruction (*)
4. Insulation and tracing of oil tubing to injection points (**), especially for mineral oil, keeping a temperature as per manual documentation
5. Double check-valve at lubricator outlet and cylinder inlet (i.e. enhanced reliability a safety)

Special Application: Nuovo Pignone Hyper-Compressor

Frame Safety Valves and Vent Manifold

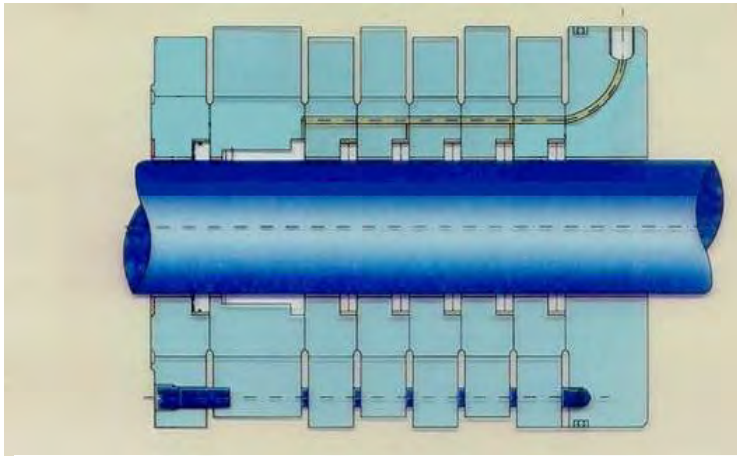
Gas entering accidentally into the frame can be vented by safety valves to a manifold at safe location. This precaution limits frame overpressure



... enhancing SAFETY

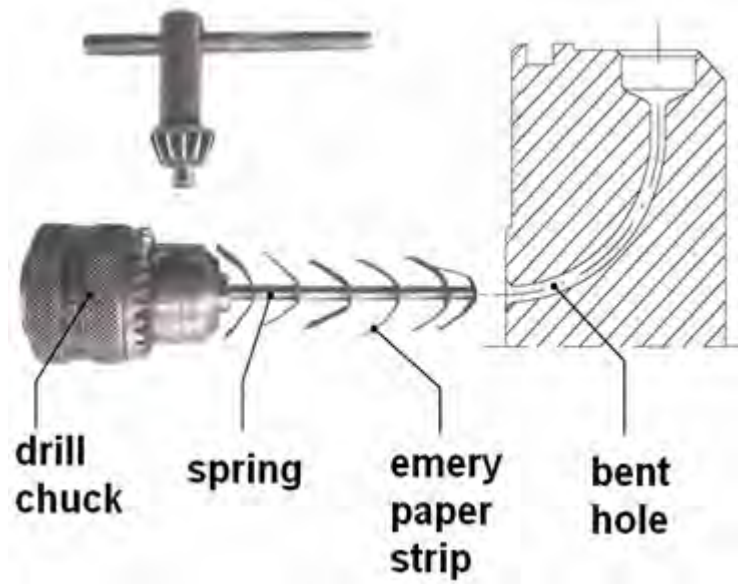
Special Application: Nuovo Pignone Hyper-Compressor

Maintenance and Repair : Packing Stack Reconditioning



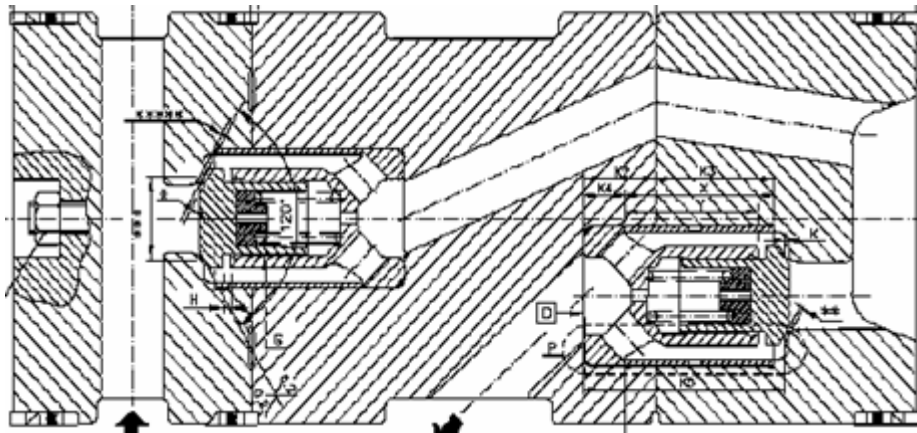
The most critical maintenance activity

- Cleaning and visual inspection
- NDT
 - Dye penetrant
 - Magnetic particle /Fluorescent Inspection
- Reconditioning of:
 - Planarity and sealing surfaces
 - Seal housing geometry
 - Oil duct honing
 - Radiusing / Polishing
 - Shot peening
 - Silver coating



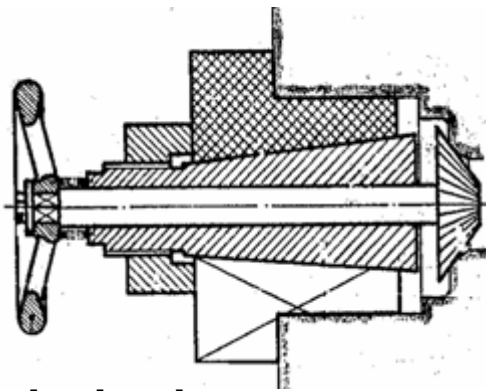
Special Application: Nuovo Pignone Hyper-Compressor

Maintenance and Repair: Cylinder valves Reconditioning



Special tools and the skilled operator's care are needed to restore original features

- Spacer adjustment to take care of material removed for surface re-lapping
- External seal housing geometry renewed
- Polishing of critical areas



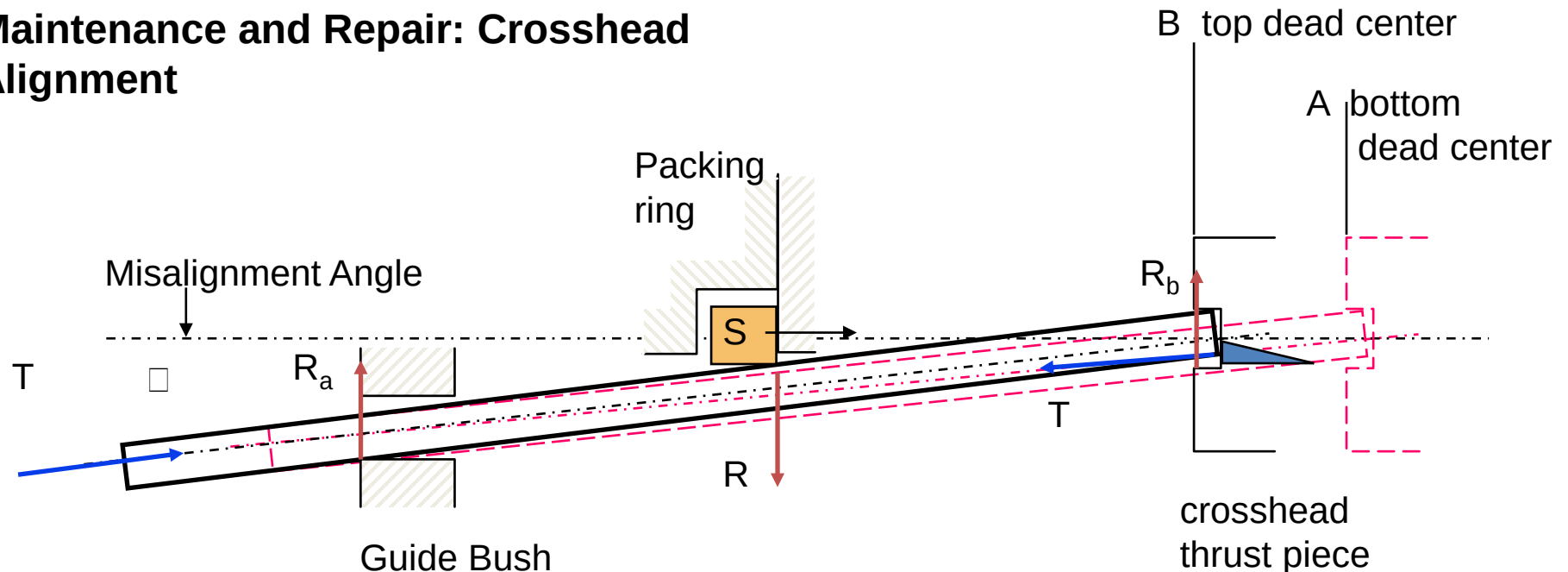
Typical valve seat reconditioning device

[illegible]

- 39

Special Application: Nuovo Pignone Hyper-Compressor

Maintenance and Repair: Crosshead Alignment



Tungsten Carbide Plunger Alignment is critical !

Tungsten Carbide Plungers are extremely hard, but extremely brittle as well. Frame Architecture driven by Minimization of Misalignment and Consequent Bending

Here some important information :

- No re-alignment required for cylinder replacement
- After any maintenance to crosshead sliding shoes, an alignment check must be done
- A special service with laser instrument is now available to make easier and faster this activity

Special Application: Nuovo Pignone Hyper-Compressor



8PK Crankgear Assembly

Special Application: Nuovo Pignone Hyper-Compressor



Crosshead

Special Application: Nuovo Pignone Hyper-Compressor



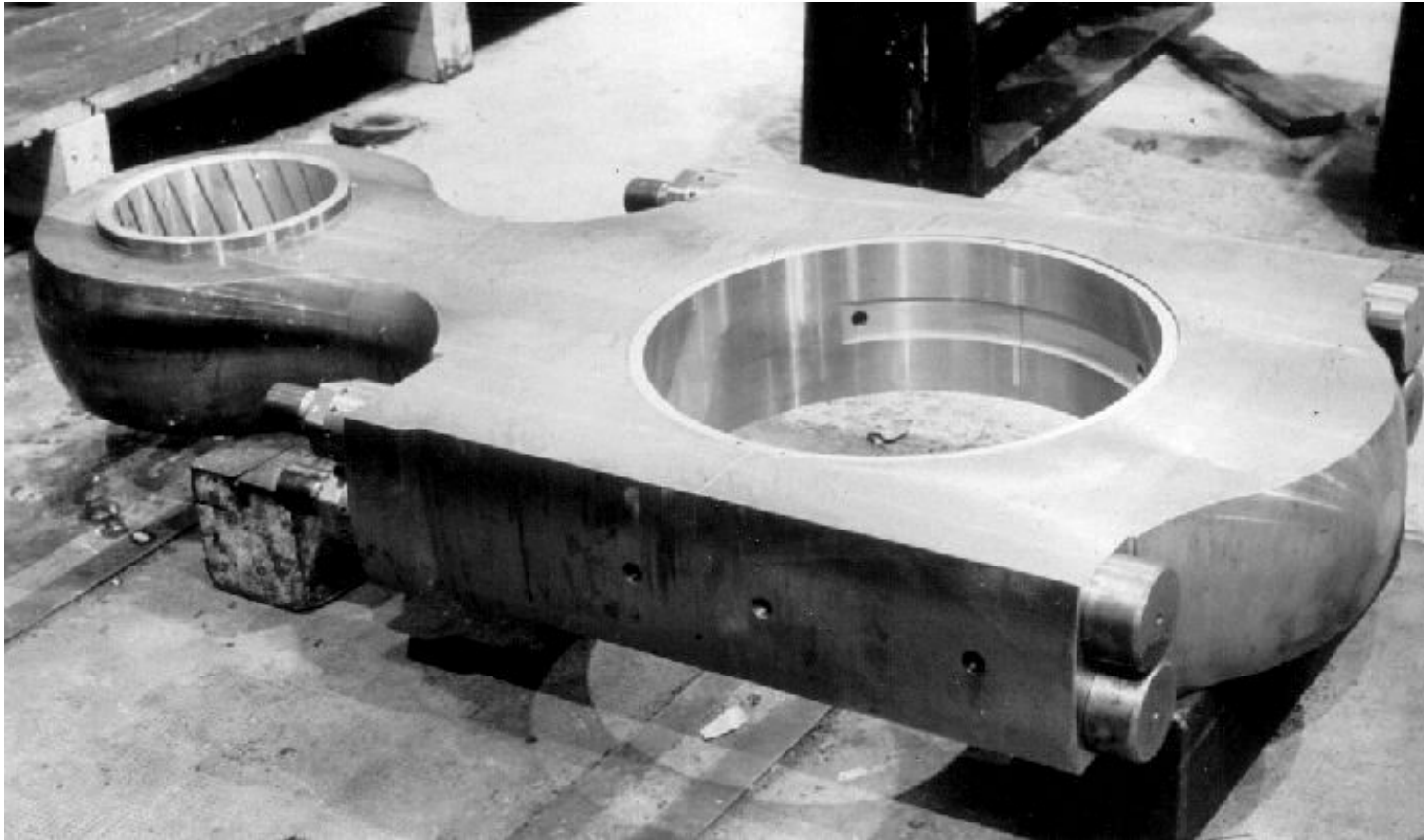
Crankshaft

Special Application: Nuovo Pignone Hyper-Compressor



Crankshaft Lube Oil Hole

Special Application: Nuovo Pignone Hyper-Compressor



Connecting Rod

Special Application: Nuovo Pignone Hyper-Compressor



Cylinder Hydraulic Tightening

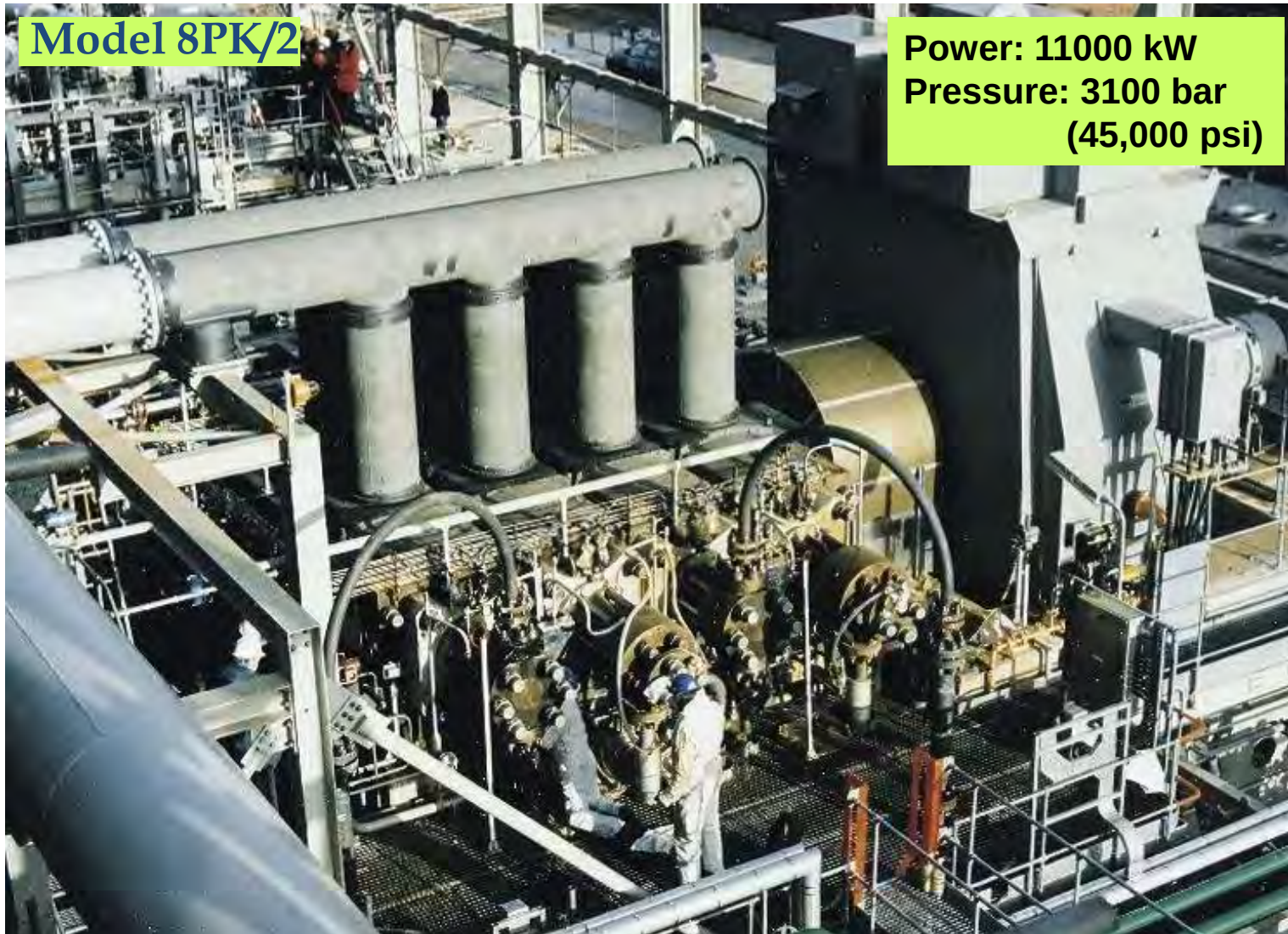
Special Application: Nuovo Pignone Hyper-Compressor

Model 6PK/2

**Power : 4100 kW
Pressure: 1860 bar
(27,000 psi)**



Special Application: Nuovo Pignone Hyper-Compressor



Special Application: Nuovo Pignone Hyper-Compressor



Power: 10,000 kW
Pressure: 2,200 bar
(32,000 psi)

Special Application: Nuovo Pignone Hyper-Compressor



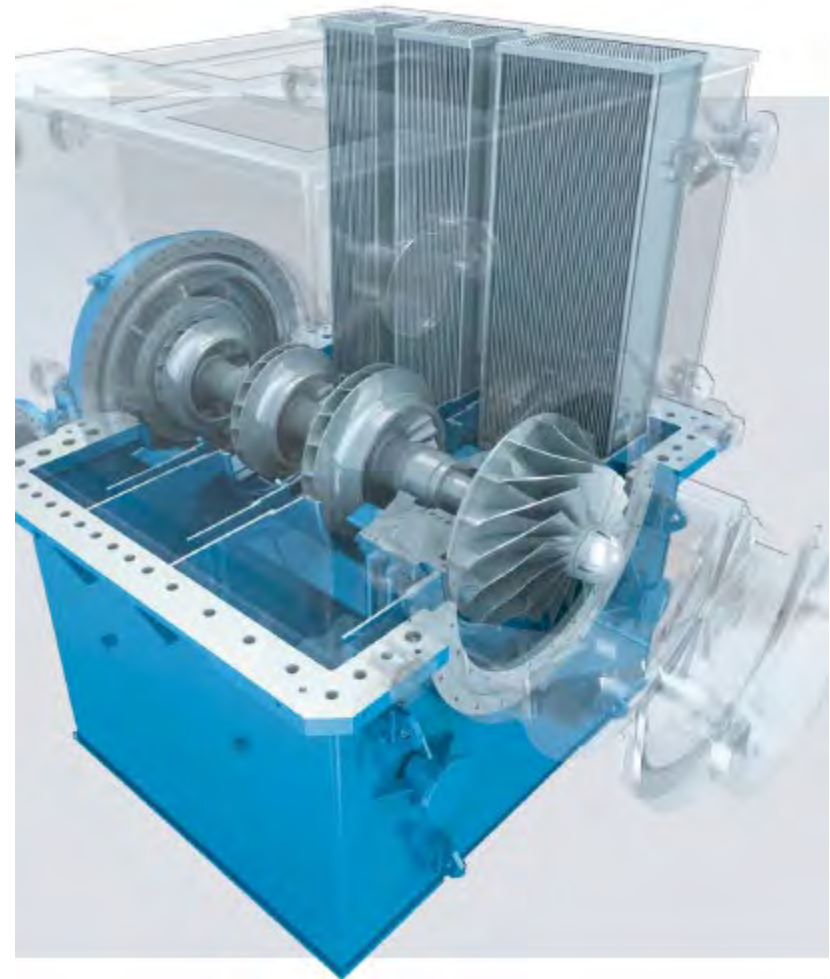
Special Application: Nuovo Pignone Hyper-Compressor

Model 12PK/2

Power: 14500 kW
Pressure: 3100 bar
(45,000 psi)



Iso Thermal Compressor



End of presentation

THANK YOU

SKG 12 Mechanical Engineering

DOMI of Compressor and Blower (7of7)

- Compressor component, maintenance and troubleshoot

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Compressor component, maintenance and troubleshoot

- Identify the problem and correct it.

Compressor Operation, Maintenance and Troubleshooting

Case study: Typical failure for compressor

Compressor Troubleshooting

The following chart list is the common failures, their possible causes and recommended remedies

Trouble	Possible Cause	Remedy
1. Vibration or abnormal compressor noise	Misalignment	Remove coupling, have driver run by itself. If it's rotate without vibration, the trouble could be caused by misalignment.
	Coupling failure	Check condition of coupling
	Compressor rotor unbalance	Check to see if dirt is the cause and if necessary, rebalance
	Stresses transmitted by gas pipes to casing resulting in misalignments	Gas pipes should be properly anchored down to prevent excessive stresses on compressor casing. Sufficient elasticity should be given to gas pipes to allow for heat expansion
	Unbalance coupling	Disassemble coupling and check for unbalance
	Surge	See that compressor operating condition are kept far from surge point
	Machines working in the vicinity of compressor	Insulate their foundation and increase elasticity of any connecting pipes

Compressor Troubleshooting (continue)

Trouble	Possible Cause	Remedy
2. Thrust bearing failure	Excessive axial thrust	Make sure coupling is clean and assembled so that excessive thrust is not transmitted from the coupled driver to the compressor
	Improper lubrication	Make sure recommended oil is used. Periodically check that there is no water or dirt in the oil
3. Journal bearing failure	Improper lubrication	Make sure recommended oil is used. Periodically check that there is no water or dirt in the oil
	Misalignment	Check alignment and if necessary, correct
	Bearing clearance incorrect	Check clearance and if necessary, correct
	Compressor or coupling unbalance	Refer to vibration section

Compressor Troubleshooting (continue)

Trouble	Possible Cause	Remedy
4. Problems in the seal system	Misalignment and vibration	Refer to vibration section
	High seal oil temperature	a. Set point on level controller is set too high. Lower set point. b. Level control valve is stuck closed or plugged.
	Dirty oil	Check condition of the filter and replace dirty cartridges. Check pipe cleanliness
	Ring clearance incorrect	Check clearance and if necessary, correct
	Insufficient oil pressure	Check that the reference gas pressure does not drop below minimum
	Low level in head tank	a. Set point on level controller on head is set too low. Raise set point. b. Level control valve stuck open. Stroke valve. Seal oil pump failure. Start stand by pump. Excessive seal leakage. Flow out dump line from one or both drain pots is high – traps are cycling time frequent – possible seal ring wear -
	High level in head tank	a. Set point on level controller is set too high. Lower set point. b. Level control valve is stuck closed or plugged. Stroke valve. Repair Valve if it will not stroke.

Compressor Troubleshooting (continue)

Trouble	Possible Cause	Remedy
5. High compressor discharge temperature	Suction temperature has risen.	Check temperature of inlet gas, and correct if it is up
	Compression ratio has risen.	Calculate compression ratio. Determine cause of increase – suction pressure is down and/or discharge pressure is up – and correct.
	Fouled impellers or other internals.	Inspect and repair.
6. Low compressor capacity	Low speed	Check speed and increase to that required.
	Too much recycle gas	Check surge or recycle controller and reset set point above surge point.
	High compression ratio. Suction pressure is low and/ or discharge pressure is up.	a. Calculate compression ration to ensure it is up. b. Determine cause of low suction pressure or high discharge pressure and correct it.
	Fouled impellers or diffusers, this condition usually builds slowly over a period of time.	Conduct an internal inspection, verify and repair.

Compressor Troubleshooting (continue)

Trouble	Possible Cause	Remedy
7. Problems in the lube oil system	Low oil pressure	a. Low level in oil reservoir. Check level and add oil. b. Oil pressure regulator set too low. Raise pressure setting. c. Worn bearings. Unit vibrates. Drain oil from bottom of tank and check for bearing cuttings. Conduct oil analysis. Inspect bearings and repair. d. Oil pump failure. Start stand-by pump. Check automatic start up. e. High oil temperature. See below. f. Filter is plugged. High pressure drop indicated. Switch and change filter elements in first filter.
	High oil temperature	a. Heater in oil tank is on. Turn it off. b. Cooling fluid flow to oil cooler is low. Fan problem. Temperature Control Valve [TCV] problem. Check inlet and outlet temperature of oil and cooling fluids, and check pressure drop of fluids.

Case study: Typical failure for compressor

Dynamic / Kinetic Compressor

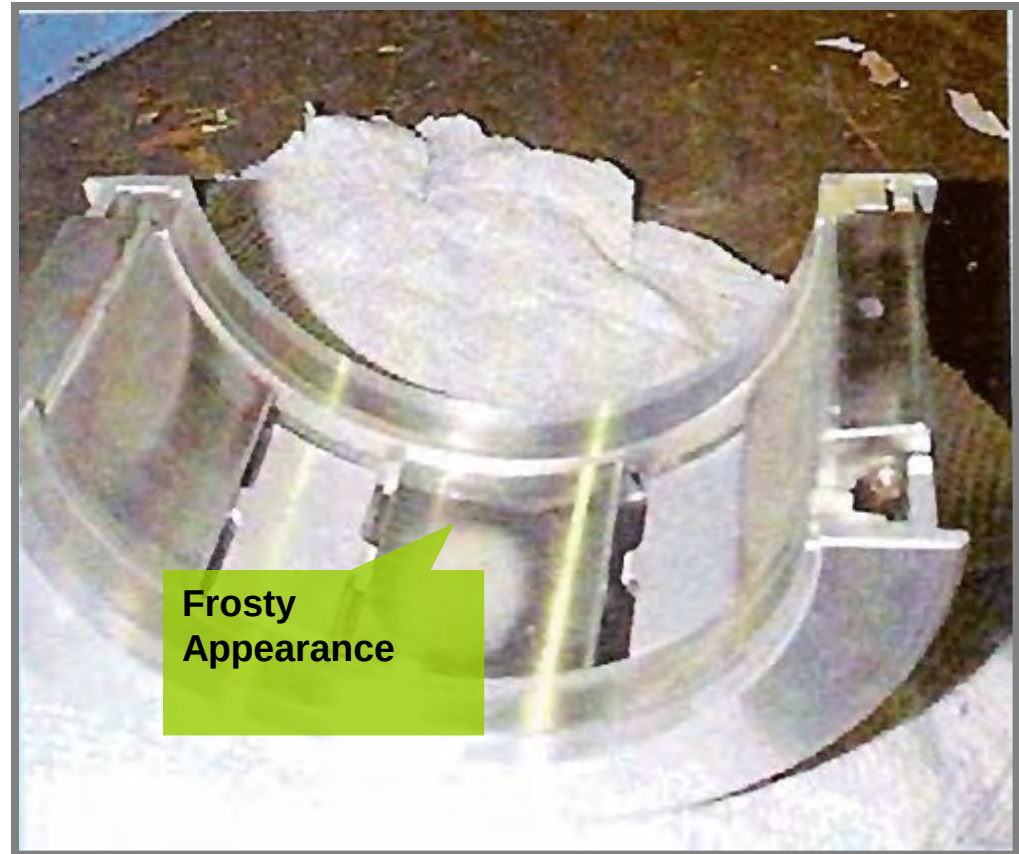
Case 1

- Recycle gas compressor in Diesel Hydro treating plant.
- Multistage steam turbine driven 3.8 Mw GHH make machine.
- Machine pulled out of service due to high bearing temperature on one of the bearing.
- Bearing & journal found severely damaged.
- Bearing replaced & journal coated with HVOF.
- Machine put in service & run on operating speed.
- Bearing temperatures are again high.
- Decided to open machine & check bearings.

Case study: Typical failure for compressor (continued)

Case 1 (continue)

- Electric discharge due to Residual Magnetism.
- Other sources
 - In electrical machinery due to current passing.
 - Static discharge in turbine due to water droplet erosion



Case study: Typical failure for compressor (continue)

Case 1 (continue)

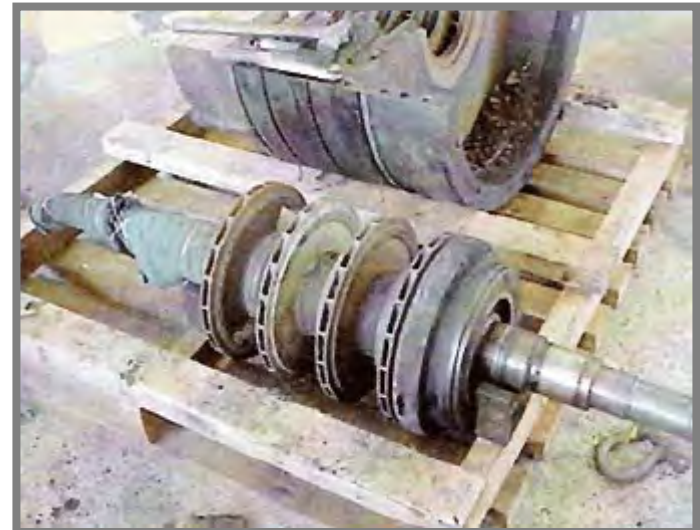
- Conclusion :
 - Bearing damaged by magnetic currents.
 - Repair introduced magnetism in rotor.
- Lessons learnt
 - Include demagnetizing of rotor in repair procedure following any repair on rotor.
 - As a minimum check residual magnetism at repair shop.
 - Please monitor slow-roll vector during start-up after repair.

Case study: Typical failure for compressor (continue)

>>Dynamic / Kinetic Compressor

Case 2

- Refinery Plat former compressor .
- Compressor rotor removed & spare unit installed in July 1999.
- Removed rotor left in workshop prior to shipping to OEM for repair of coupling thread.
- No other rotor damage , except level of fouling



Condition of rotor on withdrawal

Case study: Typical failure for compressor (continue)

>>Dynamic / Kinetic Compressor

Case 2 (continue)

Fouling levels were high , as noted on removal of rotor.

No further action taken.



Impellers Materials :

1st : 13 Cr Stainless steel.

2nd : 2,3,4 Alloy carbon steel

Case study: Typical failure for compressor (continue)

>>Dynamic / Kinetic Compressor

Case 2 (continue)

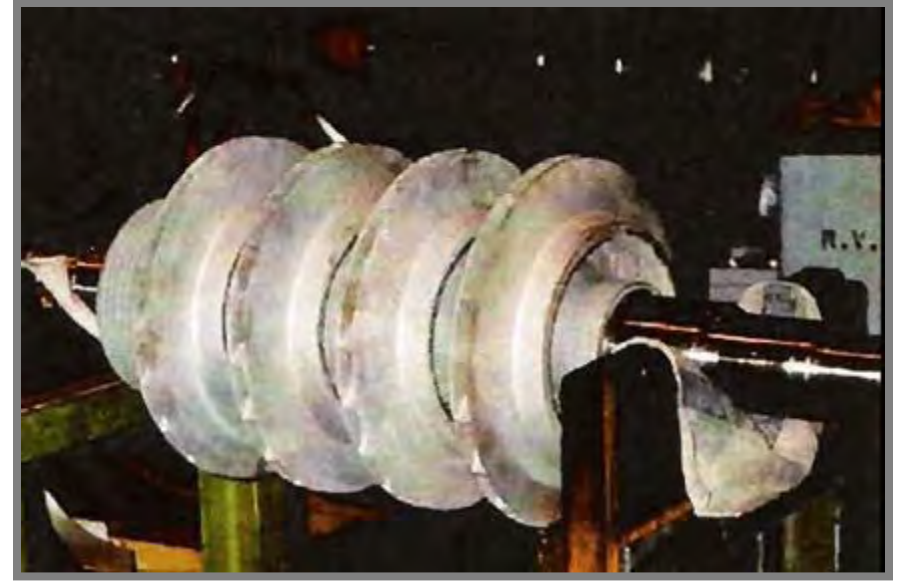


Condition of rotor after few days , preparatory to shipping

Case study: Typical failure for compressor (continue)

>>Dynamic / Kinetic Compressor

Case 2 (continue)



Condition of rotor on receipt by OEM and after grit blast ,
December 1999

Case study: Typical failure for compressor (continued)

Case 2 (continue)



Details of Impeller #2 & 3

Case 2 (continue)

- Conclusion
 - Severe pitting on impeller internal & external surfaces.
 - Weld repair not possible.
 - Risk of fracture judged too dangerous for high speed balancing.
 - Decided rotor to be scrapped.
 - Cost of new rotor : US\$ 400,000.
- Lessons learnt:
 - Irrespective of rotor condition, carryout basic cleaning of rotor & remove deposits.
 - Isolate the rotor from atmospheric moisture
 - Apply rust preventive & cover rotor.
 - Put in rotor container & purge with nitrogen.

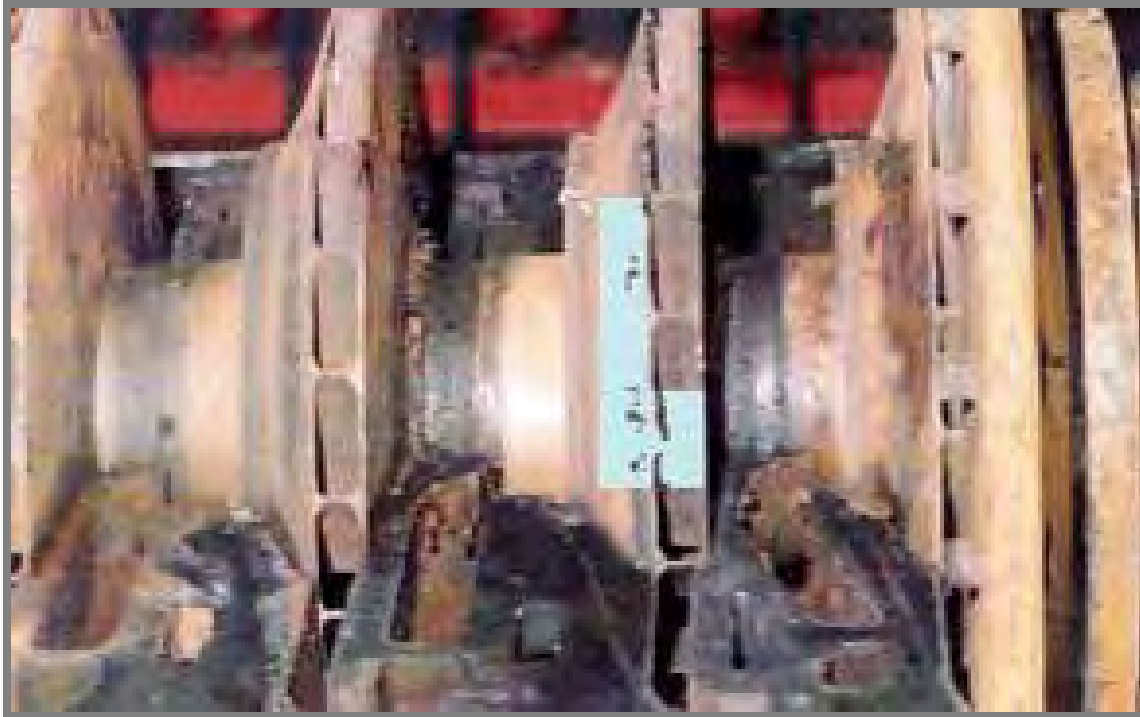
Case 3

- Fuel gas compressor at Hydrogen making unit of refinery tripped on high vibration.
- On opening compressor found heavily fouled as expected.
- Deposits in last stages found to be gummy.
- On analysis deposits found to be NH_4Cl & PCA in last stages.
- Compressor receives feed from various units :
 - Unsaturated gases from Coker, Aromatics & FCC.

Case study: Typical failure for compressor (continue)

>>Dynamic / Kinetic Compressor

Case 3 (continue)



Fouling levels were high , as noted on removal of rotor.

Case study: Typical failure for compressor

>>Positive displacement compressor

Case 1

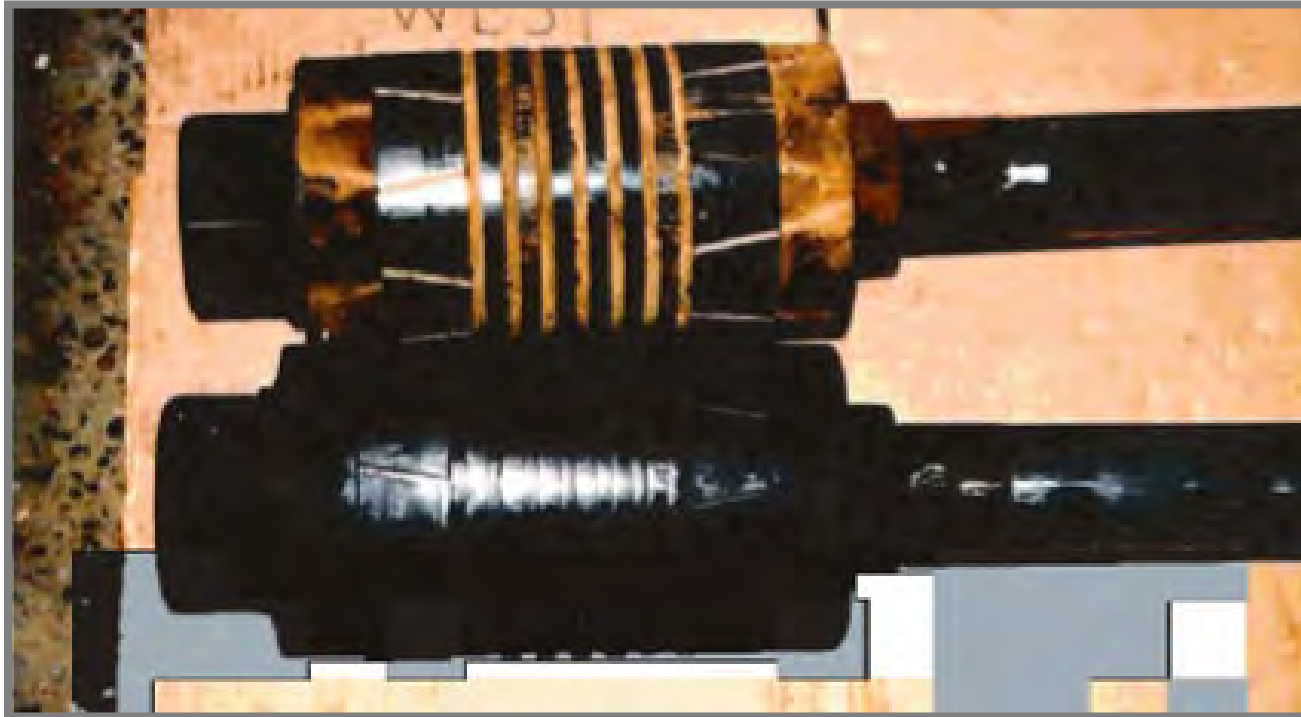
- Failed piston (aluminum)



Large bore aluminum pistons have a very poor fatigue resistance



Case 2



Lower piston fouled because of oil migration past failed compartment seal.

Case study: Typical failure for compressor (continued)

Case 3



Hydrogen Recycle gas compressor – 1st stage piston rod failure

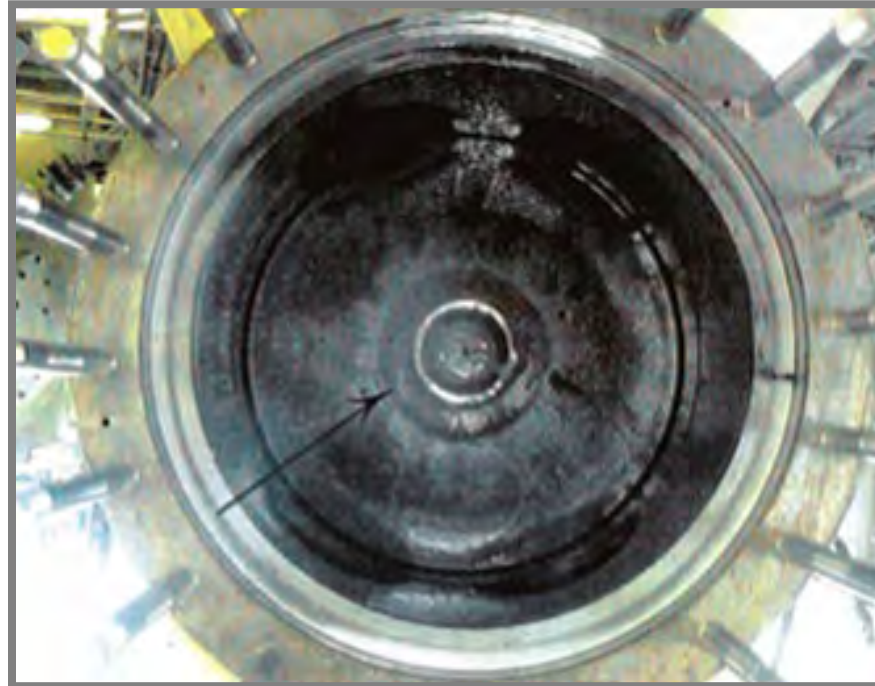
Case study: Typical failure for compressor (continue)

>>Positive displacement compressor

Case 3 (continue)



Evidence of impact



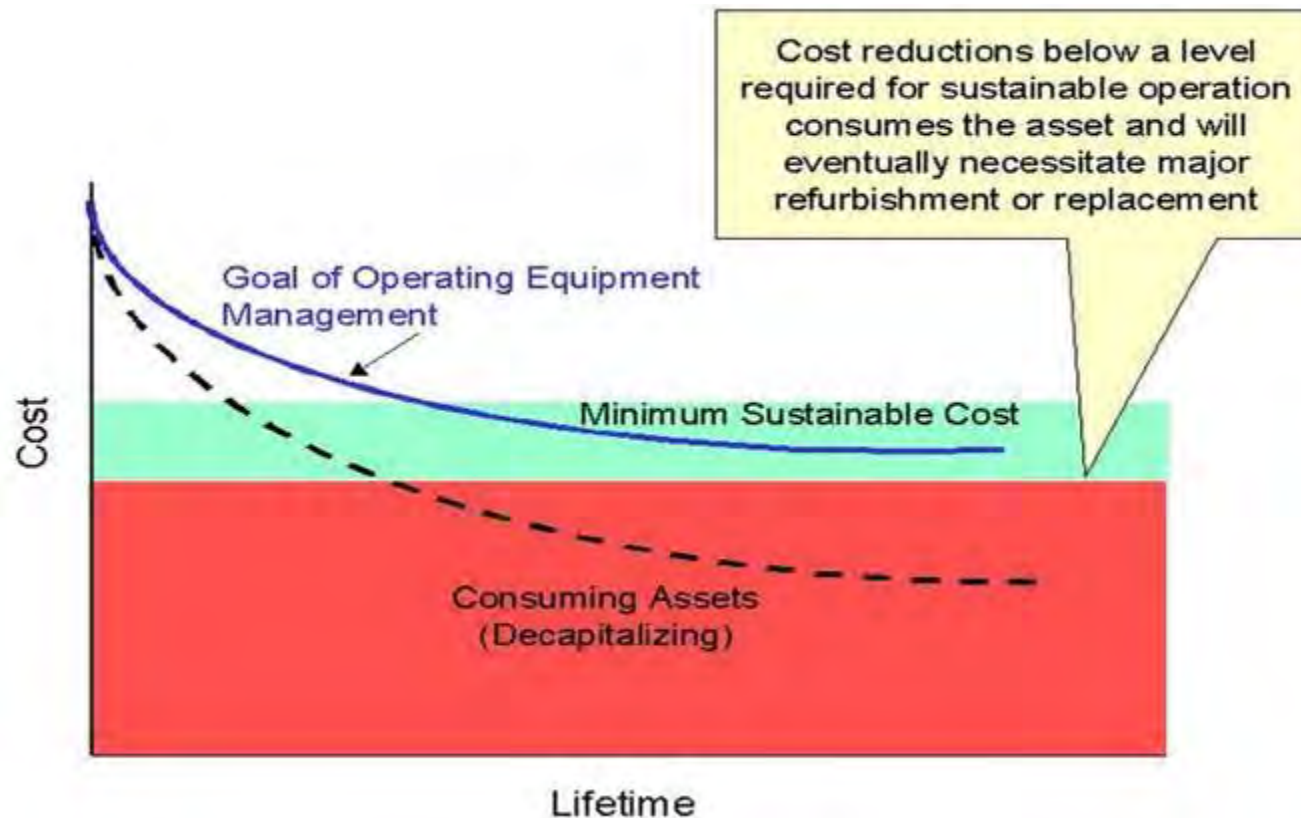
Liquid Impressions

- Conclusion
failure caused by liquid carry over.

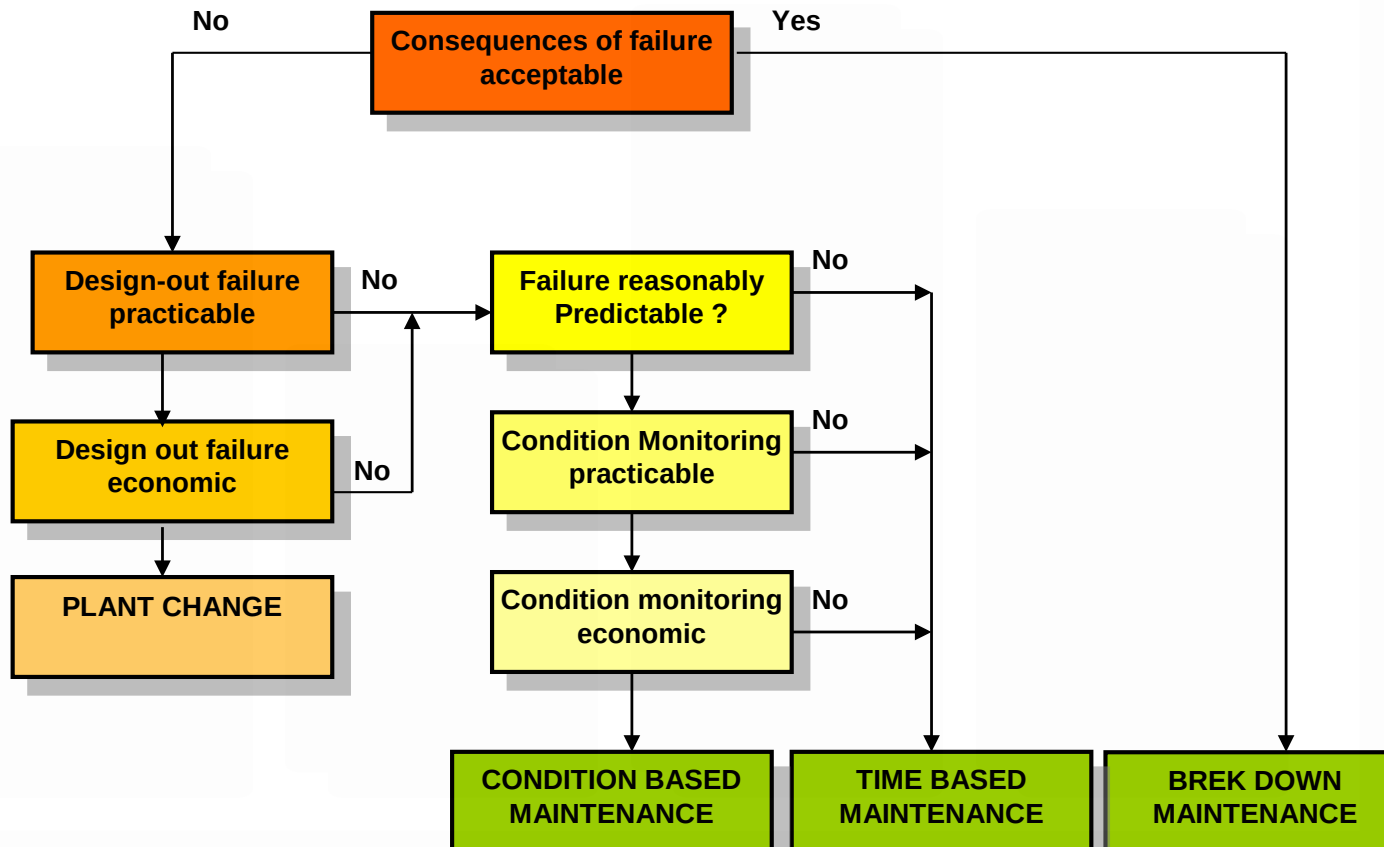
Compressor Maintenance

Compressor Maintenance

“ Maintenance is the single largest controllable expenditure in the plant “
-Dupont Company

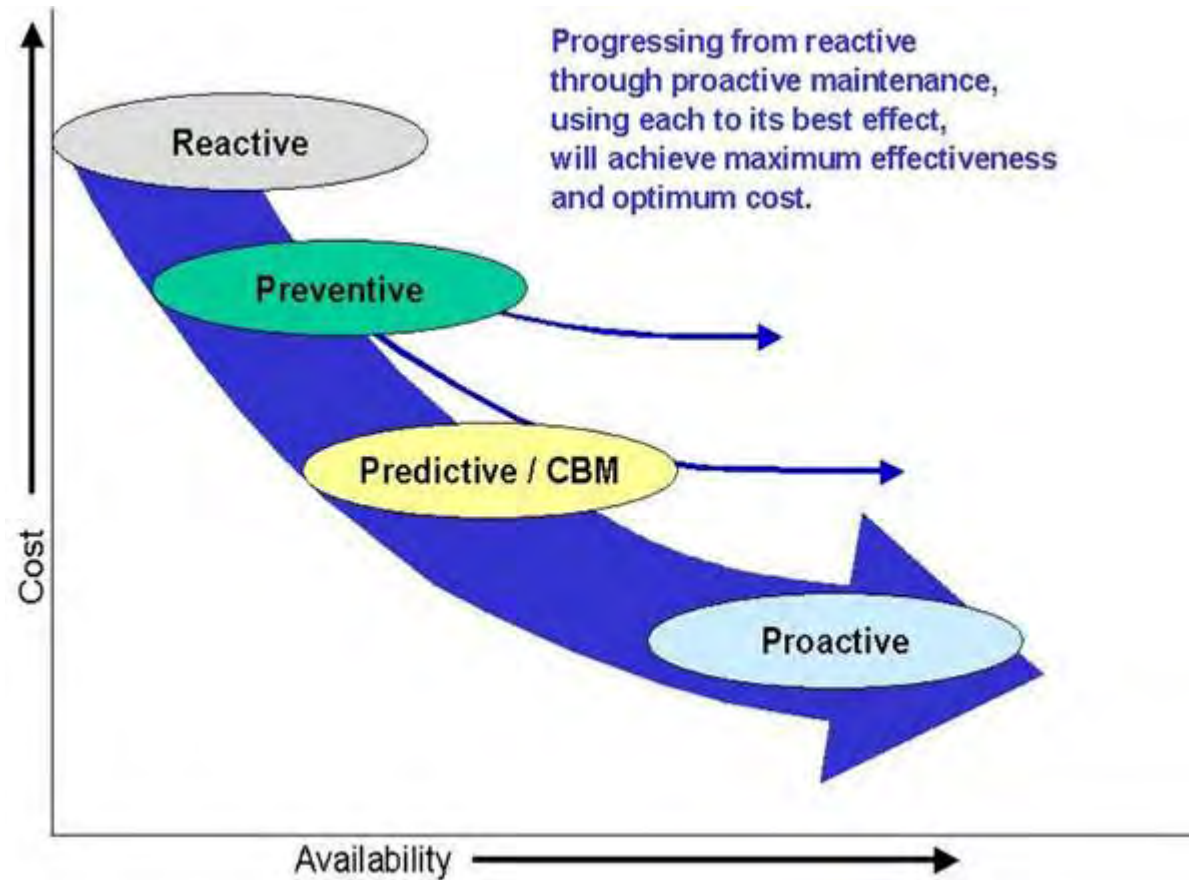


Compressor Maintenance (continue)



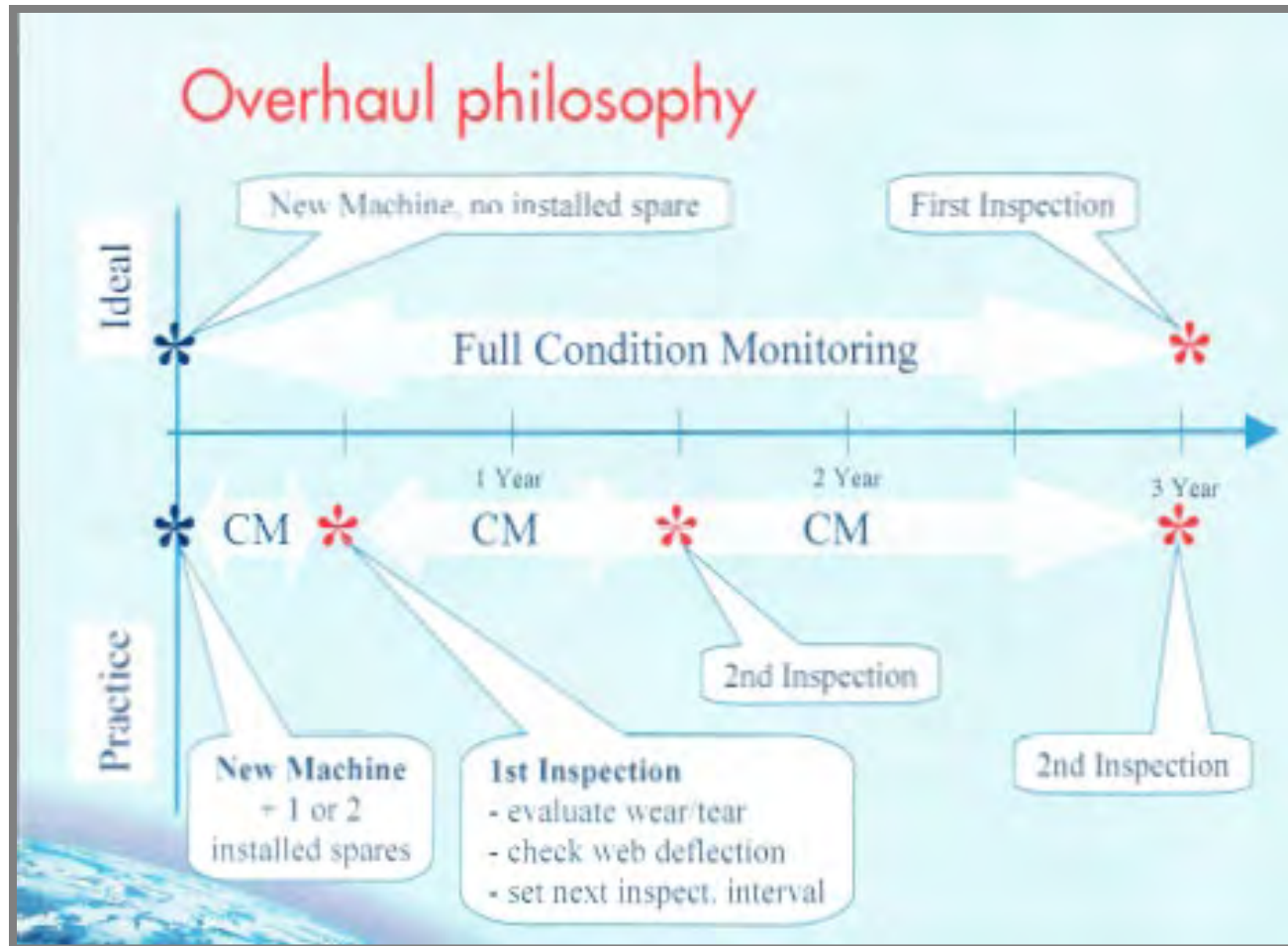
Selection of maintenance Philosophy

Compressor Maintenance (continue)

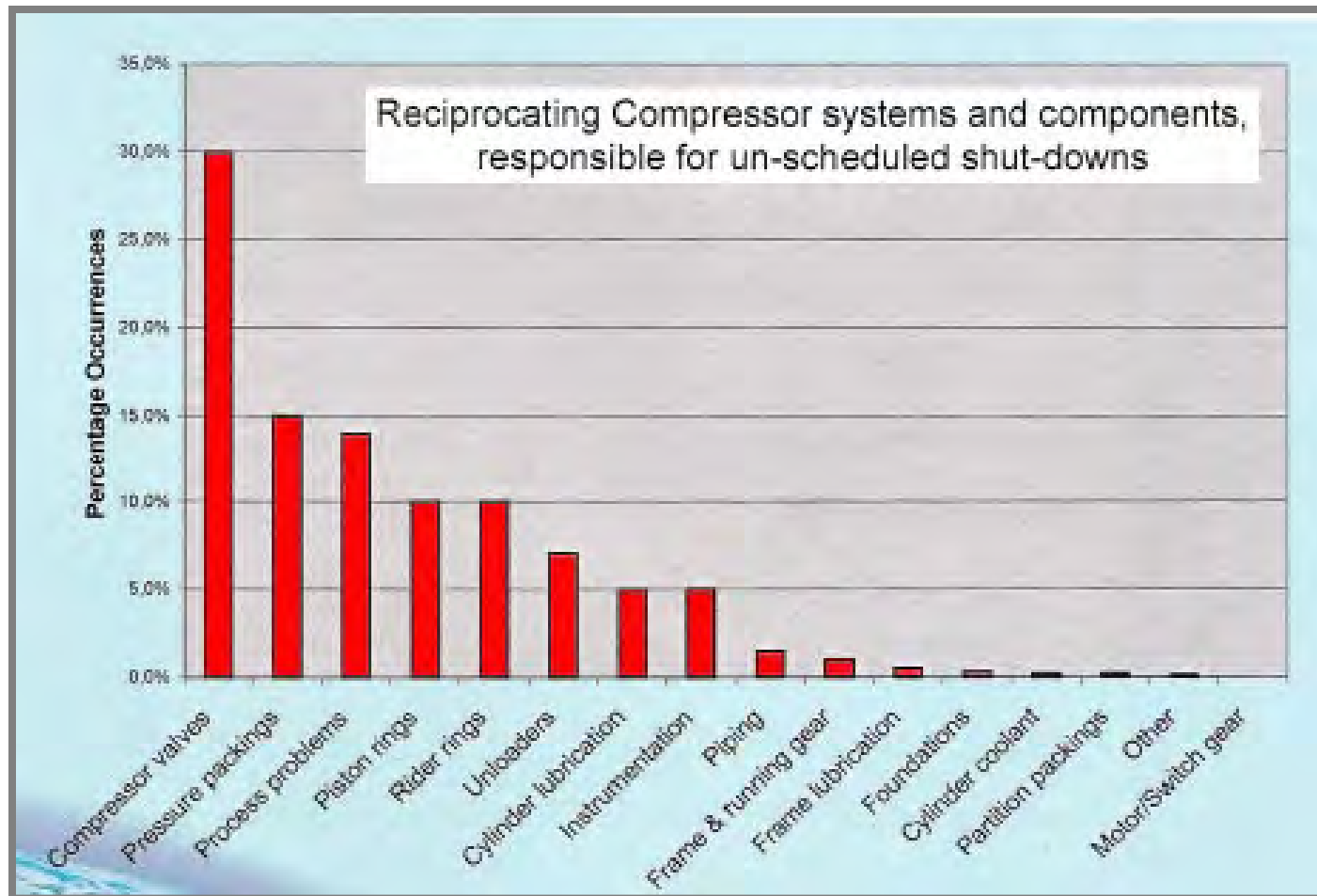


Maintenance Philosophy

Compressor Maintenance (continue)



Compressor Maintenance (continue)



Compressor Maintenance (continue)

Classification of wearing parts

Slow or non Wearing parts	Average Life
• Small / Big end bearings	5-10 Years
• Cross head	10-15 Years
• Cross head pin	5-10 Years
• Main bearings	10-15 Years
• Cylinder cover bolts	10-15 Years
• Piston liners (Ni -Resist)	5 Years
• Piston liner (Grey Iron)	10 Years or more

Compressor Maintenance (continue)

Classification of wearing parts (continue)

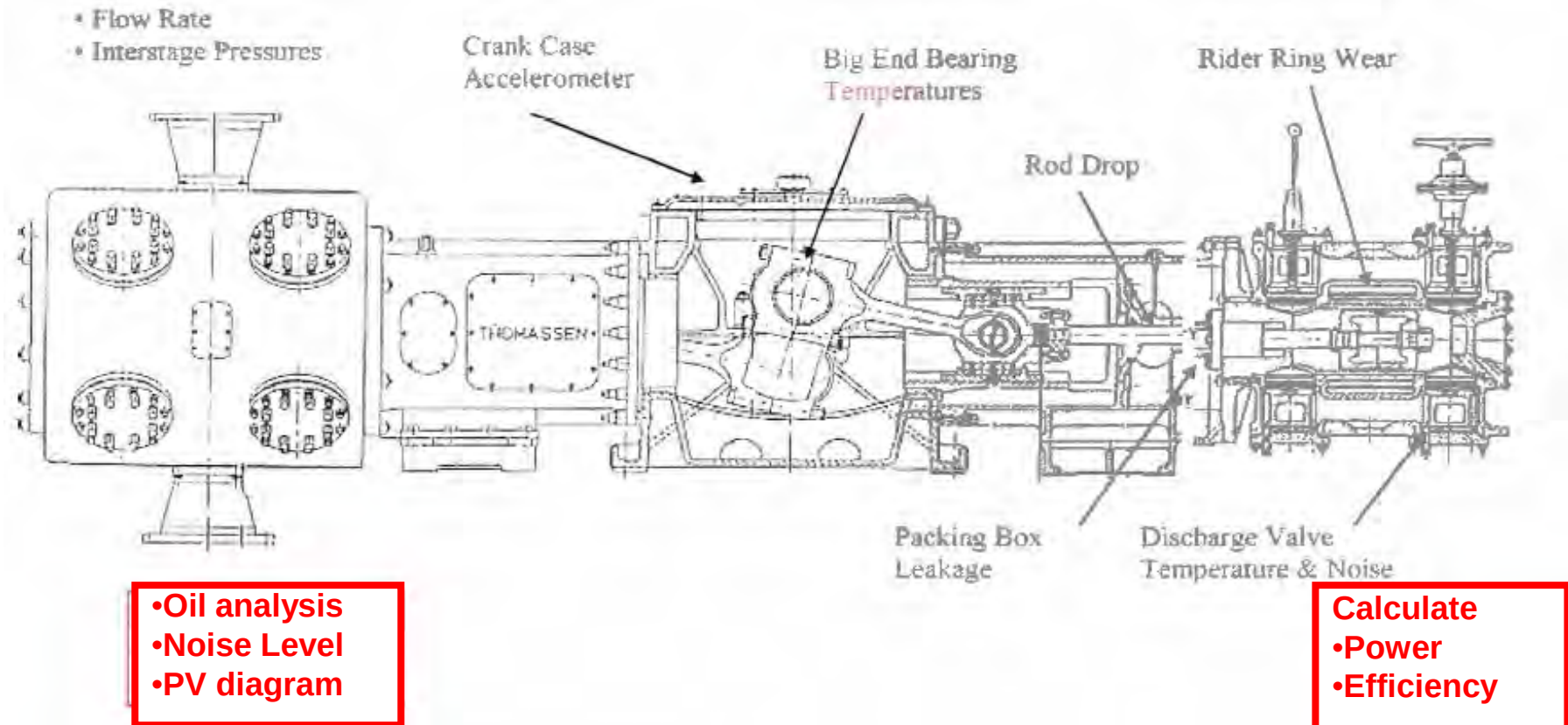
Fast Wearing parts	Average Life
• Rod Packing	< 4 Years
• Rider/ Piston ring	< 4 Years
• Valves	1-3 Years
• Piston rod	2-10 Years
• Wiper ring	< 4 Years
• Compartment seals	< 4 Years
• Packing box	3-5 Years

Breakdown maintenance

Time based maintenance

Condition based maintenance

Condition based maintenance



What to monitor on reciprocating compressor?

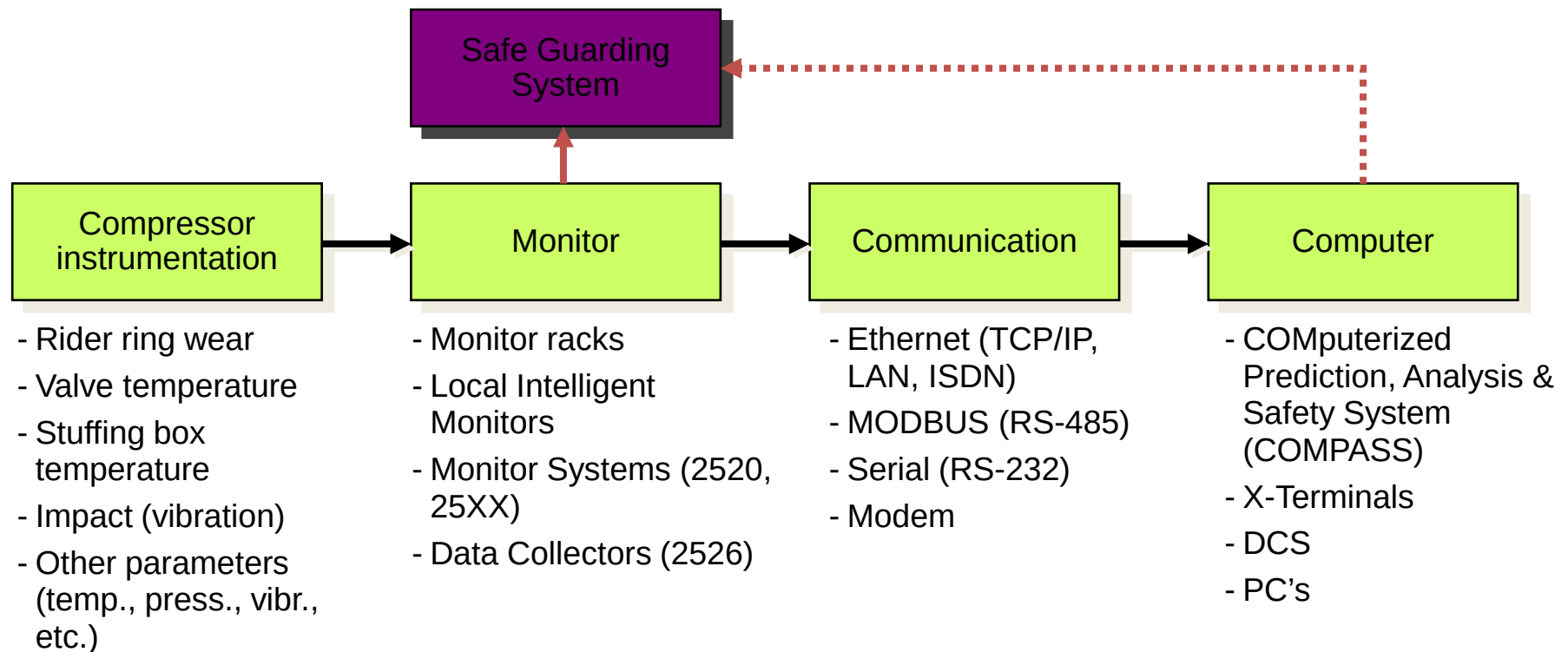
Condition based maintenance

Example: Thomason Compression Systems Monitoring.

Remote Machine Protection, Condition and Performance Monitoring Systems for Reciprocating Compressors

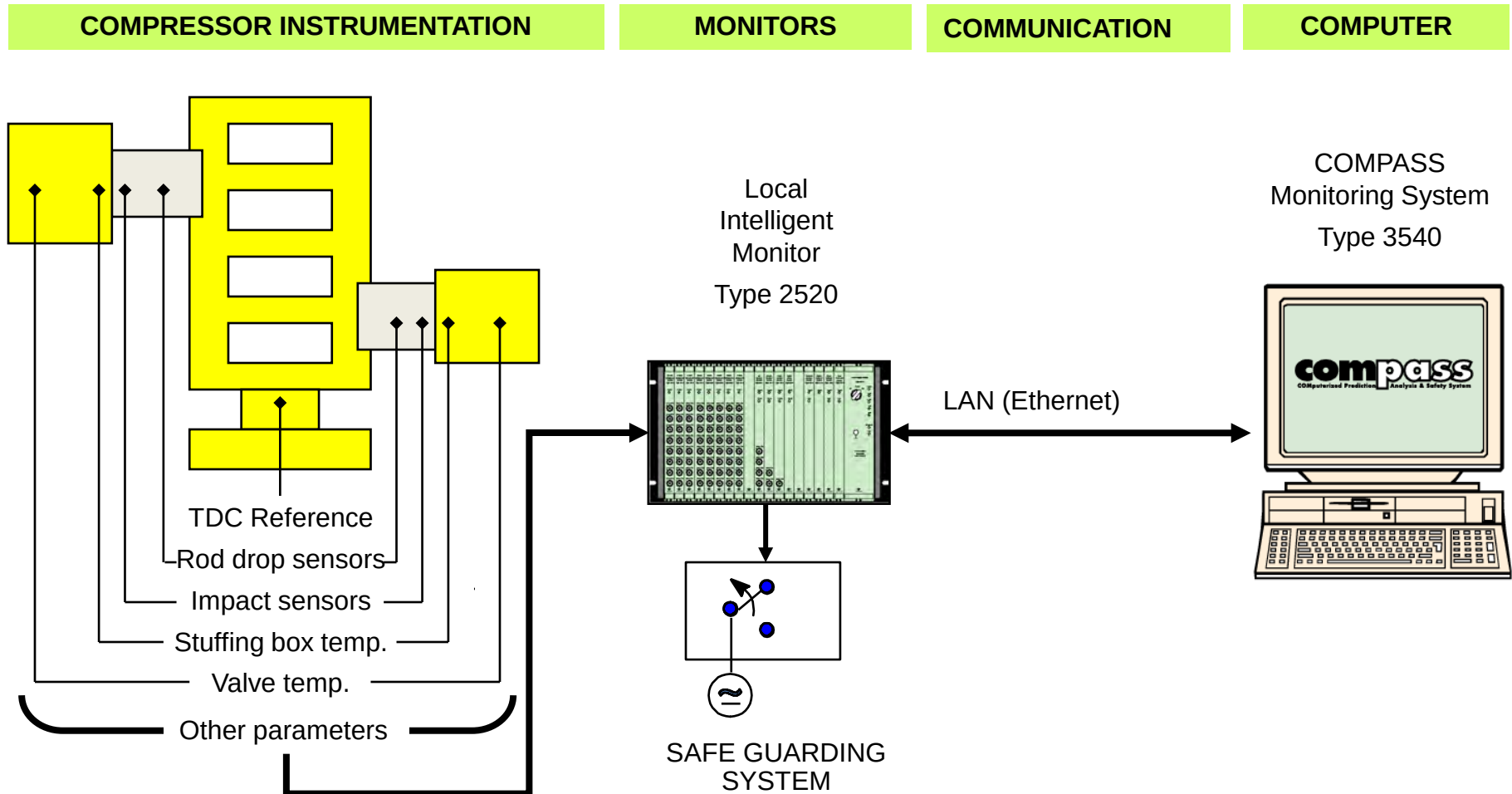


Machine Monitoring general



Condition based maintenance

Machine Monitoring set-up (example)



Machine monitoring

① Protective or Safety Monitoring

Monitoring of the results of actual machine problems.

② Potential Failure Mode Monitoring

a. Predictive or Condition Monitoring

Focuses on causes of potential machine problems.

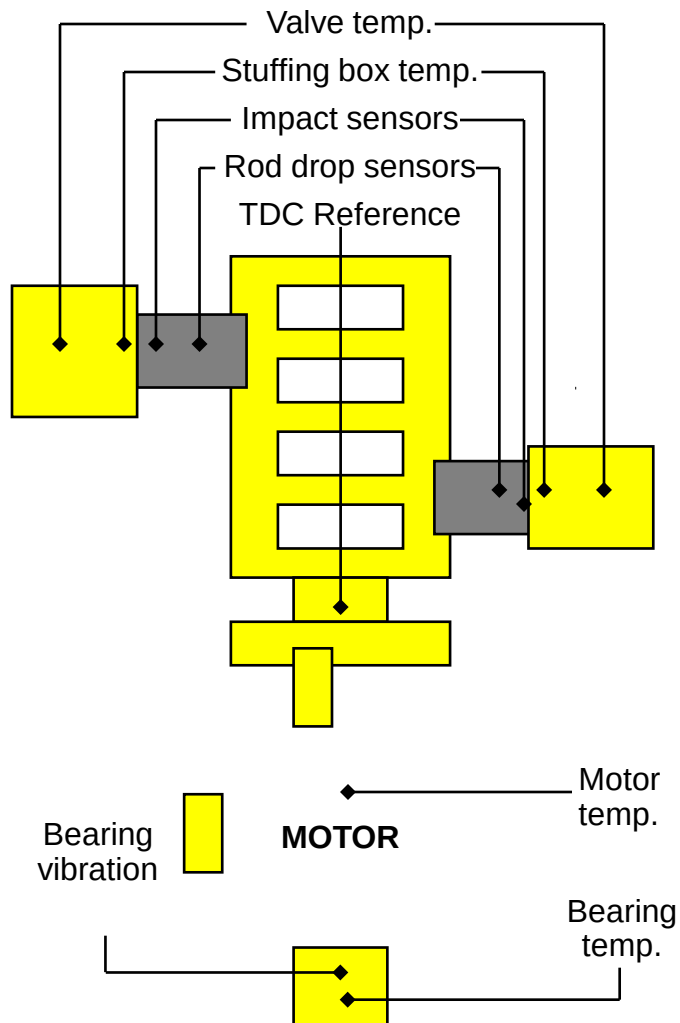
b. Function or Performance Monitoring

Detection of changes in efficiency of the machine.

③ Automatic Machine Diagnoses

Analyzing signals & data to advise on potential machine or performance problems.

Compressor instrumentation



Basic measurements:

- Rider ring wear (rod drop)
- Impact (vibration)
- Valve temperature
- Stuffing box temperature

Additional measurements:

- Bearing (oil) temperature
- Vibration of prime mover
- Temperature of prime mover
- Cooling water temperature
- Cylinder pressures
- Vibration of piping
- Other

Determination of rider ring wear (rod drop)

Rider ring wear is generally determined by means of rod drop measurement, where the gap between the piston rod and an eddy-current probe is measured. The following methods for rod drop measurement are used:

Averaged DC Voltage measurement:

- average level of the gap between probe and rod
- comparing the actual absolute value with the 'new state'
- simple (one probe required, direct indication)
- used in the past and in simple monitoring systems

One point cyclic DC Voltage measurement:

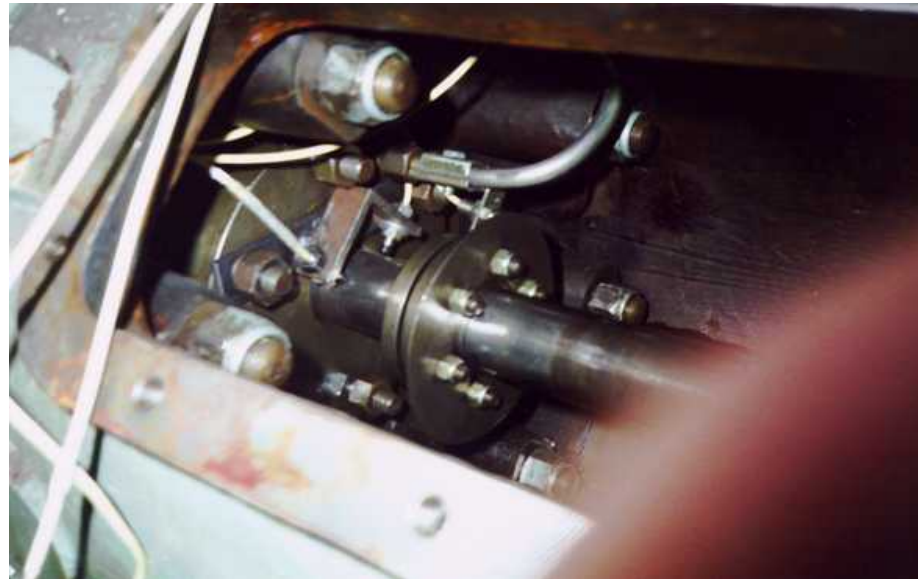
- level of the gap at **one angular** location of the shaft
- comparing the actual absolute value with the 'new state'
- min. one rod drop and a TDC ref. probe required
- used nowadays in many monitoring systems

Two point cyclic DC Voltage measurement:

- level of the gap at **two angular** locations of the shaft
- comparing the actual **relative** value with the 'new state'
- min. one rod drop and a TDC ref. probe required
- used by **TCS and B&K** in their monitoring systems

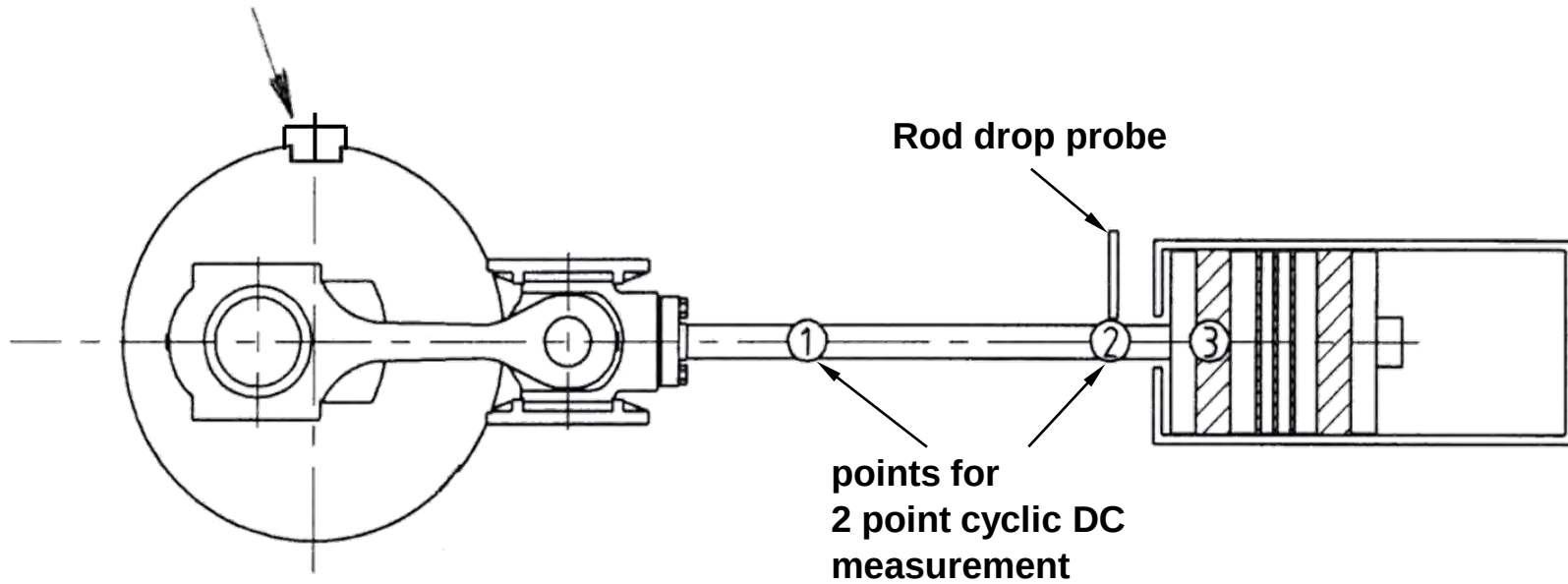
Condition based maintenance

Test compressor



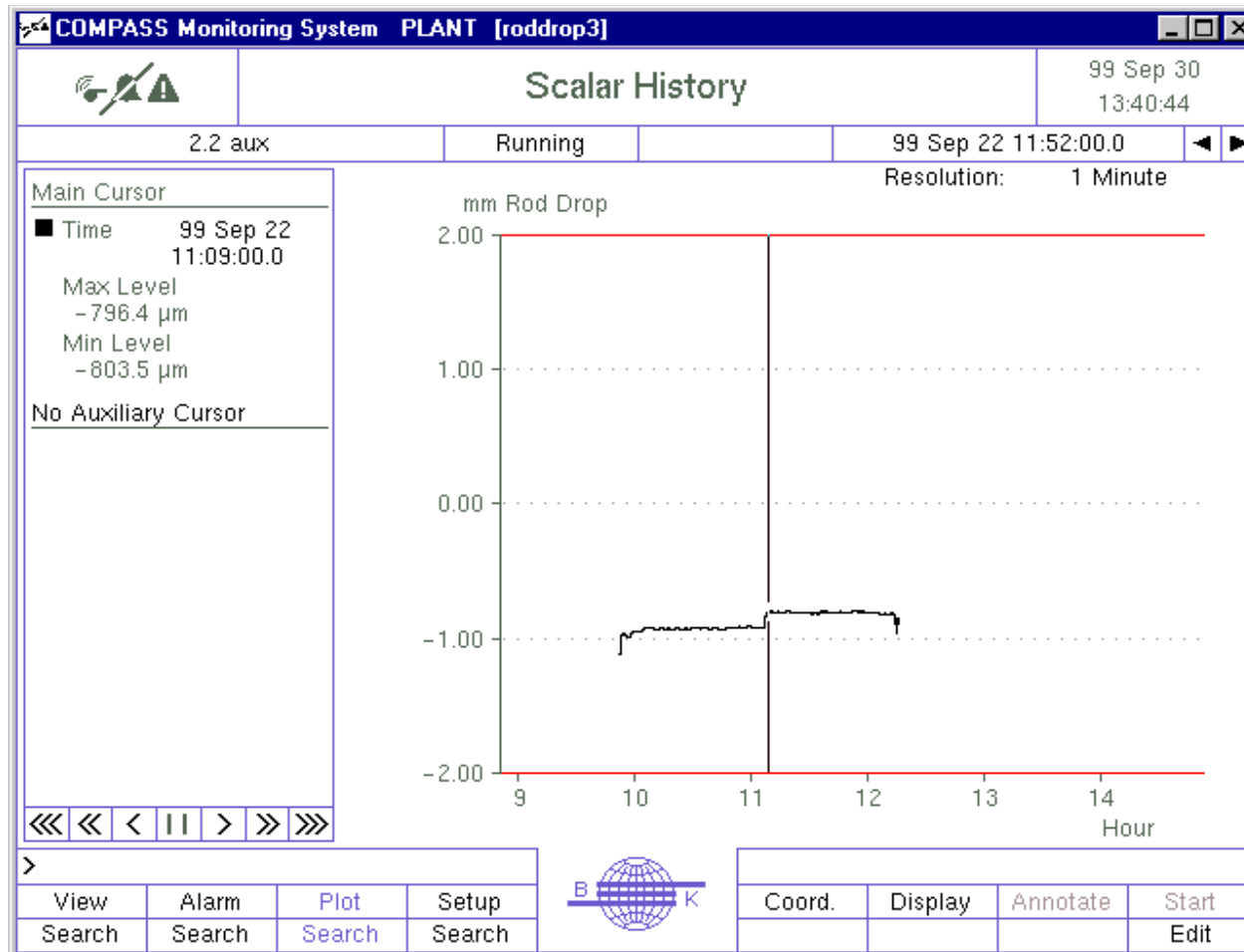
Rod drop, 2 point cyclic DC measurement

TDC probe reference signal
(one pulse per revolution)



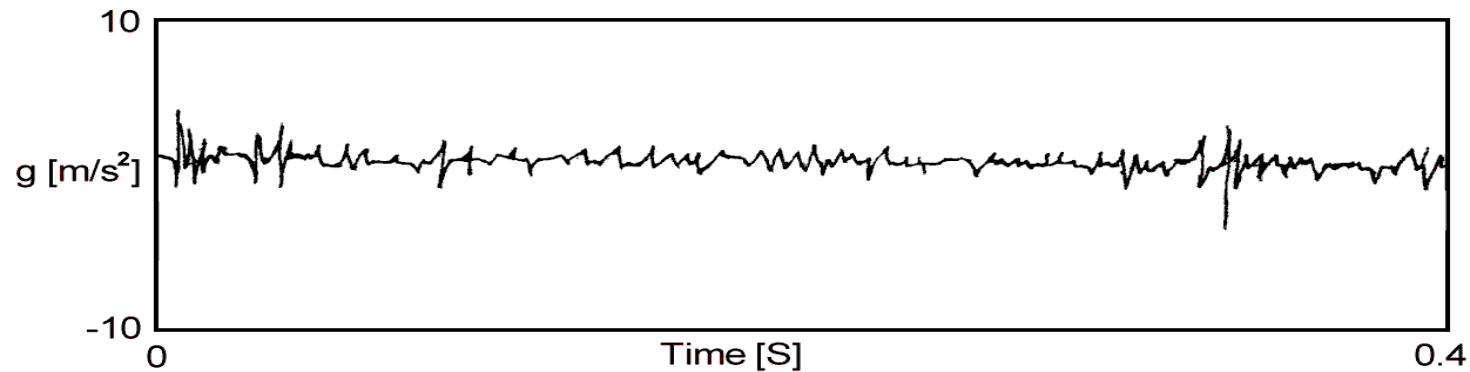
Condition based maintenance

Rider ring wear indication by COMPASS™

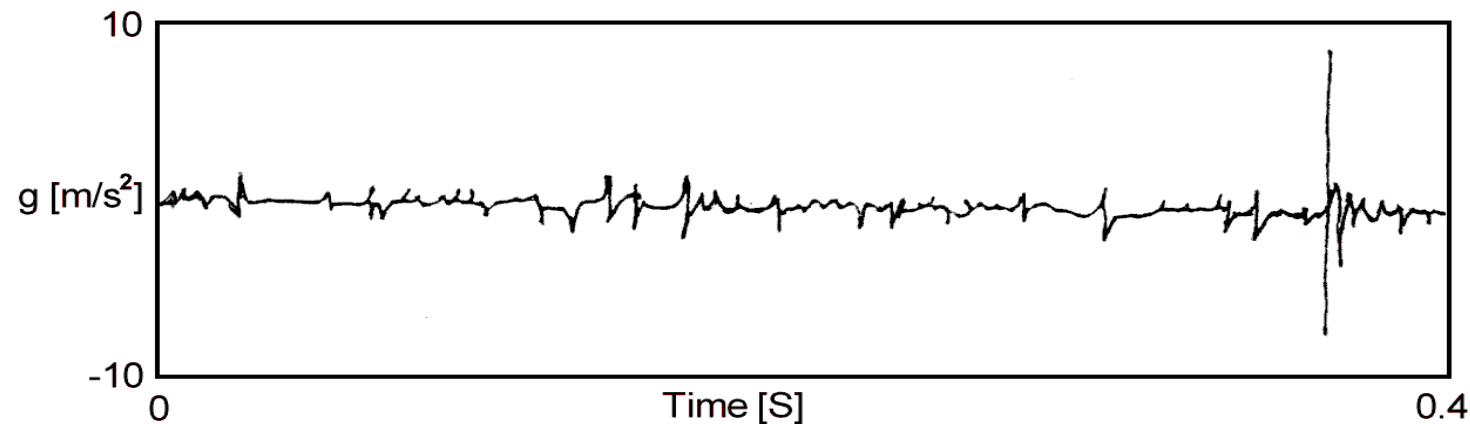


Condition based maintenance

Typical vibration output on a scope



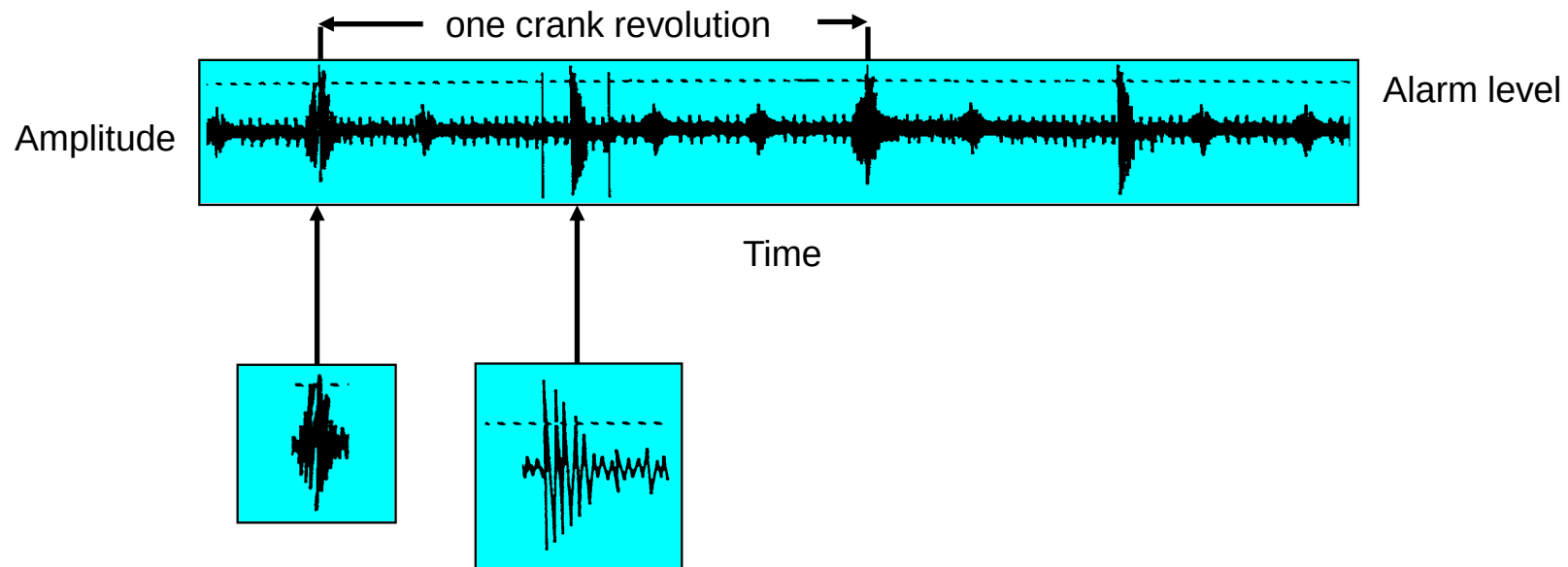
Acceleration time wave, unloaded



Acceleration time wave, loaded

Condition based maintenance

Typical impact output on a scope

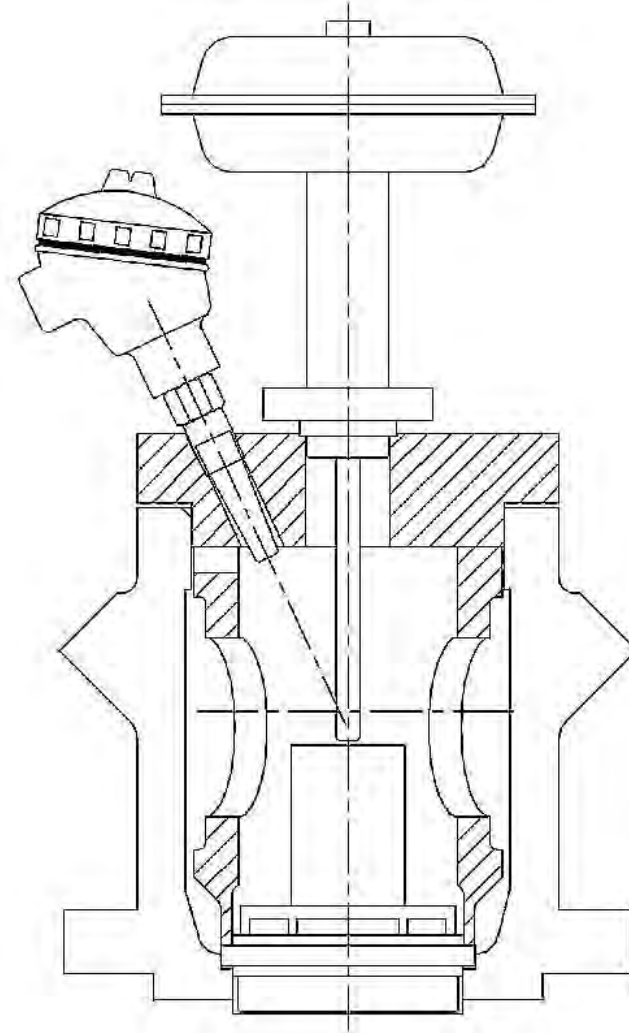


Condition based maintenance

Valve temperature measurement

One RTD or thermocouple per valve

Temperature measured absolute or relative

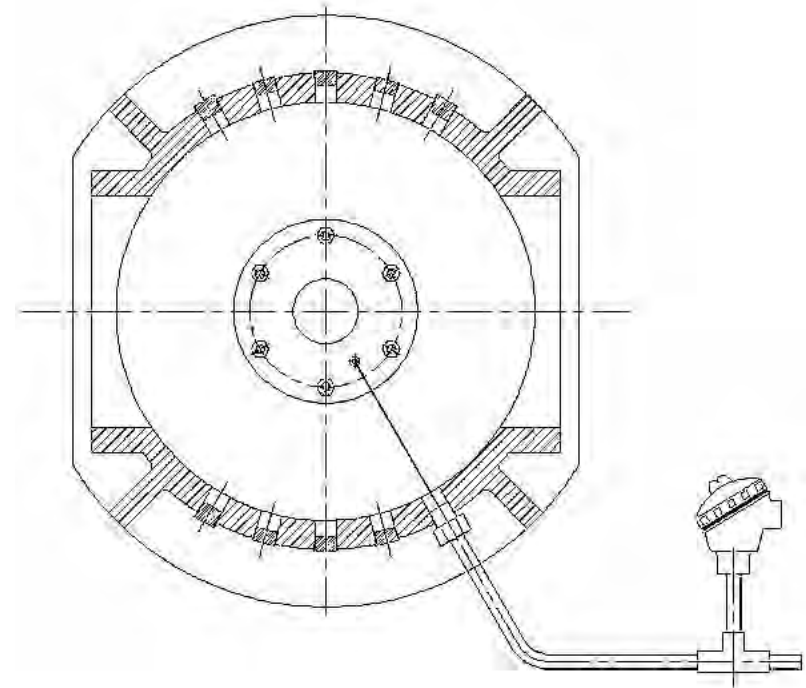


Condition based maintenance

Vent gas temperature measurement

One RTD or thermocouple per vent connection

Temperature measured absolute or relative



End of presentation

THANK YOU